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(54) **ALPHA-AMYLASE VARIANTS AND
POLYNUCLEOTIDES ENCODING SAME**

(71) Applicant: **Novozymes A/S**, Bagsvaerd (DK)

(72) Inventors: **Carsten Andersen**, Vaerloese (DK);
Chakshumathi Ghadiyaram,
Bangalore (IN); **Rajendra**
Kulothungan Sainathan, Bangalore
(IN); **Padmavathi Balumuri**, Chennai
(IN); **Padma Venkatachalam Iyer**,
Mumbai (IN); **Iben Damager**, Valby
(DK); **Astrid Munch**, Frederiksberg
(DK); **Sohel Dalal**, Ahmedabad (IN)

(73) Assignee: **Novozymes A/S**, Bagsvaerd (DK)

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None

See application file for complete search history.

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Primary Examiner — Manjunath N Rao

Assistant Examiner — Jae W Lee

(74) *Attorney, Agent, or Firm* — Yoshimi D. Barron

(57) **ABSTRACT**

The present invention relates to alpha-amylase variants. The present invention also relates to polynucleotides encoding the variants; nucleic acid constructs, vectors, and host cells comprising the polynucleotides; and methods of using the variants.

27 Claims, No Drawings

Specification includes a Sequence Listing.

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ALPHA-AMYLASE VARIANTS AND POLYNUCLEOTIDES ENCODING SAME

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a divisional of U.S. application Ser. No. 17/702,431 filed on Mar. 23, 2022 which is a divisional of U.S. application Ser. No. 16/847,773 filed on Apr. 14, 2020 now U.S. Pat. No. 11,319,509 which is a divisional of U.S. application Ser. No. 15/571,219 filed May 9, 2016, now U.S. Pat. No. 10,647,946 which is a 35 U.S.C. 371 national application of PCT/EP2016/060266 filed May 9, 2016, which claims priority or the benefit under 35 U.S.C. 119 of Indian application no. 2335/CHE/2015 filed May 8, 2015, the contents of which are fully incorporated herein by reference.

REFERENCE TO A SEQUENCE LISTING

This application contains a Sequence Listing in computer readable form, which is incorporated herein by reference. This application contains a Sequence Listing in computer readable form. The contents of the electronic sequence listing created on Mar. 5, 2024, named SQ.xml and 26,105 bytes in size, is hereby incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

Field of the Invention

The present invention relates to alpha-amylase variants, polynucleotides encoding the variants, methods of producing the variants, and methods of using the variants.

Description of the Related Art

Alpha-amylases have for many years been used in laundry where it is well-known that alpha-amylases have a beneficial effect in removal of starch containing, or starch-based, stains.

WO95/26397 discloses alkaline *Bacillus* amylases having good wash performance measured at temperatures in the range of 30-60° C.

WO00/60060 and WO00/60058 discloses further bacterial alpha-amylases having good wash performance.

In recent years there has been a desire to reduce the temperature of the laundry in order to reduce the energy consumption. Lowering the temperature in laundry often means that the performance of the detergent composition and the enzyme is reduced and a lower wash performance is therefore obtained at low temperature. It is therefore desired to find new alpha-amylases having good wash performance at low temperature. Accordingly, it is an object of the present invention to provide alpha-amylases which have good wash performance at low temperature, such as at 15° C. Thus, the present invention provides such further improved alpha-amylase variants with improved properties compared to its parent.

SUMMARY OF THE INVENTION

The present invention relates to an alpha-amylase variant of a parent polypeptide having alpha-amylase activity, wherein the variant is a fusion polypeptide which has an improved wash performance at low temperature, and wherein said variant has alpha-amylase activity.

The present invention also relates to polynucleotides encoding the variants; nucleic acid constructs, vectors, and host cells comprising the polynucleotides; and methods of producing the variants.

The present invention also relates to methods of improving wash performance of a parent alpha-amylase. Conventions for Designation of Variants

For purposes of the present invention, the mature polypeptide disclosed in SEQ ID NO: 7 is used to determine the corresponding amino acid residue in another alpha-amylase. Alternatively, the mature polypeptide disclosed in SEQ ID NO: 13 and 14 may be used to determine the corresponding amino acid residue in another alpha-amylase. The amino acid sequence of another alpha-amylase is aligned with the mature polypeptide disclosed in SEQ ID NO: 7, and based on the alignment, the amino acid position number corresponding to any amino acid residue in the mature polypeptide disclosed in any of SEQ ID NOs: 13 and 14 is determined using the Needleman-Wunsch algorithm (Needleman and Wunsch, 1970, *J. Mol. Biol.* 48: 443-453) as implemented in the Needle program of the EMBOSS package (EMBOSS: The European Molecular Biology Open Software Suite, Rice et al., 2000, *Trends Genet.* 16: 276-277), preferably version 5.0.0 or later. The parameters used are gap open penalty of 10, gap extension penalty of 0.5, and the EBLOSUM62 (EMBOSS version of BLOSUM62) substitution matrix.

Identification of the corresponding amino acid residue in another alpha-amylase can be determined by an alignment of multiple polypeptide sequences using several computer programs including, but not limited to, MUSCLE (multiple sequence comparison by log-expectation; version 3.5 or later; Edgar, 2004, *Nucleic Acids Research* 32: 1792-1797), MAFFT (version 6.857 or later; Katoh and Kuma, 2002, *Nucleic Acids Research* 30: 3059-3066; Katoh et al., 2005, *Nucleic Acids Research* 33: 511-518; Katoh and Toh, 2007, *Bioinformatics* 23: 372-374; Katoh et al., 2009, *Methods in Molecular Biology* 537: 39-64; Katoh and Toh, 2010, *Bioinformatics* 26: 1899-1900), and EMBOSS EMMA employing ClustalW (1.83 or later; Thompson et al., 1994, *Nucleic Acids Research* 22: 4673-4680), using their respective default parameters.

When the other enzyme has diverged from the mature polypeptide of SEQ ID NO: 7 such that traditional sequence-based comparison fails to detect their relationship (Lindahl and Elofsson, 2000, *J. Mol. Biol.* 295: 613-615), other pairwise sequence comparison algorithms can be used. Greater sensitivity in sequence-based searching can be attained using search programs that utilize probabilistic representations of polypeptide families (profiles) to search databases. For example, the PSI-BLAST program generates profiles through an iterative database search process and is capable of detecting remote homologs (Atschul et al., 1997, *Nucleic Acids Res.* 25: 3389-3402). Even greater sensitivity can be achieved if the family or superfamily for the polypeptide has one or more representatives in the protein structure databases. Programs such as GenTHREADER (Jones, 1999, *J. Mol. Biol.* 287: 797-815; McGuffin and Jones, 2003, *Bioinformatics* 19: 874-881) utilize information from a variety of sources (PSI-BLAST, secondary structure prediction, structural alignment profiles, and solvation potentials) as input to a neural network that predicts the structural fold for a query sequence. Similarly, the method of Gough et al., 2000, *J. Mol. Biol.* 313: 903-919, can be used to align a sequence of unknown structure with the superfamily models present in the SCOP database. These alignments can in turn be used to generate homology models

for the polypeptide, and such models can be assessed for accuracy using a variety of tools developed for that purpose.

For proteins of known structure, several tools and resources are available for retrieving and generating structural alignments. For example the SCOP superfamilies of proteins have been structurally aligned, and those alignments are accessible and downloadable. Two or more protein structures can be aligned using a variety of algorithms such as the distance alignment matrix (Holm and Sander, 1998, *Proteins* 33: 88-96) or combinatorial extension (Shindyalov and Bourne, 1998, *Protein Engineering* 11: 739-747), and implementation of these algorithms can additionally be utilized to query structure databases with a structure of interest in order to discover possible structural homologs (e.g., Holm and Park, 2000, *Bioinformatics* 16: 566-567).

In describing the variants of the present invention, the nomenclature described below is adapted for ease of reference. The accepted IUPAC single letter or three letter amino acid abbreviation is employed.

Substitutions: For an amino acid substitution, the following nomenclature is used: Original amino acid, position, substituted amino acid. Accordingly, the substitution of threonine at position 226 with alanine is designated as "Thr226Ala" or "T226A". Multiple mutations are separated by addition marks ("+"), e.g., "Gly205Arg+Ser411Phe" or "G205R+S411F", representing substitutions at positions 205 and 411 of glycine (G) with arginine (R) and serine (S) with phenylalanine (F), respectively.

Deletions: For an amino acid deletion, the following nomenclature is used: Original amino acid, position, *. Accordingly, the deletion of glycine at position 195 is designated as "Gly195*" or "G195*". Multiple deletions are separated by addition marks ("+"), e.g., "Gly195*+Ser411*" or "G195*+S411*".

Multiple modifications: Variants comprising multiple modifications are separated by addition marks ("+"), e.g., "Arg170Tyr+Gly195Glu" or "R170Y+G195E" representing a substitution of arginine and glycine at positions 170 and 195 with tyrosine and glutamic acid, respectively.

Different modifications: Where different modifications can be introduced at a position, the different alterations are separated by a comma, e.g., "Arg170Tyr,Glu" represents a substitution of arginine at position 170 with tyrosine or glutamic acid. Thus, "Tyr167Gly,Ala+Arg170Gly,Ala" designates the following variants:

"Tyr167Gly+Arg170Gly", "Tyr167Gly+Arg170Ala", "Tyr167Ala+Arg170Gly", and "Tyr167Ala+Arg170Ala".

DETAILED DESCRIPTION OF THE INVENTION

The present invention relates to a set of sequences. For ease, these are listed below;

SEQ ID NO: 1 is the nucleotide sequence of an alpha-amylase (AAI10)

SEQ ID NO: 2 is the nucleotide sequence of an alpha-amylase originating from *Alicyclobacillus* sp.

SEQ ID NO: 3 is the nucleotide sequence of an alpha-amylase originating from *Bacillus amyloliquefaciens*.

SEQ ID NO: 4 is the amino acid sequence of AAI10 including the signal peptide.

SEQ ID NO: 5 is the amino acid sequence of *Alicyclobacillus* sp. including the signal peptide.

SEQ ID NO: 6 is the amino acid sequence of *Bacillus amyloliquefaciens* including the signal peptide.

SEQ ID NO: 7 is the amino acid sequence of the mature AAI10

SEQ ID NO: 8 is the amino acid sequence of the mature *Alicyclobacillus* sp.

SEQ ID NO: 9 is the amino acid sequence of the mature *Bacillus amyloliquefaciens*

SEQ ID NO: 10 is the amino acid sequence corresponding to the A and B domains of AAI10.

SEQ ID NO: 11 is the amino acid sequence corresponding to the C domain of *Alicyclobacillus* sp.

SEQ ID NO: 12 is the amino acid sequence corresponding to the C domain of *Bacillus amyloliquefaciens*

SEQ ID NO: 13 is the amino acid sequence of the fusion polypeptide consisting of the A and B domain of AAI10 and the C domain of *Alicyclobacillus* sp.

SEQ ID NO: 14 is the amino acid sequence of the fusion polypeptide consisting of the A and the B domains of AAI10 and the C domain of *Bacillus amyloliquefaciens*

SEQ ID NO: 15 is the amino acid sequence of the mature AAI10 alpha-amylase comprising a double deletion at positions corresponding to positions 182 and 183 of SEQ ID NO: 7

Variants of the Invention

In one aspect, the present invention relates to variants of a parent polypeptide having alpha-amylase activity. Thus, in particular aspect, the present invention relates to variant of a parent polypeptide having alpha-amylase activity, wherein said variant is a fusion polypeptide which has an improved wash performance at low temperature, and wherein said variant has alpha-amylase activity.

It has been shown that variants according to the present invention have improved performance, such as wash performance, at a low temperature, in particular at 15° C., in liquid detergent compositions.

The term "variant" as used herein, refers to a polypeptide having alpha-amylase activity comprising an alteration, i.e., a substitution, insertion, and/or deletion, at one or more (e.g., several) positions. A substitution means replacement of the amino acid occupying a position with a different amino acid; a deletion means removal of the amino acid occupying a position; and an insertion means adding an amino acid adjacent to and immediately following the amino acid occupying a position, all as defined above. The variants of the present invention have at least 20%, e.g., at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, or at least 100% of the alpha-amylase activity of the polypeptide of SEQ ID NOS: 7, 13, 14, 15, or the mature polypeptide of SEQ ID NO: 4. The variants of the present invention have at least 65%, such as at least 70%, such as at least 75%, such as at least 80%, such as at least 85%, such as at least 90%, such as at least 95%, such as at least 97%, such as at least 99%, but less than 100% sequence identity to any one of SEQ ID NOS: 4, 7, 13, 14, or 15

The term "parent polypeptide" as used herein, refers to a polypeptide to which an alteration is made to produce enzyme variants. The polypeptide having any of SEQ ID NOS: 4, 7, 13, 14, and 15 may e.g. be a parent for the variants of the present invention. Any polypeptide having at least 65%, such as at least 70%, such as at least 75%, such as at least 80%, such as at least 85%, such as at least 90%, such as at least 95%, such as at least 97%, such as at least 99%, sequence identity to any one of SEQ ID NOS: 4, 7, 13, 14, or 15 may also be a parent polypeptide for the variants of the present invention.

The parent polypeptide may be a fusion polypeptide or cleavable fusion polypeptide. Such fusion polypeptide may consist of a subsequence of one parent polypeptide and a

subsequence of second parent polypeptide. In particular, a fusion polypeptide may consist of an A and B domain of one species of alpha-amylase and a C domain of another species of alpha-amylase, and thereby providing a parent polypeptide to generate a variant of the present invention. Techniques for producing fusion polypeptides are known in the art, and include ligating the coding sequences encoding the polypeptides so that they are in frame and that expression of the fusion polypeptide is under control of the same promoter(s) and terminator. Fusion polypeptides may also be constructed using intein technology in which fusion polypeptides are created post-translationally (Cooper et al., 1993, *EMBO J.* 12: 2575-2583; Dawson et al., 1994, *Science* 266: 776-779).

A fusion polypeptide may further comprise a cleavage site between the two polypeptides. Upon secretion of the fusion protein, the site is cleaved releasing the two polypeptides. Examples of cleavage sites include, but are not limited to, the sites disclosed in Martin et al., 2003, *J. Ind. Microbiol. Biotechnol.* 3: 568-576; Svetina et al., 2000, *J. Biotechnol.* 76: 245-251; Rasmussen-Wilson et al., 1997, *Appl. Environ. Microbiol.* 63: 3488-3493; Ward et al., 1995, *Biotechnology* 13: 498-503; and Contreras et al., 1991, *Biotechnology* 9: 378-381; Eaton et al., 1986, *Biochemistry* 25: 505-512; Collins-Racie et al., 1995, *Biotechnology* 13: 982-987; Carter et al., 1989, *Proteins: Structure, Function, and Genetics* 6: 240-248; and Stevens, 2003, *Drug Discovery World* 4: 35-48.

The parent polypeptide may be obtained from microorganisms of any genus. For purposes of the present invention, the term "obtained from" as used herein in connection with a given source shall mean that the parent polypeptide encoded by a polynucleotide is produced by the source or by a strain in which the polynucleotide from the source has been inserted.

In some aspects of the present invention, the parent polypeptide is *Bacillus* sp. alpha-amylase, e.g., the alpha-amylase of SEQ ID NOs: 4, the mature polypeptide thereof, i.e. SEQ ID NO: 7. In particular aspects, the parent polypeptide is a fusion protein, and originates from both a *Bacillus* sp and a *Alicyclobacillus* sp., e.g. the alpha-amylase of SEQ ID NO: 13. In other aspects, the parent polypeptide is a fusion protein and originates from two different *Bacillus* sps., e.g. resulting in the alpha-amylase of SEQ ID NO: 14.

It will be understood that for the aforementioned species, the invention encompasses both the perfect and imperfect states, and other taxonomic equivalents, e.g., anamorphs, regardless of the species name by which they are known. Those skilled in the art will readily recognize the identity of appropriate equivalents.

Strains of these species are readily accessible to the public in a number of culture collections, such as the American Type Culture Collection (ATCC), Deutsche Sammlung von Mikroorganismen und Zellkulturen GmbH (DSMZ), Centraalbureau Voor Schimmelcultures (CBS), and Agricultural Research Service Patent Culture Collection, Northern Regional Research Center (NRRL).

The parent polypeptide may be identified and obtained from other sources including microorganisms isolated from nature (e.g., soil, composts, water, etc.) or DNA samples obtained directly from natural materials (e.g., soil, composts, water, etc.) using the above-mentioned probes. Techniques for isolating microorganisms and DNA directly from natural habitats are well known in the art. A polynucleotide encoding a parent polypeptide may then be obtained by similarly screening a genomic DNA or cDNA library of another microorganism or mixed DNA sample. Once a

polynucleotide encoding a parent polypeptide has been detected with the probe(s), the polynucleotide can be isolated or cloned by utilizing techniques that are known to those of ordinary skill in the art (see, e.g., Sambrook et al., 1989, *supra*).

In one embodiment, the parent polypeptide comprises or consists of the amino acid sequence set forth in SEQ ID NO: 7, 13, or 15. In another embodiment, the parent polypeptide comprises or consists of the mature polypeptide of SEQ ID NO: 4. In particular, the parent polypeptide may comprise or consist of amino acids 1 to 485 of SEQ ID NO: 4.

In another embodiment, the parent polypeptide is a fragment of the mature polypeptide of SEQ ID NO: 4 containing at least 350, such as at least 390 amino acid residues, e.g., at least 395 and at least 397 amino acid residues of SEQ ID NO: 4.

In another embodiment, the parent polypeptide is an allelic variant of the mature polypeptide of SEQ ID NO: 4.

The term "sequence identity" as used herein, refers to the relatedness between two amino acid sequences or between two nucleotide sequences. For purposes of the present invention, the sequence identity between two amino acid sequences is determined using the Needleman-Wunsch algorithm (Needleman and Wunsch, 1970, *J. Mol. Biol.* 48: 443-453) as implemented in the Needle program of the EMBOSS package (EMBOSS: The European Molecular Biology Open Software Suite, Rice et al., 2000, *Trends Genet.* 16: 276-277), preferably version 5.0.0 or later. The parameters used are gap open penalty of 10, gap extension penalty of 0.5, and the EBLOSUM62 (EMBOSS version of BLOSUM62) substitution matrix. The output of Needle labeled "longest identity" (obtained using the—nobrief option) is used as the percent identity and is calculated as follows:

$$\frac{(\text{Identical Residues} \times 100)}{(\text{Length of Alignment} - \text{Total Number of Gaps in Alignment})}$$

For purposes of the present invention, the sequence identity between two deoxyribonucleotide sequences is determined using the Needleman-Wunsch algorithm (Needleman and Wunsch, 1970, *supra*) as implemented in the Needle program of the EMBOSS package (EMBOSS: The European Molecular Biology Open Software Suite, Rice et al., 2000, *supra*), preferably version 5.0.0 or later. The parameters used are gap open penalty of 10, gap extension penalty of 0.5, and the EDNAFULL (EMBOSS version of NCBI NUC4.4) substitution matrix. The output of Needle labeled "longest identity" (obtained using the—nobrief option) is used as the percent identity and is calculated as follows:

$$\frac{(\text{Identical Deoxyribonucleotides} \times 100)}{(\text{Length of Alignment} - \text{Total Number of Gaps in Alignment})}$$

The term "alpha-amylase activity" as used herein, refers to the activity of alpha-1,4-glucan-4-glucanohydrolases, E.C. 3.2.1.1, which constitute a group of enzymes, catalyzing hydrolysis of starch and other linear and branched 1,4-glucosidic oligo- and polysaccharides. Thus, the term "alpha-amylase" as used herein, refers to an enzyme that has alpha-amylase activity (Enzyme Class; EC 3.2.1.1) that hydrolyses alpha bonds of large, alpha-linked polysaccharides, such as starch and glycogen, yielding glucose and maltose. The terms "alpha-amylase" and "amylase" may be used interchangeably and constitute the same meaning and purpose within the scope of the present invention. For purposes of the present invention, alpha-amylase activity is determined according to the procedure described in the Examples. In one embodiment, the variants of the present

invention have at least 20%, e.g., at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, or at least 100% of the alpha-amylase activity of the polypeptide of SEQ ID NOs: 7, 13, 14, or 15; or the mature polypeptide of SEQ ID NO: 4.

The term “fusion polypeptide” as used herein, refers to a polypeptide which comprises amino acid sequences originating from more than one, such as two, three, or four, species. Such a fusion polypeptide may have been generated by molecular techniques well-known to the skilled person. A fusion polypeptide of the present invention has alpha-amylase activity. In particular, a fusion polypeptide of the present invention, comprises e.g. an A and B domain of one alpha-amylase and a C domain from another alpha-amylase.

The term “wash performance” as used herein, refers to an enzyme’s ability to remove starch or starch-containing stains present on the object to be cleaned during e.g. laundry or hard surface cleaning, such as dish wash. The term “wash performance” includes cleaning in general e.g. hard surface cleaning as in dish wash, but also wash performance on textiles such as laundry, and also industrial and institutional cleaning. The wash performance may be quantified by calculating the so-called Intensity value, and results may be displayed as “Improvement Factor” (IF). Wash performance may be determined as in described in the Examples herein.

The term “Intensity value” as used herein, refers to the wash performance measured as the brightness expressed as the intensity of the light reflected from the sample when illuminated with white light. When the sample is stained the intensity of the reflected light is lower, than that of a clean sample. Therefore the intensity of the reflected light can be used to measure wash performance, where a higher intensity value correlates with higher wash performance.

Color measurements are made with a professional flatbed scanner (Kodak iQsmart, Kodak) used to capture an image of the washed textile.

To extract a value for the light intensity from the scanned images, 24-bit pixel values from the image are converted into values for red, green and blue (RGB). The intensity value (Int) is calculated by adding the RGB values together as vectors and then taking the length of the resulting vector:

$$Int = \sqrt{r^2 + g^2 + b^2}$$

The term “improved wash performance” as used herein, refers to an improvement of the wash performance of an alpha-amylase of the present invention relative to the wash performance of the parent polypeptide. Improved wash performance may be measured by comparing of the so-called Intensity value and calculating the Improvement Factor (IF). The improved wash performance is determined according to the section “Wash performance of alpha-amylases using Automatic Mechanical Stress Assay” and using model detergent J at 15° C., 30° C. or 40° C. Other model detergents may be used, such as Model detergent A.

Thus, in one embodiment, the improved wash performance is determined by a method comprising the steps of:

- washing a fabric stained with starch with an alpha-amylase variant and a parent polypeptide sample added, respectively, to a model detergent composition, such as Model A or Model J;
- measuring the intensity of light reflected from the sample when illuminated with white light; and

- optionally, calculating the improvement factor (IF) as the ratio of delta intensity of the alpha-amylase sample over the delta intensity of the parent polypeptide sample.

In one embodiment, the improved wash performance is determined by a method comprising the steps of:

- washing a fabric stained with starch with an alpha-amylase variant and a parent polypeptide sample added, respectively, to a model detergent composition, such as Model A or Model J for 20 minutes at 15° C., 30° C., and 40° C.;
- measuring the intensity of light reflected from the sample when illuminated with white light; and
- optionally, calculating the improvement factor (IF) as the ration of delta intensity of the alpha-amylase sample over the delta intensity of the parent polypeptide sample.

The term “low temperature” as used herein, refers to is a temperature of 5-40° C., such as 5-35° C., preferably 5-30° C., more preferably 5-25° C., more preferably 5-20° C., most preferably 5-15° C., and in particular 5-10° C. In a preferred embodiment, “Low temperature” is a temperature of 10-35° C., preferably 10-30° C., more preferably 10-25° C., most preferably 10-20° C., and in particular 10-15° C. Most preferred, low temperature means 15° C.

Thus, in one embodiment, the low temperature is 5° C. to 40° C., preferably 10° C. to 30°, event more preferred 15° C. to 20° C.

In one embodiment, the improved wash performance has an Improvement Factor (IF) of ≥ 1 when compared to said parent alpha-amylase activity.

The term “Improvement Factor” as used herein, refers to a quantitative way of calculating the improvement of a particular property of a variant according to the present invention. Determination of the Improvement Factor may be according to the following formula:

$$\frac{\text{Intensity value of variant} - \text{Intensity value of blank}}{\text{Intensity value of parent} - \text{Intensity value of blank}}$$

Other formulas may be used to determine the Improvement Factor. The skilled person knows the presently presented formula as well as alternative ways of calculating the Improvement Factor.

According to the present invention, a value of 1.0 corresponds to the performance observed for the parent polypeptide. A value above 1.0 indicates an improvement of performance of the variant tested compared to the parent polypeptide. Accordingly, any value of ≥ 1.0 is indicative for improvement of property, such as performance, of the variant compared to the parent polypeptide.

According to the present invention, a variant showing improvement of property under at least one condition tested, is considered a variant having improved property as compared to the parent polypeptide.

In one embodiment, said parent polypeptide comprises an A and a B domain comprising the amino acid sequence of SEQ ID NO: 10, or an A and a B domain having at least 80%, such as 85%, such as 90%, such as 95%, such as 97% sequence identity to SEQ ID NO: 10, and a C domain of a low temperature alpha-amylase.

The terms “A domain”, “B domain” and “C domain” as used herein, refers to three distinct domains A, B and C, all part of the alpha-amylase structure, see, e.g., Machius et al., 1995, *J. Mol. Biol.* 246: 545-559. Thus, an alpha-amylase,

such as a parent polypeptide and a variant according to the invention, may comprise both an A, B, and C domain. The term “domain” means a region of a polypeptide that in itself forms a distinct and independent substructure of the whole molecule. Alpha-amylases consist of a beta/alpha-8 barrel harboring the active site residues, which is denoted the A domain, a rather long loop between the beta-sheet 3 and alpha-helix 3, which is denoted the B domain (together; “A and B domain”), and a C-domain and in some cases also a carbohydrate binding domain (e.g., WO 2005/001064; Machius et al., supra).

The domains of an alpha-amylase may be determined by structure analysis such as using crystallographically techniques. An alternative method for determining the domains of an alpha-amylase is by sequence alignment of the amino acid sequence of the alpha-amylase with another alpha-amylase for which the domains have been determined. The sequence that aligns with, e.g., the C-domain sequence in the alpha-amylase for which the C-domain has been determined can be considered the C domain for the given alpha-amylase.

The term “A and B domain” as used herein, refers to two domains of an alpha-amylase taken as one unit, whereas the C domain is another unit of the alpha-amylases. Thus, the amino acid sequence of the “A and B domain” is understood as one sequence or one part of a sequence of an alpha-amylase comprising an “A and B domain” and other domains (such as the C domain). Thus, the term “the A and B domain domain has at least 65% sequence identity to SEQ ID NO: 10” means that the amino acid sequence that form the A and B domain has at least 65% sequence identity to SEQ ID NO: 10. As used herein, the “A and B domain” of a parent polypeptide corresponds to amino acids 1 to 399 of SEQ ID NO: 7, or to amino acids 1 to 397 of SEQ ID NO: 15. The A and B domain of a parent polypeptide may have the amino acid sequence of SEQ ID NO: 10.

The term “C domain” as used herein, refers to a domain of an alpha-amylase as one unit. The “C domain” of an alpha-amylase corresponds to amino acids 400 to 485 of SEQ ID NO: 7, or to amino acid 398 to 483 of SEQ ID NO: 15. Thus, the C domain of an alpha amylase may be found by alignment of said alpha-amylase with the polypeptide of SEQ ID NO: 7 or 15. The part of said alpha-amylase that aligns with amino acids 400 to 485 of SEQ ID NO: 7 or to amino acids 398 to 483 of SEQ ID NO: 15 is according to the present invention “the C domain” of the alpha-amylase. The C domain of a parent polypeptide may have the amino acid sequence of SEQ ID NOs: 11 or 12.

The term “C domain of a low temperature alpha-amylase” as used herein, refers to the C domain of an alpha-amylase showing particular properties at low temperatures. Non-limiting examples of such low temperature alpha-amylases are those originating from *Alicyclobacillus* sp. as well as from *Bacillus amyloliquefaciens* (BAN).

In one embodiment, the parent polypeptide comprises a C domain comprising the amino acid sequence of SEQ ID NOs: 11 or 12, or a C domain having at 80%, such as 85%, such as 90%, such as 95%, such as 97%, sequence identity to SEQ ID NO: 11 or 12.

In one embodiment, the parent polypeptide comprises the amino acid sequence of SEQ ID NOs: 13 or 14, or a sequence having at least 80%, such as 85%, such as 90%, such as 95%, such as 97% sequence identity to SEQ ID NOs: 13 or 14.

As stated elsewhere herein, the parent polypeptide may be any polypeptide having alpha-amylase activity and at least 65% sequence identity to any one of the amino acid sequences as set forth in SEQ ID NOs: 13 and 14. In certain

embodiments, the parent polypeptide comprises the amino acid sequences as set forth in SEQ ID NOs: 10, 11, and 12.

In one embodiment, the parent polypeptide a sequence identity to the amino acid sequence set forth in SEQ ID NOs: 13 or 14 of at least 60%, e.g., at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100%, and which parent polypeptide has alpha-amylase activity. In one embodiment, the amino acid sequence of the parent polypeptide differs by up to 10 amino acids, e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10, from the amino acid sequences set forth in SEQ ID NOs: 7, 13, and 14.

The present invention provides variants of such parent polypeptides.

The variants according to the present invention may advantageously comprise alterations in specific positions of the A and B domain. Thus, in a particular embodiment, the variant comprises amino acid alterations in amino acid positions corresponding the amino acid sequence set forth in SEQ ID NO: 10, or a sequence having at least 65% sequence identity to the amino acid sequence set forth in SEQ ID NO: 10.

In particular, the present invention relates to an embodiment, the A and B domains comprise a modification in one or more of the following positions corresponding to positions: H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, A60, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, G196, A204, V206, P211, I214, V215, L217, L219, A225, L228, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, A265, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, N295, Q299, S304, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, and Q395 of SEQ ID NOs: 7 or 10.

Such variants have shown to have improved wash performance in at least one assay wherein the temperature under which the variants have been tested was a low temperature, such as 15° C., 30° and 40° C. Specific conditions may be seen in the Examples herein.

The term “modification” as used herein, refers to both substitutions and deletions of amino acid within the amino acid sequence of a polypeptide. The terms “alteration” and “modification” may be used interchangeably herein. This should not be understood as any limitation and thus, the terms constitute the same meaning and purpose unless explicitly stated otherwise.

The term “corresponding to” as used herein, refers to way of determining the specific amino acid of a sequence wherein reference is made to a specific amino acid sequence. E.g. for the purposes of the present invention, when references are made to specific amino acid positions, the skilled person would be able to align another amino acid sequence to said amino acid sequence that reference has been made to, in order to determine which specific amino acid may be of interest in said another amino acid sequence. Alignment of another amino acid sequence with e.g. the sequence as set forth in SEQ ID NO: 4, 7, or any other sequence listed herein, has been described elsewhere herein. Alternative alignment methods may be used, and are well-known for the skilled person.

In a particular embodiment, the variant comprises an alteration in two, three, four, five, six, seven, eight, nine, ten, eleven, twelfth, thirteen, fourteen, fifteen, or sixteen posi-

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tions corresponding to the following positions; H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, A60, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, G196, A204, V206, P211, I214, V215, L217, L219, A225, L228, L235, V238, K242, S244, M246, M248, L250, G255,

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Q256, N260, A263, V264, A265, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, S304, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, and Q395 of SEQ ID NOs: 7 or 10.

In a particular embodiment, the variant comprises an alteration in the following positions;

H1 + T5	T40 + L217	N54 + T285	Q125 + G255	G184 + P211	M248 + Q299
H1 + G7	T40 + L219	N54 + M286	Q125 + Q256	G184 + I214	M248 + S304
H1 + Q11	T40 + L228	N54 + F289	Q125 + A263	G184 + V215	M248 + R320
H1 + N16	T40 + L235	N54 + V291	Q125 + V264	G184 + L217	M248 + H321
H1 + V17	T40 + V238	N54 + Q299	Q125 + A265	G184 + L219	M248 + S323
H1 + Q32	T40 + M246	N54 + S304	Q125 + Y267	G184 + L228	M248 + H324
H1 + A37	T40 + M248	N54 + R320	Q125 + N270	G184 + L235	M248 + F328
H1 + T40	T40 + L250	N54 + H321	Q125 + G273	G184 + V238	M248 + T334
H1 + P45	T40 + G255	N54 + S323	Q125 + S280	G184 + M246	M248 + D337
H1 + W48	T40 + Q256	N54 + H324	Q125 + T285	G184 + M248	M248 + Q345
H1 + G50	T40 + A263	N54 + F328	Q125 + M286	G184 + L250	M248 + G346
H1 + T51	T40 + V264	N54 + T334	Q125 + F289	G184 + G255	M248 + G348
H1 + N54	T40 + A265	N54 + D337	Q125 + V291	G184 + Q256	M248 + T355
H1 + V56	T40 + Y267	N54 + Q345	Q125 + Q299	G184 + A263	M248 + S376
H1 + A60	T40 + N270	N54 + G346	Q125 + S304	G184 + V264	M248 + D377
H1 + K72	T40 + G273	N54 + G348	Q125 + R320	G184 + A265	M248 + D379
H1 + R87	T40 + S280	N54 + T355	Q125 + H321	G184 + Y267	M248 + Y382
H1 + Q98	T40 + T285	N54 + S376	Q125 + S323	G184 + N270	M248 + S383
H1 + M105	T40 + M286	N54 + D377	Q125 + H324	G184 + G273	M248 + Q385
H1 + G109	T40 + F289	N54 + D379	Q125 + F328	G184 + S280	M248 + K391
H1 + F113	T40 + V291	N54 + Y382	Q125 + T334	G184 + T285	M248 + K393
H1 + R116	T40 + Q299	N54 + S383	Q125 + D337	G184 + M286	M248 + Q395
H1 + Q118	T40 + S304	N54 + Q385	Q125 + Q345	G184 + F289	L250 + G255
H1 + Q125	T40 + R320	N54 + K391	Q125 + G346	G184 + V291	L250 + Q256
H1 + G133	T40 + H321	N54 + K393	Q125 + G348	G184 + Q299	L250 + A263
H1 + T134	T40 + S323	N54 + Q395	Q125 + T355	G184 + S304	L250 + V264
H1 + W140	T40 + H324	V56 + A60	Q125 + S376	G184 + R320	L250 + A265
H1 + G142	T40 + F328	V56 + K72	Q125 + D377	G184 + H321	L250 + Y267
H1 + G149	T40 + T334	V56 + R87	Q125 + D379	G184 + S323	L250 + N270
H1 + T165	T40 + D337	V56 + Q98	Q125 + Y382	G184 + H324	L250 + G273
H1 + W167	T40 + Q345	V56 + M105	Q125 + S383	G184 + F328	L250 + S280
H1 + R171	T40 + G346	V56 + G109	Q125 + Q385	G184 + T334	L250 + T285
H1 + Q172	T40 + G348	V56 + F113	Q125 + K391	G184 + D337	L250 + M286
H1 + L173	T40 + T355	V56 + R116	Q125 + K393	G184 + Q345	L250 + F289
H1 + A174	T40 + S376	V56 + Q118	Q125 + Q395	G184 + G346	L250 + V291
H1 + G184	T40 + D377	V56 + Q125	G133 + T134	G184 + G348	L250 + Q299
H1 + T193	T40 + D379	V56 + G133	G133 + W140	G184 + T355	L250 + S304
H1 + N195	T40 + Y382	V56 + T134	G133 + G142	G184 + S376	L250 + R320
H1 + G196	T40 + S383	V56 + W140	G133 + G149	G184 + D377	L250 + H321
H1 + A204	T40 + Q385	V56 + G142	G133 + T165	G184 + D379	L250 + S323
H1 + V206	T40 + K391	V56 + G149	G133 + W167	G184 + Y382	L250 + H324
H1 + P211	T40 + K393	V56 + T165	G133 + R171	G184 + S383	L250 + F328
H1 + I214	T40 + Q395	V56 + W167	G133 + Q172	G184 + Q385	L250 + T334
H1 + V215	P45 + W48	V56 + R171	G133 + L173	G184 + K391	L250 + D337
H1 + L217	P45 + G50	V56 + Q172	G133 + A174	G184 + K393	L250 + Q345
H1 + L219	P45 + T51	V56 + L173	G133 + G184	G184 + Q395	L250 + G346
H1 + L228	P45 + N54	V56 + A174	G133 + T193	T193 + N195	L250 + G348
H1 + L235	P45 + V56	V56 + G184	G133 + N195	T193 + G196	L250 + T355
H1 + V238	P45 + A60	V56 + T193	G133 + G196	T193 + A204	L250 + S376
H1 + M246	P45 + K72	V56 + N195	G133 + A204	T193 + V206	L250 + D377
H1 + M248	P45 + R87	V56 + G196	G133 + V206	T193 + P211	L250 + D379
H1 + L250	P45 + Q98	V56 + A204	G133 + P211	T193 + I214	L250 + Y382
H1 + G255	P45 + M105	V56 + V206	G133 + I214	T193 + V215	L250 + S383
H1 + Q256	P45 + G109	V56 + P211	G133 + V215	T193 + L217	L250 + Q385
H1 + A263	P45 + F113	V56 + I214	G133 + L217	T193 + L219	L250 + K391
H1 + V264	P45 + R116	V56 + V215	G133 + L219	T193 + L228	L250 + K393
H1 + A265	P45 + Q118	V56 + L217	G133 + L228	T193 + L235	L250 + Q395
H1 + Y267	P45 + Q125	V56 + L219	G133 + L235	T193 + V238	G255 + Q256
H1 + N270	P45 + G133	V56 + L228	G133 + V238	T193 + M246	G255 + A263
H1 + G273	P45 + T134	V56 + L235	G133 + M246	T193 + M248	G255 + V264
H1 + S280	P45 + W140	V56 + V238	G133 + M248	T193 + L250	G255 + A265
H1 + T285	P45 + G142	V56 + M246	G133 + L250	T193 + G255	G255 + Y267
H1 + M286	P45 + G149	V56 + M248	G133 + G255	T193 + Q256	G255 + N270
H1 + F289	P45 + T165	V56 + L250	G133 + Q256	T193 + A263	G255 + G273
H1 + V291	P45 + W167	V56 + G255	G133 + A263	T193 + V264	G255 + S280
H1 + Q299	P45 + R171	V56 + Q256	G133 + V264	T193 + A265	G255 + T285
H1 + S304	P45 + Q172	V56 + A263	G133 + A265	T193 + Y267	G255 + M286
H1 + R320	P45 + L173	V56 + V264	G133 + Y267	T193 + N270	G255 + F289
H1 + H321	P45 + A174	V56 + A265	G133 + N270	T193 + G273	G255 + V291
H1 + S323	P45 + G184	V56 + Y267	G133 + G273	T193 + S280	G255 + Q299
H1 + H324	P45 + T193	V56 + N270	G133 + S280	T193 + T285	G255 + S304

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H1 + F328	P45 + N195	V56 + G273	G133 + T285	T193 + M286	G255 + R320
H1 + T334	P45 + G196	V56 + S280	G133 + M286	T193 + F289	G255 + H321
H1 + D337	P45 + A204	V56 + T285	G133 + F289	T193 + V291	G255 + S323
H1 + Q345	P45 + V206	V56 + M286	G133 + V291	T193 + Q299	G255 + H324
H1 + G346	P45 + P211	V56 + F289	G133 + Q299	T193 + S304	G255 + F328
H1 + G348	P45 + I214	V56 + V291	G133 + S304	T193 + R320	G255 + T334
H1 + T355	P45 + V215	V56 + Q299	G133 + R320	T193 + H321	G255 + D337
H1 + S376	P45 + L217	V56 + S304	G133 + H321	T193 + S323	G255 + Q345
H1 + D377	P45 + L219	V56 + R320	G133 + S323	T193 + H324	G255 + G346
H1 + D379	P45 + L228	V56 + H321	G133 + H324	T193 + F328	G255 + G348
H1 + Y382	P45 + L235	V56 + S323	G133 + F328	T193 + T334	G255 + T355
H1 + S383	P45 + V238	V56 + H324	G133 + T334	T193 + D337	G255 + S376
H1 + Q385	P45 + M246	V56 + F328	G133 + D337	T193 + Q345	G255 + D377
H1 + K391	P45 + M248	V56 + T334	G133 + Q345	T193 + G346	G255 + D379
H1 + K393	P45 + L250	V56 + D337	G133 + G346	T193 + G348	G255 + Y382
H1 + Q395	P45 + G255	V56 + Q345	G133 + G348	T193 + T355	G255 + S383
T5 + G7	P45 + Q256	V56 + G346	G133 + T355	T193 + S376	G255 + Q385
T5 + Q11	P45 + A263	V56 + G348	G133 + S376	T193 + D377	G255 + K391
T5 + N16	P45 + V264	V56 + T355	G133 + D377	T193 + D379	G255 + K393
T5 + V17	P45 + A265	V56 + S376	G133 + D379	T193 + Y382	G255 + Q395
T5 + Q32	P45 + Y267	V56 + D377	G133 + Y382	T193 + S383	Q256 + A263
T5 + A37	P45 + N270	V56 + D379	G133 + S383	T193 + Q385	Q256 + V264
T5 + T40	P45 + G273	V56 + Y382	G133 + Q385	T193 + K391	Q256 + A265
T5 + P45	P45 + S280	V56 + S383	G133 + K391	T193 + K393	Q256 + Y267
T5 + W48	P45 + T285	V56 + Q385	G133 + K393	T193 + Q395	Q256 + N270
T5 + G50	P45 + M286	V56 + K391	G133 + Q395	N195 + G196	Q256 + G273
T5 + T51	P45 + F289	V56 + K393	T134 + W140	N195 + A204	Q256 + S280
T5 + N54	P45 + V291	V56 + Q395	T134 + G142	N195 + V206	Q256 + T285
T5 + V56	P45 + Q299	A60 + K72	T134 + G149	N195 + P211	Q256 + M286
T5 + A60	P45 + S304	A60 + R87	T134 + T165	N195 + I214	Q256 + F289
T5 + K72	P45 + R320	A60 + Q98	T134 + W167	N195 + V215	Q256 + V291
T5 + R87	P45 + H321	A60 + M105	T134 + R171	N195 + L217	Q256 + Q299
T5 + Q98	P45 + S323	A60 + G109	T134 + Q172	N195 + L219	Q256 + S304
T5 + M105	P45 + H324	A60 + F113	T134 + L173	N195 + L228	Q256 + R320
T5 + G109	P45 + F328	A60 + R116	T134 + A174	N195 + L235	Q256 + H321
T5 + F113	P45 + T334	A60 + Q118	T134 + G184	N195 + V238	Q256 + S323
T5 + R116	P45 + D337	A60 + Q125	T134 + T193	N195 + M246	Q256 + H324
T5 + Q118	P45 + Q345	A60 + G133	T134 + N195	N195 + M248	Q256 + F328
T5 + Q125	P45 + G346	A60 + T134	T134 + G196	N195 + L250	Q256 + T334
T5 + G133	P45 + G348	A60 + W140	T134 + A204	N195 + G255	Q256 + D337
T5 + T134	P45 + T355	A60 + G142	T134 + V206	N195 + Q256	Q256 + Q345
T5 + W140	P45 + S376	A60 + G149	T134 + P211	N195 + A263	Q256 + G346
T5 + G142	P45 + D377	A60 + T165	T134 + I214	N195 + V264	Q256 + G348
T5 + G149	P45 + D379	A60 + W167	T134 + V215	N195 + A265	Q256 + T355
T5 + T165	P45 + Y382	A60 + R171	T134 + L217	N195 + Y267	Q256 + S376
T5 + W167	P45 + S383	A60 + Q172	T134 + L219	N195 + N270	Q256 + D377
T5 + R171	P45 + Q385	A60 + L173	T134 + L228	N195 + G273	Q256 + D379
T5 + Q172	P45 + K391	A60 + A174	T134 + L235	N195 + S280	Q256 + Y382
T5 + L173	P45 + K393	A60 + G184	T134 + V238	N195 + T285	Q256 + S383
T5 + A174	P45 + Q395	A60 + T193	T134 + M246	N195 + M286	Q256 + Q385
T5 + G184	W48 + G50	A60 + N195	T134 + M248	N195 + F289	Q256 + K391
T5 + T193	W48 + T51	A60 + G196	T134 + L250	N195 + V291	Q256 + K393
T5 + N195	W48 + N54	A60 + A204	T134 + G255	N195 + Q299	Q256 + Q395
T5 + G196	W48 + V56	A60 + V206	T134 + Q256	N195 + S304	A263 + V264
T5 + A204	W48 + A60	A60 + P211	T134 + A263	N195 + R320	A263 + A265
T5 + V206	W48 + K72	A60 + I214	T134 + V264	N195 + H321	A263 + Y267
T5 + P211	W48 + R87	A60 + V215	T134 + A265	N195 + S323	A263 + N270
T5 + I214	W48 + Q98	A60 + L217	T134 + Y267	N195 + H324	A263 + G273
T5 + V215	W48 + M105	A60 + L219	T134 + N270	N195 + F328	A263 + S280
T5 + L217	W48 + G109	A60 + L228	T134 + G273	N195 + T334	A263 + T285
T5 + L219	W48 + F113	A60 + L235	T134 + S280	N195 + D337	A263 + M286
T5 + L228	W48 + R116	A60 + V238	T134 + T285	N195 + Q345	A263 + F289
T5 + L235	W48 + Q118	A60 + M246	T134 + M286	N195 + G346	A263 + V291
T5 + V238	W48 + Q125	A60 + M248	T134 + F289	N195 + G348	A263 + Q299
T5 + M246	W48 + G133	A60 + L250	T134 + V291	N195 + T355	A263 + S304
T5 + M248	W48 + T134	A60 + G255	T134 + Q299	N195 + S376	A263 + R320
T5 + L250	W48 + W140	A60 + Q256	T134 + S304	N195 + D377	A263 + H321
T5 + G255	W48 + G142	A60 + A263	T134 + R320	N195 + D379	A263 + S323
T5 + Q256	W48 + G149	A60 + V264	T134 + H321	N195 + Y382	A263 + H324
T5 + A263	W48 + T165	A60 + A265	T134 + S323	N195 + S383	A263 + F328
T5 + V264	W48 + W167	A60 + Y267	T134 + H324	N195 + Q385	A263 + T334
T5 + A265	W48 + R171	A60 + N270	T134 + F328	N195 + K391	A263 + D337
T5 + Y267	W48 + Q172	A60 + G273	T134 + T334	N195 + K393	A263 + Q345
T5 + N270	W48 + L173	A60 + S280	T134 + D337	N195 + Q395	A263 + G346
T5 + G273	W48 + A174	A60 + T285	T134 + Q345	G196 + A204	A263 + G348
T5 + S280	W48 + G184	A60 + M286	T134 + G346	G196 + V206	A263 + T355
T5 + T285	W48 + T193	A60 + F289	T134 + G348	G196 + P211	A263 + S376
T5 + M286	W48 + N195	A60 + V291	T134 + T355	G196 + I214	A263 + D377
T5 + F289	W48 + G196	A60 + Q299	T134 + S376	G196 + V215	A263 + D379

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T5 + V291	W48 + A204	A60 + S304	T134 + D377	G196 + L217	A263 + Y382
T5 + Q299	W48 + V206	A60 + R320	T134 + D379	G196 + L219	A263 + S383
T5 + S304	W48 + P211	A60 + H321	T134 + Y382	G196 + L228	A263 + Q385
T5 + R320	W48 + I214	A60 + S323	T134 + S383	G196 + L235	A263 + K391
T5 + H321	W48 + V215	A60 + H324	T134 + Q385	G196 + V238	A263 + K393
T5 + S323	W48 + L217	A60 + F328	T134 + K391	G196 + M246	A263 + Q395
T5 + H324	W48 + L219	A60 + T334	T134 + K393	G196 + M248	V264 + A265
T5 + F328	W48 + L228	A60 + D337	T134 + Q395	G196 + L250	V264 + Y267
T5 + T334	W48 + L235	A60 + Q345	W140 + G142	G196 + G255	V264 + N270
T5 + D337	W48 + V238	A60 + G346	W140 + G149	G196 + Q256	V264 + G273
T5 + Q345	W48 + M246	A60 + G348	W140 + T165	G196 + A263	V264 + S280
T5 + G346	W48 + M248	A60 + T355	W140 + W167	G196 + V264	V264 + T285
T5 + G348	W48 + L250	A60 + S376	W140 + R171	G196 + A265	V264 + M286
T5 + T355	W48 + G255	A60 + D377	W140 + Q172	G196 + Y267	V264 + F289
T5 + S376	W48 + Q256	A60 + D379	W140 + L173	G196 + N270	V264 + V291
T5 + D377	W48 + A263	A60 + Y382	W140 + A174	G196 + G273	V264 + Q299
T5 + D379	W48 + V264	A60 + S383	W140 + G184	G196 + S280	V264 + S304
T5 + Y382	W48 + A265	A60 + Q385	W140 + T193	G196 + T285	V264 + R320
T5 + S383	W48 + Y267	A60 + K391	W140 + N195	G196 + M286	V264 + H321
T5 + Q385	W48 + N270	A60 + K393	W140 + G196	G196 + F289	V264 + S323
T5 + K391	W48 + G273	A60 + Q395	W140 + A204	G196 + V291	V264 + H324
T5 + K393	W48 + S280	K72 + R87	W140 + V206	G196 + Q299	V264 + F328
T5 + Q395	W48 + T285	K72 + Q98	W140 + P211	G196 + S304	V264 + T334
G7 + Q11	W48 + M286	K72 + M105	W140 + I214	G196 + R320	V264 + D337
G7 + N16	W48 + F289	K72 + G109	W140 + V215	G196 + H321	V264 + Q345
G7 + V17	W48 + V291	K72 + F113	W140 + L217	G196 + S323	V264 + G346
G7 + Q32	W48 + Q299	K72 + R116	W140 + L219	G196 + H324	V264 + G348
G7 + A37	W48 + S304	K72 + Q118	W140 + L228	G196 + F328	V264 + T355
G7 + T40	W48 + R320	K72 + Q125	W140 + L235	G196 + T334	V264 + S376
G7 + P45	W48 + H321	K72 + G133	W140 + V238	G196 + D337	V264 + D377
G7 + W48	W48 + S323	K72 + T134	W140 + M246	G196 + Q345	V264 + D379
G7 + G50	W48 + H324	K72 + W140	W140 + M248	G196 + G346	V264 + Y382
G7 + T51	W48 + F328	K72 + G142	W140 + L250	G196 + G348	V264 + S383
G7 + N54	W48 + T334	K72 + G149	W140 + G255	G196 + T355	V264 + Q385
G7 + V56	W48 + D337	K72 + T165	W140 + Q256	G196 + S376	V264 + K391
G7 + A60	W48 + Q345	K72 + W167	W140 + A263	G196 + D377	V264 + K393
G7 + K72	W48 + G346	K72 + R171	W140 + V264	G196 + D379	V264 + Q395
G7 + R87	W48 + G348	K72 + Q172	W140 + A265	G196 + Y382	A265 + Y267
G7 + Q98	W48 + T355	K72 + L173	W140 + Y267	G196 + S383	A265 + N270
G7 + M105	W48 + S376	K72 + A174	W140 + N270	G196 + Q385	A265 + G273
G7 + G109	W48 + D377	K72 + G184	W140 + G273	G196 + K391	A265 + S280
G7 + F113	W48 + D379	K72 + T193	W140 + S280	G196 + K393	A265 + T285
G7 + R116	W48 + Y382	K72 + N195	W140 + T285	G196 + Q395	A265 + M286
G7 + Q118	W48 + S383	K72 + G196	W140 + M286	A204 + V206	A265 + F289
G7 + Q125	W48 + Q385	K72 + A204	W140 + F289	A204 + P211	A265 + V291
G7 + G133	W48 + K391	K72 + V206	W140 + V291	A204 + I214	A265 + Q299
G7 + T134	W48 + K393	K72 + P211	W140 + Q299	A204 + V215	A265 + S304
G7 + W140	W48 + Q395	K72 + I214	W140 + S304	A204 + L217	A265 + R320
G7 + G142	G50 + T51	K72 + V215	W140 + R320	A204 + L219	A265 + H321
G7 + G149	G50 + N54	K72 + L217	W140 + H321	A204 + L228	A265 + S323
G7 + T165	G50 + V56	K72 + L219	W140 + S323	A204 + L235	A265 + H324
G7 + W167	G50 + A60	K72 + L228	W140 + H324	A204 + V238	A265 + F328
G7 + R171	G50 + K72	K72 + L235	W140 + F328	A204 + M246	A265 + T334
G7 + Q172	G50 + R87	K72 + V238	W140 + T334	A204 + M248	A265 + D337
G7 + L173	G50 + Q98	K72 + M246	W140 + D337	A204 + L250	A265 + Q345
G7 + A174	G50 + M105	K72 + M248	W140 + Q345	A204 + G255	A265 + G346
G7 + G184	G50 + G109	K72 + L250	W140 + G346	A204 + Q256	A265 + G348
G7 + T193	G50 + F113	K72 + G255	W140 + G348	A204 + A263	A265 + T355
G7 + N195	G50 + R116	K72 + Q256	W140 + T355	A204 + V264	A265 + S376
G7 + G196	G50 + Q118	K72 + A263	W140 + S376	A204 + A265	A265 + D377
G7 + A204	G50 + Q125	K72 + V264	W140 + D377	A204 + Y267	A265 + D379
G7 + V206	G50 + G133	K72 + A265	W140 + D379	A204 + N270	A265 + Y382
G7 + P211	G50 + T134	K72 + Y267	W140 + Y382	A204 + G273	A265 + S383
G7 + I214	G50 + W140	K72 + N270	W140 + S383	A204 + S280	A265 + Q385
G7 + V215	G50 + G142	K72 + G273	W140 + Q385	A204 + T285	A265 + K391
G7 + L217	G50 + G149	K72 + S280	W140 + K391	A204 + M286	A265 + K393
G7 + L219	G50 + T165	K72 + T285	W140 + K393	A204 + F289	A265 + Q395
G7 + L228	G50 + W167	K72 + M286	W140 + Q395	A204 + V291	Y267 + N270
G7 + L235	G50 + R171	K72 + F289	G142 + G149	A204 + Q299	Y267 + G273
G7 + V238	G50 + Q172	K72 + V291	G142 + T165	A204 + S304	Y267 + S280
G7 + M246	G50 + L173	K72 + Q299	G142 + W167	A204 + R320	Y267 + T285
G7 + M248	G50 + A174	K72 + S304	G142 + R171	A204 + H321	Y267 + M286
G7 + L250	G50 + G184	K72 + R320	G142 + Q172	A204 + S323	Y267 + F289
G7 + G255	G50 + T193	K72 + H321	G142 + L173	A204 + H324	Y267 + V291
G7 + Q256	G50 + N195	K72 + S323	G142 + A174	A204 + F328	Y267 + Q299
G7 + A263	G50 + G196	K72 + H324	G142 + G184	A204 + T334	Y267 + S304
G7 + V264	G50 + A204	K72 + F328	G142 + T193	A204 + D337	Y267 + R320
G7 + A265	G50 + V206	K72 + T334	G142 + N195	A204 + Q345	Y267 + H321
G7 + Y267	G50 + P211	K72 + D337	G142 + G196	A204 + G346	Y267 + S323

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G7 + N270	G50 + I214	K72 + Q345	G142 + A204	A204 + G348	Y267 + H324
G7 + G273	G50 + V215	K72 + G346	G142 + V206	A204 + T355	Y267 + F328
G7 + S280	G50 + L217	K72 + G348	G142 + P211	A204 + S376	Y267 + T334
G7 + T285	G50 + L219	K72 + T355	G142 + I214	A204 + D377	Y267 + D337
G7 + M286	G50 + L228	K72 + S376	G142 + V215	A204 + D379	Y267 + Q345
G7 + F289	G50 + L235	K72 + D377	G142 + L217	A204 + Y382	Y267 + G346
G7 + V291	G50 + V238	K72 + D379	G142 + L219	A204 + S383	Y267 + G348
G7 + Q299	G50 + M246	K72 + Y382	G142 + L228	A204 + Q385	Y267 + T355
G7 + S304	G50 + M248	K72 + S383	G142 + L235	A204 + K391	Y267 + S376
G7 + R320	G50 + L250	K72 + Q385	G142 + V238	A204 + K393	Y267 + D377
G7 + H321	G50 + G255	K72 + K391	G142 + M246	A204 + Q395	Y267 + D379
G7 + S323	G50 + Q256	K72 + K393	G142 + M248	V206 + P211	Y267 + Y382
G7 + H324	G50 + A263	K72 + Q395	G142 + L250	V206 + I214	Y267 + S383
G7 + F328	G50 + V264	R87 + Q98	G142 + G255	V206 + V215	Y267 + Q385
G7 + T334	G50 + A265	R87 + M105	G142 + Q256	V206 + L217	Y267 + K391
G7 + D337	G50 + Y267	R87 + G109	G142 + A263	V206 + L219	Y267 + K393
G7 + Q345	G50 + N270	R87 + F113	G142 + V264	V206 + L228	Y267 + Q395
G7 + G346	G50 + G273	R87 + R116	G142 + A265	V206 + L235	N270 + G273
G7 + G348	G50 + S280	R87 + Q118	G142 + Y267	V206 + V238	N270 + S280
G7 + T355	G50 + T285	R87 + Q125	G142 + N270	V206 + M246	N270 + T285
G7 + S376	G50 + M286	R87 + G133	G142 + G273	V206 + M248	N270 + M286
G7 + D377	G50 + F289	R87 + T134	G142 + S280	V206 + L250	N270 + F289
G7 + D379	G50 + V291	R87 + W140	G142 + T285	V206 + G255	N270 + V291
G7 + Y382	G50 + Q299	R87 + G142	G142 + M286	V206 + Q256	N270 + Q299
G7 + S383	G50 + S304	R87 + G149	G142 + F289	V206 + A263	N270 + S304
G7 + Q385	G50 + R320	R87 + T165	G142 + V291	V206 + V264	N270 + R320
G7 + K391	G50 + H321	R87 + W167	G142 + Q299	V206 + A265	N270 + H321
G7 + K393	G50 + S323	R87 + R171	G142 + S304	V206 + Y267	N270 + S323
G7 + Q395	G50 + H324	R87 + Q172	G142 + R320	V206 + N270	N270 + H324
Q11 + N16	G50 + F328	R87 + L173	G142 + H321	V206 + G273	N270 + F328
Q11 + V17	G50 + T334	R87 + A174	G142 + S323	V206 + S280	N270 + T334
Q11 + Q32	G50 + D337	R87 + G184	G142 + H324	V206 + T285	N270 + D337
Q11 + A37	G50 + Q345	R87 + T193	G142 + F328	V206 + M286	N270 + Q345
Q11 + T40	G50 + G346	R87 + N195	G142 + T334	V206 + F289	N270 + G346
Q11 + P45	G50 + G348	R87 + G196	G142 + D337	V206 + V291	N270 + G348
Q11 + W48	G50 + T355	R87 + A204	G142 + Q345	V206 + Q299	N270 + T355
Q11 + G50	G50 + S376	R87 + V206	G142 + G346	V206 + S304	N270 + S376
Q11 + T51	G50 + D377	R87 + P211	G142 + G348	V206 + R320	N270 + D377
Q11 + N54	G50 + D379	R87 + I214	G142 + T355	V206 + H321	N270 + D379
Q11 + V56	G50 + Y382	R87 + V215	G142 + S376	V206 + S323	N270 + Y382
Q11 + A60	G50 + S383	R87 + L217	G142 + D377	V206 + H324	N270 + S383
Q11 + K72	G50 + Q385	R87 + L219	G142 + D379	V206 + F328	N270 + Q385
Q11 + R87	G50 + K391	R87 + L228	G142 + Y382	V206 + T334	N270 + K391
Q11 + Q98	G50 + K393	R87 + L235	G142 + S383	V206 + D337	N270 + K393
Q11 + M105	G50 + Q395	R87 + V238	G142 + Q385	V206 + Q345	N270 + Q395
Q11 + G109	T51 + N54	R87 + M246	G142 + K391	V206 + G346	G273 + S280
Q11 + F113	T51 + V56	R87 + M248	G142 + K393	V206 + G348	G273 + T285
Q11 + R116	T51 + A60	R87 + L250	G142 + Q395	V206 + T355	G273 + M286
Q11 + Q118	T51 + K72	R87 + G255	G149 + T165	V206 + S376	G273 + F289
Q11 + Q125	T51 + R87	R87 + Q256	G149 + W167	V206 + D377	G273 + V291
Q11 + G133	T51 + Q98	R87 + A263	G149 + R171	V206 + D379	G273 + Q299
Q11 + T134	T51 + M105	R87 + V264	G149 + Q172	V206 + Y382	G273 + S304
Q11 + W140	T51 + G109	R87 + A265	G149 + L173	V206 + S383	G273 + R320
Q11 + G142	T51 + F113	R87 + Y267	G149 + A174	V206 + Q385	G273 + H321
Q11 + G149	T51 + R116	R87 + N270	G149 + G184	V206 + K391	G273 + S323
Q11 + T165	T51 + Q118	R87 + G273	G149 + T193	V206 + K393	G273 + H324
Q11 + W167	T51 + Q125	R87 + S280	G149 + N195	V206 + Q395	G273 + F328
Q11 + R171	T51 + G133	R87 + T285	G149 + G196	P211 + I214	G273 + T334
Q11 + Q172	T51 + T134	R87 + M286	G149 + A204	P211 + V215	G273 + D337
Q11 + L173	T51 + W140	R87 + F289	G149 + V206	P211 + L217	G273 + Q345
Q11 + A174	T51 + G142	R87 + V291	G149 + P211	P211 + L219	G273 + G346
Q11 + G184	T51 + G149	R87 + Q299	G149 + I214	P211 + L228	G273 + G348
Q11 + T193	T51 + T165	R87 + S304	G149 + V215	P211 + L235	G273 + T355
Q11 + N195	T51 + W167	R87 + R320	G149 + L217	P211 + V238	G273 + S376
Q11 + G196	T51 + R171	R87 + H321	G149 + L219	P211 + M246	G273 + D377
Q11 + A204	T51 + Q172	R87 + S323	G149 + L228	P211 + M248	G273 + D379
Q11 + V206	T51 + L173	R87 + H324	G149 + L235	P211 + L250	G273 + Y382
Q11 + P211	T51 + A174	R87 + F328	G149 + V238	P211 + G255	G273 + S383
Q11 + I214	T51 + G184	R87 + T334	G149 + M246	P211 + Q256	G273 + Q385
Q11 + V215	T51 + T193	R87 + D337	G149 + M248	P211 + A263	G273 + K391
Q11 + L217	T51 + N195	R87 + Q345	G149 + L250	P211 + V264	G273 + K393
Q11 + L219	T51 + G196	R87 + G346	G149 + G255	P211 + A265	G273 + Q395
Q11 + L228	T51 + A204	R87 + G348	G149 + Q256	P211 + Y267	S280 + T285
Q11 + L235	T51 + V206	R87 + T355	G149 + A263	P211 + N270	S280 + M286
Q11 + V238	T51 + P211	R87 + S376	G149 + V264	P211 + G273	S280 + F289
Q11 + M246	T51 + I214	R87 + D377	G149 + A265	P211 + S280	S280 + V291
Q11 + M248	T51 + V215	R87 + D379	G149 + Y267	P211 + T285	S280 + Q299
Q11 + L250	T51 + L217	R87 + Y382	G149 + N270	P211 + M286	S280 + S304
Q11 + G255	T51 + L219	R87 + S383	G149 + G273	P211 + F289	S280 + R320

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Q11 + Q256	T51 + L228	R87 + Q385	G149 + S280	P211 + V291	S280 + H321
Q11 + A263	T51 + L235	R87 + K391	G149 + T285	P211 + Q299	S280 + S323
Q11 + V264	T51 + V238	R87 + K393	G149 + M286	P211 + S304	S280 + H324
Q11 + A265	T51 + M246	R87 + Q395	G149 + F289	P211 + R320	S280 + F328
Q11 + Y267	T51 + M248	Q98 + M105	G149 + V291	P211 + H321	S280 + T334
Q11 + N270	T51 + L250	Q98 + G109	G149 + Q299	P211 + S323	S280 + D337
Q11 + G273	T51 + G255	Q98 + F113	G149 + S304	P211 + H324	S280 + Q345
Q11 + S280	T51 + Q256	Q98 + R116	G149 + R320	P211 + F328	S280 + G346
Q11 + T285	T51 + A263	Q98 + Q118	G149 + H321	P211 + T334	S280 + G348
Q11 + M286	T51 + V264	Q98 + Q125	G149 + S323	P211 + D337	S280 + T355
Q11 + F289	T51 + A265	Q98 + G133	G149 + H324	P211 + Q345	S280 + S376
Q11 + V291	T51 + Y267	Q98 + T134	G149 + F328	P211 + G346	S280 + D377
Q11 + Q299	T51 + N270	Q98 + W140	G149 + T334	P211 + G348	S280 + D379
Q11 + S304	T51 + G273	Q98 + G142	G149 + D337	P211 + T355	S280 + Y382
Q11 + R320	T51 + S280	Q98 + G149	G149 + Q345	P211 + S376	S280 + S383
Q11 + H321	T51 + T285	Q98 + T165	G149 + G346	P211 + D377	S280 + Q385
Q11 + S323	T51 + M286	Q98 + W167	G149 + G348	P211 + D379	S280 + K391
Q11 + H324	T51 + F289	Q98 + R171	G149 + T355	P211 + Y382	S280 + K393
Q11 + F328	T51 + V291	Q98 + Q172	G149 + S376	P211 + S383	S280 + Q395
Q11 + T334	T51 + Q299	Q98 + L173	G149 + D377	P211 + Q385	T285 + M286
Q11 + D337	T51 + S304	Q98 + A174	G149 + D379	P211 + K391	T285 + F289
Q11 + Q345	T51 + R320	Q98 + G184	G149 + Y382	P211 + K393	T285 + V291
Q11 + G346	T51 + H321	Q98 + T193	G149 + S383	P211 + Q395	T285 + Q299
Q11 + G348	T51 + S323	Q98 + N195	G149 + Q385	I214 + V215	T285 + S304
Q11 + T355	T51 + H324	Q98 + G196	G149 + K391	I214 + L217	T285 + R320
Q11 + S376	T51 + F328	Q98 + A204	G149 + K393	I214 + L219	T285 + H321
Q11 + D377	T51 + T334	Q98 + V206	G149 + Q395	I214 + L228	T285 + S323
Q11 + D379	T51 + D337	Q98 + P211	T165 + W167	I214 + L235	T285 + H324
Q11 + Y382	T51 + Q345	Q98 + I214	T165 + R171	I214 + V238	T285 + F328
Q11 + S383	T51 + G346	Q98 + V215	T165 + Q172	I214 + M246	T285 + T334
Q11 + Q385	T51 + G348	Q98 + L217	T165 + L173	I214 + M248	T285 + D337
Q11 + K391	T51 + T355	Q98 + L219	T165 + A174	I214 + L250	T285 + Q345
Q11 + K393	T51 + S376	Q98 + L228	T165 + G184	I214 + G255	T285 + G346
Q11 + Q395	T51 + D377	Q98 + L235	T165 + T193	I214 + Q256	T285 + G348
N16 + V17	T51 + D379	Q98 + V238	T165 + N195	I214 + A263	T285 + T355
N16 + Q32	T51 + Y382	Q98 + M246	T165 + G196	I214 + V264	T285 + S376
N16 + A37	T51 + S383	Q98 + M248	T165 + A204	I214 + A265	T285 + D377
N16 + T40	T51 + Q385	Q98 + L250	T165 + V206	I214 + Y267	T285 + D379
N16 + P45	T51 + K391	Q98 + G255	T165 + P211	I214 + N270	T285 + Y382
N16 + W48	T51 + K393	Q98 + Q256	T165 + I214	I214 + G273	T285 + S383
N16 + G50	T51 + Q395	Q98 + A263	T165 + V215	I214 + S280	T285 + Q385
N16 + T51	N54 + V56	Q98 + V264	T165 + L217	I214 + T285	T285 + K391
N16 + N54	N54 + A60	Q98 + A265	T165 + L219	I214 + M286	T285 + K393
N16 + V56	N54 + K72	Q98 + Y267	T165 + L228	I214 + F289	T285 + Q395
N16 + A60	N54 + R87	Q98 + N270	T165 + L235	I214 + V291	M286 + F289
N16 + K72	N54 + Q98	Q98 + G273	T165 + V238	I214 + Q299	M286 + V291
N16 + R87	N54 + M105	Q98 + S280	T165 + M246	I214 + S304	M286 + Q299
N16 + Q98	N54 + G109	Q98 + T285	T165 + M248	I214 + R320	M286 + S304
N16 + M105	N54 + F113	Q98 + M286	T165 + L250	I214 + H321	M286 + R320
N16 + G109	N54 + R116	Q98 + F289	T165 + G255	I214 + S323	M286 + H321
N16 + F113	N54 + Q118	Q98 + V291	T165 + Q256	I214 + H324	M286 + S323
N16 + R116	N54 + Q125	Q98 + Q299	T165 + A263	I214 + F328	M286 + H324
N16 + Q118	N54 + G133	Q98 + S304	T165 + V264	I214 + T334	M286 + F328
N16 + Q125	N54 + T134	Q98 + R320	T165 + A265	I214 + D337	M286 + T334
N16 + G133	N54 + W140	Q98 + H321	T165 + Y267	I214 + Q345	M286 + D337
N16 + T134	N54 + G142	Q98 + S323	T165 + N270	I214 + G346	M286 + Q345
N16 + W140	N54 + G149	Q98 + H324	T165 + G273	I214 + G348	M286 + G346
N16 + G142	N54 + T165	Q98 + F328	T165 + S280	I214 + T355	M286 + G348
N16 + G149	N54 + W167	Q98 + T334	T165 + T285	I214 + S376	M286 + T355
N16 + T165	N54 + R171	Q98 + D337	T165 + M286	I214 + D377	M286 + S376
N16 + W167	N54 + Q172	Q98 + Q345	T165 + F289	I214 + D379	M286 + D377
N16 + R171	N54 + L173	Q98 + G346	T165 + V291	I214 + Y382	M286 + D379
N16 + Q172	N54 + A174	Q98 + G348	T165 + Q299	I214 + S383	M286 + Y382
N16 + L173	N54 + G184	Q98 + T355	T165 + S304	I214 + Q385	M286 + S383
N16 + A174	N54 + T193	Q98 + S376	T165 + R320	I214 + K391	M286 + Q385
N16 + G184	N54 + N195	Q98 + D377	T165 + H321	I214 + K393	M286 + K391
N16 + T193	N54 + G196	Q98 + D379	T165 + S323	I214 + Q395	M286 + K393
N16 + N195	N54 + A204	Q98 + Y382	T165 + H324	V215 + L217	M286 + Q395
N16 + G196	N54 + V206	Q98 + S383	T165 + F328	V215 + L219	F289 + V291
N16 + A204	N54 + P211	Q98 + Q385	T165 + T334	V215 + L228	F289 + Q299
N16 + V206	N54 + I214	Q98 + K391	T165 + D337	V215 + L235	F289 + S304
N16 + P211	N54 + V215	Q98 + K393	T165 + Q345	V215 + V238	F289 + R320
N16 + I214	N54 + L217	Q98 + Q395	T165 + G346	V215 + M246	F289 + H321
N16 + V215	N54 + L219	M105 + G109	T165 + G348	V215 + M248	F289 + S323
N16 + L217	N54 + L228	M105 + F113	T165 + T355	V215 + L250	F289 + H324
N16 + L219	N54 + L235	M105 + R116	T165 + S376	V215 + G255	F289 + F328
N16 + L228	N54 + V238	M105 + Q118	T165 + D377	V215 + Q256	F289 + T334
N16 + L235	N54 + M246	M105 + Q125	T165 + D379	V215 + A263	F289 + D337
N16 + V238	N54 + M248	M105 + G133	T165 + Y382	V215 + V264	F289 + Q345

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N16 + M246	N54 + L250	M105 + T134	T165 + S383	V215 + A265	F289 + G346
N16 + M248	N54 + G255	M105 + W140	T165 + Q385	V215 + Y267	F289 + G348
N16 + L250	N54 + Q256	M105 + G142	T165 + K391	V215 + N270	F289 + T355
N16 + G255	N54 + A263	M105 + G149	T165 + K393	V215 + G273	F289 + S376
N16 + Q256	N54 + V264	M105 + T165	T165 + Q395	V215 + S280	F289 + D377
N16 + A263	N54 + A265	M105 + W167	W167 + R171	V215 + T285	F289 + D379
N16 + V264	N54 + Y267	M105 + R171	W167 + Q172	V215 + M286	F289 + Y382
N16 + A265	N54 + N270	M105 + Q172	W167 + L173	V215 + F289	F289 + S383
N16 + Y267	N54 + G273	M105 + L173	W167 + A174	V215 + V291	F289 + Q385
N16 + N270	N54 + S280	M105 + A174	W167 + G184	V215 + Q299	F289 + K391
N16 + G273	Q118 + V215	M105 + G184	W167 + T193	V215 + S304	F289 + K393
N16 + S280	Q118 + L217	M105 + T193	W167 + N195	V215 + R320	F289 + Q395
N16 + T285	Q118 + L219	M105 + N195	W167 + G196	V215 + H321	V291 + Q299
N16 + M286	Q118 + L228	M105 + G196	W167 + A204	V215 + S323	V291 + S304
N16 + F289	Q118 + L235	M105 + A204	W167 + V206	V215 + H324	V291 + R320
N16 + V291	Q118 + V238	M105 + V206	W167 + P211	V215 + F328	V291 + H321
N16 + Q299	Q118 + M246	M105 + P211	W167 + I214	V215 + T334	V291 + S323
N16 + S304	Q118 + M248	M105 + I214	W167 + V215	V215 + D337	V291 + H324
N16 + R320	Q118 + L250	M105 + V215	W167 + L217	V215 + Q345	V291 + F328
N16 + H321	Q118 + G255	M105 + L217	W167 + L219	V215 + G346	V291 + T334
N16 + S323	Q118 + Q256	M105 + L219	W167 + L228	V215 + G348	V291 + D337
N16 + H324	Q118 + A263	M105 + L228	W167 + L235	V215 + T355	V291 + Q345
N16 + F328	Q118 + V264	M105 + L235	W167 + V238	V215 + S376	V291 + G346
N16 + T334	Q118 + A265	M105 + V238	W167 + M246	V215 + D377	V291 + G348
N16 + D337	Q118 + Y267	M105 + M246	W167 + M248	V215 + D379	V291 + T355
N16 + Q345	Q118 + N270	M105 + M248	W167 + L250	V215 + Y382	V291 + S376
N16 + G346	Q118 + G273	M105 + L250	W167 + G255	V215 + S383	V291 + D377
N16 + G348	Q118 + S280	M105 + G255	W167 + Q256	V215 + Q385	V291 + D379
N16 + T355	Q118 + T285	M105 + Q256	W167 + A263	V215 + K391	V291 + Y382
N16 + S376	Q118 + M286	M105 + A263	W167 + V264	V215 + K393	V291 + S383
N16 + D377	Q118 + F289	M105 + V264	W167 + A265	V215 + Q395	V291 + Q385
N16 + D379	Q118 + V291	M105 + A265	W167 + Y267	L217 + L219	V291 + K391
N16 + Y382	Q118 + Q299	M105 + Y267	W167 + N270	L217 + L228	V291 + K393
N16 + S383	Q118 + S304	M105 + N270	W167 + G273	L217 + L235	V291 + Q395
N16 + Q385	Q118 + R320	M105 + G273	W167 + S280	L217 + V238	Q299 + S304
N16 + K391	Q118 + H321	M105 + S280	W167 + T285	L217 + M246	Q299 + R320
N16 + K393	Q118 + S323	M105 + T285	W167 + M286	L217 + M248	Q299 + H321
N16 + Q395	Q118 + H324	M105 + M286	W167 + F289	L217 + L250	Q299 + S323
V17 + Q32	Q118 + F328	M105 + F289	W167 + V291	L217 + G255	Q299 + H324
V17 + A37	Q118 + T334	M105 + V291	W167 + Q299	L217 + Q256	Q299 + F328
V17 + T40	Q118 + D337	M105 + Q299	W167 + S304	L217 + A263	Q299 + T334
V17 + P45	Q118 + Q345	M105 + S304	W167 + R320	L217 + V264	Q299 + D337
V17 + W48	Q118 + G346	M105 + R320	W167 + H321	L217 + A265	Q299 + Q345
V17 + G50	Q118 + G348	M105 + H321	W167 + S323	L217 + Y267	Q299 + G346
V17 + T51	Q118 + T355	M105 + S323	W167 + H324	L217 + N270	Q299 + G348
V17 + N54	Q118 + S376	M105 + H324	W167 + F328	L217 + G273	Q299 + T355
V17 + V56	Q118 + D377	M105 + F328	W167 + T334	L217 + S280	Q299 + S376
V17 + A60	Q118 + D379	M105 + T334	W167 + D337	L217 + T285	Q299 + D377
V17 + K72	Q118 + Y382	M105 + D337	W167 + Q345	L217 + M286	Q299 + D379
V17 + R87	Q118 + S383	M105 + Q345	W167 + G346	L217 + F289	Q299 + Y382
V17 + Q98	Q118 + Q385	M105 + G346	W167 + G348	L217 + V291	Q299 + S383
V17 + M105	Q118 + K391	M105 + G348	W167 + T355	L217 + Q299	Q299 + Q385
V17 + G109	Q118 + K393	M105 + T355	W167 + S376	L217 + S304	Q299 + K391
V17 + F113	Q118 + Q395	M105 + S376	W167 + D377	L217 + R320	Q299 + K393
V17 + R116	Q125 + G133	M105 + D377	W167 + D379	L217 + H321	Q299 + Q395
V17 + Q118	Q125 + T134	M105 + D379	W167 + Y382	L217 + S323	S304 + R320
V17 + Q125	Q125 + W140	M105 + Y382	W167 + S383	L217 + H324	S304 + H321
V17 + G133	Q125 + G142	M105 + S383	W167 + Q385	L217 + F328	S304 + S323
V17 + T134	Q125 + G149	M105 + Q385	W167 + K391	L217 + T334	S304 + H324
V17 + W140	Q125 + T165	M105 + K391	W167 + K393	L217 + D337	S304 + F328
V17 + G142	Q125 + W167	M105 + K393	W167 + Q395	L217 + Q345	S304 + T334
V17 + G149	Q125 + R171	M105 + Q395	R171 + Q172	L217 + G346	S304 + D337
V17 + T165	Q125 + Q172	G109 + F113	R171 + L173	L217 + G348	S304 + Q345
V17 + W167	Q125 + L173	G109 + R116	R171 + A174	L217 + T355	S304 + G346
V17 + R171	Q125 + A174	G109 + Q118	R171 + G184	L217 + S376	S304 + G348
V17 + Q172	Q125 + G184	G109 + Q125	R171 + T193	L217 + D377	S304 + T355
V17 + L173	Q125 + T193	G109 + G133	R171 + N195	L217 + D379	S304 + S376
V17 + A174	Q125 + N195	G109 + T134	R171 + G196	L217 + Y382	S304 + D377
V17 + G184	Q125 + G196	G109 + W140	R171 + A204	L217 + S383	S304 + D379
V17 + T193	Q125 + A204	G109 + G142	R171 + V206	L217 + Q385	S304 + Y382
V17 + N195	Q125 + V206	G109 + G149	R171 + P211	L217 + K391	S304 + S383
V17 + G196	Q125 + P211	G109 + T165	R171 + I214	L217 + K393	S304 + Q385
V17 + A204	Q125 + I214	G109 + W167	R171 + V215	L217 + Q395	S304 + K391
V17 + V206	Q125 + V215	G109 + R171	R171 + L217	L219 + L228	S304 + K393
V17 + P211	Q125 + L217	G109 + Q172	R171 + L219	L219 + L235	S304 + Q395
V17 + I214	Q125 + L219	G109 + L173	R171 + L228	L219 + V238	R320 + H321
V17 + V215	Q125 + L228	G109 + A174	R171 + L235	L219 + M246	R320 + S323
V17 + L217	Q125 + L235	G109 + G184	R171 + V238	L219 + M248	R320 + H324
V17 + L219	Q125 + V238	G109 + T193	R171 + M246	L219 + L250	R320 + F328

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V17 + L228	Q125 + M246	G109 + N195	R171 + M248	L219 + G255	R320 + T334
V17 + L235	Q125 + M248	G109 + G196	R171 + L250	L219 + Q256	R320 + D337
V17 + V238	Q125 + L250	G109 + A204	R171 + G255	L219 + A263	R320 + Q345
V17 + M246	A37 + Q395	G109 + V206	R171 + Q256	L219 + V264	R320 + G346
V17 + M248	T40 + P45	G109 + P211	R171 + A263	L219 + A265	R320 + G348
V17 + L250	T40 + W48	G109 + I214	R171 + V264	L219 + Y267	R320 + T355
V17 + G255	T40 + G50	G109 + V215	R171 + A265	L219 + N270	R320 + S376
V17 + Q256	T40 + T51	G109 + L217	R171 + Y267	L219 + G273	R320 + D377
V17 + A263	T40 + N54	G109 + L219	R171 + N270	L219 + S280	R320 + D379
V17 + V264	T40 + V56	G109 + L228	R171 + G273	L219 + T285	R320 + Y382
V17 + A265	T40 + A60	G109 + L235	R171 + S280	L219 + M286	R320 + S383
V17 + Y267	T40 + K72	G109 + V238	R171 + T285	L219 + F289	R320 + Q385
V17 + N270	T40 + R87	G109 + M246	R171 + M286	L219 + V291	R320 + K391
V17 + G273	T40 + Q98	G109 + M248	R171 + F289	L219 + Q299	R320 + K393
V17 + S280	T40 + M105	G109 + L250	R171 + V291	L219 + S304	R320 + Q395
V17 + T285	T40 + G109	G109 + G255	R171 + Q299	L219 + R320	H321 + S323
V17 + M286	T40 + F113	G109 + Q256	R171 + S304	L219 + H321	H321 + H324
V17 + F289	T40 + R116	G109 + A263	R171 + R320	L219 + S323	H321 + F328
V17 + V291	T40 + Q118	G109 + V264	R171 + H321	L219 + H324	H321 + T334
V17 + Q299	T40 + Q125	G109 + A265	R171 + S323	L219 + F328	H321 + D337
V17 + S304	T40 + G133	G109 + Y267	R171 + H324	L219 + T334	H321 + Q345
V17 + R320	T40 + T134	G109 + N270	R171 + F328	L219 + D337	H321 + G346
V17 + H321	T40 + W140	G109 + G273	R171 + T334	L219 + Q345	H321 + G348
V17 + S323	T40 + G142	G109 + S280	R171 + D337	L219 + G346	H321 + T355
V17 + H324	T40 + G149	G109 + T285	R171 + Q345	L219 + G348	H321 + S376
V17 + F328	T40 + T165	G109 + M286	R171 + G346	L219 + T355	H321 + D377
V17 + T334	T40 + W167	G109 + F289	R171 + G348	L219 + S376	H321 + D379
V17 + D337	T40 + R171	G109 + V291	R171 + T355	L219 + D377	H321 + Y382
V17 + Q345	T40 + Q172	G109 + Q299	R171 + S376	L219 + D379	H321 + S383
V17 + G346	T40 + L173	G109 + S304	R171 + D377	L219 + Y382	H321 + Q385
V17 + G348	T40 + A174	G109 + R320	R171 + D379	L219 + S383	H321 + K391
V17 + T355	T40 + G184	G109 + H321	R171 + Y382	L219 + Q385	H321 + K393
V17 + S376	T40 + T193	G109 + S323	R171 + S383	L219 + K391	H321 + Q395
V17 + D377	T40 + N195	G109 + H324	R171 + Q385	L219 + K393	S323 + H324
V17 + D379	T40 + G196	G109 + F328	R171 + K391	L219 + Q395	S323 + F328
V17 + Y382	T40 + A204	G109 + T334	R171 + K393	L228 + L235	S323 + T334
V17 + S383	T40 + V206	G109 + D337	R171 + Q395	L228 + V238	S323 + D337
V17 + Q385	T40 + P211	G109 + Q345	Q172 + L173	L228 + M246	S323 + Q345
V17 + K391	T40 + I214	G109 + G346	Q172 + A174	L228 + M248	S323 + G346
V17 + K393	T40 + V215	G109 + G348	Q172 + G184	L228 + L250	S323 + G348
V17 + Q395	H1 + Q169	G109 + T355	Q172 + T193	L228 + G255	S323 + T355
Q32 + A37	H1 + A186	G109 + S376	Q172 + N195	L228 + Q256	S323 + S376
Q32 + T40	H1 + E190	G109 + D377	Q172 + G196	L228 + A263	S323 + D377
Q32 + P45	H1 + A225	G109 + D379	Q172 + A204	L228 + V264	S323 + D379
Q32 + W48	H1 + K242	G109 + Y382	Q172 + V206	L228 + A265	S323 + Y382
Q32 + G50	H1 + S244	G109 + S383	Q172 + P211	L228 + Y267	S323 + S383
Q32 + T51	H1 + N260	G109 + Q385	Q172 + I214	L228 + N270	S323 + Q385
Q32 + N54	H1 + K269	G109 + K391	Q172 + V215	L228 + G273	S323 + K391
Q32 + V56	H1 + W284	G109 + K393	Q172 + L217	L228 + S280	S323 + K393
Q32 + A60	H1 + Y295	G109 + Q395	Q172 + L219	L228 + T285	S323 + Q395
Q32 + K72	H1 + V326	F113 + R116	Q172 + L228	L228 + M286	H324 + F328
Q32 + R87	T5 + Q169	F113 + Q118	Q172 + L235	L228 + F289	H324 + T334
Q32 + Q98	T5 + A186	F113 + Q125	Q172 + V238	L228 + V291	H324 + D337
Q32 + M105	T5 + E190	F113 + G133	Q172 + M246	L228 + Q299	H324 + Q345
Q32 + G109	T5 + A225	F113 + T134	Q172 + M248	L228 + S304	H324 + G346
Q32 + F113	T5 + K242	F113 + W140	Q172 + L250	L228 + R320	H324 + G348
Q32 + R116	T5 + S244	F113 + G142	Q172 + G255	L228 + H321	H324 + T355
Q32 + Q118	T5 + N260	F113 + G149	Q172 + Q256	L228 + S323	H324 + S376
Q32 + Q125	T5 + K269	F113 + T165	Q172 + A263	L228 + H324	H324 + D377
Q32 + G133	T5 + W284	F113 + W167	Q172 + V264	L228 + F328	H324 + D379
Q32 + T134	T5 + Y295	F113 + R171	Q172 + A265	L228 + T334	H324 + Y382
Q32 + W140	T5 + V326	F113 + Q172	Q172 + Y267	L228 + D337	H324 + S383
Q32 + G142	G7 + Q169	F113 + L173	Q172 + N270	L228 + Q345	H324 + Q385
Q32 + G149	G7 + A186	F113 + A174	Q172 + G273	L228 + G346	H324 + K391
Q32 + T165	G7 + E190	F113 + G184	Q172 + S280	L228 + G348	H324 + K393
Q32 + W167	G7 + A225	F113 + T193	Q172 + T285	L228 + T355	H324 + Q395
Q32 + R171	G7 + K242	F113 + N195	Q172 + M286	L228 + S376	F328 + T334
Q32 + Q172	G7 + S244	F113 + G196	Q172 + F289	L228 + D377	F328 + D337
Q32 + L173	G7 + N260	F113 + A204	Q172 + V291	L228 + D379	F328 + Q345
Q32 + A174	G7 + K269	F113 + V206	Q172 + Q299	L228 + Y382	F328 + G346
Q32 + G184	G7 + W284	F113 + P211	Q172 + S304	L228 + S383	F328 + G348
Q32 + T193	G7 + Y295	F113 + I214	Q172 + R320	L228 + Q385	F328 + T355
Q32 + N195	G7 + V326	F113 + V215	Q172 + H321	L228 + K391	F328 + S376
Q32 + G196	Q11 + Q169	F113 + L217	Q172 + S323	L228 + K393	F328 + D377
Q32 + A204	Q11 + A186	F113 + L219	Q172 + H324	L228 + Q395	F328 + D379
Q32 + V206	Q11 + E190	F113 + L228	Q172 + F328	L235 + V238	F328 + Y382
Q32 + P211	Q11 + A225	F113 + L235	Q172 + T334	L235 + M246	F328 + S383
Q32 + I214	Q11 + K242	F113 + V238	Q172 + D337	L235 + M248	F328 + Q385
Q32 + V215	Q11 + S244	F113 + M246	Q172 + Q345	L235 + L250	F328 + K391

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Q32 + L217	Q11 + N260	F113 + M248	Q172 + G346	L235 + G255	F328 + K393
Q32 + L219	Q11 + K269	F113 + L250	Q172 + G348	L235 + Q256	F328 + Q395
Q32 + L228	Q11 + W284	F113 + G255	Q172 + T355	L235 + A263	T334 + D337
Q32 + L235	Q11 + Y295	F113 + Q256	Q172 + S376	L235 + V264	T334 + Q345
Q32 + V238	Q11 + V326	F113 + A263	Q172 + D377	L235 + A265	T334 + G346
Q32 + M246	N16 + Q169	F113 + V264	Q172 + D379	L235 + Y267	T334 + G348
Q32 + M248	N16 + A186	F113 + A265	Q172 + Y382	L235 + N270	T334 + T355
Q32 + L250	N16 + E190	F113 + Y267	Q172 + S383	L235 + G273	T334 + S376
Q32 + G255	N16 + A225	F113 + N270	Q172 + Q385	L235 + S280	T334 + D377
Q32 + Q256	N16 + K242	F113 + G273	Q172 + K391	L235 + T285	T334 + D379
Q32 + A263	N16 + S244	F113 + S280	Q172 + K393	L235 + M286	T334 + Y382
Q32 + V264	N16 + N260	F113 + T285	Q172 + Q395	L235 + F289	T334 + S383
Q32 + A265	N16 + K269	F113 + M286	L173 + A174	L235 + V291	T334 + Q385
Q32 + Y267	N16 + W284	F113 + F289	L173 + G184	L235 + Q299	T334 + K391
Q32 + N270	N16 + Y295	F113 + V291	L173 + T193	L235 + S304	T334 + K393
Q32 + G273	N16 + V326	F113 + Q299	L173 + N195	L235 + R320	T334 + Q395
Q32 + S280	V17 + Q169	F113 + S304	L173 + G196	L235 + H321	D337 + Q345
Q32 + T285	V17 + A186	F113 + R320	L173 + A204	L235 + S323	D337 + G346
Q32 + M286	V17 + E190	F113 + H321	L173 + V206	L235 + H324	D337 + G348
Q32 + F289	V17 + A225	F113 + S323	L173 + P211	L235 + F328	D337 + T355
Q32 + V291	V17 + K242	F113 + H324	L173 + I214	L235 + T334	D337 + S376
Q32 + Q299	V17 + S244	F113 + F328	L173 + V215	L235 + D337	D337 + D377
Q32 + S304	V17 + N260	F113 + T334	L173 + L217	L235 + Q345	D337 + D379
Q32 + R320	V17 + K269	F113 + D337	L173 + L219	L235 + G346	D337 + Y382
Q32 + H321	V17 + W284	F113 + Q345	L173 + L228	L235 + G348	D337 + S383
Q32 + S323	V17 + Y295	F113 + G346	L173 + L235	L235 + T355	D337 + Q385
Q32 + H324	V17 + V326	F113 + G348	L173 + V238	L235 + S376	D337 + K391
Q32 + F328	Q32 + Q169	F113 + T355	L173 + M246	L235 + D377	D337 + K393
Q32 + T334	Q32 + A186	F113 + S376	L173 + M248	L235 + D379	D337 + Q395
Q32 + D337	Q32 + E190	F113 + D377	L173 + L250	L235 + Y382	Q345 + G346
Q32 + Q345	Q32 + A225	F113 + D379	L173 + G255	L235 + S383	Q345 + G348
Q32 + G346	Q32 + K242	F113 + Y382	L173 + Q256	L235 + Q385	Q345 + T355
Q32 + G348	Q32 + S244	F113 + S383	L173 + A263	L235 + K391	Q345 + S376
Q32 + T355	Q32 + N260	F113 + Q385	L173 + V264	L235 + K393	Q345 + D377
Q32 + S376	Q32 + K269	F113 + K391	L173 + A265	L235 + Q395	Q345 + D379
Q32 + D377	Q32 + W284	F113 + K393	L173 + Y267	V238 + M246	Q345 + Y382
Q32 + D379	Q32 + Y295	F113 + Q395	L173 + N270	V238 + M248	Q345 + S383
Q32 + Y382	Q32 + V326	R116 + Q118	L173 + G273	V238 + L250	Q345 + Q385
Q32 + S383	A37 + Q169	R116 + Q125	L173 + S280	V238 + G255	Q345 + K391
Q32 + Q385	A37 + A186	R116 + G133	L173 + T285	V238 + Q256	Q345 + K393
Q32 + K391	A37 + E190	R116 + T134	L173 + M286	V238 + A263	Q345 + Q395
Q32 + K393	A37 + A225	R116 + W140	L173 + F289	V238 + V264	G346 + G348
Q32 + Q395	A37 + K242	R116 + G142	L173 + V291	V238 + A265	G346 + T355
A37 + T40	A37 + S244	R116 + G149	L173 + Q299	V238 + Y267	G346 + S376
A37 + P45	A37 + N260	R116 + T165	L173 + S304	V238 + N270	G346 + D377
A37 + W48	A37 + K269	R116 + W167	L173 + R320	V238 + G273	G346 + D379
A37 + G50	A37 + W284	R116 + R171	L173 + H321	V238 + S280	G346 + Y382
A37 + T51	A37 + Y295	R116 + Q172	L173 + S323	V238 + T285	G346 + S383
A37 + N54	A37 + V326	R116 + L173	L173 + H324	V238 + M286	G346 + Q385
A37 + V56	T40 + Q169	R116 + A174	L173 + F328	V238 + F289	G346 + K391
A37 + A60	T40 + A186	R116 + G184	L173 + T334	V238 + V291	G346 + K393
A37 + K72	T40 + E190	R116 + T193	L173 + D337	V238 + Q299	G346 + Q395
A37 + R87	T40 + A225	R116 + N195	L173 + Q345	V238 + S304	G348 + T355
A37 + Q98	T40 + K242	R116 + G196	L173 + G346	V238 + R320	G348 + S376
A37 + M105	T40 + S244	R116 + A204	L173 + G348	V238 + H321	G348 + D377
A37 + G109	T40 + N260	R116 + V206	L173 + T355	V238 + S323	G348 + D379
A37 + F113	T40 + K269	R116 + P211	L173 + S376	V238 + H324	G348 + Y382
A37 + R116	T40 + W284	R116 + I214	L173 + D377	V238 + F328	G348 + S383
A37 + Q118	T40 + Y295	R116 + V215	L173 + D379	V238 + T334	G348 + Q385
A37 + Q125	T40 + V326	R116 + L217	L173 + Y382	V238 + D337	G348 + K391
A37 + G133	P45 + Q169	R116 + L219	L173 + S383	V238 + Q345	G348 + K393
A37 + T134	P45 + A186	R116 + L228	L173 + Q385	V238 + G346	G348 + Q395
A37 + W140	P45 + E190	R116 + L235	L173 + K391	V238 + G348	T355 + S376
A37 + G142	P45 + A225	R116 + V238	L173 + K393	V238 + T355	T355 + D377
A37 + G149	P45 + K242	R116 + M246	L173 + Q395	V238 + S376	T355 + D379
A37 + T165	P45 + S244	R116 + M248	A174 + G184	V238 + D377	T355 + Y382
A37 + W167	P45 + N260	R116 + L250	A174 + T193	V238 + D379	T355 + S383
A37 + R171	P45 + K269	R116 + G255	A174 + N195	V238 + Y382	T355 + Q385
A37 + Q172	P45 + W284	R116 + Q256	A174 + G196	V238 + S383	T355 + K391
A37 + L173	P45 + Y295	R116 + A263	A174 + A204	V238 + Q385	T355 + K393
A37 + A174	P45 + V326	R116 + V264	A174 + V206	V238 + K391	T355 + Q395
A37 + G184	W48 + Q169	R116 + A265	A174 + P211	V238 + K393	S376 + D377
A37 + T193	W48 + A186	R116 + Y267	A174 + I214	V238 + Q395	S376 + D379
A37 + N195	W48 + E190	R116 + N270	A174 + V215	M246 + M248	S376 + Y382
A37 + G196	W48 + A225	R116 + G273	A174 + L217	M246 + L250	S376 + S383
A37 + A204	W48 + K242	R116 + S280	A174 + L219	M246 + G255	S376 + Q385
A37 + V206	W48 + S244	R116 + T285	A174 + L228	M246 + Q256	S376 + K391
A37 + P211	W48 + N260	R116 + M286	A174 + L235	M246 + A263	S376 + K393
A37 + I214	W48 + K269	R116 + F289	A174 + V238	M246 + V264	S376 + Q395

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A37 + V215	W48 + W284	R116 + V291	A174 + M246	M246 + A265	D377 + D379
A37 + L217	W48 + Y295	R116 + Q299	A174 + M248	M246 + Y267	D377 + Y382
A37 + L219	W48 + V326	R116 + S304	A174 + L250	M246 + N270	D377 + S383
A37 + L228	G50 + Q169	R116 + R320	A174 + G255	M246 + G273	D377 + Q385
A37 + L235	G50 + A186	R116 + H321	A174 + Q256	M246 + S280	D377 + K391
A37 + V238	G50 + E190	R116 + S323	A174 + A263	M246 + T285	D377 + K393
A37 + M246	G50 + A225	R116 + H324	A174 + V264	M246 + M286	D377 + Q395
A37 + M248	G50 + K242	R116 + F328	A174 + A265	M246 + F289	D379 + Y382
A37 + L250	G50 + S244	R116 + T334	A174 + Y267	M246 + V291	D379 + S383
A37 + G255	G50 + N260	R116 + D337	A174 + N270	M246 + Q299	D379 + Q385
A37 + Q256	G50 + K269	R116 + Q345	A174 + G273	M246 + S304	D379 + K391
A37 + A263	G50 + W284	R116 + G346	A174 + S280	M246 + R320	D379 + K393
A37 + V264	G50 + Y295	R116 + G348	A174 + T285	M246 + H321	D379 + Q395
A37 + A265	G50 + V326	R116 + T355	A174 + M286	M246 + S323	Y382 + S383
A37 + Y267	T51 + Q169	R116 + S376	A174 + F289	M246 + H324	Y382 + Q385
A37 + N270	T51 + A186	R116 + D377	A174 + V291	M246 + F328	Y382 + K391
A37 + G273	T51 + E190	R116 + D379	A174 + Q299	M246 + T334	Y382 + K393
A37 + S280	T51 + A225	R116 + Y382	A174 + S304	M246 + D337	Y382 + Q395
A37 + T285	T51 + K242	R116 + S383	A174 + R320	M246 + Q345	S383 + Q385
A37 + M286	T51 + S244	R116 + Q385	A174 + H321	M246 + G346	S383 + K391
A37 + F289	T51 + N260	R116 + K391	A174 + S323	M246 + G348	S383 + K393
A37 + V291	T51 + K269	R116 + K393	A174 + H324	M246 + T355	S383 + Q395
A37 + Q299	T51 + W284	R116 + Q395	A174 + F328	M246 + S376	Q385 + K391
A37 + S304	T51 + Y295	Q118 + Q125	A174 + T334	M246 + D377	Q385 + K393
A37 + R320	T51 + V326	Q118 + G133	A174 + D337	M246 + D379	Q385 + Q395
A37 + H321	N54 + Q169	Q118 + T134	A174 + Q345	M246 + Y382	K391 + K393
A37 + S323	N54 + A186	Q118 + W140	A174 + G346	M246 + S383	K391 + Q395
A37 + H324	N54 + E190	Q118 + G142	A174 + G348	M246 + Q385	K393 + Q395
A37 + F328	N54 + A225	Q118 + G149	A174 + T355	M246 + K391	M248 + F289
A37 + T334	N54 + K242	Q118 + T165	A174 + S376	M246 + K393	M248 + V291
A37 + D337	N54 + S244	Q118 + W167	A174 + D377	M246 + Q395	G109 + Q169
A37 + Q345	N54 + N260	Q118 + R171	A174 + D379	M248 + L250	G109 + A186
A37 + G346	N54 + K269	Q118 + Q172	A174 + Y382	M248 + G255	G109 + E190
A37 + G348	N54 + W284	Q118 + L173	A174 + S383	M248 + Q256	G109 + A225
A37 + T355	N54 + Y295	Q118 + A174	A174 + Q385	M248 + A263	G109 + K242
A37 + S376	N54 + V326	Q118 + G184	A174 + K391	M248 + V264	G109 + S244
A37 + D377	V56 + Q169	Q118 + T193	A174 + K393	M248 + A265	G109 + N260
A37 + D379	V56 + A186	Q118 + N195	A174 + Q395	M248 + Y267	G109 + K269
A37 + Y382	V56 + E190	Q118 + G196	G184 + T193	M248 + N270	G109 + W284
A37 + S383	V56 + A225	Q118 + A204	G184 + N195	M248 + G273	G109 + Y295
A37 + Q385	V56 + K242	Q118 + V206	G184 + G196	M248 + S280	G109 + V326
A37 + K391	V56 + S244	Q118 + P211	G184 + A204	M248 + T285	F113 + Q169
A37 + K393	V56 + N260	Q118 + I214	G184 + V206	M248 + M286	F113 + A186
A60 + Q169	V56 + K269	G133 + Q169	W167 + Q169	R171 + A186	F113 + A225
A60 + A186	V56 + W284	G133 + A186	W167 + A186	R171 + E190	F113 + E190
A60 + E190	V56 + Y295	G133 + E190	W167 + E190	R171 + A225	F113 + K242
A60 + A225	V56 + V326	G133 + A225	W167 + A225	R171 + K242	F113 + S244
A60 + K242	G184 + A186	G133 + K242	W167 + K242	R171 + S244	F113 + N260
A60 + S244	G184 + E190	G133 + S244	W167 + S244	R171 + N260	F113 + K269
A60 + N260	G184 + A225	G133 + N260	W167 + N260	R171 + K269	F113 + W284
A60 + K269	G184 + K242	G133 + K269	W167 + K269	R171 + W284	F113 + Y295
A60 + W284	G184 + S244	G133 + W284	W167 + W284	R171 + Y295	F113 + V326
A60 + Y295	G184 + N260	G133 + Y295	W167 + Y295	R171 + V326	R116 + Q169
A60 + V326	G184 + K269	G133 + V326	W167 + V326	Q172 + A186	R116 + A186
K72 + Q169	G184 + W284	T134 + Q169	Q169 + A186	Q172 + E190	R116 + E190
K72 + A186	G184 + Y295	T134 + A186	Q169 + E190	Q172 + A225	R116 + A225
K72 + E190	G184 + V326	T134 + E190	Q169 + L173	Q172 + K242	R116 + K242
K72 + A225	A186 + E190	T134 + A225	Q169 + A174	Q172 + S244	R116 + S244
K72 + K242	A186 + T193	T134 + K242	Q169 + G184	Q172 + N260	R116 + N260
K72 + S244	A186 + N195	T134 + S244	Q169 + T193	Q172 + K269	R116 + K269
K72 + N260	A186 + G196	T134 + N260	Q169 + N195	Q172 + W284	R116 + W284
K72 + K269	A186 + A204	T134 + K269	Q169 + G196	Q172 + Y295	R116 + Y295
K72 + W284	A186 + V206	T134 + W284	Q169 + A204	Q172 + V326	R116 + V326
K72 + Y295	A186 + P211	T134 + Y295	Q169 + V206	L173 + A186	Q118 + Q169
K72 + V326	A186 + I214	T134 + V326	Q169 + P211	L173 + E190	Q118 + A186
R87 + Q169	A186 + V215	W140 + Q169	Q169 + I214	L173 + A225	Q118 + A225
R87 + A186	A186 + L217	W140 + A186	Q169 + V215	L173 + K242	Q118 + K242
R87 + E190	A186 + L219	W140 + E190	Q169 + L217	L173 + S244	Q118 + S244
R87 + A225	A186 + A225	W140 + A225	Q169 + L219	L173 + N260	Q118 + N260
R87 + K242	A186 + L228	W140 + K242	Q169 + A225	L173 + K269	Q118 + K269
R87 + S244	A186 + L235	W140 + S244	Q169 + L228	L173 + W284	Q118 + W284
R87 + N260	A186 + V238	W140 + N260	Q169 + L235	L173 + Y295	Q118 + Y295
R87 + K269	A186 + K242	W140 + K269	Q169 + V238	L173 + V326	Q118 + V326
R87 + W284	A186 + S244	W140 + W284	Q169 + K242	A174 + A186	Q125 + Q169
R87 + Y295	A186 + M246	W140 + Y295	Q169 + S244	A174 + E190	Q125 + A186
R87 + V326	A186 + M248	W140 + V326	Q169 + M246	A174 + A225	Q125 + E190
Q98 + Q169	A186 + L250	G142 + Q169	Q169 + M248	A174 + K242	Q125 + A225
Q98 + A186	A186 + G255	G142 + A186	Q169 + L250	A174 + S244	Q125 + K242
Q98 + E190	A186 + Q256	G142 + E190	Q169 + G255	A174 + N260	Q125 + S244

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Q98 + A225	A186 + N260	G142 + A225	Q169 + Q256	A174 + K269	Q125 + N260
Q98 + K242	A186 + A263	G142 + K242	Q169 + N260	A174 + W284	Q125 + K269
Q98 + S244	A186 + V264	G142 + S244	Q169 + A263	A174 + Y295	Q125 + W284
Q98 + N260	A186 + A265	G142 + N260	Q169 + V264	A174 + V326	Q125 + Y295
Q98 + K269	A186 + Y267	G142 + K269	Q169 + A265	T193 + A225	Q125 + V326
Q98 + W284	A186 + K269	G142 + W284	Q169 + Y267	T193 + K242	V215 + A225
Q98 + Y295	A186 + N270	G142 + Y295	Q169 + K269	T193 + S244	V215 + K242
Q98 + V326	A186 + G273	G142 + V326	Q169 + N270	T193 + N260	V215 + S244
M105 + Q169	A186 + S280	G149 + Q169	Q169 + G273	T193 + K269	V215 + N260
	A186 + W284		Q169 + S280	T193 + W284	V215 + K269
M105 + A186	A186 + T285	G149 + A186	Q169 + W284	T193 + Y295	V215 + W284
M105 + E190	A186 + M286	G149 + E190	Q169 + T285	T193 + V326	V215 + Y295
M105 + A225	A186 + F289	G149 + A225	Q169 + M286	N195 + A225	V215 + V326
M105 + K242	A186 + V291	G149 + K242	Q169 + F289	N195 + K242	L217 + A225
M105 + S244	A186 + Y295	G149 + S244	Q169 + V291	N195 + S244	L217 + K242
M105 + N260	A186 + Q299	G149 + N260	Q169 + Y295	N195 + N260	L217 + S244
M105 + K269	A186 + S304	G149 + K269	Q169 + Q299	N195 + K269	L217 + N260
M105 + W284	A186 + R320	G149 + W284	Q169 + S304	N195 + W284	L217 + K269
M105 + Y295	A186 + H321	G149 + Y295	Q169 + R320	N195 + Y295	L217 + W284
M105 + V326	A186 + S323	G149 + V326	Q169 + H321	N195 + V326	L217 + Y295
A225 + L228	A186 + H324	T165 + Q169	Q169 + S323	G196 + A225	L217 + V326
A225 + L235	A186 + V326	T165 + A186	Q169 + H324	G196 + K242	L219 + A225
A225 + V238	A186 + F328	T165 + E190	Q169 + V326	G196 + K242	L219 + K242
A225 + K242	A186 + T334	T165 + A225	Q169 + F328	G196 + S244	L219 + S244
A225 + S244	A186 + D337	T165 + K242	Q169 + T334	G196 + N260	L219 + N260
A225 + M246	A186 + Q345	T165 + S244	Q169 + D337	G196 + K269	L219 + K269
A225 + M248	A186 + G346	T165 + N260	Q169 + Q345	G196 + W284	L219 + W284
A225 + L250	A186 + G348	T165 + K269	Q169 + G346	G196 + Y295	L219 + Y295
A225 + G255	A186 + T355	T165 + W284	Q169 + G348	G196 + V326	L219 + V326
A225 + Q256	A186 + S376	T165 + Y295	Q169 + T355	A204 + A225	M246 + N260
A225 + N260	A186 + D377	T165 + V326	Q169 + S376	A204 + K242	M246 + K269
A225 + A263	A186 + D379	L228 + K242	Q169 + D377	A204 + S244	M246 + W284
A225 + V264	A186 + Y382	L228 + S244	Q169 + D379	A204 + N260	M246 + Y295
A225 + A265	A186 + S383	L228 + N260	Q169 + Y382	A204 + K269	M246 + V326
A225 + Y267	A186 + Q385	L228 + K269	Q169 + S383	A204 + W284	M248 + N260
A225 + K269	A186 + K391	L228 + W284	Q169 + Q385	A204 + Y295	M248 + K269
A225 + N270	A186 + K393	L228 + Y295	Q169 + K391	A204 + V326	M248 + W284
A225 + G273	A186 + Q395	L228 + V326	Q169 + K393	V206 + A225	M248 + Y295
A225 + S280	E190 + T193	L235 + K242	Q169 + Q395	A206 + K242	M248 + V326
A225 + W284	E190 + N195	L235 + S244	Q256 + N260	A206 + S244	L250 + N260
A225 + T285	E190 + G196	L235 + N260	G256 + K269	A206 + N260	L250 + K269
A225 + M286	E190 + A204	L235 + K269	Q256 + W284	A206 + K269	L250 + W284
A225 + F289	E190 + V206	L235 + W284	Q256 + Y295	A206 + W284	L250 + Y295
A225 + V291	E190 + P211	L235 + Y295	Q256 + V326	A206 + Y295	L250 + V326
A225 + Y295	E190 + I214	L235 + V326	N260 + A263	A206 + V326	G255 + N260
A225 + Q299	E190 + V215	V238 + K242	N260 + V264	P211 + A225	G255 + K269
A225 + S304	E190 + L217	V238 + S244	N260 + A265	P211 + K242	G255 + W284
A225 + R320	E190 + L219	V238 + N260	N260 + Y267	P211 + S244	G255 + Y295
A225 + H321	E190 + A225	V238 + K269	N260 + K269	P211 + N206	G255 + V326
A225 + S323	E190 + L228	V238 + W284	N260 + N270	P211 + K269	N270 + W284
A225 + H324	E190 + L235	V238 + Y295	N260 + G273	P211 + W284	N270 + Y295
A225 + V326	E190 + V238	V238 + V326	N260 + S280	P211 + Y295	N270 + V326
A225 + F328	E190 + K242	K242 + S244	N260 + W284	P211 + V326	G273 + W284
A225 + T334	E190 + S244	K242 + M246	N260 + T285	I214 + A225	G273 + Y295
A225 + D337	E190 + M246	K242 + M248	N260 + M286	I214 + K242	G273 + V326
A225 + Q345	E190 + M248	K242 + L250	N260 + F289	I214 + S244	S280 + W284
A225 + G346	E190 + L250	K242 + G255	N260 + V291	I214 + N260	S280 + Y295
A225 + G348	E190 + G255	K242 + Q256	N260 + Y295	I214 + K269	S280 + V326
A225 + T355	E190 + Q256	K242 + N260	N260 + Q299	I214 + W284	W284 + T285
A225 + S376	E190 + N260	K242 + A263	N260 + S304	I214 + Y295	W284 + M286
A225 + D377	E190 + A263	K242 + V264	N260 + R320	I214 + V326	W284 + F289
A225 + D379	E190 + V264	K242 + A265	N260 + H321	K269 + N270	W284 + V291
A225 + Y382	E190 + A265	K242 + Y267	N260 + S323	K269 + G273	W284 + Y295
A225 + S383	E190 + Y267	K242 + K269	N260 + H324	K269 + S280	W284 + Q299
A225 + Q385	E190 + K269	K242 + N270	N260 + V326	K269 + W284	W284 + S304
A225 + K391	E190 + N270	K242 + G273	N260 + F328	K269 + T285	W284 + R320
A225 + K393	E190 + G273	K242 + S280	N260 + T334	K269 + M286	W284 + H321
A225 + Q395	E190 + S280	K242 + W284	N260 + D337	K269 + F289	W284 + S323
A263 + K269	E190 + W284	K242 + T285	N260 + Q345	K269 + V291	W284 + H324
A263 + W284	E190 + T285	K242 + M286	N260 + G346	K269 + Y295	W284 + V326
A263 + Y295	E190 + M286	K242 + F289	N260 + G348	K269 + Q299	W284 + F328
A263 + V326	E190 + F289	K242 + V291	N260 + T355	K269 + S304	W284 + T334
V264 + K269	E190 + V291	K242 + Y295	N260 + S376	K269 + R320	W284 + D337
V264 + W284	E190 + Y295	K242 + Q299	N260 + D377	K269 + H321	W284 + Q345
V264 + Y295	E190 + Q299	K242 + S304	N260 + D379	K269 + S323	W284 + G346
V264 + V326	E190 + S304	K242 + R320	N260 + Y382	K269 + H324	W284 + G348
A265 + K269	E190 + R320	K242 + H321	N260 + S383	K269 + V326	W284 + T355
A265 + W284	E190 + H321	K242 + S323	N260 + Q385	K269 + F328	W284 + S376
A265 + Y295	E190 + S323	K242 + H324	N260 + K391	K269 + T334	W284 + D377

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A265 + V326	E190 + H324	K242 + V326	N260 + K393	K269 + D337	W284 + D379
A265 + K269	E190 + V326	K242 + F328	N260 + Q395	K269 + Q345	W284 + Y382
A265 + W284	E190 + F328	K242 + T334	Y295 + S376	K269 + G346	W284 + S383
A265 + Y295	E190 + T334	K242 + D337	V326 + S376	K269 + G348	W284 + Q385
A265 + V326	E190 + D337	K242 + Q345	V326 + D377	K269 + T355	W284 + K391
Y295 + D377	E190 + Q345	K242 + G346	V326 + D379	K269 + S376	W284 + K393
Y295 + D379	E190 + G346	K242 + G348	V326 + Y382	K269 + D377	W284 + Q395
Y295 + Y382	E190 + G348	K242 + T355	V326 + S383	K269 + D379	T285 + Y295
Y295 + S383	E190 + T355	K242 + S376	V326 + Q385	K269 + Y382	T285 + V326
Y295 + Q385	E190 + S376	K242 + D377	V326 + K391	K269 + S383	M286 + Y295
Y295 + K391	E190 + D377	K242 + D379	V326 + K393	K269 + Q385	M286 + V326
Y295 + K393	E190 + D379	K242 + Y382	V326 + Q395	K269 + K391	F289 + Y295
Y295 + Q395	E190 + Y382	K242 + S383	S244 + K393	K269 + K393	F289 + V326
Q299 + V326	E190 + S383	K242 + Q385	S244 + Q395	K269 + Q395	V291 + Y295
S304 + V326	E190 + Q385	K242 + K391	S244 + Y267	S244 + R320	V294 + V326
R320 + V326	E190 + K391	K242 + K393	S244 + K269	S244 + H321	Y295 + Q299
H321 + V326	E190 + K393	K242 + Q395	S244 + N270	S244 + S323	Y295 + S304
S323 + V326	E190 + Q395	S244 + M246	S244 + G273	S244 + H324	Y295 + R320
H324 + V326	S244 + F289	S244 + M248	S244 + S280	S244 + V326	Y295 + H321
V326 + F328	S244 + V291	S244 + L250	S244 + W284	S244 + F328	Y295 + S323
V326 + T334	S244 + Y295	S244 + G255	S244 + T285	S244 + T334	Y295 + H324
V326 + D337	S244 + Q299	S244 + Q256	S244 + S376	S244 + D337	Y295 + V326
V326 + Q345	S244 + S304	S244 + N260	S244 + D377	S244 + Q345	Y295 + F328
V326 + G346	S244 + K391	S244 + A263	S244 + D379	S244 + G346	Y295 + T334
V326 + G348	S244 + M286	S244 + V264	S244 + Y382	S244 + G348	Y295 + F328
V326 + T355	Y295 + Q345	S244 + A265	S244 + S383	S244 + T355	Y295 + T334
Y295 + G348	Y295 + G346		S244 + Q385		Y295 + D337
Y295 + T355					

In one particular embodiment, the modifications in one or more positions correspond to the following; H1*, T5K, G7K, G7A, Q11H, N16S, N16H, V17L, Q32S, A37H, A37M, A37V, A37S, A37Y, A37R, A37L, T40D, T40G, T40K, P45A, W48G, W48Y, W48F, G50A, T51K, T51E, T51 G, T51A, T51 S, T51 G, T51 D, N54S, V56T, A60V, K72R, K72H, K72S, K72Q, K72E, K72N, K72A, K72M, R87S, Q98S, Q98A, M105F, M105I, M105V, M105L, G109A, G109S, F113W, F113S, F113N, F113Y, F113R, F113L, F113Q, R116Q, R116V, R116K, R116W, R116L, R116A, R116H, R116M, R116E, R116S, R116I, R116G, Q118N, Q118K, Q118G, Q118S, Q118F, Q118R, Q125P, Q125K, Q125A, Q125T, N125D, G133S, F133Q, T134E, W140Y, G142T, G149Q, T165S, T165G, T165V, W167I, W167G, W167F, W167R, W167S, W167H, W167L, W167M, W167Y, Q169E, R171H, Q172G, Q172R, Q172N, Q172D, Q172Y, Q172M, Q172S, Q172T, Q172K, Q172H, Q172E, L173V, L173G, L173H, L173A, L173I, L173P, L173T, L173F, L173M, A174S, A740, A174P, A174M, A174T, A174H, A174K, A174G, A174Q, A174N, A174V, A174L, G184T, A186D, A186N, A186E, A186Q, A186H, E190P, T193K, N195F, G196R, A204T, A204V, A204S, A204G, V206L, V206S, V206Y, P211D, I214G, I214H, I214S, I214T, I214L, I214E, I214W, V215T, L217T, L217Q, L219V, L219H, A225V, L228I, L235V, L235A, V238A, V238T, V238G, K242Q, S244Q, M246L, M246A, M246I, M246F, M246V, M246S, M248T, L250I, L250V, L250T, L250A, L250F, L250M, G255N, G255A, G255S, N260G, A263G, V264I, V264T, A265G, Y267I, Y267M, Y267H, Y267L, K269Q, N270G, N270T, G273R, S280W, S280L, S280T, S280A, S280K, S280Q, W284H, T285L, T285Q, M286F, M286L, A288L, A288V, F289I, F289L, V291G, V291A, V291T, Y295N, Y295F, Q299V, S304N, S304R, R320A, R320V, H321Y, S323N, H324L, V326L, F328L, F328M, F328V, F328I, T334S, D337H, Q345D, G346T, G346D, G346P, G348S, G348P, T355L, T355F, S376H, S376T, S376V, D377H, D377S, D377A, D377Q, D379S, D379G, D379R, D379A, Y382M, Y382I, Y382L, Y382F, S383G, S383A, Q385L, K391A, K391Y, K391V, K391M, K391E, K391D, K391R, K391H, K391W, K391I,

K391Q, K391L, K393Y, K393R, K393Q, K393S, and Q395P, wherein said positions correspond to positions of SEQ ID NOs: 7 or 10.

The inventors of the present invention have identified that these specific alterations within the A and B domain (as defined by SEQ ID NO: 10) of the amino acid sequence as set forth in SEQ ID NOs: 7 and 10, are particularly relevant for improving the performance of a variant alpha-amylase having at least 65% sequence identity to the parent polypeptide.

In one embodiment, the variant comprises an alteration in two, three, four, five, six, seven, eight, nine, ten, eleven, twelfth, thirteen, fourteen, fifteen, sixteen, or seventeen positions corresponding to the following positions; H1*, T5K, G7K, G7A, Q11H, N16S, N16H, V17L, Q32S, A37H, A37M, A37V, A37S, A37Y, A37R, A37L, T40D, T40G, T40K, P45A, W48G, W48Y, W48F, G50A, T51K, T51E, T51G, T51A, T51S, T51G, T51D, N54S, V56T, A60V, K72R, K72H, K72S, K72Q, K72E, K72N, K72A, K72M, R87S, Q98S, Q98A, M105F, M105I, M105V, M105L, G109A, G109S, F113W, F113S, F113N, F113Y, F113R, F113L, F113Q, R116Q, R116V, R116K, R116W, R116L, R116A, R116H, R116M, R116E, R116S, R116I, R116G, Q118N, Q118K, Q118G, Q118S, Q118F, Q118R, Q125P, Q125K, Q125A, Q125T, N125D, G133S, F133Q, T134E, W140Y, G142T, G149Q, T165S, T165G, T165V, W167I, W167G, W167F, W167R, W167S, W167H, W167L, W167M, W167Y, Q169E, R171H, Q172G, Q172R, Q172N, Q172D, Q172Y, Q172M, Q172S, Q172T, Q172K, Q172H, Q172E, L173V, L173G, L173H, L173A, L173I, L173P, L173T, L173F, L173M, A174S, A174D, A174P, A174M, A174T, A174H, A174K, A174G, A174Q, A174N, A174V, A174L, G184T, A186D, A186N, A186E, A186Q, A186H, E190P, T193K, N195F, G196R, A204T, A204V, A204S, A204G, V206L, V206S, V206Y, P211D, I214G, I214H, I214S, I214T, I214L, I214E, I214W, V215T, L217T, L217Q, L219V, L219H, A225V, L228I, L235V, L235A, V238A, V238T, V238G, K242Q, S244Q, M246L, M246A, M246I, M246F, M246V, M246S, M248T, L250I, L250V, L250T, L250A, L250F, L250M, G255N, G255A, G255S,

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Q256A, N260G, A263G, V264I, V264T, A265G, Y267I, Y267M, Y267H, Y267L, K269Q, N270G, N270T, G273R, S280W, S280L, S280T, S280A, S280K, S280Q, W284H, T285L, T285Q, M286F, M286L, A288L, A288V, F289I, F289L, V291G, V291A, V291T, Y295N, Y295F, Q299V, S304N, S304R, R320A, R320V, H321Y, S323N, H324L, V326L, F328L, F328M, F328V, F328I, T334S, D337H, Q345D, G346T, G346D, G346P, G348S, G348P, T355L, T355F, S376H, S376T, S376V, D377H, D377S, D377A, D377Q, D379S, D379G, D379R, D379A, Y382M, Y382I, Y382L, Y382F, S383G, S383A, Q385L, K391A, K391Y, K391V, K391M, K391E, K391D, K391R, K391H, K391W, K391I, K391Q, K391L, K393Y, K393R, K393Q, K393S, and Q395P, wherein said positions correspond to positions of SEQ ID NOs: 7 or 10.

The C domain of a variant according to the present invention may also comprise alterations. Thus, in one embodiment, the variant comprises an alteration in one or more amino acid positions corresponding to the amino acid positions of the amino acid sequence set forth in SEQ ID NOs: 11 or 12, or an amino acid sequence having at least 65% sequence identity to an amino acid sequence as set forth in SEQ ID NOs: 11 or 12.

In one particular embodiment, the C domain comprises an alteration in one or more of the following positions corresponding to positions; A420, G423, T444, A445, Q449,

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T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13 or I405, A421, A422, A428, R439 G448, W467, D476, and G477 of SEQ ID NO: 14.

In one embodiment, the variant comprises an alteration in two, three, four, five, six, seven, or eight positions corresponding to the following positions; A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13 or I405, A421, A422, A428, R439 G448, W467, D476, and G477 of SEQ ID NO: 14.

The inventors of the present invention have identified that specific amino acid substitutions within the C domain of parent polypeptide have an improved effect on performance of the variants. Thus, in a particular embodiment, the alteration in one or more positions correspond to the following T400P, H402R, A420Q, A420S, A420K, A420L, G423H, T444Q, T444S, T444D, T444Y, T444H, T444V, T444R, T444A, A445Q, Q449T, T459N, P473R, P473A, P473G, P473T, P473K, C474V, G476K, G477A, G477Q, K484A, K484G, K484P, K484E, K484Q, and K484S, wherein said positions correspond to positions of SEQ ID NO: 13; or I405L, A421H, A422P, A428T, G448D, D476K, D476G, D476N, D476Y, G477S, G477A, G477T, and G477Q, wherein said positions correspond to positions of SEQ ID NO: 14.

In a particular embodiment, the variant comprises a modification in the following positions;

H1 + T5	T40 + A174	R87 + N270	W140 + K393	V206 + V215	Y267 + C474
H1 + G7	T40 + G184	R87 + G273	W140 + Q395	V206 + L217	Y267 + G476
H1 + Q11	T40 + T193	R87 + S280	W140 + A420	V206 + L219	Y267 + G477
H1 + N16	T40 + N195	R87 + T285	W140 + G423	V206 + L235	Y267 + K484
H1 + V17	T40 + A204	R87 + M286	W140 + T444	V206 + V238	N270 + G273
H1 + Q32	T40 + V206	R87 + F289	W140 + A445	V206 + M246	N270 + S280
H1 + A37	T40 + P211	R87 + V291	W140 + Q449	V206 + M248	N270 + T285
H1 + T40	T40 + I214	R87 + Q299	W140 + T459	V206 + L250	N270 + M286
H1 + P45	T40 + V215	R87 + R320	W140 + P473	V206 + G255	N270 + F289
H1 + W48	T40 + L217	R87 + H321	W140 + C474	V206 + Q256	N270 + V291
H1 + G50	T40 + L219	R87 + S323	W140 + G476	V206 + A263	N270 + Q299
H1 + T51	T40 + L235	R87 + H324	W140 + G477	V206 + V264	N270 + R320
H1 + N54	T40 + V238	R87 + F328	W140 + K484	V206 + Y267	N270 + H321
H1 + V56	T40 + M246	R87 + T334	G142 + G149	V206 + N270	N270 + S323
H1 + K72	T40 + M248	R87 + D337	G142 + T165	V206 + G273	N270 + H324
H1 + R87	T40 + L250	R87 + Q345	G142 + W167	V206 + S280	N270 + F328
H1 + Q98	T40 + G255	R87 + G346	G142 + R171	V206 + T285	N270 + T334
H1 + M105	T40 + Q256	R87 + G348	G142 + Q172	V206 + M286	N270 + D337
H1 + G109	T40 + A263	R87 + T355	G142 + L173	V206 + F289	N270 + Q345
H1 + F113	T40 + V264	R87 + S376	G142 + A174	V206 + V291	N270 + G346
H1 + R116	T40 + Y267	R87 + D377	G142 + G184	V206 + Q299	N270 + G348
H1 + Q118	T40 + N270	R87 + D379	G142 + T193	V206 + R320	N270 + T355
H1 + Q125	T40 + G273	R87 + Y382	G142 + N195	V206 + H321	N270 + S376
H1 + G133	T40 + S280	R87 + S383	G142 + A204	V206 + S323	N270 + D377
H1 + T134	T40 + T285	R87 + Q385	G142 + V206	V206 + H324	N270 + D379
H1 + W140	T40 + M286	R87 + K391	G142 + P211	V206 + F328	N270 + Y382
H1 + G142	T40 + F289	R87 + K393	G142 + I214	V206 + T334	N270 + S383
H1 + G149	T40 + V291	R87 + Q395	G142 + V215	V206 + D337	N270 + Q385
H1 + T165	T40 + Q299	R87 + A420	G142 + L217	V206 + Q345	N270 + K391
H1 + W167	T40 + R320	R87 + G423	G142 + L219	V206 + G346	N270 + K393
H1 + R171	T40 + H321	R87 + T444	G142 + L235	V206 + G348	N270 + Q395
H1 + Q172	T40 + S323	R87 + A445	G142 + V238	V206 + T355	N270 + A420
H1 + L173	T40 + H324	R87 + Q449	G142 + M246	V206 + S376	N270 + G423
H1 + A174	T40 + F328	R87 + T459	G142 + M248	V206 + D377	N270 + T444
H1 + G184	T40 + T334	R87 + P473	G142 + L250	V206 + D379	N270 + A445
H1 + T193	T40 + D337	R87 + C474	G142 + G255	V206 + Y382	N270 + Q449
H1 + N195	T40 + Q345	R87 + G476	G142 + Q256	V206 + S383	N270 + T459
H1 + A204	T40 + G346	R87 + G477	G142 + A263	V206 + Q385	N270 + P473
H1 + V206	T40 + G348	R87 + K484	G142 + V264	V206 + K391	N270 + C474
H1 + P211	T40 + T355	Q98 + M105	G142 + Y267	V206 + K393	N270 + G476
H1 + I214	T40 + S376	Q98 + G109	G142 + N270	V206 + Q395	N270 + G477
H1 + V215	T40 + D377	Q98 + F113	G142 + G273	V206 + A420	N270 + K484
H1 + L217	T40 + D379	Q98 + R116	G142 + S280	V206 + G423	G273 + S280
H1 + L219	T40 + Y382	Q98 + Q118	G142 + T285	V206 + T444	G273 + T285
H1 + L235	T40 + S383	Q98 + Q125	G142 + M286	V206 + A445	G273 + M286
H1 + V238	T40 + Q385	Q98 + G133	G142 + F289	V206 + Q449	G273 + F289
H1 + M246	T40 + K391	Q98 + T134	G142 + V291	V206 + T459	G273 + V291
H1 + M248	T40 + K393	Q98 + W140	G142 + Q299	V206 + P473	G273 + Q299

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H1 + L250	T40 + Q395	Q98 + G142	G142 + R320	V206 + C474	G273 + R320
H1 + G255	T40 + A420	Q98 + G149	G142 + H321	V206 + G476	G273 + H321
H1 + Q256	T40 + G423	Q98 + T165	G142 + S323	V206 + G477	G273 + S323
H1 + A263	T40 + T444	Q98 + W167	G142 + H324	V206 + K484	G273 + H324
H1 + V264	T40 + A445	Q98 + R171	G142 + F328	P211 + I214	G273 + F328
H1 + Y267	T40 + Q449	Q98 + Q172	G142 + T334	P211 + V215	G273 + T334
H1 + N270	T40 + T459	Q98 + L173	G142 + D337	P211 + L217	G273 + D337
H1 + G273	T40 + P473	Q98 + A174	G142 + Q345	P211 + L219	G273 + Q345
H1 + S280	T40 + C474	Q98 + G184	G142 + G346	P211 + L235	G273 + G346
H1 + T285	T40 + G476	Q98 + T193	G142 + G348	P211 + V238	G273 + G348
H1 + M286	T40 + G477	Q98 + N195	G142 + T355	P211 + M246	G273 + T355
H1 + F289	T40 + K484	Q98 + A204	G142 + S376	P211 + M248	G273 + S376
H1 + V291	P45 + W48	Q98 + V206	G142 + D377	P211 + L250	G273 + D377
H1 + Q299	P45 + G50	Q98 + P211	G142 + D379	P211 + G255	G273 + D379
H1 + R320	P45 + T51	Q98 + I214	G142 + Y382	P211 + Q256	G273 + Y382
H1 + H321	P45 + N54	Q98 + V215	G142 + S383	P211 + A263	G273 + S383
H1 + S323	P45 + V56	Q98 + L217	G142 + Q385	P211 + V264	G273 + Q385
H1 + H324	P45 + K72	Q98 + L219	G142 + K391	P211 + Y267	G273 + K391
H1 + F328	P45 + R87	Q98 + L235	G142 + K393	P211 + N270	G273 + K393
H1 + T334	P45 + Q98	Q98 + V238	G142 + Q395	P211 + G273	G273 + Q395
H1 + D337	P45 + M105	Q98 + M246	G142 + A420	P211 + S280	G273 + A420
H1 + Q345	P45 + G109	Q98 + M248	G142 + G423	P211 + T285	G273 + G423
H1 + G346	P45 + F113	Q98 + L250	G142 + T444	P211 + M286	G273 + T444
H1 + G348	P45 + R116	Q98 + G255	G142 + A445	P211 + F289	G273 + A445
H1 + T355	P45 + Q118	Q98 + Q256	G142 + Q449	P211 + V291	G273 + Q449
H1 + S376	P45 + Q125	Q98 + A263	G142 + T459	P211 + Q299	G273 + T459
H1 + D377	P45 + G133	Q98 + V264	G142 + P473	P211 + R320	G273 + P473
H1 + D379	P45 + T134	Q98 + Y267	G142 + C474	P211 + H321	G273 + C474
H1 + Y382	P45 + W140	Q98 + N270	G142 + G476	P211 + S323	G273 + G476
H1 + S383	P45 + G142	Q98 + G273	G142 + G477	P211 + H324	G273 + G477
H1 + Q385	P45 + G149	Q98 + S280	G142 + K484	P211 + F328	G273 + K484
H1 + K391	P45 + T165	Q98 + T285	G149 + T165	P211 + T334	S280 + T285
H1 + K393	P45 + W167	Q98 + M286	G149 + W167	P211 + D337	S280 + M286
H1 + Q395	P45 + R171	Q98 + F289	G149 + R171	P211 + Q345	S280 + F289
H1 + A420	P45 + Q172	Q98 + V291	G149 + Q172	P211 + G346	S280 + V291
H1 + G423	P45 + L173	Q98 + Q299	G149 + L173	P211 + G348	S280 + Q299
H1 + T444	P45 + A174	Q98 + R320	G149 + A174	P211 + T355	S280 + R320
H1 + A445	P45 + G184	Q98 + H321	G149 + G184	P211 + S376	S280 + H321
H1 + Q449	P45 + T193	Q98 + S323	G149 + T193	P211 + D377	S280 + S323
H1 + T459	P45 + N195	Q98 + H324	G149 + N195	P211 + D379	S280 + H324
H1 + P473	P45 + A204	Q98 + F328	G149 + A204	P211 + Y382	S280 + F328
H1 + C474	P45 + V206	Q98 + T334	G149 + V206	P211 + S383	S280 + T334
H1 + G476	P45 + P211	Q98 + D337	G149 + P211	P211 + Q385	S280 + D337
H1 + G477	P45 + I214	Q98 + Q345	G149 + I214	P211 + K391	S280 + Q345
H1 + K484	P45 + V215	Q98 + G346	G149 + V215	P211 + K393	S280 + G346
T5 + G7	P45 + L217	Q98 + G348	G149 + L217	P211 + Q395	S280 + G348
T5 + Q11	P45 + L219	Q98 + T355	G149 + L219	P211 + A420	S280 + T355
T5 + N16	P45 + L235	Q98 + S376	G149 + L235	P211 + G423	S280 + S376
T5 + V17	P45 + V238	Q98 + D377	G149 + V238	P211 + T444	S280 + D377
T5 + Q32	P45 + M246	Q98 + D379	G149 + M246	P211 + A445	S280 + D379
T5 + A37	P45 + M248	Q98 + Y382	G149 + M248	P211 + Q449	S280 + Y382
T5 + T40	P45 + L250	Q98 + S383	G149 + L250	P211 + T459	S280 + S383
T5 + P45	P45 + G255	Q98 + Q385	G149 + G255	P211 + P473	S280 + Q385
T5 + W48	P45 + Q256	Q98 + K391	G149 + Q256	P211 + C474	S280 + K391
T5 + G50	P45 + A263	Q98 + K393	G149 + A263	P211 + G476	S280 + K393
T5 + T51	P45 + V264	Q98 + Q395	G149 + V264	P211 + G477	S280 + Q395
T5 + N54	P45 + Y267	Q98 + A420	G149 + Y267	P211 + K484	S280 + A420
T5 + V56	P45 + N270	Q98 + G423	G149 + N270	I214 + V215	S280 + G423
T5 + K72	P45 + G273	Q98 + T444	G149 + G273	I214 + L217	S280 + T444
T5 + R87	P45 + S280	Q98 + A445	G149 + S280	I214 + L219	S280 + A445
T5 + Q98	P45 + T285	Q98 + Q449	G149 + T285	I214 + L235	S280 + Q449
T5 + M105	P45 + M286	Q98 + T459	G149 + M286	I214 + V238	S280 + T459
T5 + G109	P45 + F289	Q98 + P473	G149 + F289	I214 + M246	S280 + P473
T5 + F113	P45 + V291	Q98 + C474	G149 + V291	I214 + M248	S280 + C474
T5 + R116	P45 + Q299	Q98 + G476	G149 + Q299	I214 + L250	S280 + G476
T5 + Q118	P45 + R320	Q98 + G477	G149 + R320	I214 + G255	S280 + G477
T5 + Q125	P45 + H321	Q98 + K484	G149 + H321	I214 + Q256	S280 + K484
T5 + G133	P45 + S323	M105 + G109	G149 + S323	I214 + A263	T285 + M286
T5 + T134	P45 + H324	M105 + F113	G149 + H324	I214 + V264	T285 + F289
T5 + W140	P45 + F328	M105 + R116	G149 + F328	I214 + Y267	T285 + V291
T5 + G142	P45 + T334	M105 + Q118	G149 + T334	I214 + N270	T285 + Q299
T5 + G149	P45 + D337	M105 + Q125	G149 + D337	I214 + G273	T285 + R320
T5 + T165	P45 + Q345	M105 + G133	G149 + Q345	I214 + S280	T285 + H321
T5 + W167	P45 + G346	M105 + T134	G149 + G346	I214 + T285	T285 + S323
T5 + R171	P45 + G348	M105 + W140	G149 + G348	I214 + M286	T285 + H324
T5 + Q172	P45 + T355	M105 + G142	G149 + T355	I214 + F289	T285 + F328
T5 + L173	P45 + S376	M105 + G149	G149 + S376	I214 + V291	T285 + T334
T5 + A174	P45 + D377	M105 + T165	G149 + D377	I214 + Q299	T285 + D337
T5 + G184	P45 + D379	M105 + W167	G149 + D379	I214 + R320	T285 + Q345

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T5 + T193	P45 + Y382	M105 + R171	G149 + Y382	I214 + H321	T285 + G346
T5 + N195	P45 + S383	M105 + Q172	G149 + S383	I214 + S323	T285 + G348
T5 + A204	P45 + Q385	M105 + L173	G149 + Q385	I214 + H324	T285 + T355
T5 + V206	P45 + K391	M105 + A174	G149 + K391	I214 + F328	T285 + S376
T5 + P211	P45 + K393	M105 + G184	G149 + K393	I214 + T334	T285 + D377
T5 + I214	P45 + Q395	M105 + T193	G149 + Q395	I214 + D337	T285 + D379
T5 + V215	P45 + A420	M105 + N195	G149 + A420	I214 + Q345	T285 + Y382
T5 + L217	P45 + G423	M105 + A204	G149 + G423	I214 + G346	T285 + S383
T5 + L219	P45 + T444	M105 + V206	G149 + T444	I214 + G348	T285 + Q385
T5 + L235	P45 + A445	M105 + P211	G149 + A445	I214 + T355	T285 + K391
T5 + V238	P45 + Q449	M105 + I214	G149 + Q449	I214 + S376	T285 + K393
T5 + M246	P45 + T459	M105 + V215	G149 + T459	I214 + D377	T285 + Q395
T5 + M248	P45 + P473	M105 + L217	G149 + P473	I214 + D379	T285 + A420
T5 + L250	P45 + C474	M105 + L219	G149 + C474	I214 + Y382	T285 + G423
T5 + G255	P45 + G476	M105 + L235	G149 + G476	I214 + S383	T285 + T444
T5 + Q256	P45 + G477	M105 + V238	G149 + G477	I214 + Q385	T285 + A445
T5 + A263	P45 + K484	M105 + M246	G149 + K484	I214 + K391	T285 + Q449
T5 + V264	W48 + G50	M105 + M248	T165 + W167	I214 + K393	T285 + T459
T5 + Y267	W48 + T51	M105 + L250	T165 + R171	I214 + Q395	T285 + P473
T5 + N270	W48 + N54	M105 + G255	T165 + Q172	I214 + A420	T285 + C474
T5 + G273	W48 + V56	M105 + Q256	T165 + L173	I214 + G423	T285 + G476
T5 + S280	W48 + K72	M105 + A263	T165 + A174	I214 + T444	T285 + G477
T5 + T285	W48 + R87	M105 + V264	T165 + G184	I214 + A445	T285 + K484
T5 + M286	W48 + Q98	M105 + Y267	T165 + T193	I214 + Q449	M286 + F289
T5 + F289	W48 + M105	M105 + N270	T165 + N195	I214 + T459	M286 + V291
T5 + V291	W48 + G109	M105 + G273	T165 + A204	I214 + P473	M286 + Q299
T5 + Q299	W48 + F113	M105 + S280	T165 + V206	I214 + C474	M286 + R320
T5 + R320	W48 + R116	M105 + T285	T165 + P211	I214 + G476	M286 + H321
T5 + H321	W48 + Q118	M105 + M286	T165 + I214	I214 + G477	M286 + S323
T5 + S323	W48 + Q125	M105 + F289	T165 + V215	I214 + K484	M286 + H324
T5 + H324	W48 + G133	M105 + V291	T165 + L217	V215 + L217	M286 + F328
T5 + F328	W48 + T134	M105 + Q299	T165 + L219	V215 + L219	M286 + T334
T5 + T334	W48 + W140	M105 + R320	T165 + L235	V215 + L235	M286 + D337
T5 + D337	W48 + G142	M105 + H321	T165 + V238	V215 + V238	M286 + Q345
T5 + Q345	W48 + G149	M105 + S323	T165 + M246	V215 + M246	M286 + G346
T5 + G346	W48 + T165	M105 + H324	T165 + M248	V215 + M248	M286 + G348
T5 + G348	W48 + W167	M105 + F328	T165 + L250	V215 + L250	M286 + T355
T5 + T355	W48 + R171	M105 + T334	T165 + G255	V215 + G255	M286 + S376
T5 + S376	W48 + Q172	M105 + D337	T165 + Q256	V215 + Q256	M286 + D377
T5 + D377	W48 + L173	M105 + Q345	T165 + A263	V215 + A263	M286 + D379
T5 + D379	W48 + A174	M105 + G346	T165 + V264	V215 + V264	M286 + Y382
T5 + Y382	W48 + G184	M105 + G348	T165 + Y267	V215 + Y267	M286 + S383
T5 + S383	W48 + T193	M105 + T355	T165 + N270	V215 + N270	M286 + Q385
T5 + Q385	W48 + N195	M105 + S376	T165 + G273	V215 + G273	M286 + K391
T5 + K391	W48 + A204	M105 + D377	T165 + S280	V215 + S280	M286 + K393
T5 + K393	W48 + V206	M105 + D379	T165 + T285	V215 + T285	M286 + Q395
T5 + Q395	W48 + P211	M105 + Y382	T165 + M286	V215 + M286	M286 + A420
T5 + A420	W48 + I214	M105 + S383	T165 + F289	V215 + F289	M286 + G423
T5 + G423	W48 + V215	M105 + Q385	T165 + V291	V215 + V291	M286 + T444
T5 + T444	W48 + L217	M105 + K391	T165 + Q299	V215 + Q299	M286 + A445
T5 + A445	W48 + L219	M105 + K393	T165 + R320	V215 + R320	M286 + Q449
T5 + Q449	W48 + L235	M105 + Q395	T165 + H321	V215 + H321	M286 + T459
T5 + T459	W48 + V238	M105 + A420	T165 + S323	V215 + S323	M286 + P473
T5 + P473	W48 + M246	M105 + G423	T165 + H324	V215 + H324	M286 + C474
T5 + C474	W48 + M248	M105 + T444	T165 + F328	V215 + F328	M286 + G476
T5 + G476	W48 + L250	M105 + A445	T165 + T334	V215 + T334	M286 + G477
T5 + G477	W48 + G255	M105 + Q449	T165 + D337	V215 + D337	M286 + K484
T5 + K484	W48 + Q256	M105 + T459	T165 + Q345	V215 + Q345	F289 + V291
G7 + Q11	W48 + A263	M105 + P473	T165 + G346	V215 + G346	F289 + Q299
G7 + N16	W48 + V264	M105 + C474	T165 + G348	V215 + G348	F289 + R320
G7 + V17	W48 + Y267	M105 + G476	T165 + T355	V215 + T355	F289 + H321
G7 + Q32	W48 + N270	M105 + G477	T165 + S376	V215 + S376	F289 + S323
G7 + A37	W48 + G273	M105 + K484	T165 + D377	V215 + D377	F289 + H324
G7 + T40	W48 + S280	G109 + F113	T165 + D379	V215 + D379	F289 + F328
G7 + P45	W48 + T285	G109 + R116	T165 + Y382	V215 + Y382	F289 + T334
G7 + W48	W48 + M286	G109 + Q118	T165 + S383	V215 + S383	F289 + D337
G7 + G50	W48 + F289	G109 + Q125	T165 + Q385	V215 + Q385	F289 + Q345
G7 + T51	W48 + V291	G109 + G133	T165 + K391	V215 + K391	F289 + G346
G7 + N54	W48 + Q299	G109 + T134	T165 + K393	V215 + K393	F289 + G348
G7 + V56	W48 + R320	G109 + W140	T165 + Q395	V215 + Q395	F289 + T355
G7 + K72	W48 + H321	G109 + G142	T165 + A420	V215 + A420	F289 + S376
G7 + R87	W48 + S323	G109 + G149	T165 + G423	V215 + G423	F289 + D377
G7 + Q98	W48 + H324	G109 + T165	T165 + T444	V215 + T444	F289 + D379
G7 + M105	W48 + F328	G109 + W167	T165 + A445	V215 + A445	F289 + Y382
G7 + G109	W48 + T334	G109 + R171	T165 + Q449	V215 + Q449	F289 + S383
G7 + F113	W48 + D337	G109 + Q172	T165 + T459	V215 + T459	F289 + Q385
G7 + R116	W48 + Q345	G109 + L173	T165 + P473	V215 + P473	F289 + K391
G7 + Q118	W48 + G346	G109 + A174	T165 + C474	V215 + C474	F289 + K393
G7 + Q125	W48 + G348	G109 + G184	T165 + G476	V215 + G476	F289 + Q395

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G7 + G133	W48 + T355	G109 + T193	T165 + G477	V215 + G477	F289 + A420
G7 + T134	W48 + S376	G109 + N195	T165 + K484	V215 + K484	F289 + G423
G7 + W140	W48 + D377	G109 + A204	W167 + R171	L217 + L219	F289 + T444
G7 + G142	W48 + D379	G109 + V206	W167 + Q172	L217 + L235	F289 + A445
G7 + G149	W48 + Y382	G109 + P211	W167 + L173	L217 + V238	F289 + Q449
G7 + T165	W48 + S383	G109 + I214	W167 + A174	L217 + M246	F289 + T459
G7 + W167	W48 + Q385	G109 + V215	W167 + G184	L217 + M248	F289 + P473
G7 + R171	W48 + K391	G109 + L217	W167 + T193	L217 + L250	F289 + C474
G7 + Q172	W48 + K393	G109 + L219	W167 + N195	L217 + G255	F289 + G476
G7 + L173	W48 + Q395	G109 + L235	W167 + A204	L217 + Q256	F289 + G477
G7 + A174	W48 + A420	G109 + V238	W167 + V206	L217 + A263	F289 + K484
G7 + G184	W48 + G423	G109 + M246	W167 + P211	L217 + V264	V291 + Q299
G7 + T193	W48 + T444	G109 + M248	W167 + I214	L217 + Y267	V291 + R320
G7 + N195	W48 + A445	G109 + L250	W167 + V215	L217 + N270	V291 + H321
G7 + A204	W48 + Q449	G109 + G255	W167 + L217	L217 + G273	V291 + S323
G7 + V206	W48 + T459	G109 + Q256	W167 + L219	L217 + S280	V291 + H324
G7 + P211	W48 + P473	G109 + A263	W167 + L235	L217 + T285	V291 + F328
G7 + I214	W48 + C474	G109 + V264	W167 + V238	L217 + M286	V291 + T334
G7 + V215	W48 + G476	G109 + Y267	W167 + M246	L217 + F289	V291 + D337
G7 + L217	W48 + G477	G109 + N270	W167 + M248	L217 + V291	V291 + Q345
G7 + L219	W48 + K484	G109 + G273	W167 + L250	L217 + Q299	V291 + G346
G7 + L235	G50 + T51	G109 + S280	W167 + G255	L217 + R320	V291 + G348
G7 + V238	G50 + N54	G109 + T285	W167 + Q256	L217 + H321	V291 + T355
G7 + M246	G50 + V56	G109 + M286	W167 + A263	L217 + S323	V291 + S376
G7 + M248	G50 + K72	G109 + F289	W167 + V264	L217 + H324	V291 + D377
G7 + L250	G50 + R87	G109 + V291	W167 + Y267	L217 + F328	V291 + D379
G7 + G255	G50 + Q98	G109 + Q299	W167 + N270	L217 + T334	V291 + Y382
G7 + Q256	G50 + M105	G109 + R320	W167 + G273	L217 + D337	V291 + S383
G7 + A263	G50 + G109	G109 + H321	W167 + S280	L217 + Q345	V291 + Q385
G7 + V264	G50 + F113	G109 + S323	W167 + T285	L217 + G346	V291 + K391
G7 + Y267	G50 + R116	G109 + H324	W167 + M286	L217 + G348	V291 + K393
G7 + N270	G50 + Q118	G109 + F328	W167 + F289	L217 + T355	V291 + Q395
G7 + G273	G50 + Q125	G109 + T334	W167 + V291	L217 + S376	V291 + A420
G7 + S280	G50 + G133	G109 + D337	W167 + Q299	L217 + D377	V291 + G423
G7 + T285	G50 + T134	G109 + Q345	W167 + R320	L217 + D379	V291 + T444
G7 + M286	G50 + W140	G109 + G346	W167 + H321	L217 + Y382	V291 + A445
G7 + F289	G50 + G142	G109 + G348	W167 + S323	L217 + S383	V291 + Q449
G7 + V291	G50 + G149	G109 + T355	W167 + H324	L217 + Q385	V291 + T459
G7 + Q299	G50 + T165	G109 + S376	W167 + F328	L217 + K391	V291 + P473
G7 + R320	G50 + W167	G109 + D377	W167 + T334	L217 + K393	V291 + C474
G7 + H321	G50 + R171	G109 + D379	W167 + D337	L217 + Q395	V291 + G476
G7 + S323	G50 + Q172	G109 + Y382	W167 + Q345	L217 + A420	V291 + G477
G7 + H324	G50 + L173	G109 + S383	W167 + G346	L217 + G423	V291 + K484
G7 + F328	G50 + A174	G109 + Q385	W167 + G348	L217 + T444	Q299 + R320
G7 + T334	G50 + G184	G109 + K391	W167 + T355	L217 + A445	Q299 + H321
G7 + D337	G50 + T193	G109 + K393	W167 + S376	L217 + Q449	Q299 + S323
G7 + Q345	G50 + N195	G109 + Q395	W167 + D377	L217 + T459	Q299 + H324
G7 + G346	G50 + A204	G109 + A420	W167 + D379	L217 + P473	Q299 + F328
G7 + G348	G50 + V206	G109 + G423	W167 + Y382	L217 + C474	Q299 + T334
G7 + T355	G50 + P211	G109 + T444	W167 + S383	L217 + G476	Q299 + D337
G7 + S376	G50 + I214	G109 + A445	W167 + Q385	L217 + G477	Q299 + Q345
G7 + D377	G50 + V215	G109 + Q449	W167 + K391	L217 + K484	Q299 + G346
G7 + D379	G50 + L217	G109 + T459	W167 + K393	L219 + L235	Q299 + G348
G7 + Y382	G50 + L219	G109 + P473	W167 + Q395	L219 + V238	Q299 + T355
G7 + S383	G50 + L235	G109 + C474	W167 + A420	L219 + M246	Q299 + S376
G7 + Q385	G50 + V238	G109 + G476	W167 + G423	L219 + M248	Q299 + D377
G7 + K391	G50 + M246	G109 + G477	W167 + T444	L219 + L250	Q299 + D379
G7 + K393	G50 + M248	G109 + K484	W167 + A445	L219 + G255	Q299 + Y382
G7 + Q395	G50 + L250	F113 + R116	W167 + Q449	L219 + Q256	Q299 + S383
G7 + A420	G50 + G255	F113 + Q118	W167 + T459	L219 + A263	Q299 + Q385
G7 + G423	G50 + Q256	F113 + Q125	W167 + P473	L219 + V264	Q299 + K391
G7 + T444	G50 + A263	F113 + G133	W167 + C474	L219 + Y267	Q299 + K393
G7 + A445	G50 + V264	F113 + T134	W167 + G476	L219 + N270	Q299 + Q395
G7 + Q449	G50 + Y267	F113 + W140	W167 + G477	L219 + G273	Q299 + A420
G7 + T459	G50 + N270	F113 + G142	W167 + K484	L219 + S280	Q299 + G423
G7 + P473	G50 + G273	F113 + G149	R171 + Q172	L219 + T285	Q299 + T444
G7 + C474	G50 + S280	F113 + T165	R171 + L173	L219 + M286	Q299 + A445
G7 + G476	G50 + T285	F113 + W167	R171 + A174	L219 + F289	Q299 + Q449
G7 + G477	G50 + M286	F113 + R171	R171 + G184	L219 + V291	Q299 + T459
G7 + K484	G50 + F289	F113 + Q172	R171 + T193	L219 + Q299	Q299 + P473
Q11 + N16	G50 + V291	F113 + L173	R171 + N195	L219 + R320	Q299 + C474
Q11 + V17	G50 + Q299	F113 + A174	R171 + A204	L219 + H321	Q299 + G476
Q11 + Q32	G50 + R320	F113 + G184	R171 + V206	L219 + S323	Q299 + G477
Q11 + A37	G50 + H321	F113 + T193	R171 + P211	L219 + H324	Q299 + K484
Q11 + T40	G50 + S323	F113 + N195	R171 + I214	L219 + F328	R320 + H321
Q11 + P45	G50 + H324	F113 + A204	R171 + V215	L219 + T334	R320 + S323
Q11 + W48	G50 + F328	F113 + V206	R171 + L217	L219 + D337	R320 + H324
Q11 + G50	G50 + T334	F113 + P211	R171 + L219	L219 + Q345	R320 + F328
Q11 + T51	G50 + D337	F113 + I214	R171 + L235	L219 + G346	R320 + T334

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Q11 + N54	G50 + Q345	F113 + V215	R171 + V238	L219 + G348	R320 + D337
Q11 + V56	G50 + G346	F113 + L217	R171 + M246	L219 + T355	R320 + Q345
Q11 + K72	G50 + G348	F113 + L219	R171 + M248	L219 + S376	R320 + G346
Q11 + R87	G50 + T355	F113 + L235	R171 + L250	L219 + D377	R320 + G348
Q11 + Q98	G50 + S376	F113 + V238	R171 + G255	L219 + D379	R320 + T355
Q11 + M105	G50 + D377	F113 + M246	R171 + Q256	L219 + Y382	R320 + S376
Q11 + G109	G50 + D379	F113 + M248	R171 + A263	L219 + S383	R320 + D377
Q11 + F113	G50 + Y382	F113 + L250	R171 + V264	L219 + Q385	R320 + D379
Q11 + R116	G50 + S383	F113 + G255	R171 + Y267	L219 + K391	R320 + Y382
Q11 + Q118	G50 + Q385	F113 + Q256	R171 + N270	L219 + K393	R320 + S383
Q11 + Q125	G50 + K391	F113 + A263	R171 + G273	L219 + Q395	R320 + Q385
Q11 + G133	G50 + K393	F113 + V264	R171 + S280	L219 + A420	R320 + K391
Q11 + T134	G50 + Q395	F113 + Y267	R171 + T285	L219 + G423	R320 + K393
Q11 + W140	G50 + A420	F113 + N270	R171 + M286	L219 + T444	R320 + Q395
Q11 + G142	G50 + G423	F113 + G273	R171 + F289	L219 + A445	R320 + A420
Q11 + G149	G50 + T444	F113 + S280	R171 + V291	L219 + Q449	R320 + G423
Q11 + T165	G50 + A445	F113 + T285	R171 + Q299	L219 + T459	R320 + T444
Q11 + W167	G50 + Q449	F113 + M286	R171 + R320	L219 + P473	R320 + A445
Q11 + R171	G50 + T459	F113 + F289	R171 + H321	L219 + C474	R320 + Q449
Q11 + Q172	G50 + P473	F113 + V291	R171 + S323	L219 + G476	R320 + T459
Q11 + L173	G50 + C474	F113 + Q299	R171 + H324	L219 + G477	R320 + P473
Q11 + A174	G50 + G476	F113 + R320	R171 + F328	L219 + K484	R320 + C474
Q11 + G184	G50 + G477	F113 + H321	R171 + T334	L235 + V238	R320 + G476
Q11 + T193	G50 + K484	F113 + S323	R171 + D337	L235 + M246	R320 + G477
Q11 + N195	T51 + N54	F113 + H324	R171 + Q345	L235 + M248	R320 + K484
Q11 + A204	T51 + V56	F113 + F328	R171 + G346	L235 + L250	H321 + S323
Q11 + V206	T51 + K72	F113 + T334	R171 + G348	L235 + G255	H321 + H324
Q11 + P211	T51 + R87	F113 + D337	R171 + T355	L235 + Q256	H321 + F328
Q11 + I214	T51 + Q98	F113 + Q345	R171 + S376	L235 + A263	H321 + T334
Q11 + V215	T51 + M105	F113 + G346	R171 + D377	L235 + V264	H321 + D337
Q11 + L217	T51 + G109	F113 + G348	R171 + D379	L235 + Y267	H321 + Q345
Q11 + L219	T51 + F113	F113 + T355	R171 + Y382	L235 + N270	H321 + G346
Q11 + L235	T51 + R116	F113 + S376	R171 + S383	L235 + G273	H321 + G348
Q11 + V238	T51 + Q118	F113 + D377	R171 + Q385	L235 + S280	H321 + T355
Q11 + M246	T51 + Q125	F113 + D379	R171 + K391	L235 + T285	H321 + S376
Q11 + M248	T51 + G133	F113 + Y382	R171 + K393	L235 + M286	H321 + D377
Q11 + L250	T51 + T134	F113 + S383	R171 + Q395	L235 + F289	H321 + D379
Q11 + G255	T51 + W140	F113 + Q385	R171 + A420	L235 + V291	H321 + Y382
Q11 + Q256	T51 + G142	F113 + K391	R171 + G423	L235 + Q299	H321 + S383
Q11 + A263	T51 + G149	F113 + K393	R171 + T444	L235 + R320	H321 + Q385
Q11 + V264	T51 + T165	F113 + Q395	R171 + A445	L235 + H321	H321 + K391
Q11 + Y267	T51 + W167	F113 + A420	R171 + Q449	L235 + S323	H321 + K393
Q11 + N270	T51 + R171	F113 + G423	R171 + T459	L235 + H324	H321 + Q395
Q11 + G273	T51 + Q172	F113 + T444	R171 + P473	L235 + F328	H321 + A420
Q11 + S280	T51 + L173	F113 + A445	R171 + C474	L235 + T334	H321 + G423
Q11 + T285	T51 + A174	F113 + Q449	R171 + G476	L235 + D337	H321 + T444
Q11 + M286	T51 + G184	F113 + T459	R171 + G477	L235 + Q345	H321 + A445
Q11 + F289	T51 + T193	F113 + P473	R171 + K484	L235 + G346	H321 + Q449
Q11 + V291	T51 + N195	F113 + C474	Q172 + L173	L235 + G348	H321 + T459
Q11 + Q299	T51 + A204	F113 + G476	Q172 + A174	L235 + T355	H321 + P473
Q11 + R320	T51 + V206	F113 + G477	Q172 + G184	L235 + S376	H321 + C474
Q11 + H321	T51 + P211	F113 + K484	Q172 + T193	L235 + D377	H321 + G476
Q11 + S323	T51 + I214	R116 + Q118	Q172 + N195	L235 + D379	H321 + G477
Q11 + H324	T51 + V215	R116 + Q125	Q172 + A204	L235 + Y382	H321 + K484
Q11 + F328	T51 + L217	R116 + G133	Q172 + V206	L235 + S383	S323 + H324
Q11 + T334	T51 + L219	R116 + T134	Q172 + P211	L235 + Q385	S323 + F328
Q11 + D337	T51 + L235	R116 + W140	Q172 + I214	L235 + K391	S323 + T334
Q11 + Q345	T51 + V238	R116 + G142	Q172 + V215	L235 + K393	S323 + D337
Q11 + G346	T51 + M246	R116 + G149	Q172 + L217	L235 + Q395	S323 + Q345
Q11 + G348	T51 + M248	R116 + T165	Q172 + L219	L235 + A420	S323 + G346
Q11 + T355	T51 + L250	R116 + W167	Q172 + L235	L235 + G423	S323 + G348
Q11 + S376	T51 + G255	R116 + R171	Q172 + V238	L235 + T444	S323 + T355
Q11 + D377	T51 + Q256	R116 + Q172	Q172 + M246	L235 + A445	S323 + S376
Q11 + D379	T51 + A263	R116 + L173	Q172 + M248	L235 + Q449	S323 + D377
Q11 + Y382	T51 + V264	R116 + A174	Q172 + L250	L235 + T459	S323 + D379
Q11 + S383	T51 + Y267	R116 + G184	Q172 + G255	L235 + P473	S323 + Y382
Q11 + Q385	T51 + N270	R116 + T193	Q172 + Q256	L235 + C474	S323 + S383
Q11 + K391	T51 + G273	R116 + N195	Q172 + A263	L235 + G476	S323 + Q385
Q11 + K393	T51 + S280	R116 + A204	Q172 + V264	L235 + G477	S323 + K391
Q11 + Q395	T51 + T285	R116 + V206	Q172 + Y267	L235 + K484	S323 + K393
Q11 + A420	T51 + M286	R116 + P211	Q172 + N270	V238 + M246	S323 + Q395
Q11 + G423	T51 + F289	R116 + I214	Q172 + G273	V238 + M248	S323 + A420
Q11 + T444	T51 + V291	R116 + V215	Q172 + S280	V238 + L250	S323 + G423
Q11 + A445	T51 + Q299	R116 + L217	Q172 + T285	V238 + G255	S323 + T444
Q11 + Q449	T51 + R320	R116 + L219	Q172 + M286	V238 + Q256	S323 + A445
Q11 + T459	T51 + H321	R116 + L235	Q172 + F289	V238 + A263	S323 + Q449
Q11 + P473	T51 + S323	R116 + V238	Q172 + V291	V238 + V264	S323 + T459
Q11 + C474	T51 + H324	R116 + M246	Q172 + Q299	V238 + Y267	S323 + P473
Q11 + G476	T51 + F328	R116 + M248	Q172 + R320	V238 + N270	S323 + C474

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Q11 + G477	T51 + T334	R116 + L250	Q172 + H321	V238 + G273	S323 + G476
Q11 + K484	T51 + D337	R116 + G255	Q172 + S323	V238 + S280	S323 + G477
N16 + V17	T51 + Q345	R116 + Q256	Q172 + H324	V238 + T285	S323 + K484
N16 + Q32	T51 + G346	R116 + A263	Q172 + F328	V238 + M286	H324 + F328
N16 + A37	T51 + G348	R116 + V264	Q172 + T334	V238 + F289	H324 + T334
N16 + T40	T51 + T355	R116 + Y267	Q172 + D337	V238 + V291	H324 + D337
N16 + P45	T51 + S376	R116 + N270	Q172 + Q345	V238 + Q299	H324 + Q345
N16 + W48	T51 + D377	R116 + G273	Q172 + G346	V238 + R320	H324 + G346
N16 + G50	T51 + D379	R116 + S280	Q172 + G348	V238 + H321	H324 + G348
N16 + T51	T51 + Y382	R116 + T285	Q172 + T355	V238 + S323	H324 + T355
N16 + N54	T51 + S383	R116 + M286	Q172 + S376	V238 + H324	H324 + S376
N16 + V56	T51 + Q385	R116 + F289	Q172 + D377	V238 + F328	H324 + D377
N16 + K72	T51 + K391	R116 + V291	Q172 + D379	V238 + T334	H324 + D379
N16 + R87	T51 + K393	R116 + Q299	Q172 + Y382	V238 + D337	H324 + Y382
N16 + Q98	T51 + Q395	R116 + R320	Q172 + S383	V238 + Q345	H324 + S383
N16 + M105	T51 + A420	R116 + H321	Q172 + Q385	V238 + G346	H324 + Q385
N16 + G109	T51 + G423	R116 + S323	Q172 + K391	V238 + G348	H324 + K391
N16 + F113	T51 + T444	R116 + H324	Q172 + K393	V238 + T355	H324 + K393
N16 + R116	T51 + A445	R116 + F328	Q172 + Q395	V238 + S376	H324 + Q395
N16 + Q118	T51 + Q449	R116 + T334	Q172 + A420	V238 + D377	H324 + A420
N16 + Q125	T51 + T459	R116 + D337	Q172 + G423	V238 + D379	H324 + G423
N16 + G133	T51 + K391	R116 + Q345	Q172 + T444	V238 + Y382	H324 + T444
N16 + T134	T51 + C474	R116 + G346	Q172 + A445	V238 + S383	H324 + A445
N16 + W140	T51 + G476	R116 + G348	Q172 + Q449	V238 + Q385	H324 + Q449
N16 + G142	T51 + G477	R116 + T355	Q172 + T459	V238 + K391	H324 + T459
N16 + G149	T51 + K484	R116 + S376	Q172 + P473	V238 + K393	H324 + P473
N16 + T165	N54 + V56	R116 + D377	Q172 + C474	V238 + Q395	H324 + C474
N16 + W167	N54 + K72	R116 + D379	Q172 + G476	V238 + A420	H324 + G476
N16 + R171	N54 + R87	R116 + Y382	Q172 + G477	V238 + G423	H324 + G477
N16 + Q172	N54 + Q98	R116 + S383	Q172 + K484	V238 + T444	H324 + K484
N16 + L173	N54 + M105	R116 + Q385	L173 + A174	V238 + A445	F328 + T334
N16 + A174	N54 + G109	R116 + K391	L173 + G184	V238 + Q449	F328 + D337
N16 + G184	N54 + F113	R116 + K393	L173 + T193	V238 + T459	F328 + Q345
N16 + T193	N54 + R116	R116 + Q395	L173 + N195	V238 + P473	F328 + G346
N16 + N195	N54 + Q118	R116 + A420	L173 + A204	V238 + C474	F328 + G348
N16 + A204	N54 + Q125	R116 + G423	L173 + V206	V238 + G476	F328 + T355
N16 + V206	N54 + G133	R116 + T444	L173 + P211	V238 + G477	F328 + S376
N16 + P211	N54 + T134	R116 + A445	L173 + I214	V238 + K484	F328 + D377
N16 + I214	N54 + W140	R116 + Q449	L173 + V215	M246 + M248	F328 + D379
N16 + V215	N54 + G142	R116 + T459	L173 + L217	M246 + L250	F328 + Y382
N16 + L217	N54 + G149	R116 + P473	L173 + L219	M246 + G255	F328 + S383
N16 + L219	N54 + T165	R116 + C474	L173 + L235	M246 + Q256	F328 + Q385
N16 + L235	N54 + W167	R116 + G476	L173 + V238	M246 + A263	F328 + K391
N16 + V238	N54 + R171	R116 + G477	L173 + M246	M246 + V264	F328 + K393
N16 + M246	N54 + Q172	R116 + K484	L173 + M248	M246 + Y267	F328 + Q395
N16 + M248	N54 + L173	Q118 + Q125	L173 + L250	M246 + N270	F328 + A420
N16 + L250	N54 + A174	Q118 + G133	L173 + G255	M246 + G273	F328 + G423
N16 + G255	N54 + G184	Q118 + T134	L173 + Q256	M246 + S280	F328 + T444
N16 + Q256	N54 + T193	Q118 + W140	L173 + A263	M246 + T285	F328 + A445
N16 + A263	N54 + N195	Q118 + G142	L173 + V264	M246 + M286	F328 + Q449
N16 + V264	N54 + A204	Q118 + G149	L173 + Y267	M246 + F289	F328 + T459
N16 + Y267	N54 + V206	Q118 + T165	L173 + N270	M246 + V291	F328 + P473
N16 + N270	N54 + P211	Q118 + W167	L173 + G273	M246 + Q299	F328 + C474
N16 + G273	N54 + I214	Q118 + R171	L173 + S280	M246 + R320	F328 + G476
N16 + S280	N54 + V215	Q118 + Q172	L173 + T285	M246 + H321	F328 + G477
N16 + T285	N54 + L217	Q118 + L173	L173 + M286	M246 + S323	F328 + K484
N16 + M286	N54 + L219	Q118 + A174	L173 + F289	M246 + H324	T334 + D337
N16 + F289	N54 + L235	Q118 + G184	L173 + V291	M246 + F328	T334 + Q345
N16 + V291	N54 + V238	Q118 + T193	L173 + Q299	M246 + T334	T334 + G346
N16 + Q299	N54 + M246	Q118 + N195	L173 + R320	M246 + D337	T334 + G348
N16 + R320	N54 + M248	Q118 + A204	L173 + H321	M246 + Q345	T334 + T355
N16 + H321	N54 + L250	Q118 + V206	L173 + S323	M246 + G346	T334 + S376
N16 + S323	N54 + G255	Q118 + P211	L173 + H324	M246 + G348	T334 + D377
N16 + H324	N54 + Q256	Q118 + I214	L173 + F328	M246 + T355	T334 + D379
N16 + F328	N54 + A263	Q118 + V215	L173 + T334	M246 + S376	T334 + Y382
N16 + T334	N54 + V264	Q118 + L217	L173 + D337	M246 + D377	T334 + S383
N16 + D337	N54 + Y267	Q118 + L219	L173 + Q345	M246 + D379	T334 + Q385
N16 + Q345	N54 + N270	Q118 + L235	L173 + G346	M246 + Y382	T334 + K391
N16 + G346	N54 + G273	Q118 + V238	L173 + G348	M246 + S383	T334 + K393
N16 + G348	N54 + S280	Q118 + M246	L173 + T355	M246 + Q385	T334 + Q395
N16 + T355	N54 + T285	Q118 + M248	L173 + S376	M246 + K391	T334 + A420
N16 + S376	N54 + M286	Q118 + L250	L173 + D377	M246 + K393	T334 + G423
N16 + D377	N54 + F289	Q118 + G255	L173 + D379	M246 + Q395	T334 + T444
N16 + D379	N54 + V291	Q118 + Q256	L173 + Y382	M246 + A420	T334 + A445
N16 + Y382	N54 + Q299	Q118 + A263	L173 + S383	M246 + G423	T334 + Q449
N16 + S383	N54 + R320	Q118 + V264	L173 + Q385	M246 + T444	T334 + T459
N16 + Q385	N54 + H321	Q118 + Y267	L173 + K391	M246 + A445	T334 + P473
N16 + K391	N54 + S323	Q118 + N270	L173 + K393	M246 + Q449	T334 + C474
N16 + K393	N54 + H324	Q118 + G273	L173 + Q395	M246 + T459	T334 + G476

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N16 + Q395	N54 + F328	Q118 + S280	L173 + A420	M246 + P473	T334 + G477
N16 + A420	N54 + T334	Q118 + T285	L173 + G423	M246 + C474	T334 + K484
N16 + G423	N54 + D337	Q118 + M286	L173 + T444	M246 + G476	D337 + Q345
N16 + T444	N54 + Q345	Q118 + F289	L173 + A445	M246 + G477	D337 + G346
N16 + A445	N54 + G346	Q118 + V291	L173 + Q449	M246 + K484	D337 + G348
N16 + Q449	N54 + G348	Q118 + Q299	L173 + T459	M248 + L250	D337 + T355
N16 + T459	N54 + T355	Q118 + R320	L173 + P473	M248 + G255	D337 + S376
N16 + P473	N54 + S376	Q118 + H321	L173 + C474	M248 + Q256	D337 + D377
N16 + C474	N54 + D377	Q118 + S323	L173 + G476	M248 + A263	D337 + D379
N16 + G476	N54 + D379	Q118 + H324	L173 + G477	M248 + V264	D337 + Y382
N16 + G477	N54 + Y382	Q118 + F328	L173 + K484	M248 + Y267	D337 + S383
N16 + K484	N54 + S383	Q118 + T334	A174 + G184	M248 + N270	D337 + Q385
V17 + Q32	N54 + Q385	Q118 + D337	A174 + T193	M248 + G273	D337 + K391
V17 + A37	N54 + K391	Q118 + Q345	A174 + N195	M248 + S280	D337 + K393
V17 + T40	N54 + K393	Q118 + G346	A174 + A204	M248 + T285	D337 + Q395
V17 + P45	N54 + Q395	Q118 + G348	A174 + V206	M248 + M286	D337 + A420
V17 + W48	N54 + A420	Q118 + T355	A174 + P211	M248 + F289	D337 + G423
V17 + G50	N54 + G423	Q118 + S376	A174 + I214	M248 + V291	D337 + T444
V17 + T51	N54 + T444	Q118 + D377	A174 + V215	M248 + Q299	D337 + A445
V17 + N54	N54 + A445	Q118 + D379	A174 + L217	M248 + R320	D337 + Q449
V17 + V56	N54 + Q449	Q118 + Y382	A174 + L219	M248 + H321	D337 + T459
V17 + K72	N54 + T459	Q118 + S383	A174 + L235	M248 + S323	D337 + P473
V17 + R87	N54 + P473	Q118 + Q385	A174 + V238	M248 + H324	D337 + C474
V17 + Q98	N54 + C474	Q118 + K391	A174 + M246	M248 + F328	D337 + G476
V17 + M105	N54 + G476	Q118 + K393	A174 + M248	M248 + T334	D337 + G477
V17 + G109	N54 + G477	Q118 + Q395	A174 + L250	M248 + D337	D337 + K484
V17 + F113	N54 + K484	Q118 + A420	A174 + G255	M248 + Q345	Q345 + G346
V17 + R116	V56 + K72	Q118 + G423	A174 + Q256	M248 + G346	Q345 + G348
V17 + Q118	V56 + R87	Q118 + T444	A174 + A263	M248 + G348	Q345 + T355
V17 + Q125	V56 + Q98	Q118 + A445	A174 + V264	M248 + T355	Q345 + S376
V17 + G133	V56 + M105	Q118 + Q449	A174 + Y267	M248 + S376	Q345 + D377
V17 + T134	V56 + G109	Q118 + T459	A174 + N270	M248 + D377	Q345 + D379
V17 + W140	V56 + F113	Q118 + P473	A174 + G273	M248 + D379	Q345 + Y382
V17 + G142	V56 + R116	Q118 + C474	A174 + S280	M248 + Y382	Q345 + S383
V17 + G149	V56 + Q118	Q118 + G476	A174 + T285	M248 + S383	Q345 + Q385
V17 + T165	V56 + Q125	Q118 + G477	A174 + M286	M248 + Q385	Q345 + K391
V17 + W167	V56 + G133	Q118 + K484	A174 + F289	M248 + K391	Q345 + K393
V17 + R171	V56 + T134	Q125 + G133	A174 + V291	M248 + K393	Q345 + Q395
V17 + Q172	V56 + W140	Q125 + T134	A174 + Q299	M248 + Q395	Q345 + A420
V17 + L173	V56 + G142	Q125 + W140	A174 + R320	M248 + A420	Q345 + G423
V17 + A174	V56 + G149	Q125 + G142	A174 + H321	M248 + G423	Q345 + T444
V17 + G184	V56 + T165	Q125 + G149	A174 + S323	M248 + T444	Q345 + A445
V17 + T193	V56 + W167	Q125 + T165	A174 + H324	M248 + A445	Q345 + Q449
V17 + N195	V56 + R171	Q125 + W167	A174 + F328	M248 + Q449	Q345 + T459
V17 + A204	V56 + Q172	Q125 + R171	A174 + T334	M248 + T459	Q345 + P473
V17 + V206	V56 + L173	Q125 + Q172	A174 + D337	M248 + P473	Q345 + C474
V17 + P211	V56 + A174	Q125 + L173	A174 + Q345	M248 + C474	Q345 + G476
V17 + I214	V56 + G184	Q125 + A174	A174 + G346	M248 + G476	Q345 + G477
V17 + V215	V56 + T193	Q125 + G184	A174 + G348	M248 + G477	Q345 + K484
V17 + L217	V56 + N195	Q125 + T193	A174 + T355	M248 + K484	G346 + G348
V17 + L219	V56 + A204	Q125 + N195	A174 + S376	L250 + G255	G346 + T355
V17 + L235	V56 + V206	Q125 + A204	A174 + D377	L250 + Q256	G346 + S376
V17 + V238	V56 + P211	Q125 + V206	A174 + D379	L250 + A263	G346 + D377
V17 + M246	V56 + I214	Q125 + P211	A174 + Y382	L250 + V264	G346 + D379
V17 + M248	V56 + V215	Q125 + I214	A174 + S383	L250 + Y267	G346 + Y382
V17 + L250	V56 + L217	Q125 + V215	A174 + Q385	L250 + N270	G346 + S383
V17 + G255	V56 + L219	Q125 + L217	A174 + K391	L250 + G273	G346 + Q385
V17 + Q256	V56 + L235	Q125 + L219	A174 + K393	L250 + S280	G346 + K391
V17 + A263	V56 + V238	Q125 + L235	A174 + Q395	L250 + T285	G346 + K393
V17 + V264	V56 + M246	Q125 + V238	A174 + A420	L250 + M286	G346 + Q395
V17 + Y267	V56 + M248	Q125 + M246	A174 + G423	L250 + F289	G346 + A420
V17 + N270	V56 + L250	Q125 + M248	A174 + T444	L250 + V291	G346 + G423
V17 + G273	V56 + G255	Q125 + L250	A174 + A445	L250 + Q299	G346 + T444
V17 + S280	V56 + Q256	Q125 + G255	A174 + Q449	L250 + R320	G346 + A445
V17 + T285	V56 + A263	Q125 + Q256	A174 + T459	L250 + H321	G346 + Q449
V17 + M286	V56 + V264	Q125 + A263	A174 + P473	L250 + S323	G346 + T459
V17 + F289	V56 + Y267	Q125 + V264	A174 + C474	L250 + H324	G346 + P473
V17 + V291	V56 + N270	Q125 + Y267	A174 + G476	L250 + F328	G346 + C474
V17 + Q299	V56 + G273	Q125 + N270	A174 + G477	L250 + T334	G346 + G476
V17 + R320	V56 + S280	Q125 + G273	A174 + K484	L250 + D337	G346 + G477
V17 + H321	V56 + T285	Q125 + S280	G184 + T193	L250 + Q345	G346 + K484
V17 + S323	V56 + M286	Q125 + T285	G184 + N195	L250 + G346	G348 + T355
V17 + H324	V56 + F289	Q125 + M286	G184 + A204	L250 + G348	G348 + S376
V17 + F328	V56 + V291	Q125 + F289	G184 + V206	L250 + T355	G348 + D377
V17 + T334	V56 + Q299	Q125 + V291	G184 + P211	L250 + S376	G348 + D379
V17 + D337	V56 + R320	Q125 + Q299	G184 + I214	L250 + D377	G348 + Y382
V17 + Q345	V56 + H321	Q125 + R320	G184 + V215	L250 + D379	G348 + S383
V17 + G346	V56 + S323	Q125 + H321	G184 + L217	L250 + Y382	G348 + Q385
V17 + G348	V56 + H324	Q125 + S323	G184 + L219	L250 + S383	G348 + K391

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V17 + T355	V56 + F328	Q125 + H324	G184 + L235	L250 + Q385	G348 + K393
V17 + S376	V56 + T334	Q125 + F328	G184 + V238	L250 + K391	G348 + Q395
V17 + D377	V56 + D337	Q125 + T334	G184 + M246	L250 + K393	G348 + A420
V17 + D379	V56 + Q345	Q125 + D337	G184 + M248	L250 + Q395	G348 + G423
V17 + Y382	V56 + Q346	Q125 + Q345	G184 + L250	L250 + A420	G348 + T444
V17 + S383	V56 + G348	Q125 + G346	G184 + G255	L250 + G423	G348 + A445
V17 + Q385	V56 + T355	Q125 + G348	G184 + Q256	L250 + T444	G348 + Q449
V17 + K391	V56 + S376	Q125 + T355	G184 + A263	L250 + A445	G348 + T459
V17 + K393	V56 + D377	Q125 + S376	G184 + V264	L250 + Q449	G348 + P473
V17 + Q395	V56 + D379	Q125 + D377	G184 + Y267	L250 + T459	G348 + C474
V17 + A420	V56 + Y382	Q125 + D379	G184 + N270	L250 + P473	G348 + G476
V17 + G423	V56 + S383	Q125 + Y382	G184 + G273	L250 + C474	G348 + G477
V17 + T444	V56 + Q385	Q125 + S383	G184 + S280	L250 + G476	G348 + K484
V17 + A445	V56 + K391	Q125 + Q385	G184 + T285	L250 + G477	T355 + S376
V17 + Q449	V56 + K393	Q125 + K391	G184 + M286	L250 + K484	T355 + D377
V17 + T459	V56 + Q395	Q125 + K393	G184 + F289	G255 + Q256	T355 + D379
V17 + P473	V56 + A420	Q125 + Q395	G184 + V291	G255 + A263	T355 + Y382
V17 + C474	V56 + G423	Q125 + A420	G184 + Q299	G255 + V264	T355 + S383
V17 + G476	V56 + T444	Q125 + G423	G184 + R320	G255 + Y267	T355 + Q385
V17 + G477	V56 + A445	Q125 + T444	G184 + H321	G255 + N270	T355 + K391
V17 + K484	V56 + Q449	Q125 + A445	G184 + S323	G255 + G273	T355 + K393
Q32 + A37	V56 + T459	Q125 + Q449	G184 + H324	G255 + S280	T355 + Q395
Q32 + T40	V56 + P473	Q125 + T459	G184 + F328	G255 + T285	T355 + A420
Q32 + P45	V56 + C474	Q125 + P473	G184 + T334	G255 + M286	T355 + G423
Q32 + W48	V56 + G476	Q125 + C474	G184 + D337	G255 + F289	T355 + T444
Q32 + G50	V56 + G477	Q125 + G476	G184 + Q345	G255 + V291	T355 + A445
Q32 + T51	V56 + K484	Q125 + G477	G184 + G346	G255 + Q299	T355 + Q449
Q32 + N54	K72 + R87	Q125 + K484	G184 + G348	G255 + R320	T355 + T459
Q32 + V56	K72 + Q98	G133 + T134	G184 + T355	G255 + H321	T355 + P473
Q32 + K72	K72 + M105	G133 + W140	G184 + S376	G255 + S323	T355 + C474
Q32 + R87	K72 + G109	G133 + G142	G184 + D377	G255 + H324	T355 + G476
Q32 + Q98	K72 + F113	G133 + G149	G184 + D379	G255 + F328	T355 + G477
Q32 + M105	K72 + R116	G133 + T165	G184 + Y382	G255 + T334	T355 + K484
Q32 + G109	K72 + Q118	G133 + W167	G184 + S383	G255 + D337	S376 + D377
Q32 + F113	K72 + Q125	G133 + R171	G184 + Q385	G255 + Q345	S376 + D379
Q32 + R116	K72 + G133	G133 + Q172	G184 + K391	G255 + G346	S376 + Y382
Q32 + Q118	K72 + T134	G133 + L173	G184 + K393	G255 + G348	S376 + S383
Q32 + Q125	K72 + W140	G133 + A174	G184 + Q395	G255 + T355	S376 + Q385
Q32 + G133	K72 + G142	G133 + G184	G184 + A420	G255 + S376	S376 + K391
Q32 + T134	K72 + G149	G133 + T193	G184 + G423	G255 + D377	S376 + K393
Q32 + W140	K72 + T165	G133 + N195	G184 + T444	G255 + D379	S376 + Q395
Q32 + G142	K72 + W167	G133 + A204	G184 + A445	G255 + Y382	S376 + A420
Q32 + G149	K72 + R171	G133 + V206	G184 + Q449	G255 + S383	S376 + G423
Q32 + T165	K72 + Q172	G133 + P211	G184 + T459	G255 + Q385	S376 + T444
Q32 + W167	K72 + L173	G133 + I214	G184 + P473	G255 + K391	S376 + A445
Q32 + R171	K72 + A174	G133 + V215	G184 + C474	G255 + K393	S376 + Q449
Q32 + Q172	K72 + G184	G133 + L217	G184 + G476	G255 + Q395	S376 + T459
Q32 + L173	K72 + T193	G133 + L219	G184 + G477	G255 + A420	S376 + P473
Q32 + A174	K72 + N195	G133 + L235	G184 + K484	G255 + G423	S376 + C474
Q32 + G184	K72 + A204	G133 + V238	T193 + N195	G255 + T444	S376 + G476
Q32 + T193	K72 + V206	G133 + M246	T193 + A204	G255 + A445	S376 + G477
Q32 + N195	K72 + P211	G133 + M248	T193 + V206	G255 + Q449	S376 + K484
Q32 + A204	K72 + I214	G133 + L250	T193 + P211	G255 + T459	D377 + D379
Q32 + V206	K72 + V215	G133 + G255	T193 + I214	G255 + P473	D377 + Y382
Q32 + P211	K72 + L217	G133 + Q256	T193 + V215	G255 + C474	D377 + S383
Q32 + I214	K72 + L219	G133 + A263	T193 + L217	G255 + G476	D377 + Q385
Q32 + V215	K72 + L235	G133 + V264	T193 + L219	G255 + G477	D377 + K391
Q32 + L217	K72 + V238	G133 + Y267	T193 + L235	G255 + K484	D377 + K393
Q32 + L219	K72 + M246	G133 + N270	T193 + V238	Q256 + A263	D377 + Q395
Q32 + L235	K72 + M248	G133 + G273	T193 + M246	Q256 + V264	D377 + A420
Q32 + V238	K72 + L250	G133 + S280	T193 + M248	Q256 + Y267	D377 + G423
Q32 + M246	K72 + G255	G133 + T285	T193 + L250	Q256 + N270	D377 + T444
Q32 + M248	K72 + Q256	G133 + M286	T193 + G255	Q256 + G273	D377 + A445
Q32 + L250	K72 + A263	G133 + F289	T193 + Q256	Q256 + S280	D377 + Q449
Q32 + G255	K72 + V264	G133 + V291	T193 + A263	Q256 + T285	D377 + T459
Q32 + Q256	K72 + Y267	G133 + Q299	T193 + V264	Q256 + M286	D377 + P473
Q32 + A263	K72 + N270	G133 + R320	T193 + Y267	Q256 + F289	D377 + C474
Q32 + V264	K72 + G273	G133 + H321	T193 + N270	Q256 + V291	D377 + G476
Q32 + Y267	K72 + S280	G133 + S323	T193 + G273	Q256 + Q299	D377 + G477
Q32 + N270	K72 + T285	G133 + H324	T193 + S280	Q256 + R320	D377 + K484
Q32 + G273	K72 + M286	G133 + F328	T193 + T285	Q256 + H321	D379 + Y382
Q32 + S280	K72 + F289	G133 + T334	T193 + M286	Q256 + S323	D379 + S383
Q32 + T285	K72 + V291	G133 + D337	T193 + F289	Q256 + H324	D379 + Q385
Q32 + M286	K72 + Q299	G133 + Q345	T193 + V291	Q256 + F328	D379 + K391
Q32 + F289	K72 + R320	G133 + G346	T193 + Q299	Q256 + T334	D379 + K393
Q32 + V291	K72 + H321	G133 + G348	T193 + R320	Q256 + D337	D379 + Q395
Q32 + Q299	K72 + S323	G133 + T355	T193 + H321	Q256 + Q345	D379 + A420
Q32 + R320	K72 + H324	G133 + S376	T193 + S323	Q256 + G346	D379 + G423
Q32 + H321	K72 + F328	G133 + D377	T193 + H324	Q256 + G348	D379 + T444

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Q32 + S323	K72 + T334	G133 + D379	T193 + F328	Q256 + T355	D379 + A445
Q32 + H324	K72 + D337	G133 + Y382	T193 + T334	Q256 + S376	D379 + Q449
Q32 + F328	K72 + Q345	G133 + S383	T193 + D337	Q256 + D377	D379 + T459
Q32 + T334	K72 + G346	G133 + Q385	T193 + Q345	Q256 + D379	D379 + P473
Q32 + D337	K72 + G348	G133 + K391	T193 + G346	Q256 + Y382	D379 + C474
Q32 + Q345	K72 + T355	G133 + K393	T193 + G348	Q256 + S383	D379 + G476
Q32 + G346	K72 + S376	G133 + Q395	T193 + T355	Q256 + Q385	D379 + G477
Q32 + G348	K72 + D377	G133 + A420	T193 + S376	Q256 + K391	D379 + K484
Q32 + T355	K72 + D379	G133 + G423	T193 + D377	Q256 + K393	Y382 + S383
Q32 + S376	K72 + Y382	G133 + T444	T193 + D379	Q256 + Q395	Y382 + Q385
Q32 + D377	K72 + S383	G133 + A445	T193 + Y382	Q256 + A420	Y382 + K391
Q32 + D379	K72 + Q385	G133 + Q449	T193 + S383	Q256 + G423	Y382 + K393
Q32 + Y382	K72 + K391	G133 + T459	T193 + Q385	Q256 + T444	Y382 + Q395
Q32 + S383	K72 + K393	G133 + P473	T193 + K391	Q256 + A445	Y382 + A420
Q32 + Q385	K72 + Q395	G133 + C474	T193 + K393	Q256 + Q449	Y382 + G423
Q32 + K391	K72 + A420	G133 + G476	T193 + Q395	Q256 + T459	Y382 + T444
Q32 + K393	K72 + G423	G133 + G477	T193 + A420	Q256 + P473	Y382 + A445
Q32 + Q395	K72 + T444	G133 + K484	T193 + G423	Q256 + C474	Y382 + Q449
Q32 + A420	K72 + A445	T134 + W140	T193 + T444	Q256 + G476	Y382 + T459
Q32 + G423	K72 + Q449	T134 + G142	T193 + A445	Q256 + G477	Y382 + P473
Q32 + T444	K72 + T459	T134 + G149	T193 + Q449	Q256 + K484	Y382 + C474
Q32 + A445	K72 + P473	T134 + T165	T193 + T459	A263 + V264	Y382 + G476
Q32 + Q449	K72 + C474	T134 + W167	T193 + P473	A263 + Y267	Y382 + G477
Q32 + T459	K72 + G476	T134 + R171	T193 + C474	A263 + N270	Y382 + K484
Q32 + P473	K72 + G477	T134 + Q172	T193 + G476	A263 + G273	S383 + Q385
Q32 + C474	K72 + K484	T134 + L173	T193 + G477	A263 + S280	S383 + K391
Q32 + G476	R87 + Q98	T134 + A174	T193 + K484	A263 + T285	S383 + K393
Q32 + G477	R87 + M105	T134 + G184	N195 + A204	A263 + M286	S383 + Q395
Q32 + K484	R87 + G109	T134 + T193	N195 + V206	A263 + F289	S383 + A420
A37 + T40	R87 + F113	T134 + N195	N195 + P211	A263 + V291	S383 + G423
A37 + P45	R87 + R116	T134 + A204	N195 + I214	A263 + Q299	S383 + T444
A37 + W48	R87 + Q118	T134 + V206	N195 + V215	A263 + R320	S383 + A445
A37 + G50	R87 + Q125	T134 + P211	N195 + L217	A263 + H321	S383 + Q449
A37 + T51	R87 + G133	T134 + I214	N195 + L219	A263 + S323	S383 + T459
A37 + N54	R87 + T134	T134 + V215	N195 + L235	A263 + H324	S383 + P473
A37 + V56	R87 + W140	T134 + L217	N195 + V238	A263 + F328	S383 + C474
A37 + K72	R87 + G142	T134 + L219	N195 + M246	A263 + T334	S383 + G476
A37 + R87	R87 + G149	T134 + L235	N195 + M248	A263 + D337	S383 + G477
A37 + Q98	R87 + T165	T134 + V238	N195 + L250	A263 + Q345	S383 + K484
A37 + M105	R87 + W167	T134 + M246	N195 + G255	A263 + G346	Q385 + K391
A37 + G109	R87 + R171	T134 + M248	N195 + Q256	A263 + G348	Q385 + K393
A37 + F113	R87 + Q172	T134 + L250	N195 + A263	A263 + T355	Q385 + Q395
A37 + R116	R87 + L173	T134 + G255	N195 + V264	A263 + S376	Q385 + A420
A37 + Q118	R87 + A174	T134 + Q256	N195 + Y267	A263 + D377	Q385 + G423
A37 + Q125	R87 + G184	T134 + A263	N195 + N270	A263 + D379	Q385 + T444
A37 + G133	R87 + T193	T134 + V264	N195 + G273	A263 + Y382	Q385 + A445
A37 + T134	R87 + N195	T134 + Y267	N195 + S280	A263 + S383	Q385 + Q449
A37 + W140	R87 + A204	T134 + N270	N195 + T285	A263 + Q385	Q385 + T459
A37 + G142	R87 + V206	T134 + G273	N195 + M286	A263 + K391	Q385 + P473
A37 + G149	R87 + P211	T134 + S280	N195 + F289	A263 + K393	Q385 + C474
A37 + T165	R87 + I214	T134 + T285	N195 + V291	A263 + Q395	Q385 + G476
A37 + W167	R87 + V215	T134 + M286	N195 + Q299	A263 + A420	Q385 + G477
A37 + R171	R87 + L217	T134 + F289	N195 + R320	A263 + G423	Q385 + K484
A37 + Q172	R87 + L219	T134 + V291	N195 + H321	A263 + T444	K391 + K393
A37 + L173	R87 + L235	T134 + Q299	N195 + S323	A263 + A445	K391 + Q395
A37 + A174	R87 + V238	T134 + R320	N195 + H324	A263 + Q449	K391 + A420
A37 + G184	R87 + M246	T134 + H321	N195 + F328	A263 + T459	K391 + G423
A37 + T193	R87 + M248	T134 + S323	N195 + T334	A263 + P473	K391 + T444
A37 + N195	R87 + L250	T134 + H324	N195 + D337	A263 + C474	K391 + A445
A37 + A204	R87 + G255	T134 + F328	N195 + Q345	A263 + G476	K391 + Q449
A37 + V206	R87 + Q256	T134 + T334	N195 + G346	A263 + G477	K391 + T459
A37 + P211	R87 + A263	T134 + D337	N195 + G348	A263 + K484	K391 + P473
A37 + I214	R87 + V264	T134 + Q345	N195 + T355	V264 + Y267	K391 + C474
A37 + V215	R87 + Y267	T134 + G346	N195 + S376	V264 + N270	K391 + G476
A37 + L217	H1 + Q169	T134 + G348	N195 + D377	V264 + G273	K391 + G477
A37 + L219	H1 + A186	T134 + T355	N195 + D379	V264 + S280	K391 + K484
A37 + L235	H1 + E190	T134 + S376	N195 + Y382	V264 + T285	K393 + Q395
A37 + V238	H1 + A225	T134 + D377	N195 + S383	V264 + M286	K393 + A420
A37 + M246	H1 + K242	T134 + D379	N195 + Q385	V264 + F289	K393 + G423
A37 + M248	H1 + S244	T134 + Y382	N195 + K391	V264 + V291	K393 + T444
A37 + L250	H1 + G258	T134 + S383	N195 + K393	V264 + Q299	K393 + A445
A37 + G255	H1 + N260	T134 + Q385	N195 + Q395	V264 + R320	K393 + Q449
A37 + Q256	H1 + K269	T134 + K391	N195 + A420	V264 + H321	K393 + T459
A37 + A263	H1 + W284	T134 + K393	N195 + G423	V264 + S323	K393 + P473
A37 + V264	H1 + Y295	T134 + Q395	N195 + T444	V264 + H324	K393 + C474
A37 + Y267	H1 + S304	T134 + A420	N195 + A445	V264 + F328	K393 + G476
A37 + N270	H1 + V326	T134 + G423	N195 + Q449	V264 + T334	K393 + G477
A37 + G273	T5 + Q169	T134 + T444	N195 + T459	V264 + D337	K393 + K484
A37 + S280	T5 + A186	T134 + A445	N195 + P473	V264 + Q345	Q395 + A420

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A37 + T285	T5 + E190	T134 + Q449	N195 + C474	V264 + G346	Q395 + G423
A37 + M286	T5 + A225	T134 + T459	N195 + G476	V264 + G348	Q395 + T444
A37 + F289	T5 + K242	T134 + P473	N195 + G477	V264 + T355	Q395 + A445
A37 + V291	T5 + S244	T134 + C474	N195 + K484	V264 + S376	Q395 + Q449
A37 + Q299	T5 + G258	T134 + G476	A204 + V206	V264 + D377	Q395 + T459
A37 + R320	T5 + N260	T134 + G477	A204 + P211	V264 + D379	Q395 + P473
A37 + H321	T5 + K269	T134 + K484	A204 + I214	V264 + Y382	Q395 + C474
A37 + S323	T5 + W284	W140 + G142	A204 + V215	V264 + S383	Q395 + G476
A37 + H324	T5 + Y295	W140 + G149	A204 + L217	V264 + Q385	Q395 + G477
A37 + F328	T5 + S304	W140 + T165	A204 + L219	V264 + K391	Q395 + K484
A37 + T334	T5 + V326	W140 + W167	A204 + L235	V264 + K393	A420 + G423
A37 + D337	G7 + Q169	W140 + R171	A204 + V238	V264 + Q395	A420 + T444
A37 + Q345	G7 + A186	W140 + Q172	A204 + M246	V264 + A420	A420 + A445
A37 + G346	G7 + E190	W140 + L173	A204 + M248	V264 + G423	A420 + Q449
A37 + G348	G7 + A225	W140 + A174	A204 + L250	V264 + T444	A420 + T459
A37 + T355	G7 + K242	W140 + G184	A204 + G255	V264 + A445	A420 + P473
A37 + S376	G7 + S244	W140 + T193	A204 + Q256	V264 + Q449	A420 + C474
A37 + D377	G7 + G258	W140 + N195	A204 + A263	V264 + T459	A420 + G476
A37 + D379	G7 + N260	W140 + A204	A204 + V264	V264 + P473	A420 + G477
A37 + Y382	G7 + A265	W140 + V206	A204 + Y267	V264 + C474	A420 + K484
A37 + S383	G7 + K269	W140 + P211	A204 + N270	V264 + G476	G423 + T444
A37 + Q385	G7 + W284	W140 + I214	A204 + G273	V264 + G477	G423 + A445
A37 + K391	G7 + Y295	W140 + V215	A204 + S280	V264 + K484	G423 + Q449
A37 + K393	G7 + S304	W140 + L217	A204 + T285	Y267 + N270	G423 + T459
A37 + Q395	G7 + V326	W140 + L219	A204 + M286	Y267 + G273	G423 + P473
A37 + A420	Q11 + Q169	W140 + L235	A204 + F289	Y267 + S280	G423 + C474
A37 + G423	Q11 + A186	W140 + V238	A204 + V291	Y267 + T285	G423 + G476
A37 + T444	Q11 + E190	W140 + M246	A204 + Q299	Y267 + M286	G423 + G477
A37 + A445	Q11 + A225	W140 + M248	A204 + R320	Y267 + F289	G423 + K484
A37 + Q449	Q11 + K242	W140 + L250	A204 + H321	Y267 + V291	T444 + A445
A37 + T459	Q11 + S244	W140 + G255	A204 + S323	Y267 + Q299	T444 + Q449
A37 + P473	Q11 + G258	W140 + Q256	A204 + H324	Y267 + R320	T444 + T459
A37 + C474	Q11 + N260	W140 + A263	A204 + F328	Y267 + H321	T444 + P473
A37 + G476	Q11 + K269	W140 + V264	A204 + T334	Y267 + S323	T444 + C474
A37 + G477	Q11 + W284	W140 + Y267	A204 + D337	Y267 + H324	T444 + G476
A37 + K484	Q11 + Y295	W140 + N270	A204 + Q345	Y267 + F328	T444 + G477
T40 + P45	Q11 + S244	W140 + G273	A204 + G346	Y267 + T334	T444 + K484
T40 + W48	Q11 + V326	W140 + S280	A204 + G348	Y267 + D337	A445 + Q449
T40 + G50	N16 + Q169	W140 + T285	A204 + T355	Y267 + Q345	A445 + T459
T40 + T51	N16 + A186	W140 + M286	A204 + S376	Y267 + G346	A445 + P473
T40 + N54	N16 + E190	W140 + F289	A204 + D377	Y267 + G348	A445 + C474
T40 + V56	N16 + A225	W140 + V291	A204 + D379	Y267 + T355	A445 + G476
T40 + K72	N16 + K242	W140 + Q299	A204 + Y382	Y267 + S376	A445 + G477
T40 + R87	N16 + S244	W140 + R320	A204 + S383	Y267 + D377	A445 + K484
T40 + Q98	N16 + G258	W140 + H321	A204 + Q385	Y267 + D379	Q449 + T459
T40 + M105	N16 + N260	W140 + S323	A204 + K391	Y267 + Y382	Q449 + P473
T40 + G109	N16 + K269	W140 + H324	A204 + K393	Y267 + S383	Q449 + C474
T40 + F113	N16 + W284	W140 + F328	A204 + Q395	Y267 + Q385	Q449 + G476
T40 + R116	N16 + Y326	W140 + T334	A204 + A420	Y267 + K391	Q449 + G477
T40 + Q118	N16 + S304	W140 + D337	A204 + G423	Y267 + K393	Q449 + K484
T40 + Q125	N16 + V326	W140 + Q345	A204 + T444	Y267 + Q395	T459 + P473
T40 + G133	V17 + Q169	W140 + G346	A204 + A445	Y267 + A420	T459 + C474
T40 + T134	V17 + A186	W140 + G348	A204 + Q449	Y267 + G423	T459 + G476
T40 + W140	V17 + E190	W140 + T355	A204 + T459	Y267 + T444	T459 + G477
T40 + G142	V17 + A225	W140 + S376	A204 + P473	Y267 + A445	T459 + K484
T40 + G149	V17 + K242	W140 + D377	A204 + C474	Y267 + Q449	P473 + C474
T40 + T165	V17 + S244	W140 + D379	A204 + G476	Y267 + T459	P473 + G476
T40 + W167	V17 + G258	W140 + Y382	A204 + G477	Y267 + P473	P473 + G477
T40 + R171	V17 + N260	W140 + S383	A204 + K484	G109 + Q169	P473 + K484
T40 + Q172	V17 + K269	W140 + Q385	V206 + P211	G109 + A186	C474 + G476
T40 + L173	V17 + W284	W140 + K391	V206 + I214	G109 + E190	C474 + G477
T40 + Q169	V17 + Y295	N54 + Q169	K72 + Q169	G109 + A225	C474 + K484
T40 + A186	V17 + S304	N54 + A186	K72 + A186	G109 + K242	G476 + G477
T40 + E190	V17 + V326	N54 + E190	K72 + E190	G109 + S244	G476 + K484
T40 + A225	Q32 + Q169	N54 + A225	K72 + A225	G109 + G258	G477 + K484
T40 + K242	Q32 + A186	N54 + K242	K72 + K242	G109 + N260	G133 + Q169
T40 + S244	Q32 + E190	N54 + S244	K72 + S244	G109 + K269	G133 + A186
T40 + G258	Q32 + A225	N54 + G258	K72 + G258	G109 + W284	G133 + E190
T40 + N260	Q32 + K242	N54 + N260	K72 + N260	G109 + Y295	G133 + A225
T40 + K269	Q32 + S244	N54 + K269	K72 + K269	G109 + S304	G133 + K242
T40 + W284	Q32 + G258	N54 + W284	K72 + W284	G109 + V326	G133 + S244
T40 + Y295	Q32 + N260	N54 + Y295	K72 + Y295	F113 + Q169	G133 + G258
T40 + S304	Q32 + K269	N54 + S304	K72 + S304	F113 + A186	G133 + N260
T40 + V326	Q32 + W284	N54 + V326	K72 + V326	F113 + E190	G133 + K269
P45 + Q169	Q32 + Y295	V56 + Q169	R87 + Q169	F113 + A225	G133 + W284
P45 + A186	Q32 + S304	V56 + A186	R87 + A186	F113 + K242	G133 + Y295
P45 + E190	Q32 + V326	V56 + E190	R87 + E190	F113 + S244	G133 + S304
P45 + A225	A37 + Q169	V56 + A225	R87 + A225	F113 + G258	G133 + V326
P45 + K242	A37 + A186	V56 + K242	R87 + K242	F113 + N260	T134 + Q169

P45 + S244	A37 + E190	V56 + S244	R87 + S244	F113 + K269	T134 + A186
P45 + G258	A37 + A225	V56 + G258	R87 + G258	F113 + W284	T134 + E190
P45 + N260	A37 + K242	V56 + N260	R87 + N260	F113 + Y295	T134 + A225
P45 + K269	A37 + S244	V56 + K269	R87 + K269	F113 + S304	T134 + K242
P45 + W284	A37 + G258	V56 + W284	R87 + W284	F113 + V326	T134 + S244
P45 + Y295	A37 + N260	V56 + Y295	R87 + Y295	R116 + Q169	T134 + G258
P45 + S304	A37 + K269	V56 + S304	R87 + S304	R116 + A186	T134 + N260
P45 + V326	A37 + W284	V56 + V326	R87 + V326	R116 + E190	T134 + K269
W48 + Q169	A37 + Y295	Q169 + R171	Q98 + Q169	R116 + A225	T134 + W284
W48 + A186	A37 + S304	Q169 + Q172	Q98 + A186	R116 + K242	T134 + Y295
W48 + E190	A37 + V326	Q169 + L173	Q98 + E190	R116 + S244	T134 + S304
W48 + A225	G142 + Q169	Q169 + A174	Q98 + A225	R116 + G258	T134 + V326
W48 + K242	G142 + A186	Q169 + G184	Q98 + K242	R116 + N260	W140 + Q169
W48 + S244	G142 + E190	Q169 + A186	Q98 + S244	R116 + K269	W140 + A186
W48 + G258	G142 + A225	Q169 + E190	Q98 + G258	R116 + W284	W140 + E190
W48 + N260	G142 + K242	Q169 + T193	Q98 + N260	R116 + Y295	W140 + A225
W48 + K269	G142 + S244	Q169 + N195	Q98 + K269	R116 + S304	W140 + K242
W48 + W284	G142 + G258	Q169 + A204	Q98 + W284	R116 + V326	W140 + S244
W48 + Y295	G142 + N260	Q169 + V206	Q98 + Y295	Q118 + Q169	W140 + G258
W48 + S304	G142 + K269	Q169 + P211	Q98 + S304	Q118 + A186	W140 + N260
W48 + V326	G142 + W284	Q169 + I214	Q98 + V326	Q118 + E190	W140 + K269
G50 + Q169	G142 + Y295	Q169 + V215	M105 + Q169	Q118 + A225	W140 + W284
G50 + A186	G142 + S304	Q169 + L217	M105 + A186	Q118 + K242	W140 + Y295
G50 + E190	G142 + V326	Q169 + L219	M105 + E190	Q118 + S244	W140 + S304
G50 + A225	G149 + Q169	Q169 + A225	M105 + A225	Q118 + G258	W140 + V326
G50 + K242	G149 + A186	Q169 + L235	M105 + K242	Q118 + N260	E190 + L235
G50 + S244	G149 + E190	Q169 + V238	M105 + S244	Q118 + K269	A225 + V238
G50 + G258	G149 + A225	Q169 + K242	M105 + G258	Q118 + W284	A225 + K242
G50 + N260	G149 + K242	Q169 + S244	M105 + N260	Q118 + Y295	A225 + S244
G50 + K269	G149 + S244	Q169 + M246	M105 + K269	Q118 + S304	A225 + M246
G50 + W284	G149 + G258	Q169 + M248	M105 + W284	Q118 + V326	A225 + M248
G50 + Y295	G149 + N260	Q169 + L250	M105 + Y295	Q125 + Q169	A225 + L250
G50 + S304	G149 + K269	Q169 + G255	M105 + S304	Q125 + A186	A225 + G255
G50 + V326	G149 + W284	Q169 + Q256	M105 + V326	Q125 + E190	A225 + Q256
T51 + Q169	G149 + Y295	Q169 + G258	R171 + A186	Q125 + A225	A225 + G258
T51 + A186	G149 + S304	Q169 + N260	R171 + E190	Q125 + K242	A225 + N260
T51 + E190	G149 + V326	Q169 + A263	R171 + A225	Q125 + S244	A225 + A263
T51 + A225	T165 + Q169	Q169 + V264	R171 + K242	Q125 + G258	A225 + V264
T51 + K242	T165 + A186	Q169 + Y267	R171 + S244	Q125 + N260	A225 + Y267
T51 + S244	T165 + E190	Q169 + K269	R171 + G258	Q125 + K269	A225 + K269
T51 + G258	T165 + A225	Q169 + N270	R171 + N260	Q125 + W284	A225 + N270
T51 + N260	T165 + K242	Q169 + G273	R171 + K269	Q125 + Y295	A225 + G273
T51 + K269	T165 + S244	Q169 + S280	R171 + W284	Q125 + S304	A225 + S280
T51 + W284	T165 + G258	Q169 + W284	R171 + Y295	Q125 + V326	A225 + W284
T51 + Y295	T165 + N260	Q169 + T285	R171 + S304	G184 + A186	A225 + T285
T51 + S304	T165 + K269	Q169 + M286	R171 + V326	G184 + E190	A225 + M286
T51 + V326	T165 + W284	Q169 + F289	Q172 + A186	G184 + A225	A225 + F289
A186 + E190	T165 + Y295	Q169 + V291	Q172 + E190	G184 + K242	A225 + V291
A186 + T193	T165 + S304	Q169 + Y295	Q172 + A225	G184 + S244	A225 + Y295
A186 + N195	T165 + V326	Q169 + Q299	Q172 + K242	G184 + G258	A225 + Q299
A186 + A204	W167 + Q169	Q169 + S304	Q172 + S244	G184 + N260	A225 + S304
A186 + V206	W167 + A186	Q169 + R320	Q172 + G258	G184 + K269	A225 + R320
A186 + P211	W167 + E190	Q169 + H321	Q172 + N260	G184 + W284	A225 + H321
A186 + I214	W167 + A225	Q169 + S323	Q172 + K269	G184 + Y295	A225 + S323
A186 + V215	W167 + K242	Q169 + H324	Q172 + W284	G184 + S304	A225 + H324
A186 + L217	W167 + S244	Q169 + V326	Q172 + Y295	G184 + V326	A225 + V326
A186 + L219	W167 + G258	Q169 + F328	Q172 + S304	L235 + K242	A225 + F328
A186 + A225	W167 + N260	Q169 + T334	Q172 + V326	L235 + S244	A225 + T334
A186 + L235	W167 + K269	Q169 + D337	L173 + A186	L235 + G258	A225 + D337
A186 + V238	W167 + W284	Q169 + Q345	L173 + E190	L235 + N260	A225 + Q345
A186 + K242	W167 + Y295	Q169 + G346	L173 + A225	L235 + K269	A225 + G346
A186 + S244	W167 + S304	Q169 + G348	L173 + K242	L235 + W284	A225 + G348
A186 + M246	W167 + V326	Q169 + T355	L173 + S244	L235 + Y295	A225 + T355
A186 + M248	E190 + T193	Q169 + S376	L173 + G258	L235 + S304	A225 + S376
A186 + L250	E190 + N195	Q169 + D377	L173 + N260	L235 + V326	A225 + D377
A186 + G255	E190 + A204	Q169 + D379	L173 + K269	V238 + K242	A225 + D379
A186 + Q256	E190 + V206	Q169 + Y382	L173 + W284	V238 + S244	A225 + Y382
A186 + G258	E190 + P211	Q169 + S383	L173 + Y295	V238 + G258	A225 + S383
A186 + N260	E190 + I214	Q169 + Q385	L173 + S304	V238 + N260	A225 + Q385
A186 + A263	E190 + V215	Q169 + K391	L173 + V326	V238 + K269	A225 + K391
A186 + V264	E190 + L217	Q169 + K393	A174 + A186	V238 + W284	A225 + K393
A186 + Y267	E190 + L219	Q169 + Q395	A174 + E190	V238 + Y295	A225 + Q395
A186 + K269	E190 + A225	Q169 + A420	A174 + A225	V238 + S304	A225 + A420
A186 + N270	E190 + L235	Q169 + G423	A174 + K242	V238 + V326	A225 + G423
A186 + G273	E190 + V238	Q169 + T444	A174 + S244	K242 + S244	A225 + T444
A186 + S280	E190 + K242	Q169 + A445	A174 + G258	K242 + M246	A225 + A445
A186 + W284	E190 + S244	Q169 + Q449	A174 + N260	K242 + M248	A225 + Q449
A186 + T285	E190 + M246	Q169 + T459	A174 + K269	K242 + L250	A225 + T459
A186 + M286	E190 + M248	Q169 + P473	A174 + W284	K242 + G255	A225 + P473

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A186 + F289	E190 + L250	Q169 + C474	A174 + Y295	K242 + Q256	A225 + C474
A186 + V291	E190 + G255	Q169 + G476	A174 + S304	K242 + G258	A225 + G476
A186 + Y295	E190 + Q256	Q169 + G477	A174 + V326	K242 + N260	A225 + G477
A186 + Q299	E190 + G258	Q169 + K484	L217 + A225	K242 + A263	A225 + K484
A186 + S304	E190 + N260	T193 + A225	L217 + K242	K242 + V264	S244 + M248
A186 + R320	E190 + A263	T193 + K242	L217 + S244	K242 + Y267	S244 + L250
A186 + H321	E190 + V264	T193 + S244	L217 + G258	K242 + K269	S244 + G255
A186 + S323	E190 + Y267	T193 + G258	L217 + N260	K242 + N270	S244 + Q256
A186 + H324	E190 + K269	T193 + N260	L217 + K269	K242 + G273	S244 + G258
A186 + V326	E190 + N270	T193 + K269	L217 + W284	K242 + S280	S244 + N260
A186 + F328	E190 + G273	T193 + W284	L217 + Y295	K242 + W284	S244 + A263
A186 + T334	E190 + S280	T193 + Y295	L217 + S304	K242 + T285	S244 + V264
A186 + D337	E190 + W284	T193 + S304	L217 + V326	K242 + M286	S244 + Y267
A186 + Q345	E190 + T285	T193 + V326	L219 + A225	K242 + F289	S244 + K269
A186 + G346	E190 + M286	N195 + A225	L219 + K242	K242 + V291	S244 + N270
A186 + G348	E190 + F289	N195 + K242	L219 + S244	K242 + Y295	S244 + G273
A186 + T355	E190 + V291	N195 + S244	L219 + G258	K242 + Q299	S244 + S280
A186 + S376	E190 + Y295	N195 + G258	L219 + N260	K242 + S304	S244 + W284
A186 + D377	E190 + Q299	N195 + N260	L219 + K269	K242 + R320	S244 + T285
A186 + D379	E190 + S304	N195 + K269	L219 + W284	K242 + H321	S244 + M286
A186 + Y382	E190 + R320	N195 + W284	L219 + Y295	K242 + S323	S244 + F289
A186 + S383	E190 + H321	N195 + Y295	L219 + S304	K242 + H324	S244 + V291
A186 + Q385	E190 + S323	N195 + S304	L219 + V326	K242 + V326	S244 + Y295
A186 + K391	E190 + H324	N195 + V326	M246 + G258	K242 + F328	S244 + Q299
A186 + K393	E190 + V326	A204 + A225	M246 + N260	K242 + T334	S244 + S304
A186 + Q395	E190 + F328	A204 + K242	M246 + K269	K242 + D337	S244 + R320
A186 + A420	E190 + T334	A204 + S244	M246 + W284	K242 + Q345	S244 + H321
A186 + G423	E190 + D337	A204 + G258	M246 + Y295	K242 + G346	S244 + S323
A186 + T444	E190 + Q345	A204 + N260	M246 + S304	K242 + G348	S244 + H324
A186 + A445	E190 + G346	A204 + K269	M246 + V326	K242 + T355	S244 + V326
A186 + Q449	E190 + G348	A204 + W284	M248 + G258	K242 + S376	S244 + F328
A186 + T459	E190 + T355	A204 + Y295	M248 + N260	K242 + D377	S244 + T334
A186 + P473	E190 + S376	A204 + S304	M248 + K269	K242 + D379	S244 + D337
A186 + C474	E190 + D377	A204 + V326	M248 + W284	K242 + Y382	S244 + Q345
A186 + G476	E190 + D379	V206 + A225	M248 + Y295	K242 + S383	S244 + G346
A186 + G477	E190 + Y382	V206 + K242	M248 + S304	K242 + Q385	S244 + G348
A186 + K484	E190 + S383	V206 + S244	M248 + V326	K242 + K391	S244 + T355
I214 + A225	E190 + Q385	V206 + G258	L250 + G258	K242 + K393	S244 + S376
I214 + K242	E190 + K391	V206 + N260	L250 + N260	K242 + Q395	S244 + D377
I214 + S244	E190 + K393	V206 + K269	L250 + K269	K242 + A420	S244 + D379
I214 + G258	E190 + Q395	V206 + W284	L250 + W284	K242 + G423	S244 + Y382
I214 + N260	E190 + A420	V206 + Y295	L250 + Y295	K242 + T444	S244 + S383
I214 + K269	E190 + G423	V206 + S304	L250 + S304	K242 + A445	S244 + Q385
I214 + W284	E190 + T444	V206 + V326	L250 + V326	K242 + Q449	S244 + K391
I214 + Y295	E190 + A445	P211 + A225	G255 + G258	K242 + T459	S244 + K393
I214 + S304	E190 + Q449	P211 + K242	G255 + N260	K242 + P473	S244 + Q395
I214 + V326	E190 + T459	P211 + S244	G255 + K269	K242 + C474	S244 + A420
V215 + A225	E190 + P473	P211 + G258	G255 + W284	K242 + G476	S244 + G423
V215 + K242	E190 + C474	P211 + N260	G255 + Y295	K242 + G477	S244 + T444
V215 + S244	E190 + G476	P211 + K269	G255 + S304	K242 + K484	S244 + A445
V215 + G258	E190 + G477	P211 + W284	G255 + V326	N260 + A263	S244 + Q449
V215 + N260	E190 + K484	P211 + Y295	Q256 + G258	N260 + V264	S244 + T459
V215 + K269	G258 + N260	P211 + S304	Q256 + N260	N260 + Y267	S244 + P473
V215 + W284	G258 + A263	P211 + V326	Q256 + K269	N260 + K269	S244 + C474
V215 + Y295	G258 + V264	A236 + K269	Q256 + W284	N260 + N270	S244 + G476
V215 + S304	G258 + Y267	A236 + W284	Q256 + Y295	N260 + G273	S244 + G477
V215 + V326	G258 + K269	A236 + Y295	Q256 + S304	N260 + S280	S244 + K484
N270 + W284	G258 + N270	A236 + S304	Q256 + V326	N260 + W284	K269 + N270
N270 + Y295	G258 + G273	A236 + V326	W284 + T285	N260 + T285	K269 + G273
N270 + S304	G258 + S280	V264 + K269	W284 + M286	N260 + M286	K269 + S280
N270 + V326	G258 + W284	V264 + W284	W284 + F289	N260 + F289	K269 + W284
G273 + W284	G258 + T285	V264 + Y295	W284 + V291	N260 + V291	K269 + T285
G273 + Y295	G258 + M286	V264 + S304	W284 + Y295	N260 + Y295	K269 + M286
G273 + S304	G258 + F289	V267 + V326	W284 + Q299	N260 + Q299	K269 + F289
G273 + V326	G258 + V291	Y267 + K269	W284 + S304	N260 + S304	K269 + V291
S280 + W284	G258 + Y295	Y267 + W284	W284 + R320	N260 + R320	K269 + Y295
S280 + Y295	G258 + Q299	Y267 + Y295	W284 + H321	N260 + H321	K269 + Q299
S280 + S304	G258 + S304	Y267 + S304	W284 + S323	N260 + S323	K269 + S304
S280 + V326	G258 + R320	Y267 + V326	W284 + H324	N260 + H324	K269 + R320
T285 + Y295	G258 + H321	A288 + F289	W284 + V326	N260 + V326	K269 + H321
T285 + S304	G258 + S323	A288 + V291	W284 + F328	N260 + F328	K269 + S323
T285 + V326	G258 + H324	A288 + Y295	W284 + T334	N260 + T334	K269 + H324
M286 + Y295	G258 + V326	A288 + Q299	W284 + D337	N260 + D337	K269 + V326
M286 + S304	G258 + F328	A288 + S304	W284 + Q345	N260 + Q345	K269 + F328
M286 + V326	G258 + T334	A288 + R320	W284 + G346	N260 + G346	K269 + T334
F289 + Y295	G258 + D337	A288 + H321	W284 + G348	N260 + G348	K269 + D337
F289 + S304	G258 + Q345	A288 + S323	W284 + T355	N260 + T355	K269 + Q345
F289 + V326	G258 + G346	A288 + H324	W284 + S376	N260 + S376	K269 + G346
V291 + Y295	G258 + G348	A288 + V326	W284 + D377	N260 + D377	K269 + G348

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V291 + S304	G258 + T355	A288 + F328	W284 + D379	N260 + D379	K269 + T355
V291 + V326	G258 + S376	A288 + T334	W284 + Y382	N260 + Y382	K269 + S376
Y295 + Q299	G258 + D377	A288 + D337	W284 + S383	N260 + S383	K269 + D377
Y295 + S304	G258 + D379	A288 + Q345	W284 + Q385	N260 + Q385	K269 + D379
Y295 + R320	G258 + Y382	A288 + G346	W284 + K391	N260 + K391	K269 + Y382
Y295 + H321	G258 + S383	A288 + G348	W284 + K393	N260 + K393	K269 + S383
Y295 + S323	G258 + Q385	A288 + T355	W284 + Q395	N260 + Q395	K269 + Q385
Y295 + H324	G258 + K391	A288 + S376	W284 + A420	N260 + A420	K269 + K391
Y295 + V326	G258 + K393	A288 + D377	W284 + G423	N260 + G423	K269 + K393
Y295 + F328	G258 + Q395	A288 + D379	W284 + T444	N260 + T444	K269 + Q395
Y295 + T334	G258 + A420	A288 + Y382	W284 + A445	N260 + A445	K269 + A420
Y295 + D337	G258 + G423	A288 + S383	W284 + Q449	N260 + Q449	K269 + G423
Y295 + Q345	G258 + T444	A288 + Q385	W284 + T459	N260 + T459	K269 + T444
Y295 + G346	G258 + A445	A288 + K391	W284 + P473	N260 + P473	K269 + A445
Y295 + G348	G258 + Q449	A288 + K393	W284 + C474	N260 + C474	K269 + Q449
Y295 + T355	G258 + T459	A288 + Q395	W284 + G476	N260 + G476	K269 + T459
Y295 + S376	G258 + P473	A288 + A420	W284 + G477	N260 + G477	K269 + P473
Y295 + D377	G258 + C474	A288 + G423	W284 + K484	N260 + K484	K269 + C474
Y295 + D379	G258 + G476	A288 + T444	S304 + V326	S304 + K393	K269 + G476
Y295 + Y382	G258 + G477	A288 + A445	S304 + Y295	S304 + Q395	K269 + G477
Y295 + S383	G258 + K484	A288 + Q449	S304 + S304	S304 + A420	K269 + K484
Y295 + Q385	Q299 + S304	A288 + T459	S304 + V326	S304 + G423	S304 + T355
Y295 + K391	Q299 + V326	A288 + P473	S304 + Q299	S304 + T444	S304 + S376
Y295 + K393	R320 + V326	A288 + C474	S304 + S304	S304 + A445	S304 + D377
Y295 + Q395	H321 + V326	A288 + G476	S304 + R320	S304 + Q449	S304 + D379
Y295 + A420	S323 + V326	A288 + G477	S304 + H321	S304 + T459	S304 + Y382
Y295 + G423	H324 + V326	A288 + K484	S304 + S323	S304 + P473	S304 + S383
Y295 + T444	Y295 + P473	S304 + Q345	S304 + H324	S304 + C474	S304 + Q385
Y295 + A445	Y295 + C474	S304 + G346	S304 + V326	S304 + G476	S304 + K391
Y295 + Q449	Y295 + G476	S304 + G348	S304 + F328	S304 + G477	S304 + D337
Y295 + T459	Y295 + G477	Y295 + K484	S304 + T334	S304 + K484	

wherein numbering is according to SEQ ID NO: 13; or

H1 + A37	N54 + F113	K72 + G109	R116 + A174	T165 + A174	N195 + V206
H1 + T40	N54 + R116	K72 + F113	R116 + G184	T165 + G184	N195 + L228
H1 + N54	N54 + N125	K72 + R116	R116 + N195	T165 + N195	N195 + G255
H1 + V56	N54 + F133	K72 + N125	R116 + G196	T165 + G196	N195 + A265
H1 + A60	N54 + T134	K72 + F133	R116 + A204	T165 + A204	N195 + S280
H1 + K72	N54 + T165	K72 + T134	R116 + V206	T165 + V206	N195 + S304
H1 + G109	N54 + Q172	K72 + T165	R116 + L228	T165 + L228	N195 + R320
H1 + F113	N54 + L173	K72 + Q172	R116 + G255	T165 + G255	N195 + H321
H1 + R116	N54 + A174	K72 + L173	R116 + A265	T165 + A265	N195 + S323
H1 + N125	N54 + G184	K72 + A174	R116 + S280	T165 + S280	N195 + K391
H1 + F133	N54 + N195	K72 + G184	R116 + S304	T165 + S304	G196 + A204
H1 + T134	N54 + G196	K72 + N195	R116 + R320	T165 + R320	G196 + V206
H1 + T165	N54 + A204	K72 + G196	R116 + H321	T165 + H321	G196 + L228
H1 + Q172	N54 + V206	K72 + A204	R116 + S323	T165 + S323	G196 + G255
H1 + L173	N54 + L228	K72 + V206	R116 + K391	T165 + K391	G196 + A265
H1 + A174	N54 + G255	K72 + L228	N125 + F133	Q172 + L173	G196 + S280
H1 + G184	N54 + A265	K72 + G255	N125 + T134	Q172 + A174	G196 + S304
H1 + N195	N54 + S280	K72 + A265	N125 + T165	Q172 + G184	G196 + R320
H1 + G196	N54 + S304	K72 + S280	N125 + Q172	Q172 + N195	G196 + H321
H1 + A204	N54 + R320	K72 + S304	N125 + L173	Q172 + G196	G196 + S323
H1 + V206	N54 + H321	K72 + R320	N125 + A174	Q172 + A204	G196 + K391
H1 + L228	N54 + S323	K72 + H321	N125 + G184	Q172 + V206	A204 + V206
H1 + G255	N54 + K391	K72 + S323	N125 + N195	Q172 + L228	A204 + L228
H1 + A265	V56 + A60	K72 + K391	N125 + G196	Q172 + G255	A204 + G255
H1 + S280	V56 + K72	G109 + F113	N125 + A204	Q172 + A265	A204 + A265
H1 + S304	V56 + G109	G109 + R116	N125 + V206	Q172 + S280	A204 + S280
H1 + R320	V56 + F113	G109 + N125	N125 + L228	Q172 + S304	A204 + S304
H1 + H321	V56 + R116	G109 + F133	N125 + G255	Q172 + R320	A204 + R320
H1 + S323	V56 + N125	G109 + T134	N125 + A265	Q172 + H321	A204 + H321
H1 + K391	V56 + F133	G109 + T165	N125 + S280	Q172 + S323	A204 + S323
A37 + T40	V56 + T134	G109 + Q172	N125 + S304	Q172 + K391	A204 + K391
A37 + N54	V56 + T165	G109 + L173	N125 + R320	L173 + A174	V206 + L228
A37 + V56	V56 + Q172	G109 + A174	N125 + H321	L173 + G184	V206 + G255
A37 + A60	V56 + L173	G109 + G184	N125 + S323	L173 + N195	V206 + A265
A37 + K72	V56 + A174	G109 + N195	N125 + K391	L173 + G196	V206 + S280
A37 + G109	V56 + G184	G109 + G196	F133 + T134	L173 + A204	V206 + S304
A37 + F113	V56 + N195	G109 + A204	F133 + T165	L173 + V206	V206 + R320
A37 + R116	V56 + G196	G109 + V206	F133 + Q172	L173 + L228	V206 + H321
A37 + N125	V56 + A204	G109 + L228	F133 + L173	L173 + G255	V206 + S323
A37 + F133	V56 + V206	G109 + G255	F133 + A174	L173 + A265	V206 + K391
A37 + T134	V56 + L228	G109 + A265	F133 + G184	L173 + S280	L228 + G255
A37 + T165	V56 + G255	G109 + S280	F133 + N195	L173 + S304	L228 + A265

-continued

A37 + Q172	V56 + A265	G109 + S304	F133 + G196	L173 + R320	L228 + S280
A37 + L173	V56 + S280	G109 + R320	F133 + A204	L173 + H321	L228 + S304
A37 + A174	V56 + S304	G109 + H321	F133 + V206	L173 + S323	L228 + R320
A37 + G184	V56 + R320	G109 + S323	F133 + L228	L173 + K391	L228 + H321
A37 + N195	V56 + H321	G109 + K391	F133 + G255	A174 + G184	L228 + S323
A37 + G196	V56 + S323	F113 + R116	F133 + A265	A174 + N195	L228 + K391
A37 + A204	V56 + K391	F113 + N125	F133 + S280	A174 + G196	G255 + A265
A37 + V206	A60 + K72	F113 + F133	F133 + S304	A174 + A204	G255 + S280
A37 + L228	A60 + G109	F113 + T134	F133 + R320	A174 + V206	G255 + S304
A37 + G255	A60 + F113	F113 + T165	F133 + H321	A174 + L228	G255 + R320
A37 + A265	A60 + R116	F113 + Q172	F133 + S323	A174 + G255	G255 + H321
A37 + S280	A60 + N125	F113 + L173	F133 + K391	A174 + A265	G255 + S323
A37 + S304	A60 + F133	F113 + A174	T134 + T165	A174 + S280	G255 + K391
A37 + R320	A60 + T134	F113 + G184	T134 + Q172	A174 + S304	A265 + S280
A37 + H321	A60 + T165	F113 + N195	T134 + L173	A174 + R320	A265 + S304
A37 + S323	A60 + Q172	F113 + G196	T134 + A174	A174 + H321	A265 + R320
A37 + K391	A60 + L173	F113 + A204	T134 + G184	A174 + S323	A265 + H321
T40 + N54	A60 + A174	F113 + V206	T134 + N195	A174 + K391	A265 + S323
T40 + V56	A60 + G184	F113 + L228	T134 + G196	G184 + N195	A265 + K391
T40 + A60	A60 + N195	F113 + G255	T134 + A204	G184 + G196	S280 + S304
T40 + K72	A60 + G196	F113 + A265	T134 + V206	G184 + A204	S280 + R320
T40 + G109	A60 + A204	F113 + S280	T134 + L228	G184 + V206	S280 + H321
T40 + F113	A60 + V206	F113 + S304	T134 + G255	G184 + L228	S280 + S323
T40 + R116	A60 + L228	F113 + R320	T134 + A265	G184 + G255	S280 + K391
T40 + N125	A60 + G255	F113 + H321	T134 + S280	G184 + A265	S304 + R320
T40 + F133	A60 + A265	F113 + S323	T134 + S304	G184 + S280	S304 + H321
T40 + T134	A60 + S280	F113 + K391	T134 + R320	G184 + S304	S304 + S323
T40 + T165	A60 + S304	R116 + N125	T134 + H321	G184 + R320	S304 + K391
T40 + Q172	A60 + R320	R116 + F133	T134 + S323	G184 + H321	R320 + H321
T40 + L173	A60 + H321	R116 + T134	T134 + K391	G184 + S323	R320 + S323
T40 + A174	A60 + S323	R116 + T165	T165 + Q172	G184 + K391	R320 + K391
T40 + G184	A60 + K391	R116 + Q172	T165 + L173	N195 + G196	H321 + S323
T40 + N195	T40 + S323	R116 + L173	H1 + Q169	N195 + A204	H321 + K391
T40 + G196	T40 + K391	T40 + A265	H1 + A186	S244 + G255	S323 + K391
T40 + A204	N54 + V56	T40 + S280	H1 + E190	S244 + G258	A265 + K269
T40 + V206	N54 + A60	T40 + S304	H1 + A225	S244 + N260	A265 + W284
T40 + L228	N54 + K72	T40 + R320	H1 + K242	S244 + A265	A265 + Y295
T40 + G255	N54 + G109	T40 + H321	H1 + S244	S244 + K269	A265 + S304
T40 + Q169	R116 + Q169	A204 + A225	H1 + G258	S244 + S280	A265 + V326
T40 + A186	R116 + A186	A204 + K242	H1 + N260	S244 + W284	K269 + S280
T40 + E190	R116 + E190	A204 + S244	H1 + K269	S244 + Y295	K269 + W284
T40 + A225	R116 + A225	A204 + G258	H1 + W284	S244 + S304	K269 + Y295
T40 + K242	R116 + K242	A204 + N260	H1 + Y295	S244 + R320	K269 + S304
T40 + S244	R116 + S244	A204 + K269	H1 + S304	S244 + H321	K269 + R320
T40 + G258	R116 + G258	A204 + W284	H1 + V326	S244 + S323	K269 + H321
T40 + N260	R116 + N260	A204 + Y295	A37 + Q169	S244 + V326	K269 + S323
T40 + K269	R116 + K269	A204 + S304	A37 + A186	S244 + K391	K269 + V326
T40 + W284	R116 + W284	V206 + A225	A37 + E190	S244 + I405	K269 + K391
T40 + Y295	R116 + Y295	V206 + K242	A37 + A225	S244 + A421	K269 + I405
T40 + S304	R116 + S304	V206 + S244	A37 + K242	S244 + A422	K269 + A421
T40 + V326	N125 + Q169	V206 + G258	A37 + S244	S244 + A428	K269 + A422
N54 + Q169	N125 + A186	V206 + N260	A37 + G258	S244 + G448	K269 + A428
N54 + A186	N125 + E190	V206 + K269	A37 + N260	S244 + D476	K269 + G448
N54 + E190	N125 + A225	V206 + W284	A37 + K269	S244 + G477	K269 + D476
N54 + A225	N125 + K242	V206 + Y295	A37 + W284	G255 + G258	K269 + G477
N54 + K242	N125 + S244	V206 + S304	A37 + Y295	G255 + N260	S280 + W284
N54 + S244	N125 + G258	V206 + V326	A37 + S304	G255 + K269	S280 + Y295
N54 + G258	N125 + N260	A225 + L228	A37 + V326	G255 + W284	S280 + S304
N54 + N260	N125 + K269	A225 + K242	Q172 + A186	G255 + Y295	S280 + V326
N54 + K269	N125 + W284	A225 + S244	Q172 + E190	G255 + S304	W284 + Y295
N54 + W284	N125 + Y295	A225 + G255	Q172 + A225	G255 + V326	W284 + S304
N54 + Y295	N125 + S304	A225 + G258	Q172 + K242	G258 + N260	W284 + R320
N54 + S304	F133 + Q169	A225 + N260	Q172 + S244	G258 + A265	W284 + H321
N54 + V326	F133 + A186	A225 + A265	Q172 + G258	G258 + K269	W284 + S323
V56 + Q169	F133 + E190	A225 + K269	Q172 + N260	G258 + S280	W284 + V326
V56 + A186	F133 + A225	A225 + S280	Q172 + K269	G258 + W284	W284 + K391
V56 + E190	F133 + K242	A225 + W284	Q172 + W284	G258 + Y295	W284 + I405
V56 + A225	F133 + S244	A225 + Y295	Q172 + Y295	G258 + S304	W284 + A421
V56 + K242	F133 + G258	A225 + S304	Q172 + S304	G258 + R320	W284 + A422
V56 + S244	F133 + N260	A225 + R320	L173 + A186	G258 + H321	W284 + A428
V56 + G258	F133 + K269	A225 + H321	L173 + E190	G258 + S323	W284 + G448
V56 + N260	F133 + W284	A225 + S323	L173 + A225	G258 + V326	W284 + D476
V56 + K269	F133 + Y295	A225 + V326	L173 + K242	G258 + K391	W284 + G477
V56 + W284	F133 + S304	A225 + K391	L173 + S244	G258 + I405	Y295 + S304
V56 + Y295	T134 + Q169	A225 + I405	L173 + G258	G258 + A421	Y295 + R320
V56 + S304	T134 + A186	A225 + A421	L173 + N260	G258 + A422	Y295 + H321
V56 + V326	T134 + E190	A225 + A422	L173 + K269	G258 + A428	Y295 + S323
A60 + Q169	T134 + A225	A225 + A428	L173 + W284	G258 + G448	Y295 + V326
A60 + A186	T134 + K242	A225 + G448	L173 + Y295	G258 + D476	Y295 + K391

A60 + E190	T134 + S244	A225 + D476	L173 + S304	G258 + G477	Y295 + I405
A60 + A225	T134 + G258	A225 + G477	A174 + A186	N260 + A265	Y295 + A421
A60 + K242	T134 + N260	L228 + K242	A174 + E190	N260 + K269	Y295 + A422
A60 + S244	T134 + K269	L228 + S244	A174 + A225	N260 + S280	Y295 + A428
A60 + G258	T134 + W284	L228 + G258	A174 + K242	N260 + W284	Y295 + G448
A60 + N260	T134 + Y295	L228 + N260	A174 + S244	N260 + Y295	Y295 + D476
A60 + K269	T134 + S304	L228 + K269	A174 + G258	N260 + S304	Y295 + G477
A60 + W284	T134 + V326	L228 + W284	A174 + N260	N260 + R320	S304 + R320
A60 + Y295	T165 + Q169	L228 + Y295	A174 + K269	N260 + H321	S304 + H321
A60 + S304	T165 + A186	L228 + S304	A174 + W284	N260 + S323	S304 + S323
A60 + V326	T165 + E190	L228 + V326	A174 + Y295	N260 + V326	S304 + V326
K72 + Q169	T165 + N195	K242 + S244	A174 + S304	N260 + K391	S304 + K391
K72 + A186	T165 + A225	K242 + G255	G184 + A186	N260 + I405	S304 + I405
K72 + E190	T165 + K242	K242 + G258	G184 + E190	N260 + A421	S304 + A421
K72 + A225	T165 + S244	K242 + N260	G184 + A225	N260 + A422	S304 + A422
K72 + K242	T165 + G258	K242 + A265	G184 + K242	N260 + A428	S304 + A428
K72 + S244	T165 + N260	K242 + K269	G184 + S244	N260 + G448	S304 + G448
K72 + G258	T165 + K269	K242 + S280	G184 + G258	N260 + D476	S304 + D476
K72 + N260	T165 + W284	K242 + W284	G184 + N260	N260 + G477	S304 + G477
K72 + K269	T165 + Y295	K242 + Y295	G184 + K269	N195 + A225	R320 + H321
K72 + W284	T165 + S304	K242 + S304	G184 + W284	N195 + K242	R320 + V326
K72 + Y295	Q169 + Q172	K242 + R320	G184 + Y295	N195 + S244	H321 + V326
K72 + S304	Q169 + L173	K242 + H321	G184 + S304	N195 + G258	S323 + V326
G109 + Q169	Q169 + A174	K242 + S323	A186 + E190	N195 + N260	V326 + K391
G109 + A186	Q169 + G184	K242 + V326	A186 + N195	N195 + K269	V326 + I405
G109 + E190	Q169 + A186	K242 + K391	A186 + G196	N195 + W284	V326 + A421
G109 + A225	Q169 + E190	K242 + I405	A186 + A204	N195 + Y295	V326 + A422
G109 + K242	Q169 + N195	K242 + A421	A186 + V206	N195 + S304	V326 + A428
G109 + S244	Q169 + G196	K242 + A422	A186 + A225	N195 + V326	V326 + G448
G109 + G258	Q169 + A204	K242 + A428	A186 + L228	G196 + A225	V326 + D476
G109 + N260	Q169 + V206	K242 + G448	A186 + K242	G196 + K242	V326 + G477
G109 + K269	Q169 + A225	K242 + D476	A186 + S244	G196 + S244	E190 + Y295
G109 + W284	Q169 + L228	K242 + G477	A186 + G255	G196 + G258	E190 + S304
G109 + Y295	Q169 + K242	E190 + A204	A186 + G258	G196 + N260	E190 + R320
G109 + S304	Q169 + S244	E190 + V206	A186 + N260	G196 + K269	E190 + H321
G109 + V326	Q169 + G255	E190 + A225	A186 + A265	G196 + W284	E190 + S323
F113 + Q169	Q169 + G258	E190 + L228	A186 + K269	G196 + Y295	E190 + V326
F113 + A186	Q169 + N260	E190 + K242	A186 + S280	G196 + S304	E190 + K391
F113 + E190	Q169 + A265	E190 + S244	A186 + W284	G196 + V326	E190 + I405
F113 + A225	Q169 + K269	E190 + G255	A186 + Y295	A186 + G448	E190 + A421
F113 + K242	Q169 + S280	E190 + G258	A186 + S304	A186 + D476	E190 + A422
F113 + S244	Q169 + W284	E190 + N260	A186 + R320	A186 + G477	E190 + A428
F113 + G258	Q169 + Y295	E190 + A265	A186 + H321	E190 + N195	E190 + G448
F113 + N260	Q169 + S304	E190 + K269	A186 + S323	E190 + G196	E190 + D476
F113 + K269	Q169 + R320	E190 + S280	A186 + V326	Q169 + A422	E190 + G477
F113 + W284	Q169 + H321	E190 + W284	A186 + K391	Q169 + A428	A186 + A422
F113 + Y295	Q169 + S323	Q169 + I405	A186 + I405	Q169 + G448	A186 + A428
F113 + S304	Q169 + V326	Q169 + A421	A186 + A421	Q169 + D476	
F113 + V326	Q169 + K391			Q169 + G477	

wherein numbering is according to SEQ ID NO: 14.

Stability of a parent polypeptide may be improved by introducing a pairwise deletion within the B domain of parent polypeptide. Thus, in one embodiment, the said variant comprises

a) a deletion and/or a substitution at two or more positions corresponding to positions R181, G182, D183, and G184 of the amino acid sequence as set forth in SEQ ID NOs: 7 or 10, and

b) an alteration at one or more positions corresponding to positions H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, A204, V206, P211, I214, V215, L217, L219, A225, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of

SEQ ID NO: 13; or H1, A37, T40, N54, V56, A60, K72, G109, F113, R116, N125, F133, T134, T165, Q172, L173, A174, G184, N195, G196, A204, V206, L228, G255, A265, S280, S304, R320, H321, S323, K391, I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14.

The term "pairwise deletion" as used herein, refers to a deletion of an amino acid in two positions. Said two positions may be adjacent to one another but may also be separated by one, two, three, four, or five amino acids.

In a particular embodiment, the deletion a) is selected from the group consisting of R181+G182, R181+D183, R181+G184, G182+D183, G182+G184, or D183+G184.

In an embodiment, variant comprises one or more of the following alterations; H1*, T5K, G7K, G7A, Q11H, N16S, N16H, V17L, Q32S, A37H, A37M, A37V, A37S, A37Y, A37R, A37L, T40G, T40K, P45A, W48G, W48Y, W48F, G50A, T51K, T51E, T51G, T51A, T51S, T51G, T51D, N54S, V56T, K72R, K72H, K72S, K72Q, K72E, K72N, K72A, K72M, R87S, Q98S, Q98A, M105F, M105I, M105V, M105L, G109A, G109S, F113W, F113S, F113N, F113Y, F113R, F113L, F113Q, R116Q, R116V, R116K, R116W, R116L, R116A, R116H, R116M, R116E, R116S, R116I,

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R116G, Q118N, Q118K, Q118G, Q118S, Q118F, Q118R, Q125P, Q125K, Q125A, Q125T, G133S, T134E, W140Y, G142T, G149Q, T165S, T165G, T165V, W167I, W167G, W167F, W167R, W167S, W167H, W167L, W167M, W167Y, Q169E, R171H, Q172G, Q172R, Q172N, Q172D, Q172Y, Q172M, Q172S, Q172T, Q172K, Q172H, Q172E, L173V, L173G, L173H, L173A, L173I, L173P, L173T, L173F, L173M, A174S, A174D, A174P, A174M, A174T, A174H, A174K, A174G, A174Q, A174N, A174V, A174L, G184T, A186D, A186N, A186E, A186Q, A186H, E190P, T193K, N195F, A204T, A204V, A204S, A204G, V206L, V206S, V206Y, P211D, I214G, I214H, I214S, I214T, I214L, I214E, I214W, V215T, L217T, L217Q, L219V, L219H, A225V, L235V, L235A, V238A, V238T, V238G, K242Q, S244Q, M246L, M246A, M246I, M246F, M246V, M246S, M248T, L250I, L250V, L250T, L250A, L250F, L250M, G255N, G255A, G255S, Q256A, N260G, A263G, V264I, V264T, Y267I, Y267M, Y267H, Y267L, K269Q, N270G, N270T, G273R, S280W, S280L, S280T, S280A, S280K, S280Q, W284H, T285L, T285Q, M286F, M286L, A288L, A288V, F289I, F289L, V291G, V291A, V291T, Y295F, Y295N, Q299V, S304R, S304N, R320A, R320V, H321Y, S323N, H324L, V326L, F328L, F328M, F328V, F328I, T334S, D337H, Q345D, G346T, G346D, G346P, G348S, G348P, T355L, T355F, S376H, S376T, S376V, D377H, D377S, D377A, D377Q, D379S, D379G, D379R, D379A, Y382M, Y382I, Y382L, Y382F, S383G, S383A, Q385L, K391A, K391Y, K391V, K391M, K391E, K391D, K391R, K391H, K391W, K391I, K391Q, K391L, K393Y, K393R, K393Q, K393S, Q395P, T400P, H402R, A420Q, A420S, A420K, A420L, G423H, T444Q, T444S, T444D, T444Y, T444H, T444V, T444R, T444A, A445Q, Q449T, T459N, P473R, P473A, P473G, P473T, P473K, C474V, G476K, G477A, G477Q, K484A, K484G, K484P, K484E, K484Q, and K484S of SEQ ID NO: 13; or H1*, G7A, A37H, T40D, W48Y, W48F, N54S, V56T, A60V, K72R, Q98A, G109A, F113Q, R116H, R116V, R116Q, N125D, F133Q, T134E, T165G, Q169E, Q172D, Q172G, Q172N, L173V, A174S, G184T, A186D, A186N, A186E, A186Q, A186H, E190P, N195F, G196R, A204G, V206L, V206Y, A225V, L228I, K242Q, S244Q, G255A, G255S, N260G, A265G, K269Q, N270T, S280Q, W284H, A288L, A288V, F289L, Y295F, Y295N, S304N, S304R, S304N, R320A, H321Y, S323N, V326L, K391A, I405L, A421H, A422P, A428T, G448D, D476K, D476G, D476N, D476Y, G477S, G477A, G477T, and G477Q of SEQ ID NO: 14.

In a particular embodiment, the variant has an IF of >1 when compared to said parent polypeptide, and wherein said variant is selected from

G182*+D183*+N195F
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+G476K
 H1*+N54S+V56T+R87S+G109A+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K
 H1*+T40G+N54S+V56T+G109A+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K
 H1*+T51K+N54S+V56T+G109A+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K
 H1*+N54S+V56T+K72R+G109A+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K
 H1*+N54S+V56T+K72H+G109A+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K
 H1*+A37H+N54S+V56T+G109A+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K
 H1*+A37M+N54S+V56T+G109A+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K

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H1*+A37V+N54S+V56T+G109A+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K
 H1*+A37S+N54S+V56T+G109A+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K
 H1*+A37Y+N54S+V56T+G109A+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K
 H1*+A37R+N54S+V56T+G109A+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K
 H1*+N54S+V56T+G109A+F113W+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K
 H1*+N54S+V56T+G109A+F113S+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K
 H1*+N54S+V56T+G109A+F113N+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K
 H1*+N54S+V56T+G109A+F113Y+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K
 H1*+N54S+V56T+G109A+F113R+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K
 H1*+N54S+V56T+G109A+Q118N+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K
 H1*+N54S+V56T+G109A+R116Q+A174S+G182*+
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 H1*+N54S+V56T+G109A+Q125P+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K
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 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
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 D183*+N195F+V206L+K391A+G476K
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 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+I214G+K391A+G476K

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 D183*+N195F+V206L+K391A+G476K
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 N195F+V206L+K391A+G476K+K484S
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 H1*+G50A+N54S+V56T+G109A+F113L+R116L+
 A174S+G182*+D183*+N195F+V206L+K391A+
 G476K
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 G476K
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 D377H+K391A+G476K
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 D183*+G184T+N195F+V206L+G255A+K391A+
 G476K

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 G476K
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 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+F328L+D377H+K391A+G476K
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+F328V+D377H+K391A+G476K
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
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 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
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 H1*+N54S+V56T+K72R+G109A+T134E+A174S+
 G182*+D183*+G184T+N195F+V206L+G255A+
 K391A+G476K
 H1*+N54S+V56T+G109A+L173H+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K
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 D183*+N195F+V206L+K391A+G476K
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 D183*+N195F+V206L+K391A+G476K
 H1*+N54S+V56T+K72N+G109A+W167L+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K
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 G182*+D183*+N195F+V206L+G255A+D377H+
 K391A+G476K
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 D183*+N195F+G184T+V206L+K391A+G476K
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 G182*+D183*+N195F+V206L+G255A+D377H+
 K391A+G476K
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 G184T+N195F+V206L+R320A+S323N+K391A+
 G476K

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H1*+N54S+G109A+A174S+G182*+D183*+N195F+
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 H1*+N54S+V56T+A174S+G182*+D183*+N195F+
 V206L+D377H+K391A+G476K
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 G476K
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 G476K
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 G476K
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 K391A+G476K
 H1*+N54S+V56T+M105F+G109A+A174S+G182*+
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 G476K
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 D183*+N195F+V206L+D377H+K391A+G476K

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H1*+N54L+V56T+G109A+A174S+G182*+D183*+
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A174H+G182*+D183*+N195F+V206L+K391A+
G476K
40 H1*+N54S+V56T+G109A+L173A+A174T+G182*+
D183*+N195F+V206L+K391A+G476K
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+V291A+F328L+K391A+G476K
45 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+M246L+K391A+G476K
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+M246S+K391A+G476K
50 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+L250V+K391A+G476K
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+L250F+K391A+G476K
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+L250M+K391A+G476K
55 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+C474V+G476K
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+A204S+V206L+K391A+G476K
60 H1*+N54S+V56T+G109A+T165G+A174S+G182*+
D183*+N195F+V206L+K391A+G476K
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+T334S+K391A+G476K
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+A263G+M286F+K391A+G476K
65 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+A263G+M286L+K391A+G476K

75

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N195F+V206L+Y267L+K391A+G476K
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H1*+T51A+N54S+V56T+G109A+Q172R+A174S+
G182*+D183*+N195F+V206L+K391A+G476K
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N195F+V206L+K391L+G476K
H1*+N54S+V56T+G109A+F113Q+R116H+Q172G+
A174S+G182*+D183*+N195F+V206L+K391A+
G476K
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D183*+N195F+A204G+V206L+K391A+P473R+
G476K
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G476K
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G476K
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C474V+G476K
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G476K
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G346T+G477A+G476K
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76

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G476K
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G476K
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G476K

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78

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G476K
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D183*+N195F+V206L+K391A+T444Q+P473R+
G476K
H1*+N54S+V56T+G109A+R116H+Q172R+A174S+
G182*+D183*+N195F+A204G+V206L+K391A+
P473R+G476K
H1*+A37V+N54S+V56T+G109A+R116Q+T165G+
A174S+G182*+D183*+N195F+V206L+K391A+
G476K
H1*+A37H+N54S+V56T+G109A+R116H+T165G+
A174S+G182*+D183*+N195F+V206L+M246F+
K391A+G476K
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+K391A+G476K
H1*+G109A+G182*+D183*+N195F+V206L+K391A+
G476K
G109A+G182*+D183*+N195F+V206L+G476K
H1*+G109A+G182*+D183*+N195F+K391A
H1*+N54S+V56T+G109A+G133S+A174S+G182*+
D183*+N195F+K391A+G476H
H1*+N54S+V56T+G109A+R116A+Q172N+L173V+
A174S+G182*+D183*+N195F+V206L+K391A+
G476K
H1*+A37V+N54S+V56T+G109A+R116H+T165G+
A174S+G182*+D183*+N195F+V206L+K391A+
G476K
H1*+P45A+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+C474V+G476K
H1*+N54S+V56T+G109A+P124A+A174S+G182*+
D183*+N195F+V206L+K391A+C474V+G476K
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+P364A+K391A+C474V+G476K
H1*+N54S+V56T+G109A+Q118R+T134E+A174S+
G182*+D183*+N195F+V206L+K391A+G476K
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
G184T+N195F+V206L+K391A+G476K
H1*+G50A+T51A+N54S+V56T+G109A+A174S+
G182*+D183*+N195F+V206L+K391A+G476K
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+V264F+K391A+G476K
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+Y382L+K391A+G476K
H1*+G7K+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+G255A+Q256A+K391A+G476K
H1*+N54S+V56T+G109A+W167Y+A174S+G182*+
D183*+N195F+V206L+K391A+G476K

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H1*+N54S+V56T+G109A+W167H+A174S+G182*+
D183*+N195F+V206L+K391A+G476K
H1*+N54S+V56T+G109A+T165V+A174S+G182*+
D183*+N195F+V206L+K391A+G476K
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+G476K
H1*+N54S+V56T+G109A+A174N+G182*+D183*+
N195F+V206L+K391A+G476K
H1*+N54S+V56T+G109A+A174Q+G182*+D183*+
N195F+V206L+K391A+G476K
H1*+N16H+V17L+N54S+V56T+G109A+A174S+
G182*+D183*+N195F+V206L+Y382F+K391A+
G476K
H1*+Q32S+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+Q385L+K391A+G476K
H1*+N54S+V56T+R87S+G109A+R171H+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116H+Q169E+Q172K+
A174*+G182*+D183*+N195F+V206L+K391A+
G476K,
H1*+N54S+V56T+G109A+A174*+G182*+D183*+
N195F+V206L+K391A+G476K,
H1*+K72R+G109A+F113Q+R116Q+W167F+Q172G+
A174S+G182*+D183*+G184T+N195F+V206L+
K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+
W167F+Q172G+A174S+G182*+D183*+G184T+
N195F+V206L+P473R+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172G+A174*+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+
W167F+Q172G+A174S+G182*+D183*+G184T+
N195F+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+R116W+Q169E+Q172K+
A174*+G182*+D183*+N195F+V206L+K391A+
G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+V56T+G109A+A174S+G182*+D183*+N195F+
V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+N260G+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+G184T+N195F+V206L+K391A+
G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+Y295N+K391A+
G476K,
H1*+N54S+V56T+G109A+W167F+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+A186N+N195F+V206L+K391A+
G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+A288V+K391A+
G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+G255A+K391A+
G476K,
H1*+N54S+G109A+Q169E+Q172K+A174*+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+W48F+N54S+V56T+G109A+Q169E+Q172K+
A174*+G182*+D183*+N195F+V206L+K391A+
G476K,

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H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+V291A+K391A+
G476K,
H1*+N54S+V56T+G109A+F113Q+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+R320A+S323N+
K391A+G476K,
H1*+V56T+G109A+Q169E+Q172K+A174*+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116H+
W167F+Q172G+A174S+G182*+D183*+G184T+
N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+N270T+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+N260G+K391A+
G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172G+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+A186D+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+A225V+K391A+
G476K,
H1*+G109A+Q169E+Q172K+A174*+G182*+D183*+
N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+G255S+K391A+
G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+K269Q+K391A+
G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+G255S+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
G182*+D183*+N195F,
H1*+N54S+V56T+G109A+Q169E+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+W284H+K391A+
G476K,
H1*+N54S+G109A+A174S+G182*+D183*+N195F+
V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+S304R+K391A+
G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+
W167F+Q172G+A174S+G182*+D183*+G184T+
N195F+V206L+K391A+P473R,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+M246V+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+K391A,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+N270G+
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+Y295F+K391A+
G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+R320A+K391A+G476K,

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H1*+N54S+V56T+G109A+Q169E+Q172K+A174T+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+Q169E+Q172K+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+W140Y+Q169E+Q172K+
 A174*+G182*+D183*+N195F+V206L+K391A+
 G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A,
 H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
 G182*+D183*+N195F+V206L+F328L+K391A+
 G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+G476K,
 F113Q+R116H+D183*+G182*+N195F,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+A288L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+N270G+K391A+G476K,
 Q169E+Q172K+A174*+D183*+G182*+N195F,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K269Q+K391A+G476K,
 H1*+W48F+N54S+V56T+G109A+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 F113Q+R116Q+D183*+G182*+N195F,
 N54S+V56T+D183*+G182*+N195F,
 H1*+N54S+V56T+G109A+A174T+G182*+D183*+
 N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+R116W+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 Q169E+Q172K+G182*+D183*+N195F,
 G182*+D183*+A186D,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+V238A+K391A+G476K,
 G182*+D183*+A186N,
 H1*+N54S+V56T+G109A+A174S+V206L+G182*+
 D183*+K391A+G476K,
 H1*+N54S+V56T+Q169E+Q172K+A174*+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206F+K391A+G476K,
 H1*+N54S+V56T+G109A+G182*+D183*+N195F+
 V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+Q172K+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+Q172K+A174*+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174N+G182*+D183*+
 N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+A186N+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+F289L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+Y295F+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+V291A+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+S244Q+K391A+G476K,
 G182*+D183*+F328L,
 G182*+D183*+N195F+A288V,
 G182*+D183*+A186N+N195F,
 G182*+D183*+E190P+N195F,
 W140Y+G182*+D183*+N195F,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+A225V+K391A+G476K,
 G182*+D183*+A186D+N195F,

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H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+P473R+G476K,
 H1*+W48Y+N54S+V56T+G109A+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
 G182*+D183*+N195F+V206L+K391A+G476K,
 G182*+D183*+N195F+V238A,
 G182*+D183*+Y295N,
 Q172G+G182*+D183*,
 F113Q+G182*+D183*+N195F,
 H1*+N54S+V56T+Q98A+G109A+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 G182*+D183*+N195F+Y295F,
 G182*+D183*+A186E,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206Y+K391A+G476K,
 G182*+D183*+N195F+W284H,
 G182*+D183*+A186Q+N195F,
 K72R+G182*+D183*,
 D183*+G182*+N195F+V326L,
 D183*+G182*+A186E+N195F,
 Q169E+G182*+D183*+N195F,
 D183*+G182*+N195F+K242Q,
 G7A+G182*+D183*+N195F,
 D183*+G182*+N195F+N260G,
 D183*+G182*+R321A+S323N,
 D183*+G182*+W284H,
 H1*+D183*+G182*,
 D183*+G182*+A186H+N195F,
 D183*+G182*+F289V,
 D183*+G182*+K391A,
 D183*+G182*+S304R,
 W167F+G182*+D183*+N195F,
 F113Q+R116H+D183*+G182*+
 D183*+G182*+N270G,
 D183*+G182*+S244Q,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 G184T+N195F+V206L+K391A+G476K,
 D183*+G182*+N195F+Y295N,
 W48F+D183*+G182*+
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+G255A+K391A+G476K,
 D183*+G182*+A186Q,
 W140Y+D183*+G182*+
 D183*+G182*+K242Q,
 D183*+G182*+V206L,
 D183*+G182*+N195F+V206F,
 Q169E+D183*+G182*+
 D183*+G182*+N195F+R320A,
 G182*+D183*+G184T+N195F,
 D183*+G182*+G184T,
 N54S+V56T+D183*+G182*+
 D183*+G182*+P473R,
 W48Y+D183*+G182*+
 D183*+G182*+R323A,
 G7A+D183*+G182*+
 G109A+D183*+G182*+
 W167F+D183*+G182*+
 H1*+N54S+V56T+G109A+Q169E+Q172K+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 A174S+D183*+G182*+
 R116Q+D183*+G182*+
 D183*+G182*+N195F+V291A,
 Q172K+D183*+G182*+
 N54S+D183*+G182*+
 D183*+G182*+N195F+S323N,
 D183*+G182*+N195F+M246V,

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W48Y+D183*+G182*+N195F,
 D183*+G182*+N195F+S304R,
 V56T+D183*+G182*, wherein numbering is according
 to SEQ ID NO: 13; and
 N54S+G182*+D183*+N195F,
 G109A+G182*+D183*+N195F,
 G182*+D183*+N195F+K391A,
 G182*+D183*+N195F+G476K,
 H1*+G182*+D183*+N195F,
 N54S+G182*+D183*+N195F,
 V56T+G182*+D183*+N195F,
 G182*+D183*+N195F+I405L,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+I405L+A421H+A422P+A428T+
 D476K,
 A174S+G182*+D183*+N195F,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+I405L+A421H+A422P+A428T,
 H1*+N54S+V56T+G109A+A174S+D183*+N195F+
 V206L+K391A+I405L+A421H+A422P+A428T,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+I405L+A421H+A422P+
 A428T+D476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+I405L+A421H+A422P+
 A428T+D476G,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+I405L+A421H+A422P+
 A428T+D476K+G477S,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+I405L+A421H+A422P+
 A428T+D476K,
 H1*+N54S+V56T+K72R+G109A+W167F+A174S+
 G182*+D183*+N195F+V206L+I405L+A421+
 A422P+A428T,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+S304N+I405L+A421H+A422P+
 A428T+D476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+I405L+A421H+A422P+
 A428T+D476K+G477A,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+I405L+A421H+A422P+
 A428T+D476G+G477T,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+I405L+A421H+A422P+
 A428T+G477Q,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+I405L+A421H+A422P+
 A428T+G477A,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+I405L+A421H+A422P+
 A428T+D476K+G477Q,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+I405L+A421H+A422P+
 A428T+D476N,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+I405L+A421H+A422P+
 A428T+D476Y,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+I405L+A421H+A422P+A428T,
 H1*+N54S+V56T+G109A+Q172D+A174S+G182*+
 D183*+N195F+A204G+V206L+I405L+A421H+
 A422P+A428T,
 H1*+N54S+V56T+K72R+G109A+T134E+A174S+
 G182*+D183*+N195F+A204G+V206L+G255A+
 I405L+A421H+A422P+A428T,

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H1*+N54S+V56T+G109A+F113Q+R116H+T165G+
 Q172G+A174S+G182*+D183*+N195F+V206L+
 I405L+A421H+A422P+A428T,
 H1*+N54S+V56T+G109A+F113Q+Q172N+A174S+
 G182*+D183*+N195F+V206L+I405L+A421H+
 A422P+A428T,
 H1*+N54S+V56T+K72R+G109A+T134E+A174S+
 G182*+D183*+N195F+G196R+V206L+G255A+
 I405L+A421H+A422P+A428T+N475S,
 H1*+N54S+V56T+G109A+T134E+A174S+G182*+
 D183*+N195F+V206L+I405L+A421H+A422P+
 A428T,
 H1*+T40D+N54S+V56T+G109A+A174S+G182*+
 D183*+N195F+V206L+I405L+A421H+A422P+
 A428T,
 H1*+N54S+V56T+G109A+F113Q+N125D+A174S+
 G182*+D183*+N195F+V206L+I405L+A421H+
 A422P+A428T,
 H1*+N54S+V56T+A60V+G109A+R116Q+A174S+
 G182*+D183*+N195F+V206L+I405L+A421H+
 A422P+A428T,
 H1*+N54S+V56T+G109A+R116V+A174S+G182*+
 D183*+N195F+V206L+I405L+A421H+A422P+
 A428T,
 H1*+N54S+V56T+G109A+R116Q+G133Q+T165G+
 Q172G+A174S+G182*+D183*+N195F+V206L+
 I405L+A421H+A422P+A428T,
 H1*+N54S+V56T+G109A+F113Q+R116H+Q172G+
 A174S+G182*+D183*+N195F+V206L+I405L+
 A421H+A422P+A428T,
 H1*+N54S+V56T+G109A+F113Q+R116Q+Q172G+
 A174S+G182*+D183*+N195F+V206L+I405L+
 A421H+A422P+A428T+G448D,
 H1*+N54S+V56T+G109A+F113Q+R116H+Q172N+
 A174S+G182*+D183*+N195F+V206L+I405L+
 A421H+A422P+A428T,
 H1*+N54S+V56T+K72R+G109A+T134E+W167F+
 Q172N+A174S+G182*+D183*+N195F+V206L+
 G255A+A265G+I405L+A421H+A422P+A428T,
 H1*+N54S+V56T+K72R+G109A+R116Q+W167F+
 Q172R+L173V+A174S+G182*+D183*+N195F+
 V206L+A265G+I405L+A421H+A422P+A428T,
 H1*+N54S+V56T+K72R+G109A+R116Q+W167F+
 Q172R+A174S+G182*+D183*+N195F+V206L+
 I405L+A421H+A422P+A428T,
 H1*+N54S+V56T+K72R+G109A+F113Q+T134E+
 W167F+Q172G+A174S+G182*+D183*+N195F+
 V206L+G255A+I405L+A421H+A422P+A428T,
 H1*+N54S+V56T+K72R+G109A+F113Q+R116H+
 W167F+Q172G+L173V+A174S+G182*+D183*+
 G184T+N195F+V206L+I405L+A421H+A422P+
 A428T,
 H1*+N54S+V56T+G109A+F113Q+W167F+Q172G+
 A174S+G182*+D183*+N195F+V206L+I405L+
 A421H+A422P+A428T,
 H1*+N54S+V56T+K72R+G109A+A174S+G182*+
 D183*+N195F+V206L+G255A+I405L+A421H+
 A422P+A428T,
 H1*+N54S+V56T+G109A+F113Q+R116Q+A174S+
 G182*+D183*+N195F+V206L+I405L+A421H+
 A422P+A428T,
 H1*+N54S+V56T+K72R+G109A+F113Q+F116H+
 W167F+Q172G+L173V+A174S+G182*+D183*+
 N195F+V206L+I405L+A421H+A422P+A428T,
 H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+
 W167F+Q172G+A174S+G182*+D183*+G184T+
 N195F+V206L+I405L+A421H+A422P+A428T,

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H1*+N54S+V56T+G109A+F113Q+R116Q+W167F+
Q172G+A174S+G182*+D183*+N195F+A204G+
V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116H+W167F+
Q172G+A174S+G182*+D183*+N195F+V206L+
A265G+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+R116Q+A174S+G182*+
D183*+N195F+V206L+I405L+A421H+A422P+
A428T,
H1*+N54S+V56T+A60V+G109A+R116Q+W167F+
Q172N+L173V+A174S+G182*+D183*+N195F+
V206L+I405L+A421H+A422P+A428T,
H1*+A37H+N54S+V56T+A60V+G109A+R116Q+
T165G+A174S+G182*+D183*+N195F+V206L+
I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+A174S+M105F+G182*+
D183*+N195F+V206L+L228I+R320A+S323N+
I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+F113Q+T134E+
A174S+G182*+D183*+N195F+V206L+G255A+
I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+R116Q+V120L+
A174S+G182*+D183*+G184T+N195F+V206L+
I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116H+W167F+
Q172G+L173V+A174S+G182*+D183*+N195F+
V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116Q+Q172N+
A174S+G182*+D183*+N195F+V206L+A265G+
I405L+A421H+A422P+A428T,
H1*+A37H+N54S+V56T+A60V+G109A+R116Q+
W167F+Q172R+A174S+G182*+D183*+
N195F+V206L+I405L+A421H+A422P+A428T, wherein
numbering is according to SEQ ID NO:14.

As can be seen from the data obtained in Examples 3 and 4, all tested variants have an Improvement Factor (IF) of at least 1.0, i.e. at least on par with a parent polypeptide having SEQ ID NO: 15, when tested in either (i) a detergent Model A at 15° C. and an enzyme concentration of 0.05 mg/L, (ii) a detergent Model A at 15° C. and an enzyme concentration of 0.2 mg/L, (iii) a detergent Model J at 15° C. and an enzyme concentration of 0.05 mg/L, or (iv) a detergent Model J at 15° C. and an enzyme concentration of 0.2 mg/L. Thus, the variants of the present invention have even in different model detergent compositions, at a low temperature as well as in a low concentrations, a wash performance at least that of the parent polypeptide having SEQ ID NO: 15. Specifications of the model detergents and specific test conditions for the AMSA, can be seen in Examples 3 and 4.

In another embodiment, the variant is selected from the group consisting of;

G182*+D183*+N195F+V206F+N260P+L312I+L313I+
R320A+T355L+L389I+T400P,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+G476K,
H1*+G50A+T51A+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+G476K,
H1*+N54S+V56T+M105F+G109A+A174S+G182*+
D183*+N195F+V206L+R320A+S323N+K391A+
G476K,
H1*+T40K+N54S+V56T+G109A+G149H+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+
D183*+N195F+V206L+G255A+K391A+G476K,
H1*+N16H+V17L+N54S+V56T+G109A+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,

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H1*+Q22N+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+Q32S+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
5 H1*+H28N+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N29S+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
10 H1*+H28N+N29S+N54S+V56T+G109A+A174S+
G182*+D183*+N195F+V206L+K391A+G476K
H1*+N54S+V56T+G109A+A174S+R176K+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
R176K+G182*+D183*+N195F+V206L+K391A+
15 G476K,
H1*+N54S+V56T+G109A+A174S+N175G+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+V291A+Y295N+Q299T+K391A+
20 G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K269S+A274K+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+L235I+K391A+G476K,
25 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+L235V+V238A+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+Q365S+K391A+G476K,
30 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+H402R+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+R320S+H321N+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+K391A+G476K,
35 H1*+N54S+V56T+G109A+Q125S+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+L235V+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+A265G+K391A+G476K, H1*+
T51Q+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+Y203F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
N175Q+G182*+D183*+N195F+V206L+K391A+
45 G476K,
H1*+H28N+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+T134E+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+D377H+K391A+G476K,
H1*+G50A+T51A+N54S+V56T+G109A+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
55 H1*+G7K+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+M202V+V206L+K391A+G476K,
H1*+N54S+V56T+M105F+G109A+A174S+G182*+
D183*+N195F+V206L+M246F+K391A+G476K,
60 H1*+N16H+V17L+N54S+V56T+G109A+A174S+
G182*+D183*+N195F+V206L+D377H+K391A+
G476K,
H1*+T51G+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+Y203G+V206L+K391A+G476K,
65 H1*+T51G+N54S+V56T+G109A+G110N+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,

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H1*+N54S+V56T+G109A+Q125S+A174S+N175G+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+A265G+V291A+F328L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+M246F+A265G+V291A+F328L+Q365S+K391A+G476K,
H1*+N16Y+N54S+V56T+G109A+R116H+A174S+G182*+D183*+N195F+V206L+G337E+D377H+K391A+G476K,
H1*+19L+N54S+V56T+G109A+Q125S+V131T+A174*+G182*+D183*+V206L+Y267F+F289H+Q361E+K391A+G476K,
H1*+N54S+V56T+G109A+R116H+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+V206L+P473R+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172G+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+R116W+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+N260G+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+G184T+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+Y295N+K391A+G476K,
H1*+N54S+V56T+G109A+W167F+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+A186N+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+A288V+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+G255A+K391A+G476K,
H1*+N54S+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+W48F+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+V291A+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+A174S+G182*+D183*+N195F+V206L+K391A+G476K,

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H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+R320A+S323N+K391A+G476K,
H1*+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116H+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+N270T+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+N260G+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+A186D+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+A225V+K391A+G476K,
H1*+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+G255S+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K269Q+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G255S+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F,
H1*+N54S+V56T+G109A+Q169E+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+W284H+K391A+G476K,
H1*+N54S+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+S304R+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+V206L+K391A+P473R,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+M246V+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+N270G+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+Y295F+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+R320A+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174T+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+G182*+D183*+N195F+V206L+K391A+G476K,

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H1*+N54S+V56T+G109A+W140Y+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+F328L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G476K,
F113Q+R116H+D183*+G182*+N195F, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+A288L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+N270G+K391A+G476K,
Q169E+Q172K+A174*+D183*+G182*+N195F,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K269Q+K391A+G476K,
H1*+W48F+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
F113Q+R116Q+D183*+G182*+N195F, N54S+V56T+D183*+G182*+N195F,
H1*+N54S+V56T+G109A+A174T+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116W+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
Q169E+Q172K+G182*+D183*+N195F,
G182*+D183*+A186D,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+V238A+K391A+G476K, G182*+D183*+A186N,
H1*+N54S+V56T+G109A+A174S+V206L+G182*+D183*+K391A+G476K,
H1*+N54S+V56T+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K, H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206F+K391A+G476K,
H1*+N54S+V56T+G109A+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172K+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174N+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+A186N+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+F289L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+Y295F+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+V291A+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+S244Q+K391A+G476K,
G182*+D183*+F328L,
G182*+D183*+N195F+A288V,
G182*+D183*+A186N+N195F,
G182*+D183*+E190P+N195F,
W140Y+G182*+D183*+N195F,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+A225V+K391A+G476K,
G182*+D183*+A186D+N195F,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+P473R+G476K,
H1*+W48Y+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,

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H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
G182*+D183*+N195F+V238A,
G182*+D183*+Y295N,
Q172G+G182*+D183*,
F113Q+G182*+D183*+N195F,
H1*+N54S+V56T+Q98A+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
G182*+D183*+N195F+Y295F,
G182*+D183*+A186E,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206Y+K391A+G476K,
G182*+D183*+N195F+W284H,
G182*+D183*+A186Q+N195F,
K72R+G182*+D183*,
D183*+G182*+N195F+V326L,
D183*+G182*+A186E+N195F,
Q169E+G182*+D183*+N195F,
D183*+G182*+N195F+K242Q,
G7A+G182*+D183*+N195F,
D183*+G182*+N195F+N260G,
D183*+G182*+R321A+S323N,
D183*+G182*+W284H,
H1*+D183*+G182*,
D183*+G182*+A186H+N195F,
D183*+G182*+F289V,
D183*+G182*+K391A,
D183*+G182*+S304R,
W167F+G182*+D183*+N195F,
F113Q+R116H+D183*+G182*,
D183*+G182*+N270G,
D183*+G182*+S244Q,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+G184T+N195F+V206L+K391A+G476K,
D183*+G182*+N195F+Y295N,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G255A+K391A+G476K,
D183*+G182*+A186Q,
D183*+G182*+N195F+V206F,
D183*+G182*+N195F+R320A,
G182*+D183*+G184T+N195F,
N54S+V56T+D183*+G182*,
W48Y+D183*+G182*,
A174S+D183*+G182*,
G7A+D183*+G182*,
N54S+D183*+G182*,
D183*+G182*+N195F+M246V,
W48Y+D183*+G182*+N195F,
D183*+G182*+N195F+S304R,
wherein numbering is according to SEQ ID NO: 13, and wherein the IF is determined by use of Model A detergent.
In one embodiment, the IF is determined by use of Model A detergent at a temperature of 15° C. In a particular embodiment, the variants have an IF of >1 as compared to the parent polypeptide, wherein the IF is determined by use of Model A detergent at a temperature of 15° C. and a concentration of 0.05 mg/L. It can be seen from the Tables in the Examples herein disclosed which such variants may be.
In another embodiment, the variant has an IF of at least 1.0 when compared to the parent polypeptide when tested in Model J detergent, and wherein the variant is selected from the group consisting of:
G182*+D183*+N195F,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,

H1*+N54S+V56T+G109A+Q172G+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172R+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
5 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+D377S+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+D377G+K391A+G476K,
10 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+D377R+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+D377A+K391A+G476K,
15 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+I214G+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+S381A+K391A+G476K,
H1*+N54S+V56T+G109A+W167I+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
20 H1*+N54S+V56T+G109A+W167G+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+G346S+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
25 N195F+V206L+G346P+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+I214H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+I214S+K391A+G476K,
30 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+I214T+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+A420Q+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
35 N195F+V206L+K391A+A420S+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+A420K+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+A420L+G476K,
40 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+K391Y+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391R+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
45 N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+P473A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+P473G+G476K,
50 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+P473T+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+Q395P+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
55 N195F+V206L+K391A+T444Q+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+T444S+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+T444D+G476K,
60 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+T444Y+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+S280W+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
65 N195F+V206L+S280L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+S280T+K391A+G476K,

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H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+S280A+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+G255N+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+T285L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+T285Q+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+Q449T+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+P473K+G476K,
 H1*+N54S+V56T+K72S+G109A+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+T40K+N54S+V56T+G109A+A174S+G182*+
 D183*+N195F+V206L+G346T+K391A+G476K,
 H1*+G50A+N54S+V56T+G109A++Q172N+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+G476K+K484A,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+G476K+K484G,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+G476K+K484P,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+G476K+K484E,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+G476K+K484Q,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+G476K+K484S,
 H1*+K+G50A+N54S+V56T+G109A+W167F+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+L173V+A174S+G182*+
 D183*+N195F+A204T+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+W140Y+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+Q98S+G109A+A174S+G182*+
 D183*+N195F+V206L+R320A+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+H324L+K391A+G476K,
 H1*+N54S+V56T+G109A+G149Q+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+G346T+K391A+G476K+G477A,
 H1*+N54S+V56T+G109AG255A++A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+G50A+N54S+V56T+G109A+F113L+R116L+
 A174S+G182*+D183*+N195F+V206L+K391A+
 G476K,
 H1*+N54S+V56T+G109A+W167F+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+L173V+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+K72R+G109A+W167F+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+G476K+G477Q,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+Q345D+K391A+G476K+G477Q,
 H1*+G50A+N54S+V56T+G109A+Q172D+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+V264I+K391A+G423H+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+R320V+K391A+G476Y,
 H1*+T5K+N54S+V56T+G109A+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,

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H1*+N54S+V56T+G109A+R116M+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+Q172Y+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 5 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+I214L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+L217V+K391A+G476K,
 10 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+L217H+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+P211D+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+K391Q+G476K,
 15 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+T444H+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+T444V+G476K,
 20 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+T444R+G476K,
 H1*+G50A+N54S+V56T+G109A+L173V+A174S+
 G182*+D183*+N195F+A204V+V206L+K391A+
 G476K,
 25 H1*+G50A+N54S+V56T+G109A+F113L+W167F+
 A174S+G182*+D183*+N195F+V206L+K391A+
 G476K,
 H1*+G7K+N54S+V56T+G109A+A174S+G182*+
 D183*+N195F+V206L+R320A+K391A+G476K,
 30 H1*+N54S+V56T+G109A+Q118G+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+S280K+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+N270G+K391A+G476K,
 35 H1*+N54S+V56T+G109A+Q172D+A174S+G182*+
 D183*+N195F+V206L+G346T+K391A+G476K+
 G477A,
 H1*+N54S+V56T+G109A+Q118S+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 40 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+R320A+K391A+G476K,
 H1*+N54S+V56T+G109A+T134E+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 45 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+G337H+Q385L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+Y382M+K391A+G476K,
 H1*+N54S+V56T+K72R+G109A+A174S+G182*+
 D183*+N195F+V206L+G255A+D377H+K391A+
 G476K,
 50 H1*+N54S+V56T+K72R+G109A+T134E+A174S+
 G182*+D183*+N195F+V206L+G255A+K391A+
 G476K,
 55 H1*+N54S+V56T+K72R+M105F+G109A+A174S+
 G182*+D183*+N195F+V206L+G255A+K391A+
 G476K,
 H1*+N54S+V56T+M105F+G109A+T134E+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 60 H1*+N54S+V56T+M105F+G109A+Q118F+T134E+
 A174S+G182*+D183*+N195F+V206L+K391A+
 G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+R320A+S323N+D377H+K391A+
 G476K,
 65 H1*+T51E+N54S+V56T+G109A+A174H+G182*+
 D183*+N195F+V206L+K391A+G476K,

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H1*+N54S+V56T+G109A+L173A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+Q299V+K391A+G476K, H1*+
N54S+V56T+G109A+L173G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+T355L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+T355F+K391A+G476K,
H1*+G50A+N54S+V56T+G109A+A174S+G182*+D183*+N195F+A204T+V206L+K391A+G476K,
H1*+G50A+N54S+V56T+G109A+A174E+G182*+D183*+N195F+A204T+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+R320A+S323N+K391A+G476K,
H1*+N54S+V56T+M105I+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+M246L+K391A+G476K,
H1*+T51G+N54S+V56T+G109A+A174N+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+T51A+N54S+V56T+G109A+A174G+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174D+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+T51S+N54S+V56T+G109A+A174P+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+T51G+N54S+V56T+G109A+A174M+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72Q+G109A+W167H+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q118R+T134E+A174S+G182*+D183*+N195F+V206L+D377H+
H1*+Q11H+N54S+V56T+G109A+Q118R+T134E+A174S+G184T+G182*+D183*+N195F+V206L+D377H+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+V206L+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+Q118R+T134E+A174S+G182*+D183*+N195F+V206L+G255A+K391A+G476K,
H1*+N54S+V56T+G109A+Q118R+A174S+G182*+D183*+N195F+V206L+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+T134E+A174S+G182*+D183*+G184T+N195F+V206L+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+Q172D+A174Q+G182*+D183*+N195F+V206L+K391A+G476K+G477A,
H1*+N54S+V56T+K72A+G109A+W167F+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+W167S+G182*+D183*+N195F+A174S+V206L+K391A+G476K,
H1*+N54S+V56T+K72H+G109A+W167S+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+W167T+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72E+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,

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H1*+N54S+V56T+K72R+G109A+W167L+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+W167M+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+W167H+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+G184T+N195F+V206L+G255A+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+G184T+N195F+V206L+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G255A+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+Q118R+A174S+G182*+D183*+N195F+V206L+G255A+D377H+
H1*+N54S+V56T+G109A+A174S+G182*+D183*+G184A+T193K+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+M105F+G109A+A174S+G182*+D183*+G184T+N195F+V206L+K391A+G476K,
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H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+V238A+D377H+K391A+G476K,
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100

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101

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H1*+N54S+V56T+G109A+Q172E+L173F+A174R+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172E+L173T+A174R+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172N+L173V+A174R+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+L173V+A174R+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172E+L173M+A174H+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172E+L173F+A174P+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+W167Y+Q172E+L173V+A174H+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+L173A+A174T+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+V291A+F328L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+M246L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+M246S+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+L250V+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+L250F+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+L250M+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+C474V+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+A204S+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+T165G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,

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H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+T334S+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+A263G+M286F+K391A+G476K,
5 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+A263G+M286L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+Y267H+K391A+G476K,
10 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+Y267L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+Y267I+K391A+G476K,
H1*+T51A+N54S+V56T+G109A+Q172R+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
15 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391 L+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
20 H1*+N54S+V56T+G109A+F113Q+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109S+F113Q+R116Q+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
25 H1*+N54S+V56T+G109A+Q172D+A174S+G182*+D183*+N195F+A204G+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+A204G+V206L+G255A+K391A+P473R+G476K,
30 H1*+N54S+V56T+K72R+G109A+W167F+A174S+G182*+D183*+N195F+A204G+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+G184T+N195F+A204G+V206L+K391A+P473R+G476K,
35 H1*+N54S+V56T+K72R+G109A+W167F+Q172N+A174S+G182*+D183*+N195F+V206L+K391A+P473R+G476K,
40 H1*+N54S+V56T+G109A+W167F+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+P473G+G476K,
H1*+N54S+V56T+G109A+W167F+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+P473R+G476K,
45 H1*+N54S+V56T+K72R+G109A+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+R116H+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
50 H1*+N54S+V56T+M105F+G109A+A174S+G182*+D183*+N195F+V206L+R320A+S323N+K391A+C474V+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+G255A+K391A+C474V+G476K,
55 H1*+N54S+V56T+G109A+T134E+A174S+G182*+D183*+N195F+V206L+K391A+C474V+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+D377H+K391A+C474V+G476K,
60 H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+K391A+C474V+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+A265G+K391A+T444Q+G476K+G477A,

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H1*+A37V+N54S+V56T+G109A+W167F+Q172G+
A174S+G182*+D183*+N195F+V206L+K391A+
G346T+G477A+G476K,
H1*+N54S+V56T+G109A+R116Q+Q172D+A174S+
G182*+D183*+N195F+V206L+G346T+K391A+
T444Q+G477A+G476K,
H1*+N54S+V56T+G109A+R116Q+W167F+Q172N+
A174S+G182*+D183*+N195F+V206L+G346T+
K391A+T444Q+G477A+G476K,
H1*+N54S+V56T+G109S+W167F+Q172R+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109S+R116Q+W167F+Q172N+
A174S+G182*+D183*+N195F+V206L+K391A+
G476K,
H1*+N54S+V56T+G109S+W167F+Q172G+A174S+
G182*+D183*+N195F+V206L+K391A+
H1*+N54S+V56T+G109A+F113Q+Q172G+A174S+
G182*+D183*+N195F+V206L+K391A+T444Q+
G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172G+
A174S+G182*+D183*+N195F+V206L+K391A+
T444Q+G476K,
H1*+N54S+V56T+G109A+F113Q+A174S+G182*+
D183*+N195F+V206L+K391A+C474V+G476K,
H1*+N54S+V56T+G109A+Q172N+A174S+G182*+
D183*+N195F+V206L+K391A+C474V+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+S280Q+H321Y+K391A+C474V+
G476K,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+
G182*+D183*+N195F+V206L+G255A+K391A+
C474V+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172G+
A174S+G182*+D183*+N195F+V206L+K391A+
T444Q+P473R+G476K,
H1*+N54S+V56T+G109A+F113Q+R116Q+Q172G+
A174S+G182*+D183*+N195F+V206L+K391A+
T444Q+G476K,
H1*+N54S+V56T+G109A+R116H+W167F+Q172N+
A174S+G182*+D183*+N195F+V206L+K391A+
G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+T165G+
Q172G+A174S+G182*+D183*+N195F+V206L+
K391A+T444Q+G476K,
H1*+N54S+V56T+G109A+F113Q+R116Q+T165G+
Q172G+A174S+G182*+D183*+N195F+V206L+
K391A+T444Q+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+W167F+
Q172R+A174S+G182*+D183*+N195F+V206L+
F289I+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+Q172N+A174S+
G182*+D183*+N195F+V206L+K391A+T444Q+
G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172N+
A174S+G182*+D183*+N195F+V206L+K391A+
G476K,
H1*+N54S+V56T+G109A+F113Q+R116Q+Q172M+
A174S+G182*+D183*+N195F+V206L+K391A+
T444Q+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172M+
A174S+G182*+D183*+N195F+V206L+K391A+
G476K,
H1*+N54S+V56T+G109A+R116H+Q172M+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,

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H1*+N54S+V56T+G109A+F113Q+R116H+T165G+
Q172M+A174S+G182*+D183*+N195F+V206L+
K391A+Q395P+G476K,
H1*+N54S+V56T+G109A+F113Q+A174S+G182*+
D183*+N195F+V206L+K391A+T444Q+P473R+
G476K,
H1*+N54S+V56T+G109A+R116Q+A174S+G182*+
D183*+N195F+V206L+A265G+K391A+T444Q+
P473R+G476K,
H1*+N54S+V56T+G109A+R116Q+A174S+G182*+
D183*+N195F+V206L+K391A+T444Q+P473R+
G476K,
H1*+N54S+V56T+K72R+G109A+T134E+T165G+
A174S+G182*+D183*+N195F+V206L+G255A+
K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+W167F+
Q172R+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
H1*+N54S+V56T+G109A+R116Q+W167F+Q172R+
A174S+G182*+D183*+N195F+V206L+A265G+
K391A+P473R+G476K,
H1*+N54S+V56T+G109A+F113Q+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172N+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+T51V+N54S+V56T+G109A+Q172D+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+S280Q+H321Y+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172N+
A174S+G182*+D183*+N195F+V206L+K391A+
G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172G+
L173V+A174S+G182*+D183*+N195F+V206L+
K391A+T444Q+P473R+G476K,
H1*+N54S+V56T+G109A+R116Q+W167F+Q172N+
L173V+A174S+G182*+D183*+N195F+V206L+
K391A+T444Q+G476K,
H1*+N54S+V56T+G109A+G133Q+W167F+Q172N+
A174S+G182*+D183*+N195F+V206L+Q395P+
K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+W167F+
Q172N+L173V+A174S+G182*+D183*+N195F+
V206L+K391A+G476K,
H1*+A37V+N54S+V56T+K72R+G109A+R116H+
W167F+Q172R+A174S+G182*+D183*+G184T+
N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116H+
W167F+Q172G+L173V+A174S+G182*+D183*+
G184T+N195F+V206L+K391A+T444Q+P473R+
G476K,
H1*+N54S+V56T+G109S+R116Q+W167F+Q172R+
A174S+G182*+D183*+N195F+V206L+K391A+
G476K,
H1*+N54S+V56T+G109S+R116Q+W167F+Q172G+
A174S+G182*+D183*+N195F+V206L+K391A+
P473R+G476K,
H1*+N54S+V56T+G109S+F113Q+W167F+Q172N+
A174S+G182*+D183*+N195F+V206L+
H1*+A37V+N54S+V56T+G109S+F113Q+W167F+
Q172R+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
H1*+N54S+V56T+K72R+G109A+R116H+T134E+
W167F+Q172G+L173V+A174S+G182*+D183*+
N195F+V206L+G255A+K391A+G476K,

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H1*+N54S+V56T+K72R+G109A+R116H+T134E+
W167F+Q172G+L173V+A174S+G182*+D183*+
N195F+V206L+G255A+K391A+Q395P+T444Q+
P473R+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+
W167F+Q172G+A174S+G182*+D183*+G184T+
N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+
T134E+W167F+Q172R+A174S+G182*+D183*+
N195F+V206L+G255A+K391A+T444Q+P473R+
G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+T134E+
W167F+Q172R+A174S+G182*+D183*+N195F+
V206L+G255A+K391A+Q395P+A445Q+P473R+
G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116H+
T134E+W167F+Q172R+A174S+G182*+D183*+
N195F+V206L+G255A+K391A+Q395P+G476K,
H1*+N54S+V56T+K72R+G109A+T134E+W167F+
Q172N+A174S+G182*+D183*+N195F+V206L+
G255A+A265G+K391A+T444Q+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+G133Q+W167F+
Q172R+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+
W167F+Q172G+A174S+G182*+D183*+N195F+
A204G+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+
W167F+Q172N+L173V+A174S+G182*+D183*+
G184T+N195F+V206L+K391A+T444Q+P473R+
G476K,
H1*+A37H+N54S+V56T+G109A+F113Q+W167F+
Q172N+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
H1*+A37H+N54S+V56T+G109A+F113Q+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+W167F+Q172N+
A174S+G182*+D183*+N195F+V206L+K391A+
T459N+G476K,
H1*+N54S+V56T+G109A+F113Q+Q172R+L173V+
A174S+G182*+D183*+N195F+V206L+K391A+
T444Q+G476K,
H1*+N54S+V56T+G109A+F113Q+R116Q+A174S+
G182*+D183*+N195F+V206L+K391A+P473R+
G476K,
H1*+N54S+V56T+A60V+G109A+R116Q+W167F+
Q172N+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
H1*+N54S+V56T+A60V+G109A+R116H+Q172R+
L173V+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
H1*+A37V+N54S+V56T+A60V+G109A+R116Q+
T165G+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
H1*+A37H+N54S+V56T+A60V+G109A+R116Q+
T165G+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
H1*+N54S+V56T+A60V+G109A+R116Q+W167F+
Q172G+L173V+A174S+G182*+D183*+N195F+
V206L+K391A+G476K,
H1*+N54S+V56T+A60V+G109A+R116Q+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37H+N54S+V56T+A60V+G109A+R116Q+
W167F+Q172R+A174S+G182*+D183*+N195F+
V206L+K391A+G476K,

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H1*+A37H+N54S+V56T+A60V+G109A+R116Q+
W167F+Q172N+A174S+G182*+D183*+N195F+
V206L+K391A+G476K,
H1*+N54S+V56T+A60V+G109A+R116Q+T165G+
A174S+G182*+D183*+N195F+V206L+K391A+
T444Q+P473R+G476K,
H1*+N54S+V56T+A60V+G109A+R116Q+A174S+
G182*+D183*+N195F+V206L+K391A+T444Q+
P473R+G476K,
H1*+N54S+V56T+A60V+G109A+R116Q+W167F+
Q172N+L173V+A174S+G182*+D183*+N195F+
V206L+K391A+G476K,
H1*+N54S+V56T+A60V+G109A+R116Q+W167F+
Q172R+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
H1*+N54S+V56T+G109A+R116A+W167F+Q172R+
A174S+G182*+D183*+N195F+A204G+V206L+
K391A+P473R+G476K,
H1*+N54S+V56T+G109A+R116A+A174S+G182*+
D183*+N195F+A204G+V206L+K391A+P473R+
G476K,
H1*+N54S+V56T+G109A+R116A+T165G+A174S+
G182*+D183*+N195F+V206L+K391A+T444Q+
P473R+G476K,
H1*+N54S+V56T+G109A+G182*+D183*+N195F+
K391A+G476K,
H1*+N54S+V56T+G109A+G182*+D183*+N195F+
V206L, N54S+V56T+G109A+G182*+D183*+N195F,
N54S+V56T+G109A+A174S+G182*+D183*+N195F+
V206L+K391A+G476K,
H1*+G109A+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
H1*+N54S+V56T+G109A+G182*+D183*+N195F+
V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+G476K,
H1*+N54S+V56T+A174S+G182*+D183*+N195F+
V206L+G273R+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L,
H1*+W48G+G109A+G182*+D183*+N195F+V206L+
K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+
K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+K391A,
H1*+N54S+V56T+G109A+F113Q+Q172M+A174S+
G182*+D183*+N195F+V206L+K391A+
H1*+N54S+V56T+G109A+R116Q+A174S+G182*+
D183*+N195F+A204G+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+T134E+
A174S+G182*+D183*+N195F+V206L+G255A+
K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+T134E+
A174S+G182*+D183*+N195F+V206L+G255A+
K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+R116Q+A174S+
G182*+D183*+G184T+N195F+V206L+K391A+
G476K,
H1*+A37V+N54S+V56T+G109A+R116A+T165G+
A174S+G182*+D183*+N195F+V206L+K391A+
G476K,
H1*+A37V+N54S+V56T+G109A+R116Q+T165G+
A174S+G182*+D183*+N195F+V206L+M246F+
K391A+G476K,

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H1*+V56A+G109A+A174S+G182*+D183*+N195F+
 V206L+G476K,
 H1*+N54S+V56T+G109A+G182*+D183*+N195F+
 V206L+K391A,
 H1*+N54S+V56T+G109A+G182*+D183*+N195F+
 V206L+G476K,
 G109A+A174S+G182*+D183*+N195F+V206L+
 K391A+G476K,
 N54S+V56T+G109A+G182*+D183*+N195F+V206L+
 K391T+G476K,
 H1*+N54S+V56T+K72R+G109A+F113Q+R116H+
 A174S+G182*+D183*+G184T+N195F+V206L+
 K391A+T444Q+G476K,
 H1*+N54S+V56T+G109S+F113Q+A174S+G182*+
 D183*+N195F+V206L+K391A+T444Q+G476K,
 H1*+N54S+V56T+G109A+F113Q+R116H+W167F+
 Q172G+L173V+A174S+G182*+D183*+N195F+
 V206L+K391A+P473R+G476K,
 H1*+N54S+V56T+G109A+F113Q+R116Q+Q172N+
 A174S+G182*+D183*+N195F+V206L+A265G+
 K391A+P473R+G476K,
 H1*+N54S+V56T+G109A+F113Q+W167F+Q172N+
 A174S+G182*+D183*+N195F+A204G+V206L+
 K391A+P473R+G476K,
 H1*+N54S+V56T+G109A+F113Q+R116H+W167F+
 Q172R+A174S+G182*+D183*+N195F+A204G+
 V206L+K391A+P473R+G476K,
 H1*+N54S+V56T+G109A+R116A+A174S+G182*+
 D183*+N195F+V206L+K391A+T444Q+P473R+
 G476K,
 H1*+N54S+V56T+G109A+R116H+Q172R+A174S+
 G182*+D183*+N195F+A204G+V206L+K391A+
 P473R+G476K,
 H1*+A37V+N54S+V56T+G109A+R116Q+T165G+
 A174S+G182*+D183*+N195F+V206L+K391
 A+G476K,
 H1*+A37H+N54S+V56T+G109A+R116H+T165G+
 A174S+G182*+D183*+N195F+V206L+M246F+
 K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+K391A+G476K,
 H1*+G109A+G182*+D183*+N195F+V206L+K391A+
 G476K, G109A+G182*+D183*+N195F+V206L+
 G476K,
 H1*+G109A+G182*+D183*+N195F+K391A,
 H1*+N54S+V56T+G109A+G133S+A174S+G182*+
 D183*+N195F+K391A+G476H,
 H1*+N54S+V56T+G109A+R116A+Q172N+L173V+
 A174S+G182*+D183*+N195F+V206L+K391A+
 G476K,
 H1*+A37V+N54S+V56T+G109A+R116H+T165G+
 A174S+G182*+D183*+N195F+V206L+K391A+
 G476K,
 H1*+P45A+N54S+V56T+G109A+A174S+G182*+
 D183*+N195F+V206L+K391A+C474V+G476K,
 H1*+N54S+V56T+G109A+P124A+A174S+G182*+
 D183*+N195F+V206L+K391A+C474V+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+P364A+K391A+C474V+G476K,
 H1*+N54S+V56T+G109A+Q118R+T134E+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 G184T+N195F+V206L+K391A+G476K,
 H1*+G50A+T51A+N54S+V56T+G109A+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+V264F+K391A+G476K,

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H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+Y382L+K391A+G476K,
 H1*+G7K+N54S+V56T+G109A+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+G255A+Q256A+K391A+G476K,
 H1*+N54S+V56T+G109A+W167Y+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+W167H+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+T165V+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174N+G182*+D183*+
 N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174Q+G182*+D183*+
 N195F+V206L+K391A+G476K,
 H1*+N16H+V17L+N54S+V56T+G109A+A174S+
 G182*+D183*+N195F+V206L+Y382F+K391A+
 G476K,
 H1*+Q32S+N54S+V56T+G109A+A174S+G182*+
 D183*+N195F+V206L+Q385L+K391A+G476K,
 H1*+N54S+V56T+R87S+G109A+R171H+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 N54S+G182*+D183*+N195F,
 G109A+G182*+D183*+N195F,
 A174S+G182*+D183*+N195F,
 G182*+D183*+N195F+K391A,
 G182*+D183*+N195F+G476K,
 H1*+N54S+V56T+G109A+R116H+Q169E+Q172K+
 A174*+G182*+D183*+N195F+V206L+K391A+
 G476K,
 H1*+N54S+V56T+G109A+A174*+G182*+D183*+
 N195F+V206L+K391A+G476K,
 H1*+K72R+G109A+F113Q+R116Q+W167F+Q172G+
 A174S+G182*+D183*+G184T+N195F+V206L+
 K391A+P473R+G476K,
 H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+
 W167F+Q172G+A174S+G182*+D183*+G184T+
 N195F+V206L+P473R+G476K,
 H1*+N54S+V56T+G109A+Q169E+Q172G+A174*+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+
 W167F+Q172G+A174S+G182*+D183*+G184T+
 N195F+K391A+P473R+G476K,
 H1*+N54S+V56T+G109A+R116W+Q169E+Q172K+
 A174*+G182*+D183*+N195F+V206L+K391A+
 G476K,
 H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+V56T+G109A+A174S+G182*+D183*+N195F+
 V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+N260G+K391A+G476K,
 H1*+N54S+V56T+K72R+G109A+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
 G182*+D183*+G184T+N195F+V206L+K391A+
 G476K,
 H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
 G182*+D183*+N195F+V206L+Y295N+K391A+
 G476K,
 H1*+N54S+V56T+G109A+W167F+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,

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H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+A186N+N195F+V206L+K391A+
G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+A288V+K391A+
G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+G255A+K391A+
G476K,
H1*+N54S+G109A+Q169E+Q172K+A174*+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+W48F+N54S+V56T+G109A+Q169E+Q172K+
A174*+G182*+D183*+N195F+V206L+K391A+
G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+V291A+K391A+
G476K,
H1*+N54S+V56T+G109A+F113Q+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+R320A+S323N+
K391A+G476K,
H1*+V56T+G109A+Q169E+Q172K+A174*+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116H+
W167F+Q172G+A174S+G182*+D183*+G184T+
N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+N270T+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+N260G+K391A+
G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172G+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+A186D+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+A225V+K391A+
G476K,
H1*+G109A+Q169E+Q172K+A174*+G182*+D183*+
N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+G255S+K391A+
G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+K269Q+K391A+
G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+G255S+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
G182*+D183*+N195F,
H1*+N54S+V56T+G109A+Q169E+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+W284H+K391A+
G476K,
H1*+N54S+G109A+A174S+G182*+D183*+N195F+
V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+S304R+K391A+
G476K,

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H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+
W167F+Q172G+A174S+G182*+D183*+G184T+
N195F+V206L+K391A+P473R,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+M246V+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+K391A,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+N270G+K391A+
G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+Y295F+K391A+
G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+R320A+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174T+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+W140Y+Q169E+Q172K+
A174*+G182*+D183*+N195F+V206L+K391A+
G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+F328L+K391A+
G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+G476K,
F113Q+R116H+D183*+G182*+N195F,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+A288L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+N270G+K391A+G476K,
Q169E+Q172K+A174*+D183*+G182*+N195F,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K269Q+K391A+G476K,
H1*+W48F+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
F113Q+R116Q+D183*+G182*+N195F,
N54S+V56T+D183*+G182*+N195F,
H1*+N54S+V56T+G109A+A174T+G182*+D183*+
N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116W+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
Q169E+Q172K+G182*+D183*+N195F, G182*+
D183*+A186D,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+V238A+K391A+G476K,
G182*+D183*+A186N,
H1*+N54S+V56T+G109A+A174S+V206L+G182*+
D183*+K391A+G476K,
H1*+N54S+V56T+Q169E+Q172K+A174*+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206F+K391A+G476K,
H1*+N54S+V56T+G109A+G182*+D183*+N195F+
V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172K+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172K+A174*+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174N+G182*+D183*+
N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+A186N+G182*+
D183*+N195F+V206L+K391A+G476K,

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H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+F289L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+Y295F+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+V291A+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+S244Q+K391A+G476K, G182*+
 D183*+F328L,
 G182*+D183*+N195F+A288V,
 G182*+D183*+A186N+N195F,
 G182*+D183*+E190P+N195F,
 W140Y+G182*+D183*+N195F,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+A225V+K391A+G476K,
 G182*+D183*+A186D+N195F,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+P473R+G476K,
 H1*+W48Y+N54S+V56T+G109A+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
 G182*+D183*+N195F+V206L+K391A+G476K,
 G182*+D183*+N195F+V238A,
 G182*+D183*+Y295N,
 Q172G+G182*+D183*+
 F113Q+G182*+D183*+N195F,
 H1*+N54S+V56T+Q98A+G109A+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 G182*+D183*+N195F+Y295F,
 G182*+D183*+A186E,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206Y+K391A+G476K,
 G182*+D183*+N195F+W284H,
 G182*+D183*+A186Q+N195F,
 K72R+G182*+D183*+
 D183*+G182*+N195F+V326L,
 D183*+G182*+A186E+N195F,
 Q169E+G182*+D183*+N195F,
 D183*+G182*+N195F+K242Q,
 G7A+G182*+D183*+N195F,
 D183*+G182*+N195F+N260G,
 D183*+G182*+R321A+S323N,
 D183*+G182*+W284H,
 H1*+D183*+G182*+
 D183*+G182*+A186H+N195F,
 D183*+G182*+F289V,
 D183*+G182*+K391A,
 D183*+G182*+S304R,
 W167F+G182*+D183*+N195F,
 F113Q+R116H+D183*+G182*+
 D183*+G182*+N270G,
 D183*+G182*+S244Q,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 G184T+N195F+V206L+K391A+G476K,
 D183*+G182*+N195F+Y295N,
 W48F+D183*+G182*+
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+G255A+K391A+G476K,
 D183*+G182*+A186Q,
 W140Y+D183*+G182*+
 D183*+G182*+K242Q,
 D183*+G182*+V206L,
 D183*+G182*+N195F+V206F,
 Q169E+D183*+G182*+
 D183*+G182*+N195F+R320A,
 G182*+D183*+G184T+N195F,
 D183*+G182*+G184T,

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D183*+G182*+P473R,
 W48Y+D183*+G182*+
 D183*+G182*+R323A,
 G7A+D183*+G182*+
 5 H1*+N54S+V56T+G109A+Q169E+Q172K+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 A174S+D183*+G182*+
 D183*+G182*+N195F+S323N,
 D183*+G182*+N195F+M246V,
 10 W48Y+D183*+G182*+N195F,
 V56T+D183*+G182*+
 wherein numbering is according to SEQ ID NO: 13,
 G182*+D183*+N195F+A428T,
 G182*+D183*+N195F,
 15 H1*+G182*+D183*+N195F,
 N54S+G182*+D183*+N195F,
 V56T+G182*+D183*+N195F,
 G182*+D183*+N195F+I405L,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 20 N195F+V206L+I405L+A421H+A422P+A428T+
 D476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+I405L+A421H+A422P+A428T,
 H1*+N54S+V56T+G109A+A174S+D183*+N195F+
 25 V206L+K391A+I405L+A421H+A422P+A428T,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+I405L+A421H+A422P+
 A428T+D476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 30 N195F+V206L+K391A+I405L+A421H+A422P+
 A428T+D476G,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+I405L+A421H+A422P+
 A428T+D476K+G477S,
 35 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+I405L+A421H+A422P+
 A428T+D476K,
 H1*+N54S+V56T+K72R+G109A+W167F+A174S+
 G182*+D183*+N195F+V206L+I405L+A421H+
 40 A422P+A428T,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+S280Q+H321Y+I405L+A421H+
 A422P+A428T,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 45 N195F+V206L+S304N+I405L+A421H+A422P+
 A428T+D476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+I405L+A421H+A422P+
 A428T+D476K+G477A,
 50 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+I405L+A421H+A422P+
 A428T+D476G+G477T,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 55 N195F+V206L+K391A+I405L+A421H+A422P+
 A428T+G477Q,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+I405L+A421H+A422P+
 A428T+D476K+G477Q,
 60 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+I405L+A421H+A422P+
 A428T+D476N,
 65 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+I405L+A421H+A422P+
 A428T+D476Y,

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H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T
H1*+N54S+V56T+G109A+Q172D+A174S+G182*+D183*+N195F+A204G+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+A204G+V206L+G255A+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116H+T165G+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+Q172N+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+G196R+V206L+G255A+I405L+A421H+A422P+A428T+N475S,
H1*+N54S+V56T+G109A+T134E+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+T40D+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+N125D+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+A60V+G109A+R116Q+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+R116V+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+R116Q+G133Q+T165G+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116Q+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T+G448D,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172N+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+T134E+W167F+Q172N+A174S+G182*+D183*+N195F+V206L+G255A+A265G+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+R116Q+W167F+Q172R+L173V+A174S+G182*+D183*+N195F+V206L+A265G+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+R116Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+F113Q+T134E+W167F+Q172G+A174S+G182*+D183*+N195F+V206L+G255A+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+F113Q+R116H+W167F+Q172G+L173V+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+W167F+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+G255A+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+G255A+I405L+A421H+A422P+A428T,

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H1*+N54S+V56T+G109A+F113Q+R116Q+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+N195F+A204G+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116H+W167F+Q172G+A174S+G182*+D183*+N195F+V206L+A265G+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+R116Q+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+A60V+G109A+R116Q+W167F+Q172N+L173V+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+A37H+N54S+V56T+A60V+G109A+R116Q+T165G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+A174S+M105F+G182*+D183*+N195F+V206L+L228I+R320A+S323N+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+F113Q+T134E+A174S+G182*+D183*+N195F+V206L+G255A+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+R116Q+V120L+A174S+G182*+D183*+G184T+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116H+W167F+Q172G+L173V+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116Q+Q172N+A174S+G182*+D183*+N195F+V206L+A265G+I405L+A421H+A422P+A428T,
H1*+A37H+N54S+V56T+A60V+G109A+R116Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T, wherein numbering is according to SEQ ID NO: 14.
In one embodiment, the IF is determined by use of Model J detergent at a temperature of 15° C. In a particular embodiment, the variants have an IF of >1 as compared to the parent polypeptide, wherein the IF is determined by use of Model J detergent at a temperature of 15° C. and a concentration of 0.05 mg/L. It can be seen from the Tables in the Examples herein disclosed which such variants may be.
In another embodiment, the variant has an IF of at least >1.5 when compared to said parent polypeptide, and wherein said variant is selected from
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+R87S+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+T40G+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+T51K+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72H+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37H+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37M+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,

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H1*+A37V+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+A37S+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+A37Y+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+A37R+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113W+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113S+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113N+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113Y+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q118N+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116Q+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116V+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116K+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116W+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116L+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+G142T+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q125P+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q125K+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+S381G+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+M246T+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+V213T+K391A+G476K,
H1*+A37L+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q118K+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116A+A174S+G182*+
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D183*+N195F+V206L+K391A+G476K,
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H1*+N54S+V56T+G109A+Q172R+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
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H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+D377R+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+D377A+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+I214G+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+S381A+K391A+G476K,

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H1*+N54S+V56T+G109A+W167I+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+W167G+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
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N195F+V206L+G346S+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+G346P+K391A+G476K,
10 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
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H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+I214S+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+I214T+K391A+G476K,
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N195F+V206L+K391A+A420Q+G476K,
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H1*+N54S+V56T+G109A+A174S+G182*+D183*+
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H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+P473T+G476K,
35 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
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H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+S280A+K391A+G476K,
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N195F+V206L+G255N+K391A+G476K,
55 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+T285L+K391A+G476K,
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N195F+V206L+T285Q+K391A+G476K,
60 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+Q449T+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+P473K+G476K,
H1*+N54S+V56T+K72S+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
65 H1*+T40K+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+G346T+K391A+G476K,

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H1*+G50A+N54S+V56T+G109A++Q172N+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
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 N195F+V206L+K391A+G476K+K484A,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
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 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
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 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+G476K+K484Q,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+G476K+K484S,
 H1*+K+G50A+N54S+V56T+G109A+W167F+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+L173V+A174S+G182*+
 D183*+N195F+A204T+V206L+K391A+
 H1*+N54S+V56T+G109A+W140Y+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+Q98S+G109A+A174S+G182*+
 D183*+N195F+V206L+R320A+K391A+G476K,
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 H1*+G50A+N54S+V56T+G109A+F113L+R116L+
 A174S+G182*+D183*+N195F+V206L+K391A+
 G476K,
 H1*+N54S+V56T+G109A+W167F+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+L173V+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+K72R+G109A+W167F+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+G476K+G477Q,
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 N195F+V206L+Q345D+K391A+G476K+G477Q,
 H1*+G50A+N54S+V56T+G109A+Q172D+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+V264I+K391A+G423H+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+R320V+K391A+G476Y,
 H1*+T5K+N54S+V56T+G109A+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+R116M+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+Q172Y+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+I214L+K391A+G476K,
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 N195F+V206L+L217V+K391A+G476K,
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 N195F+V206L+L217H+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+P211D+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+K391Q+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+T444H+G476K,

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H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+T444V+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+T444R+G476K,
 5 H1*+G50A+N54S+V56T+G109A+L173V+A174S+
 G182*+D183*+N195F+A204V+V206L+K391A+
 G476K,
 H1*+G50A+N54S+V56T+G109A+F113L+W167F+
 A174S+G182*+D183*+N195F+V206L+
 10 H1*+G7K+N54S+V56T+G109A+A174S+G182*+
 D183*+N195F+V206L+R320A+K391A+G476K
 H1*+N54S+V56T+G109A+Q118G+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 15 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+S280K+K391A+G476K,
 H1*+N54S+V56T+G109A+Q172D+A174S+G182*+
 D183*+N195F+V206L+G346T+K391A+G476K+
 G477A,
 20 H1*+N54S+V56T+G109A+Q118S+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K, H1*+N54S+
 V56T+G109A+A174S+G182*+D183*+N195F+
 V206L+R320A+K391A+G476K,
 H1*+N54S+V56T+G109A+T134E+A174S+G182*+
 25 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+G337H+Q385L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+Y382M+K391A+G476K,
 30 H1*+N54S+V56T+K72R+G109A+A174S+G182*+
 D183*+N195F+V206L+G255A+D377H+K391A+
 G476K,
 H1*+N54S+V56T+K72R+G109A+T134E+A174S+
 G182*+D183*+N195F+V206L+G255A+K391A+
 35 G476K,
 H1*+N54S+V56T+K72R+M105F+G109A+A174S+
 G182*+D183*+N195F+V206L+G255A+K391A+
 G476K,
 H1*+N54S+V56T+M105F+G109A+T134E+A174S+
 40 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+M105F+G109A+Q118F+T134E+
 A174S+G182*+D183*+N195F+V206L+K391A+
 G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 45 N195F+V206L+R320A+S323N+D377H+K391A+
 G476K,
 H1*+T51E+N54S+V56T+G109A+A174H+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+L173A+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 50 N195F+V206L+Q299V+K391A+G476K,
 H1*+N54S+V56T+G109A+L173G+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 55 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+T355L+K391A+G476K,
 H1*+G50A+N54S+V56T+G109A+A174S+G182*+
 D183*+N195F+A204T+V206L+K391A+G476K,
 H1*+G50A+N54S+V56T+G109A+A174E+G182*+
 60 D183*+N195F+A204T+V206L+K391A+G476K,
 H1*+N54S+V56T+K72R+G109A+A174S+G182*+
 D183*+N195F+V206L+R320A+S323N+K391A+
 G476K,
 H1*+N54S+V56T+M105I+G109A+A174S+G182*+
 65 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+M246L+K391A+G476K,

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H1*+T51G+N54S+V56T+G109A+A174N+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174D+G182*+D183*+
 N195F+V206L+K391A+G476K,
 H1*+T51S+N54S+V56T+G109A+A174P+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+K72A+G109A+W167S+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+K72Q+G109A+W167H+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+Q118R+T134E+A174S+
 G182*+D183*+N195F+V206L+D377H+K391A+
 G476K,
 H1*+Q11H+N54S+V56T+G109A+Q118R+T134E+
 A174S+G184T+G182*+D183*+N195F+V206L+
 D377H+K391A+G476K,
 H1*+N54S+V56T+K72R+G109A+T134E+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+K72R+G109A+T134E+A174S+
 G182*+D183*+N195F+V206L+D377H+K391A+
 G476K,
 H1*+N54S+V56T+K72R+G109A+T134E+A174S+
 G182*+D183*+G184T+N195F+V206L+K391A+
 G476K,
 H1*+N54S+V56T+G109A+Q118R+T134E+A174S+
 G182*+D183*+N195F+V206L+G255A+K391A+
 G476K,
 H1*+N54S+V56T+G109A+Q118R+A174S+G182*+
 D183*+N195F+V206L+D377H+K391A+G476K,
 H1*+N54S+V56T+G109A+T134E+A174S+G182*+
 D183*+G184T+N195F+V206L+D377H+K391A+
 G476K,
 H1*+N54S+V56T+G109A+Q172D+A174Q+G182*+
 D183*+N195F+V206L+K391A+G476K+G477A,
 H1*+N54S+V56T+K72R+G109A+W167L+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+K72R+G109A+W167M+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+K72R+G109A+A174S+G182*+
 D183*+G184T+N195F+V206L+G255A+K391A+
 G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 G184T+N195F+V206L+D377H+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+G255A+D377H+K391A+G476K,
 H1*+N54S+V56T+M105F+G109A+A174S+G182*+
 D183*+G184T+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+T134E+A174S+G182*+
 D183*+N195F+V206L+R320A+S323N+
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+V291G+D377H+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+V291A+F328L+D377H+K391A+
 G476K,
 H1*+N54S+V56T+G109A+T134E+A174S+G182*+
 D183*+G184T+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+V238A+D377H+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+V238T+D377H+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+V238G+D377H+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+F328M+D377H+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+F328L+D377H+K391A+G476K,

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H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+F328V+D377H+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+F328I+D377H+K391A+G476K,
 5 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+V291T+D377H+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+V291A+D377H+K391A+G476K,
 10 H1*+N54S+V56T+K72R+G109A+T134E+A174S+
 G182*+D183*+G184T+N195F+V206L+G255A+
 K391A+G476K,
 H1*+N54S+V56T+G109A+L173H+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K, H1*+N16S+
 15 N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+K72N+G109A+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+K72N+G109A+W167L+A174S+
 20 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+K72A+G109A+W167Y+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+Q118R+T134E+A174S+
 G182*+D183*+N195F+V206L+G255A+D377H+
 25 K391A+G476K,
 H1*+N54S+V56T+K72R+G109A+A174S+G182*+
 D183*+N195F+G184T+V206L+K391A+G476K,
 H1*+N54S+V56T+K72R+G109A+Q118R+A174S+
 G182*+D183*+N195F+V206L+G255A+D377H+
 30 K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 G184T+N195F+V206L+R320A+S323N+K391A+
 G476K,
 35 H1*+N54S+G109A+A174S+G182*+D183*+N195F+
 V206L+D377H+K391A+G476K,
 H1*+N54S+V56T+A174S+G182*+D183*+N195F+
 V206L+D377H+K391A+G476K,
 H1*+N54S+V56T+G109A+G182*+D183*+N195F+
 V206L+D377H+K391A+G476K,
 40 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+D377H+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+D377H+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 45 N195F+V206L+D377H+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+D377H+K391A,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+L217T+K391A+G476K,
 50 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+D377S+K391A+G476K,
 H1*+N54S+V56T+G109A+R116E+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 55 N195F+V206L+L217Q+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+D377A+K391A+G476K,
 H1*+N54S+V56T+G109A+T134E+A174S+G182*+
 D183*+N195F+V206L+D377H+K391A+G476K,
 60 H1*+N54S+V56T+G109A+Q118R+T134E+A174S+
 G182*+D183*+G184T+N195F+V206L+K391A+
 G476K,
 H1*+N54S+V56T+K72R+G109A+Q118R+T134E+
 A174S+G182*+D183*+N195F+V206L+K391A+
 65 G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+G346P+K391A+G476K,

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H1*+N54S+V56T+G109A+R116S+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
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H1*+T51E+N54S+V56T+G109A+Q172T+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
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H1*+T51E+N54S+V56T+G109A+Q172L+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
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H1*+N54S+V56T+K72S+G109A+A174H+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72Q+G109A+A174D+G182*+D183*+N195F+V206L+K391A+G476K,

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H1*+N54S+V56T+K72R+G109A+A174D+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72Q+G109A+A174K+G182*+D183*+N195F+V206L+K391A+G476K,
5 H1*+N54S+V56T+G109A+A174G+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+A174Q+G182*+D183*+N195F+V206L+K391A+G476K,
10 H1*+N54S+V56T+K72Q+G109A+A174G+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72S+G109A+A174N+G182*+D183*+N195F+V206L+K391A+G476K,
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35 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+M246A+K391A+G476K,
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H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+M286F+D377H+K391A+G476K,
60 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+Y267I+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+L235V+D377H+K391A+G476K,
65 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+L235A+D377H+K391A+G476K,
H1*+N54S+V56T+M105L+G109A+A174S+G182*+D183*+N195F+V206L+D377H+K391A+G476K,

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H1*+N54S+V56T+M105I+G109A+A174S+G182*+D183*+N195F+V206L+D377H+K391A+G476K,
H1*+N54S+V56T+M105V+G109A+A174S+G182*+D183*+N195F+V206L+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+M246I+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+L250I+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+L250V+D377H+K391A+G476K,
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H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+M246V+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+M246S+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+A174R+G182*+D183*+N195F+V206L+K391A+G476K,
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H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391W+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391 I+G476K,
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H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391Q+G476K,
H1*+N54S+V56T+G109A+Q172H+L173A+A174P+G182*+D183*+N195F+V206L+K391A+G476K,
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H1*+N54S+V56T+G109A+A174S+G182*+D183*+G184T+N195F+V206L+G255A+K391A+G476K,
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H1*+N54S+V56T+G109A+Q118R+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
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H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+S376V+K391A+G476K,
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H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+L250V+K391A+G476K,

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N195F+V206L+L250F+K391A+G476K,
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N195F+V206L+L250M+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+C474V+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+A204S+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+T165G+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
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H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+A263G+M286L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
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H1*+N54S+V56T+G109A+A174S+G182*+D183*+
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G182*+D183*+N195F+V206L+K391A+G476K,
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N195F+V206L+K391 L+G476K,
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A174S+G182*+D183*+N195F+V206L+K391A+
G476K,
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G182*+D183*+N195F+V206L+K391A+G476K,
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G182*+D183*+N195F+V206L+K391A+G476K,
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D183*+N195F+A204G+V206L+K391A+P473R+
G476K,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+
G182*+D183*+N195F+A204G+V206L+G255A+
K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+W167F+A174S+
G182*+D183*+N195F+A204G+V206L+K391A+
G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+
D183*+G184T+N195F+A204G+V206L+K391A+
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H1*+N54S+V56T+G109A+W167F+Q172N+A174S+
G182*+D183*+N195F+V206L+K391A+P473R+
G476K,
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G182*+D183*+N195F+V206L+K391A+P473G+
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G476K,
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K391A+G476K,
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D183*+N195F+V206L+R320A+S323N+K391A+
C474V+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+
D183*+N195F+V206L+G255A+K391A+C474V+
G476K,
H1*+N54S+V56T+G109A+T134E+A174S+G182*+
D183*+N195F+V206L+K391A+C474V+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+D377H+K391A+C474V+G476K,

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H1*+N54S+V56T+K72R+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+C474V+
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
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G477A,
H1*+A37V+N54S+V56T+G109A+W167F+Q172G+
A174S+G182*+D183*+N195F+V206L+K391A+
G346T+G477A+G476K,
H1*+N54S+V56T+G109A+R116Q+Q172D+A174S+
G182*+D183*+N195F+V206L+G346T+K391A+
T444Q+G477A+G476K,
H1*+N54S+V56T+G109A+R116Q+W167F+Q172N+
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K391A+T444Q+G477A+G476K,
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G182*+D183*+N195F+V206L+K391A+G476K,
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A174S+G182*+D183*+N195F+V206L+K391A+
G476K,
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G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+Q172G+A174S+
G182*+D183*+N195F+V206L+K391A+T444Q+
G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172G+
A174S+G182*+D183*+N195F+V206L+K391A+
T444Q+G476K,
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G476K,
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C474V+G476K,
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T444Q+P473R+G476K,
H1*+N54S+V56T+G109A+F113Q+R116Q+Q172G+
A174S+G182*+D183*+N195F+V206L+K391A+
T444Q+G476K,
H1*+N54S+V56T+G109A+R116H+W167F+Q172N+
A174S+G182*+D183*+N195F+V206L+K391A+
G476K,
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K391A+T444Q+G476K,
H1*+N54S+V56T+G109A+F113Q+R116Q+T165G+
Q172G+A174S+G182*+D183*+N195F+V206L+
K391A+T444Q+G476K,
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Q172R+A174S+G182*+D183*+N195F+V206L+
F289I+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+Q172N+A174S+
G182*+D183*+N195F+V206L+K391A+T444Q+
G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172N+
A174S+G182*+D183*+N195F+V206L+K391A+
G476K,
H1*+N54S+V56T+G109A+F113Q+R116Q+Q172M+
A174S+G182*+D183*+N195F+V206L+K391A+
T444Q+G476K,

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H1*+N54S+V56T+G109A+F113Q+R116H+Q172M+
A174S+G182*+D183*+N195F+V206L+K391A+
G476K,
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G182*+D183*+N195F+V206L+K391A+G476K,
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Q172M+A174S+G182*+D183*+N195F+V206L+
K391A+Q395P+G476K,
H1*+N54S+V56T+G109A+F113Q+A174S+G182*+
D183*+N195F+V206L+K391A+T444Q+P473R+
G476K,
H1*+N54S+V56T+G109A+R116Q+A174S+G182*+
D183*+N195F+V206L+A265G+K391A+T444Q+
P473R+G476K,
H1*+N54S+V56T+G109A+R116Q+A174S+G182*+
D183*+N195F+V206L+K391A+T444Q+P473R+
G476K,
H1*+N54S+V56T+K72R+G109A+T134E+T165G+
A174S+G182*+D183*+N195F+V206L+G255A+
K391A+G476K,
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Q172R+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
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A174S+G182*+D183*+N195F+V206L+A265G+
K391A+P473R+G476K,
H1*+N54S+V56T+G109A+F113Q+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172N+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
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G182*+D183*+N195F+V206L+K391A+G476K,
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A174S+G182*+D183*+N195F+V206L+K391A+
G476K,
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L173V+A174S+G182*+D183*+N195F+V206L+
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K391A+G476K,
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Q172N+L173V+A174S+G182*+D183*+N195F+
V206L+K391A+G476K,
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W167F+Q172R+A174S+G182*+D183*+G184T+
N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116H+
W167F+Q172G+L173V+A174S+G182*+D183*+
G184T+N195F+V206L+K391A+T444Q+P473R+
G476K,
H1*+N54S+V56T+G109S+R116Q+W167F+Q172R+
A174S+G182*+D183*+N195F+V206L+K391A+
G476K,
H1*+N54S+V56T+G109S+R116Q+W167F+Q172G+
A174S+G182*+D183*+N195F+V206L+K391A+
P473R+G476K,
H1*+N54S+V56T+G109S+F113Q+W167F+Q172N+
A174S+G182*+D183*+N195F+V206L+K391A+
G476K,

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H1*+A37V+N54S+V56T+G109S+F113Q+W167F+
Q172R+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
H1*+N54S+V56T+K72R+G109A+R116H+T134E+
W167F+Q172G+L173V+A174S+G182*+D183*+
N195F+V206L+G255A+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+R116H+T134E+
W167F+Q172G+L173V+A174S+G182*+D183*+
N195F+V206L+G255A+K391A+Q395P+T444Q+
P473R+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+
W167F+Q172G+A174S+G182*+D183*+G184T+
N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+
T134E+W167F+Q172R+A174S+G182*+D183*+
N195F+V206L+G255A+K391A+T444Q+P473R+
G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+T134E+
W167F+Q172R+A174S+G182*+D183*+N195F+
V206L+G255A+K391A+Q395P+A445Q+P473R+
G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116H+
T134E+W167F+Q172R+A174S+G182*+D183*+
N195F+V206L+G255A+K391A+Q395P+G476K,
H1*+N54S+V56T+K72R+G109A+T134E+W167F+
Q172N+A174S+G182*+D183*+N195F+V206L+
G255A+A265G+K391A+T444Q+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+G133Q+W167F+
Q172R+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+
W167F+Q172G+A174S+G182*+D183*+N195F+
A204G+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+
W167F+Q172N+L173V+A174S+G182*+D183*+
G184T+N195F+V206L+K391A+T444Q+P473R+
G476K,
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Q172N+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
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G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+W167F+Q172N+
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T459N+G476K,
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T444Q+G476K,
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G182*+D183*+N195F+V206L+K391A+P473R+
G476K,
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Q172N+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
H1*+N54S+V56T+A60V+G109A+R116H+Q172R+
L173V+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
H1*+A37V+N54S+V56T+A60V+G109A+R116Q+
T165G+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
H1*+A37H+N54S+V56T+A60V+G109A+R116Q+
T165G+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
H1*+N54S+V56T+A60V+G109A+R116Q+W167F+
Q172G+L173V+A174S+G182*+D183*+N195F+
V206L+K391A+G476K,

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H1*+N54S+V56T+A60V+G109A+R116Q+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37H+N54S+V56T+A60V+G109A+R116Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37H+N54S+V56T+A60V+G109A+R116Q+W167F+Q172N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+A60V+G109A+R116Q+T165G+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+P473R+G476K,
H1*+N54S+V56T+A60V+G109A+R116Q+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+P473R+G476K,
H1*+N54S+V56T+A60V+G109A+R116Q+W167F+Q172N+L173V+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+A60V+G109A+R116Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116A+W167F+Q172R+A174S+G182*+D183*+N195F+A204G+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+R116A+A174S+G182*+D183*+N195F+A204G+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+R116A+T165G+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+P473R+G476K,
H1*+N54S+V56T+G109A+G182*+D183*+N195F+K391A+G476K,
H1*+N54S+V56T+G109A+G182*+D183*+N195F+V206L,
N54S+V56T+G109A+G182*+D183*+N195F,
N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+G476K,
H1*+N54S+V56T+A174S+G182*+D183*+N195F+V206L+G273R+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L,
H1*+W48G+G109A+G182*+D183*+N195F+V206L+K391A+G476K,
N54S+V56T+G109A+A174S+G182*+D183*+N195F+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+Q172M+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116Q+A174S+G182*+D183*+N195F+A204G+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+T134E+A174S+G182*+D183*+N195F+V206L+G255A+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+T134E+A174S+G182*+D183*+N195F+V206L+G255A+K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+R116Q+A174S+G182*+D183*+G184T+N195F+V206L+K391A+G476K,

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H1*+A37V+N54S+V56T+G109A+R116A+T165G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37V+N54S+V56T+G109A+R116Q+T165G+A174S+G182*+D183*+N195F+V206L+M246F+K391A+G476K,
H1*+V56A+G109A+A174S+G182*+D183*+N195F+V206L+G476K, H1*+N54S+V56T+G109A+G182*+D183*+N195F+V206L+K391A,
H1*+N54S+V56T+G109A+G182*+D183*+N195F+V206L+G476K,
G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
N54S+V56T+G109A+G182*+D183*+N195F+V206L+K391T+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116H+A174S+G182*+D183*+G184T+N195F+V206L+K391A+T444Q+G476K,
H1*+N54S+V56T+G109S+F113Q+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+W167F+Q172G+L173V+A174S+G182*+D183*+N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+F113Q+R116Q+Q172N+A174S+G182*+D183*+N195F+V206L+A265G+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+F113Q+W167F+Q172N+A174S+G182*+D183*+N195F+A204G+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+R116A+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+P473R+G476K,
H1*+N54S+V56T+G109A+R116H+Q172R+A174S+G182*+D183*+N195F+A204G+V206L+K391A+P473R+G476K,
H1*+A37V+N54S+V56T+G109A+R116Q+T165G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37H+N54S+V56T+G109A+R116H+T165G+A174S+G182*+D183*+N195F+V206L+M246F+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+K391A+G476K, H1*+G109A+G182*+D183*+N195F+V206L+K391A+G476K,
G109A+G182*+D183*+N195F+V206L+G476K,
H1*+G109A+G182*+D183*+N195F+K391A,
H1*+N54S+V56T+G109A+G133S+A174S+G182*+D183*+N195F+K391A+G476H,
H1*+N54S+V56T+G109A+R116A+Q172N+L173V+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37V+N54S+V56T+G109A+R116H+T165G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+P45A+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+C474V+G476K,
H1*+N54S+V56T+G109A+P124A+A174S+G182*+D183*+N195F+V206L+K391A+C474V+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+P364A+K391A+C474V+G476K,
H1*+N54S+V56T+G109A+Q118R+T134E+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+G184T+N195F+V206L+K391A+G476K,

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H1*+G50A+T51A+N54S+V56T+G109A+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+V264F+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+Y382L+K391A+G476K,
H1*+G7K+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+G255A+Q256A+K391A+G476K,
H1*+N54S+V56T+G109A+W167Y+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+W167H+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+T165V+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174N+G182*+D183*+
N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174Q+G182*+D183*+
N195F+V206L+K391A+G476K,
H1*+N16H+V17L+N54S+V56T+G109A+A174S+
G182*+D183*+N195F+V206L+Y382F+K391A+
G476K,
H1*+Q32S+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+Q385L+K391A+G476K,
H1*+N54S+V56T+R87S+G109A+R171H+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
G109A+G182*+D183*+N195F,
G182*+D183*+N195F+K391A,
H1*+N54S+V56T+G109A+R116H+Q169E+Q172K+
A174*+G182*+D183*+N195F+V206L+K391A+
G476K,
H1*+N54S+V56T+G109A+A174*+G182*+D183*+
N195F+V206L+K391A+G476K,
H1*+K72R+G109A+F113Q+R116Q+W167F+Q172G+
A174S+G182*+D183*+G184T+N195F+V206L+
K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+
W167F+Q172G+A174S+G182*+D183*+G184T+
N195F+V206L+P473R+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172G+A174*+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+
W167F+Q172G+A174S+G182*+D183*+G184T+
N195F+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+R116W+Q169E+Q172K+
A174*+G182*+D183*+N195F+V206L+K391A+
G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+V56T+G109A+A174S+G182*+D183*+N195F+
V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+N260G+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+G184T+N195F+V206L+K391A+
G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+Y295N+K391A+
G476K,
H1*+N54S+V56T+G109A+W167F+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,

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H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+A186N+N195F+V206L+K391A+
G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+A288V+K391A+
G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+G255A+K391A+
G476K,
H1*+N54S+G109A+Q169E+Q172K+A174*+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+W48F+N54S+V56T+G109A+Q169E+Q172K+
A174*+G182*+D183*+N195F+V206L+K391A+
G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+V291A+K391A+
G476K,
H1*+N54S+V56T+G109A+F113Q+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+R320A+S323N+
K391A+G476K,
H1*+V56T+G109A+Q169E+Q172K+A174*+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116H+
W167F+Q172G+A174S+G182*+D183*+G184T+
N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+N270T+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+N260G+K391A+
G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172G+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+A186D+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+A225V+K391A+
G476K,
H1*+G109A+Q169E+Q172K+A174*+G182*+D183*+
N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+G255S+K391A+
G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+K269Q+K391A+
G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+G255S+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
G182*+D183*+N195F,
H1*+N54S+V56T+G109A+Q169E+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+W284H+K391A+
G476K,
H1*+N54S+G109A+A174S+G182*+D183*+N195F+
V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+S304R+K391A+
G476K,

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H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+
W167F+Q172G+A174S+G182*+D183*+G184T+
N195F+V206L+K391A+P473R,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+M246V+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+K391A,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+N270G+K391A+
G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+Y295F+K391A+
G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+R320A+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174T+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+W140Y+Q169E+Q172K+
A174*+G182*+D183*+N195F+V206L+K391A+
G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+F328L+K391A+
G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+G476K,
F113Q+R116H+D183*+G182*+N195F, H1*+N54S+
V56T+G109A+A174S+G182*+D183*+N195F+
V206L+A288L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+N270G+K391A+G476K,
Q169E+Q172K+A174*+D183*+G182*+N195F,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K269Q+K391A+G476K,
H1*+W48F+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
F113Q+R116Q+D183*+G182*+N195F,
N54S+V56T+D183*+G182*+N195F,
H1*+N54S+V56T+G109A+A174T+G182*+D183*+
N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116W+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
Q169E+Q172K+G182*+D183*+N195F,
G182*+D183*+A186D,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+V238A+K391A+G476K,
G182*+D183*+A186N,
H1*+N54S+V56T+G109A+A174S+V206L+G182*+
D183*+K391A+G476K,
H1*+N54S+V56T+Q169E+Q172K+A174*+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206F+K391A+G476K,
H1*+N54S+V56T+G109A+G182*+D183*+N195F+
V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172K+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172K+A174*+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174N+G182*+D183*+
N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+A186N+G182*+
D183*+N195F+V206L+K391A+G476K,

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H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+F289L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+Y295F+K391A+G476K,
5 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+V291A+K391A+G476K,
G182*+D183*+A186N+N195F,
G182*+D183*+A186D+N195F,
G182*+D183*+N195F+W284H,
10 D183*+G182*+N195F+V326L,
wherein numbering is according to SEQ ID NO: 13, and
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+I405L+A421H+A422P+
A428T+D476G+G477T,
15 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+I405L+A421H+A422P+
A428T+G477Q,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+I405L+A421H+A422P+
A428T+G477A,
20 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+I405L+A421H+A422P+
A428T+G477A,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+I405L+A421H+A422P+
A428T+D476N,
25 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+I405L+A421H+A422P+
A428T+D476Y,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206I+I405L+A421H+A422P+A428T,
30 H1*+N54S+V56T+G109A+T134E+A174S+G182*+
D183*+N195F+V206L+I405L+A421H+A422P+
A428T,
H1*+T40D+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+I405L+A421H+A422P+
A428T,
35 H1*+N54S+V56T+G109A+F113Q+N125D+A174S+
G182*+D183*+N195F+V206L+I405L+A421H+
A422P+A428T,
40 H1*+N54S+V56T+A60V+G109A+R116Q+A174S+
G182*+D183*+N195F+V206L+I405L+A421H+
A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172G+
A174S+G182*+D183*+N195F+V206L+I405L+
A421H+A422P+A428T,
45 H1*+N54S+V56T+G109A+F113Q+R116H+Q172N+
A174S+G182*+D183*+N195F+V206L+I405L+
A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+T134E+W167F+
Q172N+A174S+G182*+D183*+N195F+V206L+
G255A+A265G+I405L+A421H+A422P+A428T,
50 H1*+N54S+V56T+K72R+G109A+R116Q+W167F+
Q172R+L173V+A174S+G182*+D183*+N195F+
V206L+A265G+I405L+A421H+A422P+A428T,
55 H1*+N54S+V56T+K72R+G109A+R116Q+W167F+
Q172R+A174S+G182*+D183*+N195F+V206L+
I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+F113Q+T134E+
W167F+Q172G+A174S+G182*+D183*+N195F+
V206L+G255A+I405L+A421H+A422P+A428T,
60 H1*+N54S+V56T+K72R+G109A+F113Q+R116H+
W167F+Q172G+L173V+A174S+G182*+D183*+
G184T+N195F+V206L+I405L+A421H+A422P+
A428T,
65 H1*+N54S+V56T+K72R+G109A+A174S+G182*+
D183*+N195F+V206L+G255A+I405L+A421H+
A422P+A428T,

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H1*+N54S+V56T+G109A+F113Q+R116Q+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+F113Q+F116H+W167F+Q172G+L173V+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+N195F+A204G+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116H+W167F+Q172G+A174S+G182*+D183*+N195F+V206L+A265G+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+R116Q+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+A37H+N54S+V56T+A60V+G109A+R116Q+T165G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+F113Q+T134E+A174S+G182*+D183*+N195F+V206L+G255A+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+R116Q+V120L+A174S+G182*+D183*+G184T+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116H+W167F+Q172G+L173V+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116Q+Q172N+A174S+G182*+D183*+N195F+V206L+A265G+I405L+A421H+A422P+A428T,
H1*+A37H+N54S+V56T+A60V+G109A+R116Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T, wherein numbering is according to SEQ ID NO: 14, and wherein the IF is determined by use of Model A detergent.

In one embodiment, the IF is determined by use of Model A detergent at a temperature of 15° C. In a particular embodiment, the variants have an IF of >1.5 as compared to the parent polypeptide, wherein the IF is determined by use of Model A detergent at a temperature of 15° C. and a concentration of 0.05 mg/L. It can be seen from the Tables in the Examples herein disclosed which such variants may be.

In another embodiment, the variant has an IF of at least 1.5 when compared to the parent polypeptide and when determined by use of Model J detergent, wherein the variant is selected from the group consisting of:

G182*+D183*+N195F,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+R87S+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+T40G+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+T51K+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72H+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37H+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37M+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,

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H1*+A37V+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37S+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
5 H1*+A37Y+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37R+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
10 H1*+N54S+V56T+G109A+F113W+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113S+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
15 H1*+N54S+V56T+G109A+F113Y+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113R+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
20 H1*+N54S+V56T+G109A+Q118N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116Q+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116V+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
25 H1*+N54S+V56T+G109A+R116K+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116L+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
30 H1*+N54S+V56T+G109A+G142T+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q125P+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q125K+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
35 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+S381G+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+M246T+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+V213T+K391A+G476K,
H1*+A37L+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q118K+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116H+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
50 H1*+N54S+V56T+G109A+Q125A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172R+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+D377S+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+D377G+K391A+G476K,
60 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+D377R+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+D377A+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+I214G+K391A+G476K,
65 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+S381A+K391A+G476K,

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H1*+N54S+V56T+G109A+W167I+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+W167G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G346S+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G346P+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+I214H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+I214S+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+I214T+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+A420Q+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+A420S+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+A420K+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+A420L+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+K391Y+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+K391R+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+P473A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+P473G+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+P473T+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+Q395P+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+T444S+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+T444D+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+T444Y+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+S280W+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+S280L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+S280T+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+S280A+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G255N+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+T285L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+T285Q+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+Q449T+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+P473K+G476K,
H1*+N54S+V56T+K72S+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+T40K+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G346T+K391A+G476K,

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H1*+G50A+N54S+V56T+G109A++Q172N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K+K484A,
5 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K+K484G,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K+K484P,
10 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K+K484E,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K+K484Q,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K+K484S,
15 H1*+K+G50A+N54S+V56T+G109A+W167F+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+L173V+A174S+G182*+D183*+N195F+A204T+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+W140Y+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
20 H1*+N54S+V56T+Q98S+G109A+A174S+G182*+D183*+N195F+V206L+R320A+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+H324L+K391A+G476K,
25 H1*+N54S+V56T+G109A+G149Q+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G346T+K391A+G476K+G477A,
H1*+N54S+V56T+G109AG255A++A174S+G182*+D183*+N195F+V206L+K391A+G476K,
30 H1*+G50A+N54S+V56T+G109A+F113L+R116L+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+W167F+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
35 H1*+N54S+V56T+G109A+L173V+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+W167F+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
40 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K+G477Q,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+Q345D+K391A+G476K+G477Q,
H1*+G50A+N54S+V56T+G109A+Q172D+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
45 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+V264I+K391A+G423H+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+R320V+K391A+G476Y,
50 H1*+T5K+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116M+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172Y+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
55 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+I214L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+L217V+K391A+G476K,
60 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+L217H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+P211D+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+K391Q+G476K,
65 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+T444H+G476K,

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H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+T444V+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+T444R+G476K,
H1*+G50A+N54S+V56T+G109A+L173V+A174S+G182*+D183*+N195F+A204V+V206L+K391A+G476K,
H1*+G50A+N54S+V56T+G109A+F113L+W167F+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+G7K+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+R320A+K391A+G476K,
H1*+N54S+V56T+G109A+Q118G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+S280K+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+N270G+K391A+G476K,
H1*+N54S+V56T+G109A+Q172D+A174S+G182*+D183*+N195F+V206L+G346T+K391A+G476K+G477A,
H1*+N54S+V56T+G109A+Q118S+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+R320A+K391A+G476K,
H1*+N54S+V56T+G109A+T134E+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+Y382M+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+G255A+D377H+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+V206L+G255A+K391A+G476K,
H1*+N54S+V56T+K72R+M105F+G109A+A174S+G182*+D183*+N195F+V206L+G255A+K391A+G476K,
H1*+N54S+V56T+M105F+G109A+T134E+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+M105F+G109A+Q118F+T134E+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+R320A+S323N+D377H+K391A+G476K,
H1*+T51E+N54S+V56T+G109A+A174H+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+L173A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+Q299V+K391A+G476K,
H1*+N54S+V56T+G109A+L173G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+T355L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+T355F+K391A+G476K,
H1*+G50A+N54S+V56T+G109A+A174S+G182*+D183*+N195F+A204T+V206L+K391A+G476K,
H1*+G50A+N54S+V56T+G109A+A174E+G182*+D183*+N195F+A204T+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+R320A+S323N+K391A+G476K,

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H1*+N54S+V56T+M105I+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+M246L+K391A+G476K,
5 H1*+T51G+N54S+V56T+G109A+A174N+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+T51A+N54S+V56T+G109A+A174G+G182*+D183*+N195F+V206L+K391A+G476K,
10 H1*+N54S+V56T+G109A+A174D+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+T51G+N54S+V56T+G109A+A174M+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72Q+G109A+W167H+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
15 H1*+N54S+V56T+G109A+Q118R+T134E+A174S+G182*+D183*+N195F+V206L+D377H+K391A+G476K,
H1*+Q11H+N54S+V56T+G109A+Q118R+T134E+A174S+G184T+G182*+D183*+N195F+V206L+D377H+K391A+G476K,
20 H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+V206L+D377H+K391A+G476K,
25 H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+G184T+N195F+V206L+K391A+G476K,
30 H1*+N54S+V56T+G109A+Q118R+T134E+A174S+G182*+D183*+N195F+V206L+G255A+K391A+G476K,
H1*+N54S+V56T+G109A+Q118R+A174S+G182*+D183*+N195F+V206L+D377H+K391A+G476K,
35 H1*+N54S+V56T+G109A+T134E+A174S+G182*+D183*+G184T+N195F+V206L+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+Q172D+A174Q+G182*+D183*+N195F+V206L+K391A+G476K+G477A,
40 H1*+N54S+V56T+K72A+G109A+W167F+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+W167S+G182*+D183*+N195F+A174S+V206L+K391A+G476K,
45 H1*+N54S+V56T+K72R+G109A+W167L+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+W167M+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+W167H+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
50 H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+G184T+N195F+V206L+G255A+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+G184T+N195F+V206L+K391A+G476K,
55 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G255A+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+Q118R+A174S+G182*+D183*+N195F+V206L+G255A+D377H+K391A+G476K,
60 H1*+N54S+V56T+M105F+G109A+A174S+G182*+D183*+G184T+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+T134E+A174S+G182*+D183*+N195F+V206L+R320A+S323N+K391A+G476K,
65 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+V291G+D377H+K391A+G476K,

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H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+V291A+F328L+D377H+K391A+
G476K,
H1*+V56T+G109A+A174S+G182*+D183*+N195F+
V206L+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+T134E+A174S+G182*+
D183*+G184T+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+V238A+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+V238T+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+V238G+D377H+K391A+
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+F328M+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+F328L+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+F328V+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+F328I+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+V291T+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+V291A+D377H+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+
G182*+D183*+G184T+N195F+V206L+G255A+
K391A+G476K,
H1*+N54S+V56T+G109A+L173H+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q125T+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N16S+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72N+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72N+G109A+W167L+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72A+G109A+W167Y+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q118R+T134E+A174S+
G182*+D183*+N195F+V206L+G255A+D377H+
K391A+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+
D183*+N195F+G184T+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+Q118R+A174S+
G182*+D183*+N195F+V206L+G255A+D377H+
K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
G184T+N195F+V206L+R320A+S323N+K391A+
G476K,
H1*+N54S+G109A+A174S+G182*+D183*+N195F+
V206L+D377H+K391A+G476K,
H1*+N54S+V56T+A174S+G182*+D183*+N195F+
V206L+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+G182*+D183*+N195F+
V206L+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+D377H+K391A,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+D377R+K391A+G476K,

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H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+L217T+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+D377S+K391A+G476K,
5 H1*+N54S+V56T+G109A+R116E+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+L217Q+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+G346D+K391A+G476K,
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N195F+V206L+D377A+K391A+G476K,
H1*+N54S+V56T+G109A+T134E+A174S+G182*+
D183*+N195F+V206L+D377H+K391A+G476K,
15 H1*+N54S+V56T+G109A+Q118R+T134E+A174S+
G182*+D183*+G184T+N195F+V206L+K391A+
G476K,
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20 G476K,
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N195F+V206L+G346P+K391A+G476K,
H1*+N54S+V56T+G109A+R116S+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
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N195F+V206L+D377Q+K391A+G476K,
H1*+N54S+V56T+G109A+R116I+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
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N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116G+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174T+G182*+D183*+
N195F+V206L+K391A+G476K,
35 H1*+N54S+V56T+G109A+Q118R+T134E+A174S+
G182*+D183*+G184T+N195F+V206L+G255A+
K391A+G476K,
H1*+N54S+V56T+K72R+G109A+Q118R+T134E+
A174S+G182*+D183*+N195F+V206L+G255A+
40 K391A+G476K,
H1*+T51G+N54S+V56T+G109A+Q172R+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+T51A+N54S+V56T+G109A+Q172D+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
45 H1*+T51S+N54S+V56T+G109A+Q172M+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+T51A+N54S+V56T+G109A+Q172S+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
50 H1*+T51E+N54S+V56T+G109A+Q172T+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+T51A+N54S+V56T+G109A+Q172A+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
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G182*+D183*+N195F+V206L+K391A+
55 H1*+T51G+N54S+V56T+G109A+Q172K+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+T51A+N54S+V56T+G109A+Q172G+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+T51E+N54S+V56T+G109A+Q172L+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
60 H1*+T51E+N54S+V56T+G109A+Q172R+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172K+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
65 H1*+T51D+N54S+V56T+G109A+Q172N+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,

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H1*+T51A+N54S+V56T+G109A+Q172K+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+Q172S+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72H+G109A+Q172T+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
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H1*+N54S+V56T+G109A+F113L+W167F+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
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H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+M246A+K391A+G476K,
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H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+I214W+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+Q172K+A174S+G182*+D183*+N195F+V206L+K391A+G476K,

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H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+A204T+V206L+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+A204S+V206L+D377H+K391A+G476K,
5 H1*+N54S+V56T+G109A+T165S+A174S+G182*+D183*+N195F+V206L+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+T334S+D377H+K391A+G476K,
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H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+L235A+D377H+K391A+G476K,
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H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391Y+G476K,
35 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391V+G476K,
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55 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206S+K391A+G476K,
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H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391H+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391W+G476K,
65 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391I+G476K,

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H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+L250T+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+L250A+D377H+K391A+G476K,
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N195F+V206L+L250F+D377H+K391A+
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+L250M+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391Q+G476K,
H1*+N54S+V56T+G109A+Q172H+L173A+A174P+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172E+L173A+A174R+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72Q+G109A+W167R+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
G184T+N195F+V206L+G255A+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+
D183*+G184T+N195F+V206L+D377H+K391A+
G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
G184T+N195F+V206L+G255A+D377H+K391A+
G476K,
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N195F+V206L+S376T+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
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G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172D+L173A+A174T+
G182*+D183*+N195F+V206L+K391A+G476K,
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H1*+N54S+V56T+G109A+A174S+G182*+D183*+
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N195F+V206L+V238T+K391A+G476K,
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G476K,
H1*+N54S+V56T+G109A+W167F+Q172E+L173P+
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G476K,
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G182*+D183*+N195F+V206L+K391A+G476K,
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G182*+D183*+N195F+V206L+K391A+G476K,

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H1*+N54S+V56T+G109A+L173V+A174R+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172E+L173M+A174H+
G182*+D183*+N195F+V206L+K391A+G476K,
5 H1*+N54S+V56T+G109A+Q172E+L173F+A174P+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+W167Y+Q172E+L173V+
A174H+G182*+D183*+N195F+V206L+K391A+
G476K,
10 H1*+N54S+V56T+G109A+L173A+A174T+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+V291A+F328L+K391A+G476K,
15 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+M246L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+M246S+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
20 N195F+V206L+L250V+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+L250F+K391A+G476K,
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N195F+V206L+K391A+C474V+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+A204S+V206L+K391A+G476K,
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D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+T334S+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
35 N195F+V206L+A263G+M286L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+Y267H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+Y267L+K391A+G476K,
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N195F+V206L+Y267I+K391A+G476K,
H1*+T51A+N54S+V56T+G109A+Q172R+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
45 N195F+V206L+K391 L+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172G+
A174S+G182*+D183*+N195F+V206L+K391A+
G476K,
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G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109S+F113Q+R116Q+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172D+A174S+G182*+
D183*+N195F+A204G+V206L+K391A+P473R+
55 G476K,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+
G182*+D183*+N195F+A204G+V206L+G255A+
K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+W167F+A174S+
G182*+D183*+N195F+A204G+V206L+K391A+
60 G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+
D183*+G184T+N195F+A204G+V206L+K391A+
P473R+G476K,
65 H1*+N54S+V56T+G109A+W167F+Q172N+A174S+
G182*+D183*+N195F+V206L+K391A+P473R+
G476K,

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H1*+N54S+V56T+G109A+W167F+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+P473G+G476K,
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H1*+N54S+V56T+G109A+F113Q+R116Q+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+G476K,
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H1*+N54S+V56T+G109A+G133Q+W167F+Q172N+
A174S+G182*+D183*+N195F+V206L+Q395P+
K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+W167F+
Q172N+L173V+A174S+G182*+D183*+N195F+
V206L+K391A+G476K,
H1*+A37V+N54S+V56T+K72R+G109A+R116H+
W167F+Q172R+A174S+G182*+D183*+G184T+
N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116H+
W167F+Q172G+L173V+A174S+G182*+D183*+
G184T+N195F+V206L+K391A+T444Q+P473R+
G476K,
H1*+N54S+V56T+G109S+R116Q+W167F+Q172R+
A174S+G182*+D183*+N195F+V206L+K391A+
G476K,
H1*+N54S+V56T+G109S+R116Q+W167F+Q172G+
A174S+G182*+D183*+N195F+V206L+K391A+
P473R+G476K,
H1*+N54S+V56T+G109S+F113Q+W167F+Q172N+
A174S+G182*+D183*+N195F+V206L+K391A+
G476K,
H1*+A37V+N54S+V56T+G109S+F113Q+W167F+
Q172R+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
H1*+N54S+V56T+K72R+G109A+R116H+T134E+
W167F+Q172G+L173V+A174S+G182*+D183*+
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H1*+N54S+V56T+K72R+G109A+R116H+T134E+
W167F+Q172G+L173V+A174S+G182*+D183*+
N195F+V206L+G255A+K391A+Q395P+T444Q+
P473R+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+
W167F+Q172G+A174S+G182*+D183*+G184T+
N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+
T134E+W167F+Q172R+A174S+G182*+D183*+
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G476K,
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W167F+Q172R+A174S+G182*+D183*+N195F+
V206L+G255A+K391A+Q395P+A445Q+P473R+
G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116H+
T134E+W167F+Q172R+A174S+G182*+D183*+
N195F+V206L+G255A+K391A+Q395P+G476K,
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Q172N+A174S+G182*+D183*+N195F+V206L+
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K391A+G476K,
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W167F+Q172G+A174S+G182*+D183*+N195F+
A204G+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+
W167F+Q172N+L173V+A174S+G182*+D183*+
G184T+N195F+V206L+K391A+T444Q+P473R+
G476K,
H1*+A37H+N54S+V56T+G109A+F113Q+W167F+
Q172N+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
H1*+A37H+N54S+V56T+G109A+F113Q+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+W167F+Q172N+
A174S+G182*+D183*+N195F+V206L+K391A+
T459N+G476K,

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H1*+N54S+V56T+G109A+F113Q+Q172R+L173V+
A174S+G182*+D183*+N195F+V206L+K391A+
T444Q+G476K,
H1*+N54S+V56T+G109A+F113Q+R116Q+A174S+
G182*+D183*+N195F+V206L+K391A+P473R+
G476K,
H1*+N54S+V56T+A60V+G109A+R116Q+W167F+
Q172N+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
H1*+N54S+V56T+A60V+G109A+R116H+Q172R+
L173V+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
H1*+A37V+N54S+V56T+A60V+G109A+R116Q+
T165G+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
H1*+A37H+N54S+V56T+A60V+G109A+R116Q+
T165G+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
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Q172N+L173V+A174S+G182*+D183*+N195F+
V206L+K391A+G476K,
H1*+N54S+V56T+A60V+G109A+R116Q+W167F+
Q172R+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
H1*+N54S+V56T+G109A+R116A+W167F+Q172R+
A174S+G182*+D183*+N195F+A204G+V206L+
K391A+P473R+G476K,
H1*+N54S+V56T+G109A+R116A+A174S+G182*+
D183*+N195F+A204G+V206L+K391A+P473R+
G476K,
H1*+N54S+V56T+G109A+R116A+T165G+A174S+
G182*+D183*+N195F+V206L+K391A+T444Q+
P473R+G476K,
H1*+N54S+V56T+G109A+G182*+D183*+N195F+
K391A+G476K,
H1*+N54S+V56T+G109A+G182*+D183*+N195F+
V206L,
N54S+V56T+G109A+G182*+D183*+N195F,
N54S+V56T+G109A+A174S+G182*+D183*+N195F+
V206L+K391A+G476K,
H1*+G109A+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
H1*+N54S+V56T+G109A+G182*+D183*+N195F+
V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+G476K,
H1*+N54S+V56T+A174S+G182*+D183*+N195F+
V206L+G273R+K391A+G476K,

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H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L,
H1*+W48G+G109A+G182*+D183*+N195F+V206L+
K391A+G476K,
N54S+V56T+G109A+A174S+G182*+D183*+N195F+
K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+K391A,
H1*+N54S+V56T+G109A+F113Q+Q172M+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116Q+A174S+G182*+
D183*+N195F+A204G+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+T134E+
A174S+G182*+D183*+N195F+V206L+G255A+
K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+T134E+
A174S+G182*+D183*+N195F+V206L+G255A+
K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+R116Q+A174S+
G182*+D183*+G184T+N195F+V206L+K391A+
G476K,
H1*+A37V+N54S+V56T+G109A+R116A+T165G+
A174S+G182*+D183*+N195F+V206L+K391A+
G476K,
H1*+A37V+N54S+V56T+G109A+R116Q+T165G+
A174S+G182*+D183*+N195F+V206L+M246F+
K391A+G476K,
H1*+V56A+G109A+A174S+G182*+D183*+N195F+
V206L+G476K,
H1*+N54S+V56T+G109A+G182*+D183*+N195F+
V206L+K391A,
H1*+N54S+V56T+G109A+G182*+D183*+N195F+
V206L+G476K, G109A+A174S+G182*+D183*+
N195F+V206L+K391A+G476K,
N54S+V56T+G109A+G182*+D183*+N195F+V206L+
K391T+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116H+
A174S+G182*+D183*+G184T+N195F+V206L+
K391A+T444Q+G476K,
H1*+N54S+V56T+G109S+F113Q+A174S+G182*+
D183*+N195F+V206L+K391A+T444Q+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+W167F+
Q172G+L173V+A174S+G182*+D183*+N195F+
V206L+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+F113Q+R116Q+Q172N+
A174S+G182*+D183*+N195F+V206L+A265G+
K391A+P473R+G476K,
H1*+N54S+V56T+G109A+F113Q+W167F+Q172N+
A174S+G182*+D183*+N195F+A204G+V206L+
K391A+P473R+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+W167F+
Q172R+A174S+G182*+D183*+N195F+A204G+
V206L+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+R116A+A174S+G182*+
D183*+N195F+V206L+K391A+T444Q+P473R+
G476K,
H1*+N54S+V56T+G109A+R116H+Q172R+A174S+
G182*+D183*+N195F+A204G+V206L+K391A+
P473R+G476K,
H1*+A37V+N54S+V56T+G109A+R116Q+T165G+
A174S+G182*+D183*+N195F+V206L+K391
A+G476K,
H1*+A37H+N54S+V56T+G109A+R116H+T165G+
A174S+G182*+D183*+N195F+V206L+M246F+
K391A+G476K,

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H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+K391A+G476K, H1*+G109A+G182*+
D183*+N195F+V206L+K391A+G476K,
G109A+G182*+D183*+N195F+V206L+G476K,
H1*+G109A+G182*+D183*+N195F+K391A,
H1*+N54S+V56T+G109A+G133S+A174S+G182*+
D183*+N195F+K391A+G476H,
H1*+N54S+V56T+G109A+R116A+Q172N+L173V+
A174S+G182*+D183*+N195F+V206L+K391A+
G476K,
H1*+A37V+N54S+V56T+G109A+R116H+T165G+
A174S+G182*+D183*+N195F+V206L+K391A+
G476K,
H1*+P45A+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+C474V+G476K,
H1*+N54S+V56T+G109A+P124A+A174S+G182*+
D183*+N195F+V206L+K391A+C474V+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+P364A+K391A+C474V+G476K,
H1*+N54S+V56T+G109A+Q118R+T134E+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
G184T+N195F+V206L+K391A+G476K,
H1*+G50A+T51A+N54S+V56T+G109A+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+V264F+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+Y382L+K391A+G476K,
H1*+G7K+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+G255A+Q256A+K391A+G476K,
H1*+N54S+V56T+G109A+W167Y+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+W167H+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+T165V+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174N+G182*+D183*+
N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174Q+G182*+D183*+
N195F+V206L+K391A+G476K,
H1*+N16H+V17L+N54S+V56T+G109A+A174S+
G182*+D183*+N195F+V206L+Y382F+K391A+
G476K,
H1*+Q32S+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+Q385L+K391A+G476K,
H1*+N54S+V56T+R87S+G109A+R171H+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,
N54S+G182*+D183*+N195F,
G109A+G182*+D183*+N195F,
G182*+D183*+N195F+G476K,
H1*+N54S+V56T+G109A+R116H+Q169E+Q172K+
A174*+G182*+D183*+N195F+V206L+K391A+
G476K,
H1*+N54S+V56T+G109A+A174*+G182*+D183*+
N195F+V206L+K391A+G476K,
H1*+K72R+G109A+F113Q+R116Q+W167F+Q172G+
A174S+G182*+D183*+G184T+N195F+V206L+
K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+
W167F+Q172G+A174S+G182*+D183*+G184T+
N195F+V206L+P473R+G476K,

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H1*+N54S+V56T+G109A+Q169E+Q172G+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+R116W+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206F+K391A+G476K,
H1*+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+N260G+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+G184T+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+Y295N+K391A+G476K,
H1*+N54S+V56T+G109A+W167F+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+A186N+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+A288V+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+G255A+K391A+G476K,
H1*+N54S+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+W48F+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+V291A+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+R320A+S323N+K391A+G476K,
H1*+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116H+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+N270T+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+N260G+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+A186D+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+A225V+K391A+G476K,

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H1*+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+G255S+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K269Q+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G255S+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H182*+D183*+N195F,
H1*+N54S+V56T+G109A+Q169E+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+W284H+K391A+G476K,
H1*+N54S+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+S304R+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+V206L+K391A+P473R,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+M246V+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+N270G+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+Y295F+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+R320A+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174T+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+W140Y+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+F328L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G476K, F113Q+R116H+D183*+G182*+N195F,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+A288L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+N270G+K391A+G476K, Q169E+Q172K+A174*+D183*+G182*+N195F,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K269Q+K391A+G476K,
H1*+W48F+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K, F113Q+R116Q+D183*+G182*+N195F,
N54S+V56T+D183*+G182*+N195F,
H1*+N54S+V56T+G109A+A174T+G182*+D183*+N195F+V206L+K391A+G476K,

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H1*+N54S+V56T+G109A+R116W+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
Q169E+Q172K+G182*+D183*+N195F,
G182*+D183*+A186D,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+V238A+K391A+G476K,
G182*+D183*+A186N,
H1*+N54S+V56T+G109A+A174S+V206L+G182*+
D183*+K391A+G476K,
H1*+N54S+V56T+Q169E+Q172K+A174*+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206F+K391A+G476K,
H1*+N54S+V56T+G109A+G182*+D183*+N195F+
V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172K+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172K+A174*+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174N+G182*+D183*+
N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+A186N+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+F289L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+Y295F+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+V291A+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+S244Q+K391A+G476K,
G182*+D183*+F328L,
G182*+D183*+N195F+A288V,
G182*+D183*+A186N+N195F,
G182*+D183*+E190P+N195F,
W140Y+G182*+D183*+N195F,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+A225V+K391A+G476K,
G182*+D183*+A186D+N195F,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+P473R+G476K,
H1*+W48Y+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+
G182*+D183*+N195F+V206L+K391A+G476K,
G182*+D183*+N195F+V238A,
G182*+D183*+Y295N,
Q172G+G182*+D183*+
F113Q+G182*+D183*+N195F,
H1*+N54S+V56T+Q98A+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
G182*+D183*+N195F+Y295F,
G182*+D183*+A186E,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206Y+K391A+G476K,
G182*+D183*+N195F+W284H,
G182*+D183*+A186Q+N195F,
K72R+G182*+D183*+
D183*+G182*+N195F+V326L,
D183*+G182*+A186E+N195F,
Q169E+G182*+D183*+N195F,
D183*+G182*+N195F+K242Q,
G7A+G182*+D183*+N195F,
D183*+G182*+N195F+N260G,
D183*+G182*+R321A+S323N,
D183*+G182*+W284H,
D183*+G182*+A186H+N195F,
D183*+G182*+K391A,

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wherein numbering is according to SEQ ID NO: 13,
V56T+G182*+D183*+N195F,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+I405L+A421H+A422P+A428T,
5 H1*+N54S+V56T+G109A+A174S+D183*+N195F+
V206L+K391A+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+I405L+A421H+A422P+
A428T+D476K+G477S,
10 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+I405L+A421H+A422P+
A428T+D476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+I405L+A421H+A422P+
A428T+D476G+G477T,
15 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+I405L+A421H+A422P+
A428T+G477Q,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+I405L+A421H+A422P+
A428T+G477A,
20 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+I405L+A421H+A422P+
A428T+G477Q,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+I405L+A421H+A422P+
A428T+D476K+G477Q,
25 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+I405L+A421H+A422P+
A428T+D476N,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+I405L+A421H+A422P+
A428T+D476Y,
30 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+Q172D+A174S+G182*+
D183*+N195F+A204G+V206L+I405L+A421H+
A422P+A428T,
35 H1*+N54S+V56T+K72R+G109A+T134E+A174S+
G182*+D183*+N195F+A204G+V206L+G255A+
I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116H+T165G+
Q172G+A174S+G182*+D183*+N195F+V206L+
I405L+A421H+A422P+A428T,
40 H1*+N54S+V56T+G109A+F113Q+Q172N+A174S+
G182*+D183*+N195F+V206L+I405L+A421H+
A422P+A428T,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+
G182*+D183*+N195F+G196R+V206L+G255A+
I405L+A421H+A422P+A428T+N475S,
45 H1*+N54S+V56T+G109A+T134E+A174S+G182*+
D183*+N195F+V206L+I405L+A421H+A422P+
A428T,
H1*+T40D+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+I405L+A421H+A422P+
A428T,
50 H1*+N54S+V56T+G109A+F113Q+N125D+A174S+
G182*+D183*+N195F+V206L+I405L+A421H+
A422P+A428T,
H1*+N54S+V56T+A60V+G109A+R116Q+A174S+
G182*+D183*+N195F+V206L+I405L+A421H+
A422P+A428T,
55 H1*+N54S+V56T+G109A+R116V+A174S+G182*+
D183*+N195F+V206L+I405L+A421H+A422P+
A428T,
H1*+N54S+V56T+G109A+R116Q+G133Q+T165G+
Q172G+A174S+G182*+D183*+N195F+V206L+
I405L+A421H+A422P+A428T,

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H1*+N54S+V56T+G109A+F113Q+R116H+Q172G+
A174S+G182*+D183*+N195F+V206L+I405L+
A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116Q+Q172G+
A174S+G182*+D183*+N195F+V206L+I405L+
A421H+A422P+A428T+G448D,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172N+
A174S+G182*+D183*+N195F+V206L+I405L+
A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+T134E+W167F+
Q172N+A174S+G182*+D183*+N195F+V206L+
G255A+A265G+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+R116Q+W167F+
Q172R+L173V+A174S+G182*+D183*+N195F+
V206L+A265G+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+R116Q+W167F+
Q172R+A174S+G182*+D183*+N195F+V206L+
I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+F113Q+T134E+
W167F+Q172G+A174S+G182*+D183*+N195F+
V206L+G255A+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+F113Q+R116H+
W167F+Q172G+L173V+A174S+G182*+D183*+
G184T+N195F+V206L+I405L+A421H+A422P+
A428T,
H1*+N54S+V56T+G109A+F113Q+W167F+Q172G+
A174S+G182*+D183*+N195F+V206L+I405L+
A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+
D183*+N195F+V206L+G255A+I405L+A421H+
A422P+A428T,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+
W167F+Q172G+A174S+G182*+D183*+G184T+
N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116Q+W167F+
Q172G+A174S+G182*+D183*+N195F+A204G+
V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+R116Q+A174S+G182*+
D183*+N195F+V206L+I405L+A421H+A422P+
A428T,
H1*+A37H+N54S+V56T+A60V+G109A+R116Q+
T165G+A174S+G182*+D183*+N195F+V206L+
I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+A174S+M105F+G182*+
D183*+N195F+V206L+L228I+R320A+S323N+
I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+F113Q+T134E+
A174S+G182*+D183*+N195F+V206L+G255A+
I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+R116Q+V120L+
A174S+G182*+D183*+G184T+N195F+V206L+
I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116H+W167F+
Q172G+L173V+A174S+G182*+D183*+N195F+
V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116Q+Q172N+
A174S+G182*+D183*+N195F+V206L+A265G+
I405L+A421H+A422P+A428T,
H1*+A37H+N54S+V56T+A60V+G109A+R116Q+
W167F+Q172R+A174S+G182*+D183*+N195F+
V206L+I405L+A421H+A422P+A428T, wherein num-
bering is according to SEQ ID NO: 14.

In one embodiment, the IF is determined by use of Model J detergent at a temperature of 15° C. In a particular embodiment, the variants have an IF of >1.5 as compared to the parent polypeptide, wherein the IF is determined by use of Model J detergent at a temperature of 15° C. and a

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concentration of 0.05 mg/L. It can be seen from the Tables in the Examples herein disclosed which such variants may be.

Some variants according to the invention have an even further improved wash performance indicated by an IF of at least 2.0 when compared the a parent polypeptide having an amino acid sequence set forth in SEQ ID NO: 15, and wherein the IF is determined by use of Model A detergent at a temperature of 15° C.

Thus, in one embodiment, the variant has an IF of at least >2.0 when compared to said parent polypeptide, and wherein said variant is selected from

H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+K391A+G476K,
H1*+N54S+V56T+R87S+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+T40G+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+T51K+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72H+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+A37H+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+A37M+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+A37V+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+A37S+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+A37Y+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+A37R+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113W+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113S+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116Q+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116V+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116K+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q125P+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+S381G+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+
N195F+V206L+M246T+K391A+G476K,
H1*+A37L+N54S+V56T+G109A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116H+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q125A+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172G+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172R+A174S+G182*+
D183*+N195F+V206L+K391A+G476K,

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H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+D377S+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+D377A+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+S381A+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+G346P+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+I214H+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+I214S+K391A+G476K, H1*+N54S+
 V56T+G109A+A174S+G182*+D183*+N195F+
 V206L+K391A+A420Q+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+A420S+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+A420K+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+A420L+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+K391Y+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+P473R+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+P473A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+P473G+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+T444D+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+T444Y+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+S280W+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+S280L+K391A+G476K,
 H1*+N54S+V56T+K72S+G109A+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+T40K+N54S+V56T+G109A+A174S+G182*+
 D183*+N195F+V206L+G346T+K391A+G476K,
 H1*+G50A+N54S+V56T+G109A++Q172N+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+G476K+K484S,
 H1*+K+G50A+N54S+V56T+G109A+W167F+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+L173V+A174S+G182*+
 D183*+N195F+A204T+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+W140Y+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+G346T+K391A+G476K+G477A,
 H1*+N54S+V56T+G109AG255A++A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+G50A+N54S+V56T+G109A+F113L+R116L+
 A174S+G182*+D183*+N195F+V206L+K391A+
 G476K,
 H1*+N54S+V56T+G109A+W167F+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+K72R+G109A+W167F+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+G476Y,
 H1*+N54S+V56T+G109A+R116M+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+Q172Y+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,

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H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+I214L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+L217V+K391A+G476K,
 5 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+L217H+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+P211D+K391A+G476K,
 10 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+K391Q+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+T444H+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+T444V+G476K,
 15 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+T444R+G476K,
 H1*+G50A+N54S+V56T+G109A+L173V+A174S+
 G182*+D183*+N195F+A204V+V206L+K391A+
 G476K,
 20 H1*+G7K+N54S+V56T+G109A+A174S+G182*+
 D183*+N195F+V206L+R320A+K391A+G476K
 H1*+N54S+V56T+G109A+Q172D+A174S+G182*+
 D183*+N195F+V206L+G346T+K391A+G476K+
 G477A,
 25 H1*+N54S+V56T+G109A+Q118S+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+R320A+K391A+G476K,
 H1*+N54S+V56T+G109A+T134E+A174S+G182*+
 30 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+Y382M+K391A+G476K,
 H1*+N54S+V56T+K72R+G109A+T134E+A174S+
 G182*+D183*+N195F+V206L+G255A+K391A+
 35 G476K,
 H1*+N54S+V56T+K72R+M105F+G109A+A174S+
 G182*+D183*+N195F+V206L+G255A+K391A+
 G476K,
 H1*+N54S+V56T+M105F+G109A+T134E+A174S+
 40 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+R320A+S323N+D377H+K391A+
 G476K,
 H1*+T51E+N54S+V56T+G109A+A174H+G182*+
 D183*+N195F+V206L+K391A+G476K,
 45 H1*+N54S+V56T+G109A+L173A+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+Q299V+K391A+G476K,
 50 H1*+N54S+V56T+G109A+L173G+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+T355L+K391A+G476K,
 H1*+G50A+N54S+V56T+G109A+A174S+G182*+
 D183*+N195F+A204T+V206L+K391A+G476K,
 55 H1*+N54S+V56T+K72R+G109A+A174S+G182*+
 D183*+N195F+V206L+R320A+S323N+K391A+
 G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+M246L+K391A+G476K,
 60 H1*+N54S+V56T+K72Q+G109A+W167H+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+K72R+G109A+T134E+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 65 H1*+N54S+V56T+K72R+G109A+T134E+A174S+
 G182*+D183*+N195F+V206L+D377H+K391A+
 G476K,

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H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+G184T+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q118R+T134E+A174S+G182*+D183*+N195F+V206L+G255A+K391A+G476K,
H1*+N54S+V56T+G109A+Q118R+A174S+G182*+D183*+N195F+V206L+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+T134E+A174S+G182*+D183*+G184T+N195F+V206L+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+Q172D+A174Q+G182*+D183*+N195F+V206L+K391A+G476K+G477A,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+G184T+N195F+V206L+G255A+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+G184T+N195F+V206L+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G255A+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+T134E+A174S+G182*+D183*+G184T+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+V238T+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+V291A+D377H+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+G184T+N195F+V206L+G255A+K391A+G476K,
H1*+N54S+V56T+G109A+L173H+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72N+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72N+G109A+W167L+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q118R+T134E+A174S+G182*+D183*+N195F+V206L+G255A+D377H+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+D377H+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+D377H+K391A,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+D377S+K391A+G476K,
H1*+N54S+V56T+G109A+R116E+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+T134E+A174S+G182*+D183*+N195F+V206L+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+R116S+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116I+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+T51A+N54S+V56T+G109A+Q172S+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+T51E+N54S+V56T+G109A+Q172T+A174S+G182*+D183*+N195F+V206L+K391A+G476K,

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H1*+T51S+N54S+V56T+G109A+Q172N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+T51E+N54S+V56T+G109A+Q172L+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
5 H1*+T51E+N54S+V56T+G109A+Q172R+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172K+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+Q172S+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
10 H1*+N54S+V56T+K72H+G109A+Q172T+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72Q+G109A+Q172R+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
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H1*+N54S+V56T+K72M+G109A+Q172S+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
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20 H1*+N54S+V56T+K72S+G109A+A174H+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72Q+G109A+A174D+G182*+D183*+N195F+V206L+K391A+G476K,
25 H1*+N54S+V56T+K72R+G109A+A174D+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72Q+G109A+A174K+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174G+G182*+D183*+N195F+V206L+K391A+G476K,
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H1*+N54S+V56T+K72Q+G109A+A174G+G182*+D183*+N195F+V206L+K391A+G476K,
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H1*+N54S+V56T+K72R+G109A+A174H+G182*+D183*+N195F+V206L+K391A+G476K,
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H1*+N54S+V56T+M105V+G109A+A174S+G182*+D183*+N195F+V206L+D377H+K391A+G476K,
65 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+M246I+D377H+K391A+G476K,
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 N195F+V206L+K391Y+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391V+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391M+G476K,
 H1*+N54S+V56T+G109A+T165G+A174S+G182*+
 D183*+N195F+V206L+D377H+K391A+G476K,
 H1*+N54S+V56T+G109S+A174S+G182*+D183*+
 N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174V+G182*+D183*+
 N195F+V206L+K391A+G476K,
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 G182*+D183*+N195F+V206L+K391A+G476K,
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 G476K,
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 N195F+V206L+V264T+K391A+G476K,
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 A174S+G182*+D183*+N195F+V206L+K391A+
 G476K,
 H1*+N54S+V56T+G109A+Q172N+L173T+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+Q172E+L173M+A174H+
 G182*+D183*+N195F+V206L+K391A+G476K,

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H1*+N54S+V56T+G109A+W167Y+Q172E+L173V+
 A174H+G182*+D183*+N195F+V206L+K391A+
 G476K,
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 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+V291A+F328L+K391A+G476K,
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 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
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 H1*+N54S+V56T+G109A+F113Q+R116H+Q172G+
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 G476K,
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 G476K,
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 K391A+G476K,
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 G476K,
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 N195F+V206L+D377H+K391A+C474V+G476K,
 H1*+N54S+V56T+K72R+G109A+A174S+G182*+
 D183*+N195F+V206L+K391A+C474V+G476K,

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H1*+N54S+V56T+G109A+R116Q+A174S+G182*+D183*+N195F+V206L+A265G+K391A+T444Q+P473R+G476K,
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H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+T134E+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+G255A+K391A+T444Q+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116H+T134E+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+G255A+K391A+Q395P+G476K,
H1*+N54S+V56T+K72R+G109A+T134E+W167F+Q172N+A174S+G182*+D183*+N195F+V206L+G255A+A265G+K391A+T444Q+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+G133Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+N195F+A204G+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+W167F+Q172N+L173V+A174S+G182*+D183*+G184T+N195F+V206L+K391A+T444Q+P473R+G476K,
H1*+A37H+N54S+V56T+G109A+F113Q+W167F+Q172N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+W167F+Q172N+A174S+G182*+D183*+N195F+V206L+K391A+T459N+G476K,
H1*+N54S+V56T+G109A+F113Q+R116Q+A174S+G182*+D183*+N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+A60V+G109A+R116Q+W167F+Q172N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+A60V+G109A+R116H+Q172R+L173V+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37V+N54S+V56T+A60V+G109A+R116Q+T165G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37H+N54S+V56T+A60V+G109A+R116Q+T165G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+A60V+G109A+R116Q+W167F+Q172G+L173V+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+A60V+G109A+R116Q+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37H+N54S+V56T+A60V+G109A+R116Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37H+N54S+V56T+A60V+G109A+R116Q+W167F+Q172N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+A60V+G109A+R116Q+T165G+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+P473R+G476K,
H1*+N54S+V56T+A60V+G109A+R116Q+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+P473R+G476K,
H1*+N54S+V56T+A60V+G109A+R116Q+W167F+Q172N+L173V+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+A60V+G109A+R116Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+K391A+G476K,

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H1*+N54S+V56T+G109A+R116A+W167F+Q172R+A174S+G182*+D183*+N195F+A204G+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+R116A+A174S+G182*+D183*+N195F+A204G+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+R116A+T165G+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+P473R+G476K,
H1*+N54S+V56T+G109A+G182*+D183*+N195F+K391A+G476K,
H1*+N54S+V56T+G109A+G182*+D183*+N195F+V206L,
N54S+V56T+G109A+G182*+D183*+N195F,
N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+G476K,
H1*+N54S+V56T+A174S+G182*+D183*+N195F+V206L+G273R+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L,
N54S+V56T+G109A+A174S+G182*+D183*+N195F+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+Q172M+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+T134E+A174S+G182*+D183*+N195F+V206L+G255A+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+T134E+A174S+G182*+D183*+N195F+V206L+G255A+K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+R116Q+A174S+G182*+D183*+G184T+N195F+V206L+K391A+G476K,
H1*+A37V+N54S+V56T+G109A+R116A+T165G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37V+N54S+V56T+G109A+R116Q+T165G+A174S+G182*+D183*+N195F+V206L+M246F+K391A+G476K,
H1*+V56A+G109A+A174S+G182*+D183*+N195F+V206L+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116H+A174S+G182*+D183*+G184T+N195F+V206L+K391A+T444Q+G476K,
H1*+N54S+V56T+G109S+F113Q+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+W167F+Q172G+L173V+A174S+G182*+D183*+N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+F113Q+R116Q+Q172N+A174S+G182*+D183*+N195F+V206L+A265G+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+F113Q+W167F+Q172N+A174S+G182*+D183*+N195F+A204G+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+W167F+Q172R+A174S+G182*+D183*+N195F+A204G+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+R116A+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+P473R+G476K,

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H1*+N54S+V56T+G109A+R116H+Q172R+A174S+G182*+D183*+N195F+A204G+V206L+K391A+P473R+G476K,
H1*+A37V+N54S+V56T+G109A+R116Q+T165G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37H+N54S+V56T+G109A+R116H+T165G+A174S+G182*+D183*+N195F+V206L+M246F+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+K391A+G476K,
H1*+N54S+V56T+G109A+G133S+A174S+G182*+D183*+N195F+K391A+G476H,
H1*+N54S+V56T+G109A+R116A+Q172N+L173V+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37V+N54S+V56T+G109A+R116H+T165G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+P45A+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+C474V+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+P364A+K391A+C474V+G476K,
H1*+N54S+V56T+G109A+Q118R+T134E+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+G184T+N195F+V206L+K391A+G476K,
H1*+G50A+T51A+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+V264F+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+Y382L+K391A+G476K,
H1*+G7K+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G255A+Q256A+K391A+G476K,
H1*+N54S+V56T+G109A+W167Y+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+W167H+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+T165V+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174N+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174Q+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N16H+V17L+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+Y382F+K391A+G476K,
H1*+Q32S+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+Q385L+K391A+G476K,
H1*+N54S+V56T+R87S+G109A+R171H+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116H+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+V206L+P473R+G476K,

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H1*+N54S+V56T+G109A+Q169E+Q172G+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+R116W+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+N260G+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+G184T+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+Y295N+K391A+G476K,
H1*+N54S+V56T+G109A+W167F+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+A186N+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+A288V+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+G255A+K391A+G476K,
H1*+N54S+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+W48F+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+V291A+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+R320A+S323N+K391A+G476K,
H1*+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116H+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+N270T+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+N260G+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+A186D+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+A225V+K391A+G476K,

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H1*+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+G255S+K391A+G476K,
 H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K269Q+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G255S+K391A+G476K,
 H1*+N54S+V56T+G109A+Q169E+Q172K+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 wherein numbering is according to SEQ ID NO: 13, and
 H1*+T40D+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T
 H1*+N54S+V56T+G109A+F113Q+N125D+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T
 H1*+N54S+V56T+G109A+F113Q+R116H+Q172N+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T
 H1*+N54S+V56T+K72R+G109A+T134E+W167F+Q172N+A174S+G182*+D183*+N195F+V206L+G255A+A265G+I405L+A421H+A422P+A428T
 H1*+N54S+V56T+K72R+G109A+R116Q+W167F+Q172R+L173V+A174S+G182*+D183*+N195F+V206L+A265G+I405L+A421H+A422P+A428T
 H1*+N54S+V56T+K72R+G109A+R116Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T
 H1*+N54S+V56T+K72R+G109A+F113Q+T134E+W167F+Q172G+A174S+G182*+D183*+N195F+V206L+G255A+I405L+A421H+A422P+A428T
 H1*+N54S+V56T+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+N195F+A204G+V206L+I405L+A421H+A422P+A428T
 H1*+N54S+V56T+K72R+G109A+R116Q+V120L+A174S+G182*+D183*+G184T+N195F+V206L+I405L+A421H+A422P+A428T
 H1*+N54S+V56T+G109A+F113Q+R116H+W167F+Q172G+L173V+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T
 H1*+N54S+V56T+G109A+F113Q+R116Q+Q172N+A174S+G182*+D183*+N195F+V206L+A265G+I405L+A421H+A422P+A428T
 H1*+A37H+N54S+V56T+A60V+G109A+R116Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T, wherein numbering is according to SEQ ID NO: 14, and wherein the IF is determined by use of Model A detergent.

In one embodiment, the variant has an IF of at least 2.0 when compared to the parent polypeptide, wherein the IF is determined by use of Model A detergent at a temperature of 15° C. and a concentration of 0.05 mg enzyme/L.

In one embodiment, the variant has an IF of at least 2.0 when compared to the parent polypeptide and when determined by use of Model J detergent, and wherein the variant is selected from the group consisting of:

H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+R87S+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+T40G+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+T51K+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,

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H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+K72H+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 5 H1*+A37H+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+A37M+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 10 H1*+A37V+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+A37S+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+A37Y+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 15 H1*+A37R+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+F113W+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 20 H1*+N54S+V56T+G109A+F113S+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+F113N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+F113Y+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 25 H1*+N54S+V56T+G109A+F113R+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+Q118N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 30 H1*+N54S+V56T+G109A+R116Q+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+R116V+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+R116K+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 35 H1*+N54S+V56T+G109A+Q125P+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+Q125K+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 40 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+S381G+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+M246T+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+V213T+K391A+G476K,
 45 H1*+A37L+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+R116A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 50 H1*+N54S+V56T+G109A+R116H+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+Q125A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 55 H1*+N54S+V56T+G109A+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+Q172R+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+D377S+K391A+G476K,
 60 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+D377G+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+D377R+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+D377A+K391A+G476K,
 65 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+I214G+K391A+G476K,

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H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+S381A+K391A+G476K,
 H1*+N54S+V56T+G109A+W167I+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+W167G+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+G346S+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+G346P+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+I214H+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+I214S+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+I214T+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+A420S+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+A420K+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+A420L+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+K391Y+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+K391R+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+P473R+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+P473A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+P473G+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+P473T+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+Q395P+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+T444Q+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+T444D+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+T444Y+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+S280W+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+S280L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+S280T+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+S280A+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+G255N+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+T285L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+T285Q+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+P473K+G476K,
 H1*+N54S+V56T+K72S+G109A+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+T40K+N54S+V56T+G109A+A174S+G182*+
 D183*+N195F+V206L+G346T+K391A+G476K,
 H1*+G50A+N54S+V56T+G109A++Q172N+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,

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H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+G476K+K484A,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+G476K+K484G,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+G476K+K484P,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+G476K+K484E,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+G476K+K484Q,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+G476K+K484S,
 H1*+K+G50A+N54S+V56T+G109A+W167F+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+L173V+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+W140Y+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+Q98S+G109A+A174S+G182*+
 D183*+N195F+V206L+R320A+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+H324L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+G346T+K391A+G476K+G477A,
 H1*+N54S+V56T+G109A+G255A++A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+G50A+N54S+V56T+G109A+F113L+R116L+
 A174S+G182*+D183*+N195F+V206L+K391A+
 G476K,
 H1*+N54S+V56T+G109A+W167F+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+L173V+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+K72R+G109A+W167F+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+G476K+G477Q,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+Q345D+K391A+G476K+G477Q,
 H1*+G50A+N54S+V56T+G109A+Q172D+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+V264I+K391A+G423H+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+R320V+K391A+G476Y,
 H1*+N54S+V56T+G109A+R116M+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+L217V+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+L217H+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+P211D+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+K391Q+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+T444H+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+T444V+G476K,
 H1*+G50A+N54S+V56T+G109A+L173V+A174S+
 G182*+D183*+N195F+V206L+K391A+
 G476K,
 H1*+G50A+N54S+V56T+G109A+F113L+W167F+
 A174S+G182*+D183*+N195F+V206L+K391A+
 G476K,
 H1*+G7K+N54S+V56T+G109A+A174S+G182*+
 D183*+N195F+V206L+R320A+K391A+G476K,

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H1*+N54S+V56T+G109A+Q118G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172D+A174S+G182*+D183*+N195F+V206L+G346T+K391A+G476K+G477A,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+R320A+K391A+G476K,
H1*+N54S+V56T+G109A+T134E+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+Y382M+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+G255A+D377H+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+V206L+G255A+K391A+G476K,
H1*+N54S+V56T+K72R+M105F+G109A+A174S+G182*+D183*+N195F+V206L+G255A+K391A+G476K,
H1*+N54S+V56T+M105F+G109A+T134E+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+R320A+S323N+D377H+K391A+G476K,
H1*+T51E+N54S+V56T+G109A+A174H+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+L173A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+Q299V+K391A+G476K,
H1*+N54S+V56T+G109A+L173G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+T355L+K391A+G476K,
H1*+G50A+N54S+V56T+G109A+A174S+G182*+D183*+N195F+A204T+V206L+K391A+G476K,
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H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+R320A+S323N+K391A+G476K,
H1*+T51G+N54S+V56T+G109A+A174N+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q118R+T134E+A174S+G182*+D183*+N195F+V206L+D377H+K391A+G476K,
H1*+Q11H+N54S+V56T+G109A+Q118R+T134E+A174S+G184T+G182*+D183*+N195F+V206L+D377H+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+V206L+D377H+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q118R+T134E+A174S+G182*+D183*+N195F+V206L+G255A+K391A+G476K,
H1*+N54S+V56T+G109A+Q118R+A174S+G182*+D183*+N195F+V206L+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+T134E+A174S+G182*+D183*+G184T+N195F+V206L+D377H+K391A+G476K,

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H1*+N54S+V56T+G109A+Q172D+A174Q+G182*+D183*+N195F+V206L+K391A+G476K+G477A,
H1*+N54S+V56T+K72R+G109A+W167L+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
5 H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+G184T+N195F+V206L+G255A+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+G184T+N195F+V206L+D377H+K391A+G476K,
10 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G255A+D377H+K391A+G476K,
H1*+N54S+V56T+M105F+G109A+A174S+G182*+D183*+G184T+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+T134E+A174S+G182*+D183*+N195F+V206L+R320A+S323N+K391A+G476K,
15 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+V291G+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+V291A+F328L+D377H+K391A+G476K,
20 H1*+N54S+V56T+G109A+T134E+A174S+G182*+D183*+G184T+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+V238A+D377H+K391A+G476K,
25 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+V238T+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+V238G+D377H+K391A+G476K,
30 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+F328M+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+F328I+D377H+K391A+G476K,
35 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+V291T+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+V291A+D377H+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+G184T+N195F+V206L+G255A+K391A+G476K,
40 H1*+N54S+V56T+G109A+L173H+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q125T+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
45 H1*+N16S+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72N+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q118R+T134E+A174S+G182*+D183*+N195F+V206L+G255A+D377H+K391A+G476K,
50 H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+G184T+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+Q118R+A174S+G182*+D183*+N195F+V206L+G255A+D377H+K391A+G476K,
55 H1*+N54S+V56T+G109A+A174S+G182*+D183*+G184T+N195F+V206L+R320A+S323N+K391A+G476K,
H1*+N54S+G109A+A174S+G182*+D183*+N195F+V206L+D377H+K391A+G476K,
H1*+N54S+V56T+A174S+G182*+D183*+N195F+V206L+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+G182*+D183*+N195F+V206L+D377H+K391A+G476K,
65 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+D377H+K391A+G476K,

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H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+D377H+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+D377H+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+D377H+K391A,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+L217T+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+D377S+K391A+G476K,
 H1*+N54S+V56T+G109A+R116E+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+D377A+K391A+G476K,
 H1*+N54S+V56T+G109A+T134E+A174S+G182*+
 D183*+N195F+V206L+D377H+K391A+G476K,
 H1*+N54S+V56T+G109A+Q118R+T134E+A174S+
 G182*+D183*+G184T+N195F+V206L+K391A+
 G476K,
 H1*+N54S+V56T+K72R+G109A+Q118R+T134E+
 A174S+G182*+D183*+N195F+V206L+K391A+
 G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+G346P+K391A+G476K,
 H1*+N54S+V56T+G109A+R116S+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+D377Q+K391A+G476K,
 H1*+N54S+V56T+G109A+R116I+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
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 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+K72R+G109A+Q118R+T134E+
 A174S+G182*+D183*+N195F+V206L+G255A+
 K391A+G476K,
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 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+T51S+N54S+V56T+G109A+Q172M+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+T51A+N54S+V56T+G109A+Q172S+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+T51E+N54S+V56T+G109A+Q172T+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+T51A+N54S+V56T+G109A+Q172A+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+T51S+N54S+V56T+G109A+Q172N+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+T51A+N54S+V56T+G109A+Q172G+A174S+
 G182*+D183*+N195F+V206L+K391A+
 H1*+T51E+N54S+V56T+G109A+Q172L+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+Q172K+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+K72R+G109A+Q172S+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+K72H+G109A+Q172T+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+K72R+G109A+Q172M+A174S+
 G182*+D183*+N195F+V206L+Y295N+K391A+
 G476K,
 H1*+N54S+V56T+K72Q+G109A+Q172R+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+K72R+G109A+Q172T+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+K72M+G109A+Q172S+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,

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H1*+N54S+V56T+K72Q+G109A+Q172M+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+Q172M+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 5 H1*+N54S+V56T+K72S+G109A+A174H+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+K72Q+G109A+A174D+G182*+
 D183*+N195F+V206L+K391A+G476K,
 10 H1*+N54S+V56T+K72R+G109A+A174D+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+K72Q+G109A+A174K+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174G+G182*+D183*+
 N195F+V206L+K391A+G476K,
 15 H1*+N54S+V56T+K72R+G109A+A174Q+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+K72Q+G109A+A174G+G182*+
 D183*+N195F+V206L+K391A+G476K,
 20 H1*+N54S+V56T+K72S+G109A+A174N+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+K72R+G109A+A174H+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+T444A+G476K,
 25 H1*+N54S+V56T+G109A+F113L+W167F+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+M105F+G109A+A174S+G182*+
 D183*+N195F+V206L+D377H+K391A+G476K,
 30 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+A204G+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+S376H+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+I214E+K391A+G476K,
 35 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+I214W+K391A+G476K,
 H1*+N54S+V56T+K72R+G109A+Q172K+A174S+
 G182*+D183*+N195F+V206L+K391A+
 40 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+A204T+V206L+D377H+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+A204S+V206L+D377H+K391A+G476K,
 H1*+N54S+V56T+G109A+T165S+A174S+G182*+
 D183*+N195F+V206L+D377H+K391A+G476K,
 45 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+T334S+D377H+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+A263G+D377H+K391A+G476K,
 50 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+M286F+D377H+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+L235V+D377H+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+L235A+D377H+K391A+G476K,
 55 H1*+N54S+V56T+M105L+G109A+A174S+G182*+
 D183*+N195F+V206L+D377H+K391A+G476K,
 H1*+N54S+V56T+M105I+G109A+A174S+G182*+
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 D183*+N195F+V206L+D377H+K391A+G476K,
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 N195F+V206L+M246I+D377H+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+L250I+D377H+K391A+G476K,
 65 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+L250V+D377H+K391A+G476K,

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H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391Y+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391V+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391M+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391E+G476K,
 H1*+N54S+V56T+G109A+T165G+A174S+G182*+
 D183*+N195F+V206L+D377H+K391A+G476K,
 H1*+N54S+V56T+G109S+A174S+G182*+D183*+
 N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174V+G182*+D183*+
 N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+M246F+D377H+K391A+G476K,
 H1*+N54S+V56T+G109A+A174L+G182*+D183*+
 N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206S+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391 D+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
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 N195F+V206L+K391H+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391W+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+K391 I+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
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 N195F+V206L+L250A+D377H+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+L250F+D377H+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
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 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 G184T+N195F+V206L+G255A+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 G184T+N195F+V206L+G255A+D377H+K391A+
 G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+S376T+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+S376V+K391A+G476K,
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 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+Q172D+L173A+A174T+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+Y267M+D377H+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+Y267H+D377H+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+V238A+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+V238G+K391A+G476K, H1*+
 N54S+V56T+M105V+G109A+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+V264T+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+V264I+K391A+G476K,

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H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+L235V+K391A+G476K,
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 A174S+G182*+D183*+N195F+V206L+K391A+
 G476K, 5
 H1*+N54S+V56T+G109A+W167F+Q172E+L173P+
 A174K+G182*+D183*+N195F+V206L+K391A+
 G476K,
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 G182*+D183*+N195F+V206L+K391A+G476K, 10
 H1*+N54S+V56T+G109A+Q172E+L173F+A174R+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+Q172E+L173T+A174R+
 G182*+D183*+N195F+V206L+K391A+G476K,
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 G182*+D183*+N195F+V206L+K391A+G476K, 15
 H1*+N54S+V56T+G109A+Q172E+L173M+A174H+
 G182*+D183*+N195F+V206L+K391A+G476K,
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 G182*+D183*+N195F+V206L+K391A+G476K, 20
 H1*+N54S+V56T+G109A+W167Y+Q172E+L173V+
 A174H+G182*+D183*+N195F+V206L+K391A+
 G476K,
 H1*+N54S+V56T+G109A+L173A+A174T+G182*+
 D183*+N195F+V206L+K391A+G476K, 25
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+V291A+F328L+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+M246L+K391A+G476K, 30
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 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+L250V+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+L250F+K391A+G476K, 35
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+L250M+K391A+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+L250M+K391A+G476K,
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 N195F+V206L+K391A+C474V+G476K, 40
 H1*+N54S+V56T+G109A+T165G+A174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
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 N195F+V206L+T334S+K391A+G476K,
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 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
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 H1*+N54S+V56T+G109A+A174S+G182*+D183*+
 N195F+V206L+Y267L+K391A+G476K,
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 N195F+V206L+K391 L+G476K, 50
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 A174S+G182*+D183*+N195F+V206L+K391A+
 G476K,
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 G182*+D183*+N195F+V206L+K391A+
 H1*+N54S+V56T+G109S+F113Q+R116Q+A174S+
 G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+Q172D+A174S+G182*+
 D183*+N195F+A204G+V206L+K391A+P473R+
 G476K, 60
 H1*+N54S+V56T+K72R+G109A+T134E+A174S+
 G182*+D183*+N195F+A204G+V206L+G255A+
 K391A+P473R+G476K,
 H1*+N54S+V56T+K72R+G109A+W167F+A174S+
 G182*+D183*+N195F+A204G+V206L+K391A+
 G476K, 65

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H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+G184T+N195F+A204G+V206L+K391A+P473R+G476K,
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H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+V206L+G255A+K391A+C474V+G476K,

5 H1*+N54S+V56T+G109A+F113Q+R116H+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+P473R+G476K,

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10 H1*+N54S+V56T+G109A+R116H+W167F+Q172N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,

H1*+N54S+V56T+G109A+F113Q+R116H+T165G+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+G476K,

15 H1*+N54S+V56T+G109A+F113Q+R116Q+T165G+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+G476K,

H1*+N54S+V56T+G109A+F113Q+R116H+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+F289I+K391A+G476K,

20 H1*+N54S+V56T+G109A+F113Q+Q172N+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+G476K,

25 H1*+N54S+V56T+G109A+F113Q+R116H+Q172N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,

H1*+N54S+V56T+G109A+F113Q+R116Q+Q172M+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+G476K,

30 H1*+N54S+V56T+G109A+F113Q+R116H+Q172M+A174S+G182*+D183*+N195F+V206L+K391A+G476K,

H1*+N54S+V56T+G109A+R116H+Q172M+A174S+G182*+D183*+N195F+V206L+K391A+G476K,

35 H1*+N54S+V56T+G109A+F113Q+R116H+T165G+Q172M+A174S+G182*+D183*+N195F+V206L+K391A+Q395P+G476K,

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40 H1*+N54S+V56T+G109A+R116Q+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+P473R+G476K,

45 H1*+N54S+V56T+K72R+G109A+T134E+T165G+A174S+G182*+D183*+N195F+V206L+G255A+K391A+G476K,

H1*+N54S+V56T+G109A+F113Q+R116H+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+K391A+G476K,

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H1*+N54S+V56T+G109A+F113Q+A174S+G182*+D183*+N195F+V206L+K391A+G476K,

55 H1*+N54S+V56T+G109A+Q172N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,

H1*+T51V+N54S+V56T+G109A+Q172D+A174S+G182*+D183*+N195F+V206L+K391A+G476K,

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H1*+N54S+V56T+G109A+F113Q+R116H+Q172N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,

65 H1*+N54S+V56T+G109A+F113Q+R116H+Q172G+L173V+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+P473R+G476K,

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L173V+A174S+G182*+D183*+N195F+V206L+
K391A+T444Q+G476K,
H1*+N54S+V56T+G109A+G133Q+W167F+Q172N+
A174S+G182*+D183*+N195F+V206L+Q395P+
K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+W167F+
Q172N+L173V+A174S+G182*+D183*+N195F+
V206L+K391A+G476K,
H1*+A37V+N54S+V56T+K72R+G109A+R116H+
W167F+Q172R+A174S+G182*+D183*+G184T+
N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116H+
W167F+Q172G+L173V+A174S+G182*+D183*+
G184T+N195F+V206L+K391A+T444Q+P473R+
G476K,
H1*+N54S+V56T+G109S+R116Q+W167F+Q172R+
A174S+G182*+D183*+N195F+V206L+K391A+
G476K,
H1*+N54S+V56T+G109S+R116Q+W167F+Q172G+
A174S+G182*+D183*+N195F+V206L+K391A+
P473R+G476K,
H1*+N54S+V56T+G109S+F113Q+W167F+Q172N+
A174S+G182*+D183*+N195F+V206L+K391A+
G476K,
H1*+A37V+N54S+V56T+G109S+F113Q+W167F+
Q172R+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
H1*+N54S+V56T+K72R+G109A+R116H+T134E+
W167F+Q172G+L173V+A174S+G182*+D183*+
N195F+V206L+G255A+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+R116H+T134E+
W167F+Q172G+L173V+A174S+G182*+D183*+
N195F+V206L+G255A+K391A+Q395P+T444Q+
P473R+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+
W167F+Q172G+A174S+G182*+D183*+G184T+
N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+
T134E+W167F+Q172R+A174S+G182*+D183*+
N195F+V206L+G255A+K391A+T444Q+P473R+
G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+T134E+
W167F+Q172R+A174S+G182*+D183*+N195F+
V206L+G255A+K391A+Q395P+A445Q+P473R+
G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116H+
T134E+W167F+Q172R+A174S+G182*+D183*+
N195F+V206L+G255A+K391A+Q395P+G476K,
H1*+N54S+V56T+K72R+G109A+T134E+W167F+
Q172N+A174S+G182*+D183*+N195F+V206L+
G255A+A265G+K391A+T444Q+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+G133Q+W167F+
Q172R+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+
W167F+Q172G+A174S+G182*+D183*+N195F+
A204G+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+
W167F+Q172N+L173V+A174S+G182*+D183*+
G184T+N195F+V206L+K391A+T444Q+P473R+
G476K,
H1*+A37H+N54S+V56T+G109A+F113Q+W167F+
Q172N+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
H1*+A37H+N54S+V56T+G109A+F113Q+A174S+
G182*+D183*+N195F+V206L+K391A+G476K,

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H1*+N54S+V56T+G109A+F113Q+W167F+Q172N+
A174S+G182*+D183*+N195F+V206L+K391A+
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T444Q+G476K,
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G182*+D183*+N195F+V206L+K391A+P473R+
G476K,
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Q172N+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
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L173V+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
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T165G+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
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K391A+G476K,
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Q172G+L173V+A174S+G182*+D183*+N195F+
V206L+K391A+G476K,
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G182*+D183*+N195F+V206L+K391A+G476K,
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W167F+Q172R+A174S+G182*+D183*+N195F+
V206L+K391A+G476K,
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W167F+Q172N+A174S+G182*+D183*+N195F+
V206L+K391A+G476K,
H1*+N54S+V56T+A60V+G109A+R116Q+T165G+
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T444Q+P473R+G476K,
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G182*+D183*+N195F+V206L+K391A+T444Q+
P473R+G476K,
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V206L+K391A+G476K,
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Q172R+A174S+G182*+D183*+N195F+V206L+
K391A+G476K,
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K391A+P473R+G476K,
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D183*+N195F+A204G+V206L+K391A+P473R+
G476K,
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G182*+D183*+N195F+V206L+K391A+T444Q+
P473R+G476K,
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K391A+G476K,
H1*+N54S+V56T+G109A+G182*+D183*+N195F+
V206L,
N54S+V56T+G109A+G182*+D183*+N195F,
N54S+V56T+G109A+A174S+G182*+D183*+N195F+
V206L+K391A+G476K,
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K391A+G476K,
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V206L+K391A+G476K,
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N195F+V206L+K391A,

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G109A+G182*+D183*+N195F+V206L+G476K,
H1*+N54S+V56T+G109A+G133S+A174S+G182*+D183*+N195F+K391A+G476H,

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H1*+N54S+V56T+G109A+R116A+Q172N+L173V+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
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H1*+N16H+V17L+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+Y382F+K391A+G476K,
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H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+G184T+N195F+V206L+K391A+G476K,
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Q169E+Q172K+A174*+D183*+G182*+N195F,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K269Q+K391A+G476K,
H1*+W48F+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
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N54S+V56T+D183*+G182*+N195F,
H1*+N54S+V56T+G109A+A174T+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116W+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
Q169E+Q172K+G182*+D183*+N195F,
G182*+D183*+A186D,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+V238A+K391A+G476K,
G182*+D183*+A186N,
wherein numbering is according to SEQ ID NO: 13,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,

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H1*+N54S+V56T+G109A+Q172D+A174S+G182*+D183*+N195F+A204G+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+A204G+V206L+G255A+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116H+T165G+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+G196R+V206L+G255A+I405L+A421H+A422P+A428T+N475S,
H1*+N54S+V56T+G109A+T134E+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+N125D+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+R116V+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+R116Q+G133Q+T165G+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116Q+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T+G448D,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172N+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+T134E+W167F+Q172N+A174S+G182*+D183*+N195F+V206L+G255A+A265G+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+R116Q+W167F+Q172R+L173V+A174S+G182*+D183*+N195F+V206L+A265G+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+R116Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+F113Q+T134E+W167F+Q172G+A174S+G182*+D183*+N195F+V206L+G255A+A265G+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+W167F+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+G255A+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+K72R+G109A+R116Q+V120L+A174S+G182*+D183*+G184T+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116H+W167F+Q172G+L173V+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
H1*+N54S+V56T+G109A+F113Q+R116Q+Q172N+A174S+G182*+D183*+N195F+V206L+A265G+I405L+A421H+A422P+A428T,
H1*+A37H+N54S+V56T+A60V+G109A+R116Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T, wherein numbering is according to SEQ ID NO: 14.

In one embodiment, the variant has an IF of at least 2.0 when compared to the parent polypeptide, wherein the IF is

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determined by use of Model J detergent at a temperature of 15° C. and an enzymes concentration of 0.05 mg/L.

Other variants according to the invention have an IF of at least 3.0, indicating that at least these variants have a significantly improved wash performance when compared to the parent polypeptide having an amino acid sequence set forth in SEQ ID NO: 15.

In a particular embodiment, the variant is selected from a group consisting of:

10 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+N195F+V206L+K391A+G476K,
15 H1*+N54S+V56T+G109A+R116H+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+V206L+K391A+P473R+G476K,
20 H1*+N54S+V56T+G109A+R116Q+W167F+A174S+G182*+D183*+G184T+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
25 H1*+N54S+V56T+G109A+F113Q+R116Q+Q172N+A174S+G182*+D183*+N195F+V206L+A265G+K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+G133Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
30 H1*+N54S+V56T+K72R+G109A+R116H+T134E+W167F+Q172G+L173V+A174S+G182*+D183*+N195F+V206L+G255A+K391A+G476K, and
35 H1*+N54S+V56T+K72R+G109A+R116H+T134E+W167F+Q172G+L173V+A174S+G182*+D183*+N195F+V206L+G255A+K391A+Q395P+T444Q+P473R+G476K, wherein numbering is according to SEQ ID NO: 13.

40 In another embodiment, the variant has an IF of at least 3.0 when compared to the parent polypeptide and determined by use of Model J detergent, and wherein the variant is selected from the group consisting of:

45 H1*+T51K+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37V+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113Y+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
50 H1*+N54S+V56T+G109A+R116Q+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q125A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+P473R+G476K,
55 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+P473A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+T444D+G476K,
60 H1*+N54S+V56T+K72S+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+G50A+N54S+V56T+G109A+L173V+A174S+G182*+D183*+N195F+A204V+V206L+K391A+G476K,
65 H1*+N54S+V56T+G109A+Q172D+A174S+G182*+D183*+N195F+V206L+G346T+K391A+G476K+G477A,

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H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+V206L+G255A+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+V206L+D377H+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+G184T+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+T134E+A174S+G182*+D183*+G184T+N195F+V206L+D377H+K391A+G476K,
H1*+N54S+V56T+G109A+Q172D+A174Q+G182*+D183*+N195F+V206L+K391A+G476K+G477A,
H1*+N54S+V56T+G109A+T134E+A174S+G182*+D183*+G184T+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+G184T+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+W167F+Q172E+L173P+A174K+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+L250M+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109S+F113Q+R116Q+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q172D+A174S+G182*+D183*+N195F+A204G+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+A204G+V206L+G255A+K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+W167F+A174S+G182*+D183*+N195F+A204G+V206L+K391A+G476K,
H1*+N54S+V56T+M105F+G109A+A174S+G182*+D183*+N195F+V206L+R320A+S323N+K391A+C474V+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+G255A+K391A+C474V+G476K,
H1*+N54S+V56T+G109A+T134E+A174S+G182*+D183*+N195F+V206L+K391A+C474V+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+D377H+K391A+C474V+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+K391A+C474V+G476K,
H1*+N54S+V56T+G109A+R116Q+Q172D+A174S+G182*+D183*+N195F+V206L+G346T+K391A+T444Q+G477A+G476K,
H1*+N54S+V56T+G109S+R116Q+W167F+Q172N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109S+W167F+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+G476K,

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H1*+N54S+V56T+G109A+F113Q+A174S+G182*+D183*+N195F+V206L+K391A+C474V+G476K,
H1*+N54S+V56T+G109A+Q172N+A174S+G182*+D183*+N195F+V206L+V264I+K391A+C474V+G476K,
H1*+N54S+V56T+K72R+G109A+T134E+A174S+G182*+D183*+N195F+V206L+G255A+K391A+C474V+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+P473R+G476K,
H1*+N54S+V56T+G109A+F113Q+R116Q+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+T165G+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+G476K,
H1*+N54S+V56T+G109A+F113Q+R116Q+T165G+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+G476K,
H1*+N54S+V56T+G109A+F113Q+Q172N+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172M+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+T134E+T165G+A174S+G182*+D183*+N195F+V206L+G255A+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+R116H+Q172N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116Q+W167F+Q172N+L173V+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+G476K,
H1*+N54S+V56T+G109A+G133Q+W167F+Q172N+A174S+G182*+D183*+N195F+V206L+Q395P+K391A+G476K,
H1*+A37V+N54S+V56T+K72R+G109A+R116H+W167F+Q172R+A174S+G182*+D183*+G184T+N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116H+W167F+Q172G+L173V+A174S+G182*+D183*+G184T+N195F+V206L+K391A+T444Q+P473R+G476K,
H1*+N54S+V56T+G109S+R116Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109S+R116Q+W167F+Q172G+A174S+G182*+D183*+N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+G109S+F113Q+W167F+Q172N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37V+N54S+V56T+G109S+F113Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+R116H+T134E+W167F+Q172G+L173V+A174S+G182*+D183*+N195F+V206L+G255A+K391A+Q395P+T444Q+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+V206L+K391A+P473R+G476K,

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H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+T134E+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+G255A+K391A+T444Q+P473R+G476K,
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H1*+N54S+V56T+K72R+G109A+G133Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+N195F+A204G+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+F113Q+R116Q+A174S+G182*+D183*+N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+A60V+G109A+R116Q+W167F+Q172N+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+A60V+G109A+R116H+Q172R+L173V+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
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H1*+N54S+V56T+A60V+G109A+R116Q+T165G+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+P473R+G476K,
H1*+N54S+V56T+A60V+G109A+R116Q+W167F+Q172N+L173V+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+A60V+G109A+R116Q+W167F+Q172R+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116A+A174S+G182*+D183*+N195F+A204G+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+T134E+A174S+G182*+D183*+N195F+V206L+G255A+K391A+P473R+G476K,
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H1*+N54S+V56T+K72R+G109A+F113Q+R116H+A174S+G182*+D183*+G184T+N195F+V206L+K391A+T444Q+G476K,
H1*+N54S+V56T+G109A+F113Q+A174S+G182*+D183*+N195F+V206L+K391A+T444Q+G476K,
H1*+N54S+V56T+G109A+R116H+W167F+Q172G+L173V+A174S+G182*+D183*+N195F+V206L+K391A+P473R+G476K,

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H1*+N54S+V56T+G109A+F113Q+R116Q+Q172N+A174S+G182*+D183*+N195F+V206L+A265G+K391A+P473R+G476K,
H1*+A37V+N54S+V56T+G109A+R116Q+T165G+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+A37H+N54S+V56T+G109A+R116H+T165G+A174S+G182*+D183*+N195F+V206L+M246F+K391A+G476K,
H1*+N54S+V56T+G109A+R116A+Q172N+L173V+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+P45A+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+C474V+G476K,
H1*+N54S+V56T+G109A+Q118R+T134E+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
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H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+V264F+K391A+G476K,
H1*+G7K+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+G255A+Q256A+K391A+G476K,
H1*+N54S+V56T+G109A+W167Y+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N16H+V17L+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+Y382F+K391A+G476K,
H1*+Q32S+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+Q385L+K391A+G476K,
H1*+N54S+V56T+R87S+G109A+R171H+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+R116H+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+V206L+K391A+P473R+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+V206L+P473R+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172G+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+F113Q+R116Q+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+K391A+P473R+G476K,
H1*+N54S+V56T+G109A+R116W+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206F+K391A+G476K,
H1*+V56T+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+N260G+K391A+G476K,
H1*+N54S+V56T+K72R+G109A+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+G184T+N195F+V206L+K391A+G476K,

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H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+Y295N+K391A+G476K,
 H1*+N54S+V56T+G109A+W167F+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+A186N+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+A288V+K391A+G476K,
 H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+G255A+K391A+G476K,
 H1*+N54S+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+W48F+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+V291A+K391A+G476K,
 H1*+N54S+V56T+G109A+F113Q+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+R320A+S323N+K391A+G476K,
 H1*+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+K391A+G476K,
 H1*+N54S+V56T+K72R+G109A+F113Q+R116H+W167F+Q172G+A174S+G182*+D183*+G184T+N195F+V206L+K391A+P473R+G476K,
 H1*+N54S+V56T+G109A+A174S+G182*+D183*+N195F+V206L+N270T+K391A+G476K,
 H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+G476K,
 H1*+N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+D183*+N195F+V206L+N260G+K391A+G476K,
 H1*+N54S+V56T+G109A+F113Q+R116H+A174S+G182*+D183*+N195F+V206L+K391A+G476K,
 wherein numbering is according to SEQ ID NO: 13,
 H1*+N54S+V56T+G109A+F113Q+R116H+Q172G+A174S+G182*+D183*+N195F+V206L+I405L+A421H+A422P+A428T,
 H1*+N54S+V56T+G109A+F113Q+R116Q+Q172N+A174S+G182*+D183*+N195F+V206L+A265G+I405L+A421H+A422P+A428T, wherein numbering is according to SEQ ID NO: 14.

The variants according to the present invention may further comprise one or more additional alterations than those described above at one or more (e.g., several) other positions.

The amino acid changes may be of a minor nature, that is conservative amino acid substitutions or insertions that do not significantly affect the folding and/or activity of the protein; small deletions, typically of 1-30 amino acids; small amino- or carboxyl-terminal extensions, such as an amino-terminal methionine residue; a small linker peptide of up to 20-25 residues; or a small extension that facilitates purification by changing net charge or another function, such as a poly-histidine tract, an antigenic epitope or a binding domain.

Examples of conservative substitutions are within the groups of basic amino acids (arginine, lysine and histidine), acidic amino acids (glutamic acid and aspartic acid), polar amino acids (glutamine and asparagine), hydrophobic amino

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acids (leucine, isoleucine and valine), aromatic amino acids (phenylalanine, tryptophan and tyrosine), and small amino acids (glycine, alanine, serine, threonine and methionine). Amino acid substitutions that do not generally alter specific activity are known in the art and are described, for example, by H. Neurath and R. L. Hill, 1979, In, *The Proteins*, Academic Press, New York. Common substitutions are Ala/Ser, Val/Ile, Asp/Glu, Thr/Ser, Ala/Gly, Ala/Thr, Ser/Asn, Ala/Val, Ser/Gly, Tyr/Phe, Ala/Pro, Lys/Arg, Asp/Asn, Leu/Ile, Leu/Val, Ala/Glu, and Asp/Gly.

Alternatively, the amino acid changes are of such a nature that the physico-chemical properties of the polypeptides are altered. For example, amino acid changes may improve the thermal stability of the polypeptide, alter the substrate specificity, change the pH optimum, and the like.

Essential amino acids in a polypeptide may be identified according to procedures known in the art, such as site-directed mutagenesis or alanine-scanning mutagenesis (Cunningham and Wells, 1989, *Science* 244: 1081-1085). In the latter technique, single alanine mutations are introduced at every residue in the molecule, and the resultant mutant molecules are tested for alpha-amylase activity to identify amino acid residues that are critical to the activity of the molecule. See also, Hilton et al., 1996, *J. Biol. Chem.* 271: 4699-4708. The active site of the enzyme or other biological interaction can also be determined by physical analysis of structure, as determined by such techniques as nuclear magnetic resonance, crystallography, electron diffraction, or photoaffinity labeling, in conjunction with mutation of putative contact site amino acids. See, for example, de Vos et al., 1992, *Science* 255: 306-312; Smith et al., 1992, *J. Mol. Biol.* 224: 899-904; Wlodaver et al., 1992, *FEBS Lett.* 309: 59-64. The identity of essential amino acids can also be inferred from an alignment with a related polypeptide.

The variants according to the present invention may consist of 400 to 485 amino acids, e.g., 410 to 485, 425 to 485, and 440 to 485 amino acids.

Polynucleotides

The present invention also relates to polynucleotides encoding a variant of the present invention. Thus, in particular, the present invention relates to a polynucleotide encoding a variant comprising an alteration in one or more positions corresponding to the H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, A204, V206, P211, I214, V215, L217, L219, A225, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13; or H1, A37, T40, N54, V56, A60, K72, G109, F113, R116, N125, F133, T134, T165, Q169, Q172, L173, A174, G184, A186, E190, N195, G196, A204, V206, A225, L228, K242, S244, G255, N260, A265, K269, S280, W284, S304, R320, H321, S323, V326, K391, I405, A421, A422, A428, G448, D476, and G477 of SEQ ID NO: 14.

The term "polynucleotides encoding" as used herein, refers to a polynucleotide that encodes a mature polypeptide having alpha-amylase having alpha-amylase activity. In one aspect, the polypeptide coding sequence is the nucleotide sequence set forth in SEQ ID NO: 1, 2, or 3.

In one embodiment, the polynucleotide encoding a variant according to the present invention as at least 60%, such as

at least 65%, such as at least 70%, such as at least 75%, such as at least 80%, such as at least 85%, such as at least 90%, such as at least 95%, such as at least 96%, such as at least 97%, such as at least 98%, such as at least 99% but less than 100% sequence identity to the polynucleotide of SEQ ID NOs: 1, 2 and 3.

Nucleic Acid Constructs

The present invention also relates to nucleic acid constructs comprising a polynucleotide encoding a variant of the present invention operably linked to one or more control sequences that direct the expression of the coding sequence in a suitable host cell under conditions compatible with the control sequences. Thus, in particular, the present invention relates to a nucleic acid construct comprising a polynucleotide encoding a variant comprising an alteration in one or more positions corresponding to the positions H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, A204, V206, P211, I214, V215, L217, L219, A225, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13; or H1, A37, T40, N54, V56, A60, K72, G109, F113, R116, N125, F133, T134, T165, Q169, Q172, L173, A174, G184, A186, N195, G196, A204, V206, A225, L228, K242, S244, G255, N260, A265, K269, S280, W284, Y295, S304, R320, H321, S323, V326, K391, I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14, wherein the polynucleotide is operably linked to one or more control sequences.

The term “nucleic acid construct” as used herein, refers to a nucleic acid molecule, either single- or double-stranded, which is isolated from a naturally occurring gene or is modified to contain segments of nucleic acids in a manner that would not otherwise exist in nature or which is synthetic, which comprises one or more control sequences.

The term “operably linked” as used herein, refers to a configuration in which a control sequence is placed at an appropriate position relative to the coding sequence of a polynucleotide such that the control sequence directs expression of the coding sequence.

The polynucleotide may be manipulated in a variety of ways to provide for expression of a variant. Manipulation of the polynucleotide prior to its insertion into a vector may be desirable or necessary depending on the expression vector. The techniques for modifying polynucleotides utilizing recombinant DNA methods are well known in the art.

The control sequence may be a promoter, a polynucleotide which is recognized by a host cell for expression of the polynucleotide. The promoter comprises transcriptional control sequences that mediate the expression of the variant. The promoter may be any polynucleotide that shows transcriptional activity in the host cell including mutant, truncated, and hybrid promoters, and may be obtained from genes encoding extracellular or intracellular polypeptides either homologous or heterologous to the host cell.

Expression Vectors

The present invention also relates to recombinant expression vectors comprising a polynucleotide encoding a variant of the present invention, a promoter, and transcriptional and translational stop signals. Thus, the present invention relates to an expression vector, optionally recombinant, comprising

a polynucleotide encoding a variant comprising an alteration in one or more positions corresponding to the positions H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, A204, V206, P211, I214, V215, L217, L219, A225, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13; or H1, A37, T40, N54, V56, A60, K72, G109, F113, R116, N125, F133, T134, T165, Q169, Q172, L173, A174, G184, A186, N195, G196, A204, V206, A225, L228, K242, S244, G255, N260, A265, K269, S280, W284, Y295, S304, R320, H321, S323, V326, K391, I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14, a promoter, and transcriptional and translational stop signals.

The term “expression vector” as used herein, refers to a linear or circular DNA molecule that comprises a polynucleotide encoding a variant and is operably linked to control sequences that provide for its expression.

The various nucleotide and control sequences may be joined together to produce a recombinant expression vector that may include one or more convenient restriction sites to allow for insertion or substitution of the polynucleotide encoding the variant at such sites. Alternatively, the polynucleotide may be expressed by inserting the polynucleotide or a nucleic acid construct comprising the polynucleotide into an appropriate vector for expression. In creating the expression vector, the coding sequence is located in the vector so that the coding sequence is operably linked with the appropriate control sequences for expression.

The recombinant expression vector may be any vector (e.g., a plasmid or virus) that can be conveniently subjected to recombinant DNA procedures and can bring about expression of the polynucleotide. The choice of the vector will typically depend on the compatibility of the vector with the host cell into which the vector is to be introduced. The vector may be a linear or closed circular plasmid.

The vector may be an autonomously replicating vector, i.e., a vector that exists as an extrachromosomal entity, the replication of which is independent of chromosomal replication, e.g., a plasmid, an extrachromosomal element, a minichromosome, or an artificial chromosome. The vector may contain any means for assuring self-replication. Alternatively, the vector may be one that, when introduced into the host cell, is integrated into the genome and replicated together with the chromosome(s) into which it has been integrated. Furthermore, a single vector or plasmid or two or more vectors or plasmids that together contain the total DNA to be introduced into the genome of the host cell, or a transposon, may be used.

The vector preferably contains one or more selectable markers that permit easy selection of transformed, transfected, transduced, or the like cells. A selectable marker is a gene the product of which provides for biocide or viral resistance, resistance to heavy metals, prototrophy to auxotrophs, and the like.

The skilled person would know which expression vector is the most suitable for specific expression systems. Thus, the present invention is not limited to any specific expression vector, but any expression vector comprising the poly-

nucleotide encoding a variant according to the invention is considered part of the present invention.

Host Cells

The present invention also relates to recombinant host cells, comprising a polynucleotide encoding a variant of the present invention operably linked to one or more control sequences that direct the production of a variant of the present invention. Thus, the present invention relates to a host cell, optionally a recombinant host cell, comprising a polynucleotide encoding a variant comprising an alteration in one or more positions corresponding to the positions H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, A204, V206, P211, I214, V215, L217, L219, A225, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, G476, G477, and K484 of SEQ ID NO: 13; or H1, A37, T40, N54, V56, A60, K72, G109, F113, R116, N125, F133, T134, T165, Q169, Q172, L173, A174, G184, A186, E190, N195, G196, A204, V206, A225, L228, K242, S244, G255, N260, A265, K269, S280, W284, Y295, S304, R320, H321, S323, V326, K391, I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14, wherein the method comprises the steps of a) cultivating a host cell according to the invention under conditions suitable for expression of the variant, and b) recovering the variant.

The term "host cell" as used herein, refers to any cell type that is susceptible to transformation, transfection, transduction, or the like with a nucleic acid construct or expression vector comprising a polynucleotide of the present invention. The term "host cell" encompasses any progeny of a parent cell that is not identical to the parent cell due to mutations that occur during replication.

A construct or vector comprising a polynucleotide is introduced into a host cell so that the construct or vector is maintained as a chromosomal integrant or as a self-replicating extra-chromosomal vector as described earlier. The choice of a host cell will to a large extent depend upon the gene encoding the variant and its source.

The host cell may be any cell useful in the recombinant production of a variant, e.g., a prokaryote or a eukaryote.

The prokaryotic host cell may be any Gram-positive or Gram-negative bacterium. Gram-positive bacteria include, but are not limited to, *Bacillus*, *Clostridium*, *Enterococcus*, *Geobacillus*, *Lactobacillus*, *Lactococcus*, *Oceanobacillus*, *Staphylococcus*, *Streptococcus*, and *Streptomyces*. Gram-negative bacteria include, but are not limited to, *Campylobacter*, *E. coli*, *Flavobacterium*, *Fusobacterium*, *Helicobacter*, *Ilyobacter*, *Neisseria*, *Pseudomonas*, *Salmonella*, and *Ureaplasma*.

Methods According to the Invention

The present invention also relates to methods of producing a variant, comprising: (a) cultivating a host cell of the present invention under conditions suitable for expression of the variant; and (b) recovering the variant. Thus, the present invention relates to a method of producing a variant comprising an alteration in one or more positions corresponding to the positions H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, A204, V206, P211, I214, V215,

L217, L219, A225, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13; or H1, A37, T40, N54, V56, A60, K72, G109, F113, R116, N125, F133, T134, T165, Q169, Q172, L173, A174, G184, A186, E190, N195, G196, A204, V206, A225, L228, K242, S244, G255, N260, A265, K269, S280, W284, Y295, S304, R320, H321, S323, V326, K391, I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14, wherein the method comprises the steps of a) cultivating a host cell according to the invention under conditions suitable for expression of the variant, and b) recovering the variant.

The host cells are cultivated in a nutrient medium suitable for production of the variant using methods known in the art. For example, the cell may be cultivated by shake flask cultivation, or small-scale or large-scale fermentation (including continuous, batch, fed-batch, or solid state fermentations) in laboratory or industrial fermentors performed in a suitable medium and under conditions allowing the variant to be expressed and/or isolated. The cultivation takes place in a suitable nutrient medium comprising carbon and nitrogen sources and inorganic salts, using procedures known in the art. Suitable media are available from commercial suppliers or may be prepared according to published compositions (e.g., in catalogues of the American Type Culture Collection). If the variant is secreted into the nutrient medium, the variant can be recovered directly from the medium. If the variant is not secreted, it can be recovered from cell lysates.

The variant may be detected using methods known in the art that are specific for the polypeptides having alpha amylase activity. These detection methods include, but are not limited to, use of specific antibodies, formation of an enzyme product, or disappearance of an enzyme substrate. For example, an enzyme assay may be used to determine the activity of the variant.

The variant may be recovered using methods known in the art. For example, the variant may be recovered from the nutrient medium by conventional procedures including, but not limited to, collection, centrifugation, filtration, extraction, spray-drying, evaporation, or precipitation.

The variant may be purified by a variety of procedures known in the art including, but not limited to, chromatography (e.g., ion exchange, affinity, hydrophobic, chromatofocusing, and size exclusion), electrophoretic procedures (e.g., preparative isoelectric focusing), differential solubility (e.g., ammonium sulfate precipitation), SDS-PAGE, or extraction (see, e.g., *Protein Purification*, Janson and Ryden, editors, VCH Publishers, New York, 1989) to obtain substantially pure variants.

In an alternative aspect, the variant is not recovered, but rather a host cell of the present invention expressing the variant is used as a source of the variant.

Furthermore, the present invention relates to methods for obtaining a variant, comprising introducing into a parent alpha-amylase having at least 80% sequence identity to the polypeptide of SEQ ID NO: 4

- (a) a substitution and/or deletion of two or more positions in the parent alpha-amylase said positions corresponding to positions 181, 182, 183, and 184 of the mature polypeptide of SEQ ID NO: 4; and
- (b) an alteration at one or more positions corresponding to positions H1, T5, G7, Q11, N16, V17, Q32, A37, T40,

P45, W48, G50, T51, N54, V56, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, A204, V206, P211, I214, V215, L217, L219, A225, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13; or H1, A37, T40, N54, V56, A60, K72, G109, F113, R116, N125, F133, T134, T165, Q169, Q172, L173, A174, G184, A186, E190, N195, G196, A204, V206, A225, L228, K242, S244, G255, N260, A265, K269, S280, W284, Y295, S304, R320, H321, S323, V326, K391, I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14,

wherein the resulting variant has at least 80%, such as at least 85%, such as at least 90%, such as at least 95%, such as at least 97%, but less than 100% sequence identity with the mature polypeptide of SEQ ID NO: 4, wherein said variant has alpha-amylase activity;

(c) recovering said variant.

The variants may be prepared using any mutagenesis procedure known in the art, such as site-directed mutagenesis, synthetic gene construction, semi-synthetic gene construction, random mutagenesis, shuffling, etc.

Site-directed mutagenesis is a technique in which one or more (e.g., several) mutations are introduced at one or more defined sites in a polynucleotide encoding the parent.

Site-directed mutagenesis can be accomplished in vitro by PCR involving the use of oligonucleotide primers containing the desired mutation. Site-directed mutagenesis can also be performed in vitro by cassette mutagenesis involving the cleavage by a restriction enzyme at a site in the plasmid comprising a polynucleotide encoding the parent polypeptide and subsequent ligation of an oligonucleotide containing the mutation in the polynucleotide. Usually the restriction enzyme that digests the plasmid and the oligonucleotide is the same, permitting sticky ends of the plasmid and the insert to ligate to one another. See, e.g., Scherer and Davis, 1979, *Proc. Natl. Acad. Sci. USA* 76: 4949-4955; and Barton et al., 1990, *Nucleic Acids Res.* 18: 7349-4966.

Site-directed mutagenesis can also be accomplished in vivo by methods known in the art. See, e.g., U.S. Patent Application Publication No. 2004/0171154; Storici et al., 2001, *Nature Biotechnol.* 19: 773-776; Kren et al., 1998, *Nat. Med.* 4: 285-290; and Calissano and Macino, 1996, *Fungal Genet. Newslett.* 43: 15-16.

Any site-directed mutagenesis procedure may be used in the present invention. There are many commercial kits available that can be used to prepare variants. The skilled person in the art is well-aware of such commercial kits and how to use them.

Synthetic gene construction entails in vitro synthesis of a designed polynucleotide molecule to encode a polypeptide of interest. Gene synthesis may be performed utilizing a number of techniques, such as the multiplex microchip-based technology described by Tian et al. (2004, *Nature* 432: 1050-1054) and similar technologies wherein oligonucleotides are synthesized and assembled upon photo-programmable microfluidic chips.

Single or multiple amino acid substitutions, deletions, and/or insertions may be made and tested using known

methods of mutagenesis, recombination, and/or shuffling, followed by a relevant screening procedure, such as those disclosed by Reidhaar-Olson and Sauer, 1988, *Science* 241: 53-57; Bowie and Sauer, 1989, *Proc. Natl. Acad. Sci. USA* 86: 2152-2156; WO 95/17413; or WO 95/22625. Other methods that may be used include error-prone PCR, phage display (e.g., Lowman et al., 1991, *Biochemistry* 30: 10832-10837; U.S. Pat. No. 5,223,409; WO 92/06204) and region-directed mutagenesis (Derbyshire et al., 1986, *Gene* 46: 145; Ner et al., 1988, *DNA* 7: 127).

Mutagenesis/shuffling methods may be combined with high-throughput, automated screening methods to detect activity of cloned, mutagenized polypeptides expressed by host cells (Ness et al., 1999, *Nature Biotechnology* 17: 893-896). Mutagenized DNA molecules that encode active polypeptides may be recovered from the host cells and rapidly sequenced using standard methods in the art. These methods allow the rapid determination of the importance of individual amino acid residues in a polypeptide.

Semi-synthetic gene construction is accomplished by combining aspects of synthetic gene construction, and/or site-directed mutagenesis, and/or random mutagenesis, and/or shuffling. Semi-synthetic construction is typified by a process utilizing polynucleotide fragments that are synthesized, in combination with PCR techniques. Defined regions of genes may thus be synthesized de novo, while other regions may be amplified using site-specific mutagenic primers, while yet other regions may be subjected to error-prone PCR or non-error prone PCR amplification. Polynucleotide subsequences may then be shuffled.

The present invention also relates to a method of improving wash performance of a parent polypeptide having the amino acid sequence of SEQ ID NO: 13 or 14, or having at least 80% sequence identity thereto, said method comprising the steps of:

- a) a substitution and/or deletion of two, three or four positions in the parent alpha-amylase said positions corresponding to positions 181, G182, D183, and G184 of the mature polypeptide of SEQ ID NO: 4; and
- b) an alteration at one or more positions corresponding to positions H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, A204, V206, P211, I214, V215, L217, L219, A225, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13; or H1, A37, T40, N54, V56, A60, K72, G109, F113, R116, N125, F133, T134, T165, Q169, Q172, L173, A174, G184, A186, E190, N195, G196, A204, V206, A225, L228, K242, S244, G255, N260, A265, K269, S280, W284, Y295, S304, R320, H321, S323, V326, K391, I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14,

wherein the variant has at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, at least 98%, at least 99%, but less than 100% sequence identity with the polypeptide of SEQ ID NO: 4 or 7, and wherein the variant has alpha-amylase activity and improved wash performance compared to the parent polypeptide.

In one embodiment, the variant has at least 50%, such as at least 60%, or at least 70%, or at least 80%, or at least 90%, or at least 100% of the activity of the parent polypeptide having the amino acid sequence of SEQ ID NO: 7.

In one embodiment, the activity is determined according to a Phadebas assay.

The alpha-amylase activity may be determined by a method using the Phadebas substrate (from for example Magle Life Sciences, Lund, Sweden). A Phadebas tablet includes interlinked starch polymers that are in the form of globular microspheres that are insoluble in water. A blue dye is covalently bound to these microspheres. The interlinked starch polymers in the microsphere are degraded at a speed that is proportional to the alpha-amylase activity. When the alpha-amylase degrades the starch polymers, the released blue dye is water soluble and concentration of dye can be determined by measuring absorbance at 620 nm. The concentration of blue is proportional to the alpha-amylase activity in the sample.

The variant sample to be analyzed is diluted in activity buffer with the desired pH. Two substrate tablets are suspended in 5 mL activity buffer and mixed on magnetic stirrer. During mixing of substrate transfer 150 µl to micro-titer plate (MTP) or PCR-MTP. Add 30 µl diluted amylase sample to 150 µl substrate and mix. Incubate for 15 minutes at 37° C. The reaction is stopped by adding 30 µl 1M NaOH and mix. Centrifuge MTP for 5 minutes at 4000×g. Transfer 100 µl to new MTP and measure absorbance at 620 nm.

The alpha-amylase sample should be diluted so that the absorbance at 620 nm is between 0 and 2.2, and is within the linear range of the activity assay.

Thus, in one embodiment, the activity is determined by a method comprising the steps of;

- a) incubating an alpha-amylase variant according to the invention with a dyed amylose substrate for 15 minute at 37° C.; and
- b) measuring the absorption at OD 620 nm.

In a further embodiment, the activity is determined by a method comprising the steps of;

- a) incubating an alpha-amylase variant according to the invention with a dyed amylose substrate for 15 minute at 37° C.; and
- b) centrifuging the sample;
- c) transferring the supernatant to reader plate, and measuring the absorption at OD 620 nm.

Fermentation Broth Formulations or Cell Compositions

The present invention also relates to a fermentation broth formulation or a cell composition comprising a polypeptide of the present invention. Thus, in one embodiment, the fermentation broth formulation or the cell composition comprises a polynucleotide encoding a variant comprising an alteration in one or more positions corresponding to the positions H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, A204, V206, P211, I214, V215, L217, L219, A225, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13; or H1, A37, T40, N54, V56, A60, K72, G109, F113, R116, N125, F133, T134, T165, Q169, Q172, L173, A174, G184, A186, E190, N195, G196, A204, V206, A225, L228, K242, S244, G255, N260, A265, K269, S280, W284, Y295, S304, R320, H321, S323, V326, K391, I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14.

G255, N260, A265, K269, S280, W284, Y295, S304, R320, H321, S323, V326, K391, I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14, a nucleic acid construct encoding a variant comprising an alteration in one or more positions corresponding to the positions H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, A204, V206, P211, I214, V215, L217, L219, A225, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13; or H1, A37, T40, N54, V56, A60, K72, G109, F113, R116, N125, F133, T134, T165, Q169, Q172, L173, A174, G184, A186, E190, N195, G196, A204, V206, A225, L228, K242, S244, G255, N260, A265, K269, S280, W284, Y295, S304, R320, H321, S323, V326, K391, I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14, or an expression vector encoding a variant comprising an alteration in one or more positions corresponding to the positions H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q269, R171, Q172, L173, A174, G184, A186, E190, T193, N195, A204, V206, P211, I214, V215, L217, L219, A225, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13; or H1, A37, T40, N54, V56, A60, K72, G109, F113, R116, N125, F133, T134, T165, Q169, Q172, L173, A174, G184, A186, E190, N195, G196, A204, V206, A225, L228, K242, S244, G255, N260, A265, K269, S280, W284, Y295, S304, R320, H321, S323, V326, K391, I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14. The fermentation broth product may further comprise additional ingredients used in the fermentation process, such as, for example, cells (including, the host cells containing the gene encoding the polypeptide of the present invention which are used to produce the polypeptide of interest), cell debris, biomass, fermentation media and/or fermentation products. In some embodiments, the composition is a cell-killed whole broth containing organic acid(s), killed cells and/or cell debris, and culture medium.

The term “fermentation broth” as used herein refers to a preparation produced by cellular fermentation that undergoes no or minimal recovery and/or purification. For example, fermentation broths are produced when microbial cultures are grown to saturation, incubated under carbon-limiting conditions to allow protein synthesis (e.g., expression of enzymes by host cells) and secretion into cell culture medium. The fermentation broth can contain unfractionated or fractionated contents of the fermentation materials derived at the end of the fermentation. Typically, the fermentation broth is unfractionated and comprises the spent culture medium and cell debris present after the microbial cells (e.g., filamentous fungal cells) are removed, e.g., by centrifugation. In some embodiments, the fermentation

broth contains spent cell culture medium, extracellular enzymes, and viable and/or nonviable microbial cells.

In one embodiment, the fermentation broth formulation and cell compositions comprise a first organic acid component comprising at least one 1-5 carbon organic acid and/or a salt thereof and a second organic acid component comprising at least one 6 or more carbon organic acid and/or a salt thereof. In a particular embodiment, the first organic acid component is acetic acid, formic acid, propionic acid, a salt thereof, or a mixture of two or more of the foregoing and the second organic acid component is benzoic acid, cyclohexanecarboxylic acid, 4-methylvaleric acid, phenylacetic acid, a salt thereof, or a mixture of two or more of the foregoing.

In one embodiment, the composition contains an organic acid(s), and optionally further contains killed cells and/or cell debris. In one embodiment, the killed cells and/or cell debris are removed from a cell-killed whole broth to provide a composition that is free of these components.

The fermentation broth formulations or cell compositions may further comprise a preservative and/or anti-microbial (e.g., bacteriostatic) agent, including, but not limited to, sorbitol, sodium chloride, potassium sorbate, and others known in the art.

The cell-killed whole broth or composition may comprise the unfractionated contents of the fermentation materials derived at the end of the fermentation. Typically, the cell-killed whole broth or composition comprises the spent culture medium and cell debris present after the microbial cells (e.g., filamentous fungal cells) are grown to saturation, incubated under carbon-limiting conditions to allow protein synthesis. In some embodiments, the cell-killed whole broth or composition comprises the spent cell culture medium, extracellular enzymes, and killed filamentous fungal cells. In some embodiments, the microbial cells present in the cell-killed whole broth or composition may be permeabilized and/or lysed using methods known in the art.

A whole broth or cell composition as described herein is typically a liquid, but may comprise insoluble components, such as killed cells, cell debris, culture media components, and/or insoluble enzyme(s). In some embodiments, insoluble components may be removed to provide a clarified liquid composition.

The whole broth formulations and cell compositions of the present invention may be produced by a method described in WO 90/15861 or WO 2010/096673.

Enzyme Compositions

The present invention also relates to compositions comprising a variant according to the invention. Thus, the invention relates to a composition comprising a variant comprising an alteration in one or more positions corresponding to the positions H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, A204, V206, P211, I214, V215, L217, L219, A225, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, G476, G477, and K484 of SEQ ID NO: 13; or H1, A37, T40, N54, V56, A60, K72, G109, F113, R116, N125, F133, T134, T165, Q169, Q172, L173, A174, G184, A186, E190, N195, G196, A204, V206, A225, L228, K242, S244, G255, N260, A265, K269, S280, W284, Y295,

S304, R320, H321, S323, V326, K391, I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14.

In one embodiment, the composition comprises a variant comprising

- a) a deletion and/or a substitution at two or more positions corresponding to positions R181, G182, D183, and G184 of the amino acid sequence as set forth in SEQ ID NOs: 7 or 10, and
- b) an alteration at one or more positions corresponding to positions H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, T173, A174, A186, E190, T193, N195, A204, V206, P211, V213, I214, L217, A225, L235, V238, K242, S244, M246, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, G337, Q345, G346, T355, S376, D377, S381, Y382, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13; or H1, A37, T40, N54, V56, A60, K72, G109, F113, R116, N125, F133, T134, T165, Q169, Q172, L173, A174, G184, A186, E190, N195, G196, A204, V206, A225, L228, K242, S244, G255, N260, A265, K269, S280, W284, Y295, S304, R320, H321, S323, V326, K391, I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14. Preferably, the compositions are enriched in such a variant. The term "enriched" as used herein, refers to that the alpha-amylase activity of the composition has been increased, e.g., with an enrichment factor of at least 1.1.

The compositions may comprise a variant of the present invention as the major enzymatic component, e.g., a mono-component composition. In another embodiment, the composition further comprises at least one further active component.

The term "active component" as used herein, refers to any biological or non-biological molecule which in itself is active. For example, an active component is an enzyme.

Thus, in one embodiment, the further active component is an enzyme, such as a protease, lipase, cellulose, pectate lyase and mannanase. Thus, the compositions may comprise multiple enzymatic activities, such as one or more (e.g., several) enzymes selected from the group consisting of hydrolase, isomerase, ligase, lyase, oxidoreductase, or transferase, e.g., an alpha-galactosidase, alpha-glucosidase, aminopeptidase, amylase, beta-galactosidase, beta-glucosidase, beta-xylosidase, carbohydrase, carboxypeptidase, catalase, cellobiohydrolase, cellulase, chitinase, cutinase, cyclodextrin glycosyltransferase, deoxyribonuclease, endoglucanase, esterase, glucoamylase, invertase, laccase, lipase, mannosidase, mutanase, oxidase, pectinolytic enzyme, peroxidase, phytase, polyphenoloxidase, proteolytic enzyme, ribonuclease, transglutaminase, or xylanase.

In one embodiment, the composition is a liquid laundry or liquid dish wash composition, such as an Automatic Dish Wash (ADW) liquid detergent composition, or a powder laundry, such as a soap bar, or powder dish wash composition, such as an ADW unit dose detergent composition and such as a Hand Dish Wash (HDW) detergent composition.

The choice of additional components is within the skills of the skilled person in the art and includes conventional ingredients, including the exemplary non-limiting components set forth below. The choice of components may

include, for fabric care, the consideration of the type of fabric to be cleaned, the type and/or degree of soiling, the temperature at which cleaning is to take place, and the formulation of the detergent product. Although components mentioned below are categorized by general header according to a particular functionality, this is not to be construed as a limitation, as a component may comprise additional functionalities as will be appreciated by the skilled artisan.

In one embodiment of the present invention, the variant of the present invention may be added to a detergent composition in an amount corresponding to 0.001-100 mg of protein, such as 0.01-100 mg of protein, preferably 0.005-50 mg of protein, more preferably 0.01-25 mg of protein, even more preferably 0.05-10 mg of protein, most preferably 0.05-5 mg of protein, and even most preferably 0.01-1 mg of protein per liter of wash liquor. The term "protein" in this context is contemplated to be understood to include a variant according to the present invention.

A composition for use in automatic dish wash (ADW), for example, may include 0.0001%-50%, such as 0.001%-20%, such as 0.01%-10%, such as 0.05-5% of enzyme protein by weight of the composition.

A composition for use in laundry granulation, for example, may include 0.0001%-50%, such as 0.001%-20%, such as 0.01%-10%, such as 0.05%-5% of enzyme protein by weight of the composition.

A composition for use in laundry liquid, for example, may include 0.0001%-10%, such as 0.001-7%, such as 0.1%-5% of enzyme protein by weight of the composition.

The variants of the invention as well as the further active components, such as additional enzymes, may be stabilized using conventional stabilizing agents, e.g., a polyol such as propylene glycol or glycerol, a sugar or sugar alcohol, lactic acid, boric acid, or a boric acid derivative, e.g., an aromatic borate ester, or a phenyl boronic acid derivative such as 4-formylphenyl boronic acid, and the composition may be formulated as described in, for example, WO92/19709 and WO92/19708.

In certain markets different wash conditions and, as such, different types of detergents are used. This is disclosed in e.g. EP 1 025 240. For example, In Asia (Japan) a low detergent concentration system is used, while the United States uses a medium detergent concentration system, and Europe uses a high detergent concentration system.

A low detergent concentration system includes detergents where less than about 800 ppm of detergent components are present in the wash water. Japanese detergents are typically considered low detergent concentration system as they have approximately 667 ppm of detergent components present in the wash water.

A medium detergent concentration includes detergents where between about 800 ppm and about 2000 ppm of detergent components are present in the wash water. North American detergents are generally considered to be medium detergent concentration systems as they have approximately 975 ppm of detergent components present in the wash water.

A high detergent concentration system includes detergents where greater than about 2000 ppm of detergent components are present in the wash water. European detergents are generally considered to be high detergent concentration systems as they have approximately 4500-5000 ppm of detergent components in the wash water.

Latin American detergents are generally high suds phosphate builder detergents and the range of detergents used in Latin America can fall in both the medium and high detergent concentrations as they range from 1500 ppm to 6000

ppm of detergent components in the wash water. Such detergent compositions are all embodiments of the invention.

A variant of the present invention may also be incorporated in the detergent formulations disclosed in WO97/07202, which is hereby incorporated by reference.

Examples are given herein of preferred uses of the compositions of the present invention. The dosage of the composition and other conditions under which the composition is used may be determined on the basis of methods known in the art.

In particular, a composition according to the present invention further comprises a chelator.

The term "chelator" as used herein, refers to chemicals which form molecules with certain metal ions, inactivating the ions so that they cannot react with other elements. Thus, a chelator may be defined as a binding agent that suppresses chemical activity by forming chelates.

Chelation is the formation or presence of two or more separate bindings between a ligand and a single central atom. The ligand may be any organic compound, a silicate or a phosphate. In the present context the term "chelating agents" comprises chelants, chelating agent, chelating agents, complexing agents, or sequestering agents that forms water-soluble complexes with metal ions such as calcium and magnesium. The chelate effect describes the enhanced affinity of chelating ligands for a metal ion compared to the affinity of a collection of similar nonchelating ligands for the same metal. Chelating agents having binding capacity with metal ions, in particular calcium (Ca²⁺) ions, and has been used widely in detergents and compositions in general for wash, such as laundry or dish wash. Chelating agents have however shown themselves to inhibit enzymatic activity. The term chelating agent is used in the present application interchangeably with "complexing agent" or "chelating agent" or "chelant".

Since most alpha-amylases are calcium sensitive the presence of chelating agents these may impair the enzyme activity. The calcium sensitivity of alpha-amylases can be determined by incubating a given alpha-amylase in the presence of a strong chelating agent and analyze the impact of this incubation on the activity of the alpha-amylase in question. A calcium sensitive alpha-amylase will lose a major part or all of its activity during the incubation. Chelating agent may be present in the composition in an amount from 0.0001 wt % to 20 wt %, preferably from 0.01 to 10 wt %, more preferably from 0.1 to 5 wt %.

Non-limiting examples of chelating agents are; EDTA, DTPMA, HEDP, and citrate. Thus, in one embodiment, the composition comprises a variant according to the invention and a chelating agent, such as EDTA, DTPMA, HEDP or citrate.

The term "EDTA" as used herein, refers to ethylenediamine-tetra-acetic acid which falls under the definition of "strong chelating agents".

The term "DTPMA" as used herein, refers to diethylenetriamine penta(methylene phosphonic acid). DTPMA can inhibit the scale formation of carbonate, sulfate and phosphate.

The term "HEDP" as used herein, refers to hydroxyethane diphosphonic acid, which falls under the definition of "strong chelating agents".

The chelate effect or the chelating effect describes the enhanced affinity of chelating ligands for a metal ion compared to the affinity of a collection of similar nonchelating ligands for the same metal. However, the strength of this chelate effect can be determined by various types of assays

or measure methods thereby differentiating or ranking the chelating agents according to their chelating effect (or strength).

In an assay the chelating agents may be characterized by their ability to reduce the concentration of free calcium ions (Ca^{2+}) from 2.0 mM to 0.10 mM or less at pH 8.0, e.g. by using a test based on the method described by M. K. Nagarajan et al., JAOCS, Vol. 61, no. 9 (September 1984), pp. 1475-1478.

For reference, a chelator having the same ability to reduce the concentration of free calcium ions (Ca^{2+}) from 2.0 mM to 0.10 mM at pH as EDTA at equal concentrations of the chelator are said to be strong chelators.

The composition of the present invention may be in any convenient form, e.g., a bar, a homogenous tablet, a tablet having two or more layers, a pouch having one or more compartments, a regular or compact powder, a granule, a paste, a gel, or a regular, compact or concentrated liquid. There are a number of detergent formulation forms such as layers (same or different phases), pouches, as well as forms for machine dosing unit.

Pouches can be configured as single or multicompartments. It can be of any form, shape and material which is suitable for hold the composition, e.g. without allowing the release of the composition from the pouch prior to water contact. The pouch is made from water soluble film which encloses an inner volume. Said inner volume can be divided into compartments of the pouch. Preferred films are polymeric materials preferably polymers which are formed into a film or sheet. Preferred polymers, copolymers or derivatives thereof are selected polyacrylates, and water soluble acrylate copolymers, methyl cellulose, carboxy methyl cellulose, sodium dextrin, ethyl cellulose, hydroxyethyl cellulose, hydroxypropyl methyl cellulose, malto dextrin, poly methacrylates, most preferably polyvinyl alcohol copolymers and, hydroxypropyl methyl cellulose (HPMC). Preferably the level of polymer in the film for example PVA is at least about 60%. Preferred average molecular weight will typically be about 20,000 to about 150,000. Films can also be of blend compositions comprising hydrolytically degradable and water soluble polymer blends such as polyactide and polyvinyl alcohol (known under the Trade reference M8630 as sold by Chris Craft In. Prod. Of Gary, Ind., US) plus plasticisers like glycerol, ethylene glycerol, Propylene glycol, sorbitol and mixtures thereof. The pouches can comprise a solid laundry cleaning composition or part components and/or a liquid cleaning composition or part components separated by the water soluble film. The compartment for liquid components can be different in composition than compartments containing solids. Ref: (US2009/0011970 A1).

Detergent ingredients may be separated physically from each other by compartments in water dissolvable pouches or in different layers of tablets. Thereby negative storage interaction between components may be avoided. Different dissolution profiles of each of the compartments can also give rise to delayed dissolution of selected components in the wash solution.

A liquid or gel detergent, which is not unit dosed, may be aqueous, typically containing at least 20% by weight and up to 95% water, such as up to about 70% water, up to about 65% water, up to about 55% water, up to about 45% water, up to about 35% water. Other types of liquids, including without limitation, alkanols, amines, diols, ethers and polyols may be included in an aqueous liquid or gel. An aqueous liquid or gel detergent may contain from 0-30% organic solvent. A liquid or gel detergent may be non-aqueous.

Another form of composition is in the form of a soap bar, such as a laundry soap bar, and may be used for hand washing laundry, fabrics and/or textiles. The term "soap bar" as used herein, refers to includes laundry bars, soap bars, combo bars, syndet bars and detergent bars. The types of bar usually differ in the type of surfactant they contain, and the term laundry soap bar includes those containing soaps from fatty acids and/or synthetic soaps. The laundry soap bar has a physical form which is solid and not a liquid, gel or a powder at room temperature. The term "solid" as used herein, refers to a physical form which does not significantly change over time, i.e. if a solid object (e.g. laundry soap bar) is placed inside a container, the solid object does not change to fill the container it is placed in. The bar is a solid typically in bar form but can be in other solid shapes such as round or oval.

The soap bar may also comprise complexing agents like EDTA and HEDP, perfumes and/or different type of fillers, surfactants e.g. anionic synthetic surfactants, builders, polymeric soil release agents, detergent chelators, stabilizing agents, fillers, dyes, colorants, dye transfer inhibitors, alkoxylated polycarbonates, suds suppressers, structurants, binders, leaching agents, bleaching activators, clay soil removal agents, anti-redeposition agents, polymeric dispersing agents, brighteners, fabric softeners, perfumes and/or other compounds known in the art.

The soap bar may be processed in conventional laundry soap bar making equipment such as but not limited to: mixers, plodders, e.g. a two stage vacuum plodder, extruders, cutters, logo-stampers, cooling tunnels and wrappers. The invention is not limited to preparing the soap bars by any single method. The premix of the invention may be added to the soap at different stages of the process. For example, the premix comprising a soap, an enzyme, optionally one or more additional enzymes, a protease inhibitor, and a salt of a monovalent cation and an organic anion may be prepared and the mixture may then plodded. The enzyme and optional additional enzymes may be added at the same time as an enzyme inhibitor, e.g. a protease inhibitor, for example in liquid form. Besides the mixing step and the plodding step, the process may further comprise the steps of milling, extruding, cutting, stamping, cooling and/or wrapping.

Uses

The present invention further relates to the use of a variant according to the present invention in a cleaning process such as laundry or hard surface cleaning including automated dish wash and industrial cleaning. The soils and stains that are important for cleaning are composed of many different substances, and a range of different enzymes, all with different substrate specificities, have been developed for use in detergents both in relation to laundry and hard surface cleaning, such as dishwashing. These enzymes are considered to provide an enzyme detergency benefit, since they specifically improve stain removal in the cleaning process that they are used in, compared to the same process without enzymes. Stain removing enzymes that are known in the art include enzymes such as proteases, amylases, lipases, cutinases, cellulases, endoglucanases, xyloglucanases, pectinases, pectin lyases, xanthanases, peroxidaes, haloperoxigenases, catalases and mannanases.

In one embodiment, the invention relates the use of variants of the present invention in detergent compositions, for use in cleaning hard-surfaces, such as dish wash, or in laundering or for stain removal. In another embodiment, the invention relates to the use of an alpha-amylase variant according to the invention in a cleaning process such as

laundry or hard surface cleaning including, but not limited to, dish wash and industrial cleaning. Thus, in one embodiment, the invention relates to the use of a variant comprising an alteration in one or more positions corresponding to the positions H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, A204, V206, P211, I214, V215, L217, L219, A225, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13; or H1, A37, T40, N54, V56, A60, K72, G109, F113, R116, N125, F133, T134, T165, Q169, Q172, L173, A174, G184, A186, E190, N195, G196, A204, V206, A225, L228, K242, S244, G255, N260, A265, K269, S280, W284, Y295, S304, R320, H321, S323, V326, K391, I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14 in a cleaning process such as laundry or hard surface cleaning including dish wash and industrial cleaning.

In a particular embodiment, the invention relates to the use of a variant comprising a) a substitution and/or deletion of two, three or four positions in the parent alpha-amylase said positions corresponding to positions 181, G182, D183, and G184 of the mature polypeptide of SEQ ID NO: 4; and

b) an alteration at one or more positions corresponding to positions H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, A204, V206, P211, I214, V215, L217, L219, A225, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13; or H1, A37, T40, N54, V56, A60, K72, G109, F113, R116, N125, F133, T134, T165, Q169, Q172, L173, A174, G184, A186, E190, N195, G196, A204, V206, A225, L228, K242, S244, G255, N260, A265, K269, S280, W284, Y295, S304, R320, H321, S323, V326, K391, I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14, in a cleaning process such as laundry or hard surface cleaning including dish wash and industrial cleaning.

In one embodiment of the invention relates the use of a composition according to the invention comprising a variant of the present invention together with one or more surfactants and optionally one or more detergent components, selected from the list comprising of hydrotropes, builders and co-builders, bleaching systems, polymers, fabric hueing agents and adjunct materials, or any mixture thereof in detergent compositions and in detergent applications.

A further embodiment is the use of the composition according to the invention comprising a variant of the present invention together with one or more surfactants, and one or more additional enzymes selected from the group comprising of proteases, lipases, cutinases, cellulases, endoglucanases, xyloglucanases, pectinases, pectin lyases, xan-

thanases, peroxidaes, haloperoxygenases, catalases and mannanases, or any mixture thereof in detergent compositions and in detergent applications.

In another aspect, the invention relates to a laundering process which may be for household laundering as well as industrial laundering. Furthermore, the invention relates to a process for the laundering of textiles (e.g. fabrics, garments, cloths etc.) where the process comprises treating the textile with a washing solution containing a detergent composition and an alpha-amylase of the present invention. The laundering can for example be carried out using a household or an industrial washing machine or be carried out by hand using a detergent composition containing a glucoamylase of the invention.

In another aspect, the invention relates to a dish wash process which may be for household dish wash as well as industrial dish wash. The term "dish wash" as used herein, refers to both manual dish wash and automated dish wash. Furthermore, the invention relates to a process for the washing of hard surfaces (e.g. cutlery such as knives, forks, spoons; crockery such as plates, glasses, bowls; and pans) where the process comprises treating the hard surface with a washing solution containing a detergent composition and an alpha-amylase variant of the present invention. The hard surface washing can for example be carried out using a household or an industrial dishwasher or be carried out by hand using a detergent composition containing an alpha-amylase of the invention, optionally together with one or more further enzymes selected from the group comprising of proteases, amylases, lipases, cutinases, cellulases, endoglucanases, xyloglucanases, pectinases, pectin lyases, xanthanases, peroxidaes, haloperoxygenases, catalases, mannanases, or any mixture thereof.

In a further aspect, the invention relates to a method for removing a stain from a surface comprising contacting the surface with a composition comprising an alpha-amylase of the present invention together with one or more surfactants and optionally one or more detergent components selected from the list comprising of hydrotropes, builders and co-builders, bleaching systems, polymers, fabric hueing agents and adjunct materials, or any mixture thereof in detergent compositions and in detergent applications. A further aspect is a method for removing a stain from a surface comprising contacting the surface with a composition comprising an alpha-amylase variant of the present invention together with one or more surfactants, one or more additional enzymes selected from the group comprising of proteases, lipases, cutinases, cellulases, endoglucanases, xyloglucanases, pectinases, pectin lyases, xanthanases, peroxidaes, haloperoxygenases, catalases and mannanases, or any mixture thereof in detergent compositions and in detergent applications.

The present invention is further described by the following examples that should not be construed as limiting the scope of the invention.

EXAMPLES

Example 1a: Construction of the Parent Polypeptide as Set Forth in SEQ ID NO: 13

Construction of a hybrid between the A and B domain from a polypeptide having the amino acid sequence set forth SEQ ID NO: 10 (the first parent polypeptide) and the C domain from a polypeptide having the amino acid sequence set forth SEQ ID NO: 11 (the second parent polypeptide).

Based on 3D structural alignment of the two amylases, amino acid no 1 to amino acid no 397 of the first parent

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polypeptide are defined as domain A and B, and amino acid 398 to amino acid no. 485 of the second parent polypeptide are defined as the C domain. Synthetic DNA fragments coding for part of the A domain from the first parent polypeptide and the C domain from the second parent polypeptide were designed and purchased from an external vendor. In addition to combining fragments of the two different amylases, the following stabilizing substitutions were designed into the synthetic amylase gene: G182* and D183*.

The new gene consisting of a gene fragment encoding the A and B domain of the first parent polypeptide and the C domain of the second parent polypeptide coded by the synthetic gene fragments was constructed by triple SOE (splicing by overlap extension) PCR method. The two fragments were assembled by splicing by overlap extension polymerase chain reaction and transformed into a competent *B. subtilis* host for overexpression. The resulting amylase consisting of the A and B domain from the first parent polypeptide and the C domain from the second parent polypeptide, and having the deletions of G182* and D183* is the polypeptide as set forth in SEQ ID NO: 13.

Example 1b: Construction of the Parent Polypeptide as Set Forth in SEQ ID NO: 14

Construction of a hybrid between the A and B domain from a polypeptide having the amino acid sequence set forth SEQ ID NO: 10 (the first parent polypeptide) and the C domain from a polypeptide having the amino acid sequence set forth SEQ ID NO: 12 (the third parent polypeptide).

Based on 3D structural alignment of the two amylases, amino acid no 1 to amino acid no 397 of the first parent polypeptide are defined as domain A and B, and amino acid 398 to amino acid no. 485 of the third parent polypeptide are defined as the C domain. Synthetic DNA fragments coding for part of the A domain from the first parent polypeptide and the C domain from the second parent polypeptide were designed and purchased from an external vendor. In addition to combining fragments of the two different amylases, the following stabilizing substitutions were designed into the synthetic amylase gene: G182* and D183*.

The new gene consisting of a gene fragment encoding the A and B domain of the first parent polypeptide and the C domain of the second parent polypeptide coded by the synthetic gene fragments was constructed by triple SOE (splicing by overlap extension) PCR method. A DNA fragment was amplified from an expression clone of the first amylase. The fragments were finally assembled by triple SOE and the derived PCR fragment was transformed into a suitable *B. subtilis* host and the gene integrated into the *B. subtilis* chromosome by homologous recombination into the pectate lyase (pel) locus. The gene coding for chloramphenicol acetyltransferase was used as maker (as described in (Diderichsen et al., 1993, Plasmid 30: 312-315). Chloramphenicol resistant clones were analyzed by DNA sequencing to verify the correct DNA sequence of the construct. The resulting amylase consisting of the A and B domain from the first parent polypeptide and the C domain from the second parent polypeptide, and having the deletions of G182* and D183* is the polypeptide as set forth in SEQ ID NO: 14.

Example 2: Generation of Variants According to the Invention

The variants of the present invention have been generated by site-directed mutagenesis. Genomic DNA prepared from

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the organism containing amylase gene at the Pel locus was used as template for generating the site-directed mutants.

Mutagenic forward primer and PnMi4490 (CAATC-CAAGAGAACCCTGATACGGATG—SEQ ID NO: 16) reverse primer was used to generate a ~3.8 kb fragment. This fragment was used as a megaprimer along with PnMi4491 (CGGAACGCCTGGCTGACAACACG—SEQ ID NO: 17) forward primer to get 6 kb insertion cassette. To enable integration in the Pel locus by double cross-over upon transformation, along with the amylase and cat genes, the cassette contained upstream and downstream Pel sequences at the ends. Selection was done on LB Agar containing chloramphenicol and the mutation was confirmed by DNA sequencing of amylase gene.

A library of variants were generated where different combinations of alterations in the following positions were made; H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, N54, V56, A60, K72, R87, Q98, M105, G109, F113, R116, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, Q172, L173, A174, G184, A186, E190, T193, N195, G196, A204, V206, P211, V213, I214, L217, A225, L228, L235, V238, K242, S244, M246, L2502, G255, N260, A263, V264, A265, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, S304, R320, H321, S323, H324, V326, F328, T334, G337, Q345, G346, T355, S376, D377, D381, Y382, Q385, K391, and Q395 of SEQ ID NOs: 7 or 10; A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13; and 1405, A421, A422, A428, G448, D476, and G477 of SEQ ID NO: 14.

The generated variants are listed in the following examples.

Example 3: Wash Performance of Generated Variants Using Automatic Mechanical Stress Assay (AMSA)

In order to assess the wash performance of the variants of the present invention in a detergent base composition, washing experiments may be performed using Automatic Mechanical Stress Assay (AMSA). With the AMSA test the wash performance of a large quantity of small volume enzyme-detergent solutions can be examined. The AMSA plate has a number of slots for test solutions and a lid firmly squeezing the textile swatch to be washed against all the slot openings. During the washing time, the plate, test solutions, textile and lid were vigorously shaken to bring the test solution in contact with the textile and apply mechanical stress in a regular, periodic oscillating manner. For further description see WO 02/42740, especially the paragraph Special method embodiments at page 23-24.

General Wash Performance Description

A test solution comprising water (6°dH), 0.79 g/L detergent, e.g. Model detergent J as described below, and the enzyme of the invention at concentration of 0, 0.05 or 0.2 mg enzyme protein/L, is prepared. Fabrics stained with starch (CS-28 from Center For Test materials BV, P.O. Box 120, 3133 KT, Vlaardingen, The Netherlands) was added and washed for 20 minutes at 15° C. and 30° C., or alternatively 20 minutes at 15° C. and 40° C. as specified below. After thorough rinse underrunning tap water and drying in the dark, the light intensity values of the stained fabrics were subsequently measured as a measure for wash performance. The test with 0 mg enzyme protein/L was used as a blank and corresponded to the contribution from the detergent. Preferably mechanical action is applied during the wash step, e.g. in the form of shaking, rotating or stirring the wash

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solution with the fabrics. The AMSA wash performance experiments was conducted under the experimental conditions specified below:

TABLE A

Experimental condition	
Detergent	Liquid Model detergent J (see Table B)
Detergent dosage	0.79 g/L
Test solution volume	160 micro L
pH	As is
Wash time	20 minutes
Temperature	15° C. or 30° C.
Water hardness	6° dH
Enzyme concentration in test	0.2 mg enzyme protein/L and 0.05 mg enzyme protein/L
Test material	CS-28 (Rice starch cotton)

TABLE B

Model detergent J		
Compound	Content of compound (% w/w)	% active component (% w/w)
LAS	5.15	5.00
linear alkylbenzene sulfonates		
AS	5.00	4.50
alkylbenzene sulfonates		
AEOS	14.18	10.00
alkyl ethoxy sulfate		
Coco fatty acid	1.00	1.00
AEO	5.00	5.00
alkyl ethoxylate		
MEA	0.30	0.30
monoethanolamine		
MPG	3.00	3.00
monopropylene glycol		
Ethanol	1.50	1.35
DTPA (as Na5 salt)	0.25	0.10
pentasodium diethylenetriaminepenta-acetic acid		
Sodium citrate	4.00	4.00
Sodium formate	1.00	1.00
Sodium hydroxide	0.66	0.66
H ₂ O, ion exchanged	58.95	58.95

Water hardness was adjusted to 6°dH by addition of CaCl₂, MgCl₂, and NaHCO₃ (Ca²⁺:Mg²⁺:HCO₃⁻=2:1:4.5) to the test system. After washing the textiles were flushed in tap water and dried.

TABLE C

Experimental condition	
Detergent	Liquid Model detergent A (see Table D)
Detergent dosage	3.33 g/L
Test solution volume	160 micro L
pH	As is
Wash time	20 minutes
Temperature	15° C. or 40° C.
Water hardness	15° dH
Enzyme concentration in test	0.2 mg enzyme protein/L, 0.05 mg enzyme protein/L
Test material	CS-28 (Rice starch cotton)

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TABLE D

Model detergent A		
Compound	Content of compound (% w/w)	% active component (% w/w)
LAS	12.00	11.60
(linear alkylbenzene sulfonates)		
AEOS	17.63	4.90
(alkyl ethoxy sulfate), SLES		
(sodium lauryl ether sulfate)		
Soy fatty acid	2.75	2.48
Coco fatty acid	2.75	2.80
AEO (alkyl ethoxylate)	11.00	11.00
Sodium hydroxide	1.75	1.80
Ethanol/Propan-2-ol	3.00	2.70/0.30
MPG	6.00	6.00
monopropylene glycol		
Glycerol	1.71	1.70
TEA (triethanolamine)	3.33	3.30
Sodium formate	1.00	1.00
Sodium citrate	2.00	2.00
DTMPA	0.48	0.20
(diethylenetriaminepenta-acetic acid)		
PCA	0.46	0.18
polycarboxylic acid type polymer		
Phenoxy ethanol	0.50	0.50
H ₂ O, ion exchanged	33.64	33.64

Water hardness was adjusted to 15°dH by addition of CaCl₂, MgCl₂, and NaHCO₃ (Ca²⁺:Mg²⁺:HCO₃⁻=4:1:7.5) to the test system. After washing the textiles were flushed in tap water and dried.

The wash performance was measured as the brightness expressed as the intensity of the light reflected from the sample when illuminated with white light. When the sample was stained the intensity of the reflected light was lower, than that of a clean sample. Therefore the intensity of the reflected light can be used to measure wash performance.

Color measurements were made with a professional flat-bed scanner (EPSON Expression 10000XL, EPSON) used to capture an image of the washed textile.

To extract a value for the light intensity from the scanned images, 48→24 Bit Color pixel values from the image were converted into values for red, green and blue (RGB). The intensity value (Int) is calculated by adding the RGB values together as vectors and then taking the length of the resulting vector:

$$Int = \sqrt{r^2 + g^2 + b^2}$$

The term “improved wash performance” of the present experiment is defined as displaying an alteration of the wash performance of a variant of the present invention relative to the wash performance of the polypeptide having an amino acid sequence as set forth in SEQ ID NO: 15. The alteration may e.g. be seen as increased stain removal. Improved wash performance was determined as described above. The wash performance was considered to be improved if the Improvement Factor (IF) is at least 1.0, preferably at least 1.05 in one or more of the conditions listed above; i.e. either in Model detergent A at 15° C. or 40° C., where the variant concentration was 0.05 or 0.2 mg/L or in Model detergent J at 15°

C. or 30° C., where the variant concentration was 0.05 or 0.2 mg/L. The wash conditions were as described above in Tables A and C.

The terms “Delta intensity” or “Delta intensity value” as used herein refers to the result of an intensity measurement of a test material, e.g. a swatch CS-28 (Center For Testma-
5 terials BV, P.O. Box 120, 3133 KT Vlaardingen, the Neth-

erlands). The swatch was measured with a portion of the swatch, washed under identical conditions, as background. The delta intensity is the intensity value of the test material washed with amylase subtracting the intensity value of the test material washed without amylase.

The wash performance of the variants according to the invention obtained by AMSA are the following;

Modifications in SEQ ID NO: 13	A-15- 0.05 mg/L	A-15- 0.2 mg/L	A-40- 0.05 mg/L	A-40- 0.2 mg/L	J-15- 0.05 mg/L	J-15- 0.2 mg/L	J-30- 0.05 mg/L	J-30- 0.2 mg/L
Reference	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
G182* + D183* + N195F	1.0	1.3	1.1	1.0	1.5	1.3	1.0	1.0
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.1	1.9	1.2	1.2	2.6	2.0	1.5	1.3
H1* + N54S + V56T + R87S + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.8	2.2	1.3	1.1	2.6	2.4	1.9	1.5
H1* + T40G + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.2	1.7	1.3	1.1	2.5	2.2	1.9	1.6
H1* + T51K + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.7	2.1	1.3	1.1	3.2	2.3	2.0	1.5
H1* + N54S + V56T + K72R + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.6	2.2	1.2	1.0	2.6	2.5	1.7	1.5
H1* + N54S + V56T + K72H + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.5	2.4	1.4	1.1	2.6	2.3	1.7	1.6
H1* + A37H + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.8	2.7	1.4	1.2	2.8	2.8	1.9	1.6
H1* + A37M + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.0	2.3	1.4	1.2	2.7	2.5	2.1	1.6
H1* + A37V + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	3.0	2.6	1.4	1.2	3.1	2.5	2.2	1.7
H1* + A37S + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.9	2.5	1.2	1.2	2.9	2.3	1.5	1.3
H1* + A37Y + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.6	2.0	1.3	1.2	2.5	2.1	1.5	1.3
H1* + A37R + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.2	1.9	1.3	1.2	2.5	2.1	1.4	1.3
H1* + N54S + V56T + G109A + F113W + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.8	2.2	1.3	1.2	2.7	2.1	1.5	1.2
H1* + N54S + V56T + G109A + F113S + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.3	1.9	1.3	1.1	2.4	1.7	1.3	1.2
H1* + N54S + V56T + G109A + F113N + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.0	1.9	1.2	1.2	2.3	2.1	1.4	1.2
H1* + N54S + V56T + G109A + F113Y + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.9	2.1	1.2	1.2	3.1	2.3	1.5	1.3
H1* + N54S + V56T + G109A + F113R + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.0	1.4	1.2	1.2	2.0	1.8	1.4	1.3
H1* + N54S + V56T + G109A + Q118N + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.9	1.8	1.3	1.2	2.9	2.3	1.5	1.3
H1* + N54S + V56T + G109A + R116Q + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.6	2.0	1.2	1.2	3.5	2.6	1.7	1.4
H1* + N54S + V56T + G109A + R116V + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.1	2.0	1.1	1.1	2.7	2.3	1.6	1.3
H1* + N54S + V56T + G109A + R116K + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.2	2.1	1.2	1.1	2.4	2.2	1.4	1.3
H1* + N54S + V56T + G109A + R116W + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.8	2.0	0.7	0.9	1.1	1.9	1.0	1.2
H1* + N54S + V56T + G109A + R116L + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.8	1.8	1.2	1.1	1.9	2.0	1.3	1.2
H1* + N54S + V56T + G109A + G142T + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.6	1.7	1.2	1.1	1.9	2.0	1.3	1.1
H1* + N54S + V56T + G109A + Q125P + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.1	2.1	1.2	1.1	2.4	2.0	1.4	1.3
H1* + N54S + V56T + G109A + Q125K + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.9	1.8	1.3	1.1	2.6	1.9	1.5	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + S381G + K391A + G476K	2.3	2.1	1.0	1.0	2.5	2.2	1.4	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + M246T + K391A + G476K	2.0	1.9	1.1	1.0	2.6	1.8	1.5	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + V213T + K391A + G476K	1.9	1.9	1.1	1.1	2.8	2.1	1.5	1.3
H1* + A37L + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.5	2.2	1.2	1.1	2.5	2.2	1.8	1.4
H1* + N54S + V56T + G109A + Q118K + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.8	2.0	1.2	1.1	1.5	1.6	1.4	1.2
H1* + N54S + V56T + G109A + R116A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.7	2.3	1.2	1.1	2.4	2.1	1.6	1.3

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H1* + N54S + V56T + G109A + R116H + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.9	2.4	1.3	1.2	2.4	2.3	1.9	1.5
H1* + N54S + V56T + G109A + Q125A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.1	2.0	1.1	1.2	3.1	2.1	1.5	1.2
H1* + N54S + V56T + G109A + Q172G + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.6	2.1	1.4	1.2	2.7	2.2	2.1	1.5
H1* + N54S + V56T + G109A + Q172R + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.6	2.2	1.4	1.2	2.7	2.4	2.0	1.5
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + D377S + K391A + G476K	2.4	1.7	1.3	1.2	2.8	1.8	1.7	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + D377G + K391A + G476K	1.7	1.7	1.2	1.1	2.3	1.5	1.7	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + D377R + K391A + G476K	1.9	1.5	1.1	1.1	2.6	1.6	1.4	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + D377A + K391A + G476K	2.0	1.9	1.2	1.0	2.8	1.9	1.5	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + I214G + K391A + G476K	1.6	1.6	1.3	1.1	2.9	1.7	1.5	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + S381A + K391A + G476K	2.3	1.6	1.4	1.2	2.8	1.9	1.6	1.4
H1* + N54S + V56T + G109A + W167I + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.7	1.9	1.3	1.2	2.8	1.8	1.6	1.3
H1* + N54S + V56T + G109A + W167G + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.5	1.8	1.2	1.1	2.5	1.8	1.5	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + G346S + K391A + G476K	1.7	1.8	1.3	1.2	2.8	2.0	1.6	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + G346P + K391A + G476K	2.1	1.9	1.3	1.1	2.9	2.1	1.6	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + I214H + K391A + G476K	2.2	1.7	1.3	1.1	2.2	2.1	1.7	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + I214S + K391A + G476K	2.3	1.8	1.3	1.1	2.9	2.0	1.5	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + I214T + K391A + G476K	1.9	1.7	1.3	1.2	2.4	1.7	1.6	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + A420Q + G476K	2.0	1.7	1.2	1.1	1.9	1.7	1.3	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + A420S + G476K	2.0	1.8	1.2	1.1	2.6	1.8	1.5	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + A420K + G476K	2.2	1.9	1.3	1.2	2.3	1.8	1.7	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + A420L + G476K	2.2	2.0	1.3	1.2	2.4	1.9	1.2	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + K391Y + G476K	2.0	1.9	1.3	1.2	2.9	1.9	1.1	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + K391R + G476K	1.9	1.9	1.3	1.2	2.5	1.7	1.4	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + P473R + G476K	2.2	1.9	1.2	1.2	3.4	2.0	2.1	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + P473A + G476K	2.5	1.8	1.5	1.1	3.4	1.9	1.9	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + P473G + G476K	2.5	1.9	1.3	1.2	2.8	1.9	1.5	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + P473T + G476K	1.9	1.7	1.3	1.2	2.5	1.9	1.5	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + Q395P + G476K	1.8	1.6	1.3	1.1	2.3	1.8	1.8	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + T444Q + G476K	1.7	1.8	1.3	1.2	2.8	2.0	1.4	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + T444S + G476K	1.7	1.8	1.3	1.2	2.8	2.0	1.5	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + T444D + G476K	2.3	1.8	1.3	1.3	3.0	2.3	1.7	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + T444Y + G476K	2.0	1.7	1.5	1.2	2.0	2.1	1.8	1.5
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + S280W + K391A + G476K	2.3	1.7	1.3	1.1	2.5	2.0	1.5	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + S280L + K391A + G476K	2.1	1.7	1.2	1.1	2.5	1.8	1.4	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + S280T + K391A + G476K	1.7	1.6	1.1	1.1	2.4	1.6	1.3	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + S280A + K391A + G476K	1.9	1.4	1.2	1.0	2.3	1.8	1.3	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + G255N + K391A + G476K	1.9	1.7	1.2	1.2	2.6	2.1	1.5	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + T285L + K391A + G476K	1.9	1.8	1.3	1.2	2.3	2.2	1.5	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + T285Q + K391A + G476K	1.6	1.7	1.2	1.2	2.2	2.1	1.6	1.3

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H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + Q449T + G476K	1.8	1.6	1.1	1.1	1.8	1.6	1.2	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + P473K + G476K	1.8	1.7	1.3	1.1	2.2	2.0	1.3	1.2
H1* + N54S + V56T + K72S + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.5	2.3	1.2	1.1	3.7	2.2	1.5	1.4
H1* + T40K + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + G346T + K391A + G476K	2.3	2.1	1.2	1.1	2.0	1.7	1.5	1.3
H1* + G50A + N54S + V56T + G109A + Q172N + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.4	2.1	1.2	1.2	2.0	1.9	1.4	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K + K484A	1.9	1.7	1.3	1.0	2.4	1.6	1.5	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K + K484G	1.8	1.8	1.2	1.0	2.2	1.8	1.4	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K + K484P	1.8	1.7	1.1	1.0	2.4	2.0	1.4	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K + K484E	1.9	1.4	1.2	1.1	2.5	1.7	1.3	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K + K484Q	1.7	1.5	1.1	1.0	2.0	1.3	1.3	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K + K484S	2.3	2.0	1.2	1.2	2.7	1.7	1.6	1.4
H1*K + G50A + N54S + V56T + G109A + W167F + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.1	1.9	1.1	1.1	2.2	1.7	1.3	1.2
H1* + N54S + V56T + G109A + L173V + A174S + G182* + D183* + N195F + A204T + V206L + K391A + G476K	2.2	1.8	1.1	1.1	2.2	1.7	1.1	1.1
H1* + N54S + V56T + G109A + W104Y + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.2	1.9	1.2	1.1	2.5	1.9	1.3	1.3
H1* + N54S + V56T + Q98S + G109A + A174S + G182* + D183* + N195F + V206L + R320A + K391A + G476K	1.9	1.8	1.2	1.1	2.1	1.8	1.1	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + H324L + K391A + G476K	1.9	1.7	1.1	1.1	2.1	1.5	1.2	1.1
H1* + N54S + V56T + G109A + G149Q + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.7	1.9	1.2	1.2	1.6	1.4	1.0	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + G346T + K391A + G476K + G477A	2.1	2.0	1.2	1.2	2.3	1.7	1.2	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + P473G + G476K	2.5	1.9	1.3	1.2	2.8	1.9	1.5	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + P473T + G476K	1.9	1.7	1.3	1.2	2.5	1.9	1.5	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + Q395P + G476K	1.8	1.6	1.3	1.1	2.3	1.8	1.8	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + T444Q + G476K	1.7	1.8	1.3	1.2	2.8	2.0	1.4	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + T444D + G476K	1.7	1.8	1.3	1.2	2.8	2.0	1.5	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + T444S + G476K	2.3	1.8	1.3	1.3	3.0	2.3	1.7	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + T444Y + G476K	2.0	1.7	1.5	1.2	2.0	2.1	1.8	1.5
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + S280W + K391A + G476K	2.3	1.7	1.3	1.1	2.5	2.0	1.5	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + S280L + K391A + G476K	2.1	1.7	1.2	1.1	2.5	1.8	1.4	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + S280T + K391A + G476K	1.7	1.6	1.1	1.1	2.4	1.6	1.3	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + S280A + K391A + G476K	1.9	1.4	1.2	1.0	2.3	1.8	1.3	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + G255N + K391A + G476K	1.9	1.7	1.2	1.2	2.6	2.1	1.5	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + T285L + K391A + G476K	1.9	1.8	1.3	1.2	2.3	2.2	1.5	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + T285Q + K391A + G476K	1.6	1.7	1.2	1.2	2.2	2.1	1.6	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + Q449T + G476K	1.8	1.6	1.1	1.1	1.8	1.6	1.2	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + P473K + G476K	1.8	1.7	1.3	1.1	2.2	2.0	1.3	1.2
H1* + N54S + V56T + K72S + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.5	2.3	1.2	1.1	3.7	2.2	1.5	1.4
H1* + T40K + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + G346T + K391A + G476K	2.3	2.1	1.2	1.1	2.0	1.7	1.5	1.3
H1* + G50A + N54S + V56T + G109A + Q172N + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.4	2.1	1.2	1.2	2.0	1.9	1.4	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K + K484A	1.9	1.7	1.3	1.0	2.4	1.6	1.5	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K + K484G	1.8	1.8	1.2	1.0	2.2	1.8	1.4	1.2

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H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K + K484P	1.8	1.7	1.1	1.0	2.4	2.0	1.4	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K + K484E	1.9	1.4	1.2	1.1	2.5	1.7	1.3	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K + K484Q	1.7	1.5	1.1	1.0	2.0	1.3	1.3	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K + K484S	2.3	2.0	1.2	1.2	2.7	1.7	1.6	1.4
H1*K + G50A + N54S + V56T + G109A + W167F + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.1	1.9	1.1	1.1	2.2	1.7	1.3	1.2
H1* + N54S + V56T + G109A + L173V + A174S + G182* + D183* + N195F + A204T + V206L + K391A + G476K	2.2	1.8	1.1	1.1	2.2	1.7	1.1	1.1
H1* + N54S + V56T + G109A + W140Y + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.2	1.9	1.2	1.1	2.5	1.9	1.3	1.3
H1* + N54S + V56T + Q98S + G109A + A174S + G182* + D183* + N195F + V206L + R320A + K391A + G476K	1.9	1.8	1.2	1.1	2.1	1.8	1.1	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + H324L + K391A + G476K	1.9	1.7	1.1	1.1	2.1	1.5	1.2	1.1
H1* + N54S + V56T + G109A + G149Q + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.7	1.9	1.2	1.2	1.6	1.4	1.0	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + G346T + K391A + G476K + G477A	2.1	2.0	1.2	1.2	2.3	1.7	1.2	1.2
H1* + N54S + V56T + G109AG255A ++ A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.1	2.0	1.2	1.2	2.3	1.8	1.2	1.2
H1* + G50A + N54S + V56T + G109A + F113L + R116L + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.3	2.2	1.2	1.1	2.6	2.2	1.3	1.2
H1* + N54S + V56T + G109A + W167F + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.0	1.9	1.1	1.0	2.3	2.0	1.4	1.2
H1* + N54S + V56T + G109A + L173V + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.8	1.9	1.0	1.0	2.2	1.8	1.4	1.3
H1* + N54S + V56T + K72R + G109A + W167F + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.1	1.7	1.1	1.1	2.9	2.0	1.4	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K + G477Q	1.8	1.4	1.0	1.1	2.4	1.9	1.4	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + Q345D + K391A + G476K + G477Q	1.6	1.5	1.0	1.0	2.2	1.7	1.3	1.2
H1* + G50A + N54S + V56T + G109A + Q172D + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.9	1.9	0.9	1.1	2.1	1.9	1.6	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + V264L + K391A + G423H + G476K	1.9	1.9	0.7	1.0	2.2	2.0	1.4	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + R320V + K391A + G476Y	2.1	2.0	0.8	1.0	2.1	2.1	1.5	1.3
H1* + T5K + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.7	1.8	0.9	1.0	1.8	1.7	1.4	1.2
H1* + N54S + V56T + G109A + R116M + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.9	2.2	1.2	1.1	2.8	2.1	1.2	1.1
H1* + N54S + V56T + G109A + Q172Y + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.1	1.8	1.1	1.1	1.9	1.8	1.4	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + I214L + K391A + G476K	2.0	1.7	1.1	1.1	1.9	1.7	1.3	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + L217V + K391A + G476K	2.2	1.9	1.2	1.2	2.2	2.2	1.3	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + L217H + K391A + G476K	2.2	1.7	1.2	1.2	2.3	2.1	1.4	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + P211D + K391A + G476K	2.1	2.0	1.1	1.1	2.3	2.1	1.2	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + K391Q + G476K	2.1	1.9	1.1	1.2	2.3	1.9	1.3	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + T444H + G476K	2.0	1.9	1.1	1.1	2.3	1.9	1.4	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + T444V + G476K	2.2	1.7	1.1	1.0	2.3	1.8	1.2	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + T444R + G476K	2.0	1.6	1.2	1.1	1.9	1.8	1.3	1.2
H1* + G50A + N54S + V56T + G109A + L173V + A174S + G182* + D183* + N195F + A204V + V206L + K391A + G476K	2.4	2.1	1.2	1.2	3.2	2.2	1.6	1.4
H1* + G50A + N54S + V56T + G109A + F113L + W167F + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.9	1.7	1.0	1.1	2.1	1.8	1.2	1.2
H1* + G7K + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + R320A + K391A + G476K	2.0	1.8	1.1	1.1	2.7	2.1	1.6	1.2
H1* + N54S + V56T + G109A + Q118G + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.7	1.8	1.1	1.1	2.4	1.6	1.5	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + S280K + K391A + G476K	1.6	1.6	1.0	1.1	1.6	1.6	1.3	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + N270G + K391A + G476K	1.4	1.6	1.1	1.1	1.8	1.6	1.4	1.2
H1* + N54S + V56T + G109A + Q172D + A174S + G182* + D183* + N195F + V206L + G346T + K391A + G476K + G477A	2.1	1.9	1.2	1.1	3.1	2.3	1.6	1.4

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H1* + N54S + V56T + G109A + Q118S + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.1	1.9	1.1	1.1	1.8	1.5	1.4	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + R320A + K391A + G476K	2.0	1.9	1.2	1.2	2.3	1.7	1.5	1.3
H1* + N54S + V56T + G109A + T134E + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.4	2.1	1.2	1.1	2.8	2.4	1.4	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + G337H + Q385L + K391A + G476K	1.7	1.8	1.0	1.0	1.8	1.8	1.2	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + Y382M + K391A + G476K	2.0	1.8	1.1	1.1	2.2	1.9	1.3	1.2
H1* + N54S + V56T + K72R + G109A + A174S + G182* + D183* + N195F + V206L + G255A + D377H + K391A + G476K	1.9	1.9	1.1	1.1	2.7	2.2	1.4	1.3
H1* + N54S + V56T + K72R + G109A + T134E + A174S + G182* + D183* + N195F + V206L + G255A + K391A + G476K	2.4	2.0	1.2	1.1	3.0	2.3	1.4	1.3
H1* + N54S + V56T + K72R + M105F + G109A + A174S + G182* + D183* + N195F + V206L + G255A + K391A + G476K	2.0	1.7	1.2	1.1	2.5	2.0	1.4	1.2
H1* + N54S + V56T + M105F + G109A + T134E + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.0	1.9	1.0	1.0	2.3	2.1	1.2	1.1
H1* + N54S + V56T + M105F + G109A + Q118F + T134E + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.9	1.7	1.1	1.1	1.7	1.9	1.3	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + R320A + S323N + D377H + K391A + G476K	2.0	1.9	1.1	1.0	2.3	2.0	1.4	1.3
H1* + T51E + N54S + V56T + G109A + A174H + G182* + D183* + N195F + V206L + K391A + G476K	2.3	1.9	1.2	1.1	2.6	2.2	1.6	1.4
H1* + N54S + V56T + G109A + L173A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.4	2.0	1.3	1.2	2.4	2.0	1.6	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + Q299V + K391A + G476K	2.5	2.0	1.3	1.2	2.4	1.6	1.6	1.4
H1* + N54S + V56T + G109A + L173G + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.2	2.0	1.3	1.2	2.3	1.9	1.7	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + T355L + K391A + G476K	2.1	2.0	1.3	1.2	2.4	2.0	1.5	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + T355F + K391A + G476K	1.3	1.7	1.0	1.0	1.9	1.6	1.2	1.1
H1* + G50A + N54S + V56T + G109A + A174S + G182* + D183* + N195F + A204T + V206L + K391A + G476K	2.3	2.0	1.2	1.1	2.1	1.8	1.4	1.3
H1* + G50A + N54S + V56T + G109A + A174E + G182* + D183* + N195F + A204T + V206L + K391A + G476K	1.8	1.6	1.1	1.1	2.1	1.8	1.2	1.2
H1* + N54S + V56T + K72R + G109A + A174S + G182* + D183* + N195F + V206L + R320A + S323N + K391A + G476K	2.2	1.8	1.2	1.1	2.7	2.1	1.4	1.2
H1* + N54S + V56T + M105I + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.8	1.8	1.1	1.1	1.5	1.5	1.3	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + M246L + K391A + G476K	2.0	1.8	1.1	1.0	1.8	1.6	1.3	1.2
H1* + T51G + N54S + V56T + G109A + A174N + G182* + D183* + N195F + V206L + K391A + G476K	1.6	1.5	1.0	1.0	2.1	1.7	1.3	1.1
H1* + T51A + N54S + V56T + G109A + A174G + G182* + D183* + N195F + V206L + K391A + G476K	1.4	1.5	0.9	0.9	1.6	1.5	1.3	1.0
H1* + N54S + V56T + G109A + A174D + G182* + D183* + N195F + V206L + K391A + G476K	1.9	1.5	1.0	1.0	1.5	1.5	1.1	1.0
H1* + T51S + N54S + V56T + G109A + A174P + G182* + D183* + N195F + V206L + K391A + G476K	1.7	1.7	0.9	1.0	1.1	1.2	0.8	0.9
H1* + T51G + N54S + V56T + G109A + A174M + G182* + D183* + N195F + V206L + K391A + G476K	1.3	1.5	1.0	1.0	1.6	1.5	0.9	1.1
H1* + N54S + V56T + G109A + K72A + W167R + A174S + G182* + D183* + N195F + V206L + K391A + G476K	0.9	1.2	0.5	0.7	0.7	0.8	0.4	0.7
H1* + N54S + V56T + K72A + G109A + W167S + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.6	1.8	0.4	0.8	0.7	1.2	0.6	0.8
H1* + N54S + V56T + K72R + G109A + W167R + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.1	1.1	0.8	0.8	0.7	0.8	0.7	0.7
H1* + N54S + V56T + K72Q + G109A + W167H + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.4	2.2	0.9	1.0	1.8	1.7	1.0	1.1
H1* + N54S + V56T + G109A + Q118R + T134E + A174S + G182* + D183* + N195F + V206L + D377H + K391A + G476K	1.8	1.7	1.1	1.0	2.1	1.7	1.2	1.1
H1* + Q11H + N54S + V56T + G109A + Q118R + T134E + A174S + G184T + G182* + D183* + N195F + V206L + D377H + K391A + G476K	1.9	1.6	1.1	1.0	2.2	1.6	1.3	1.3
H1* + N54S + V56T + K72R + G109A + T134E + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.3	2.1	1.2	1.1	3.2	1.9	1.3	1.3
H1* + N54S + V56T + K72R + G109A + T134E + A174S + G182* + D183* + N195F + V206L + D377H + K391A + G476K	2.2	2.0	1.1	1.0	3.8	1.9	1.3	1.2
H1* + N54S + V56T + K72R + G109A + T134E + A174S + G182* + D183* + G184T + N195F + V206L + K391A + G476K	2.2	2.0	1.2	1.1	3.9	2.4	1.4	1.3
H1* + N54S + V56T + G109A + Q118R + T134E + A174S + G182* + D183* + N195F + V206L + G255A + K391A + G476K	2.0	1.9	1.0	1.2	2.5	2.0	1.2	1.1
H1* + N54S + V56T + G109A + Q118R + A174S + G182* + D183* + N195F + V206L + D377H + K391A + G476K	2.2	1.9	1.1	1.1	2.8	1.8	1.2	1.2
H1* + N54S + V56T + G109A + T134E + A174S + G182* + D183* + G184T + N195F + V206L + D377H + K391A + G476K	2.1	2.1	1.0	1.1	3.4	2.3	1.3	1.3

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H1* + N54S + V56T + G109A + Q172D + A174Q + G182* + D183* + N195F + V206L + K391A + G476K + G477A	2.0	1.9	1.2	1.1	3.0	2.2	1.1	1.2
H1* + N54S + V56T + K72A + G109A + W167F + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.1	1.4	0.6	0.8	1.5	1.3	1.0	0.8
H1* + N54S + V56T + G109A + W167S + G182* + D183* + N195F + A174S + V206L + K391A + G476K	1.4	1.3	0.8	1.0	1.5	1.6	1.3	1.1
H1* + N54S + V56T + K72H + G109A + W167S + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.4	1.4	0.5	1.0	1.0	1.4	0.8	1.1
H1* + N54S + V56T + K72R + G109A + W167T + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.3	1.2	0.8	0.8	1.2	1.0	1.1	1.0
H1* + N54S + V56T + K72E + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.3	1.4	0.8	0.9	1.4	1.3	0.9	0.9
H1* + N54S + V56T + K72R + G109A + W167L + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.6	1.6	1.1	1.1	2.0	1.5	1.4	1.2
H1* + N54S + V56T + K72R + G109A + W167M + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.5	1.6	0.8	1.0	1.5	1.3	1.2	1.2
H1* + N54S + V56T + K72R + G109A + W167H + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.3	1.4	0.6	1.0	1.6	1.3	0.9	1.1
H1* + N54S + V56T + K72R + G109A + A174S + G182* + D183* + N195F + V206L + G255A + K391A + G476K	2.0	1.9	1.2	1.1	2.5	2.0	1.4	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + G184T + N195F + V206L + D377H + K391A + G476K	2.2	1.9	1.1	1.1	2.3	1.9	1.5	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + G255A + D377H + K391A + G476K	2.1	2.0	1.2	1.0	2.1	1.7	1.3	1.3
H1* + N54S + V56T + G109A + Q118R + A174S + G182* + D183* + N195F + V206L + G255A + D377H + K391A + G476K	1.2	1.5	1.0	1.0	1.8	1.2	1.0	1.0
H1* + N54S + V56T + G109A + A174S + G182* + D183* + G184A + T193K + N195F + V206L + K391A + G476K	1.2	1.3	0.9	1.0	1.3	1.3	1.0	1.0
H1* + N54S + V56T + M105F + G109A + A174S + G182* + D183* + G184T + N195F + V206L + K391A + G476K	1.9	2.0	1.1	1.2	2.1	1.9	1.4	1.3
H1* + N54S + V56T + G109A + T134E + A174S + G182* + D183* + N195F + V206L + R320A + S323N + K391A + G476K	1.7	2.1	1.1	1.1	2.1	2.0	1.3	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + V291G + D377H + K391A + G476K	1.7	1.8	1.2	1.1	2.1	1.9	1.4	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + V291A + F328L + D377H + K391A + G476K	1.7	1.9	1.1	1.1	2.1	1.9	1.3	1.3
H1* + V56T + G109A + A174S + G182* + D183* + N195F + V206L + D377H + K391A + G476K	1.3	1.8	1.1	1.1	1.5	1.5	1.3	1.2
H1* + N54S + V56T + G109A + T134E + A174S + G182* + D183* + G184T + N195F + V206L + K391A + G476K	2.4	2.2	1.1	1.1	3.0	2.4	1.5	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + V238A + D377H + K391A + G476K	1.7	1.8	1.2	1.1	2.1	1.9	1.4	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + V238T + D377H + K391A + G476K	2.0	1.8	1.1	1.1	2.5	1.9	1.3	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + V238G + D377H + K391A + G476K	1.9	1.6	1.1	1.1	2.3	1.8	1.2	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + F328M + D377H + K391A + G476K	1.6	1.8	1.0	1.0	2.1	1.8	1.2	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + F328L + D377H + K391A + G476K	1.6	1.7	0.9	1.0	1.7	1.5	1.0	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + F328V + D377H + K391A + G476K	1.6	1.6	0.8	1.0	1.8	1.7	1.0	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + F328I + D377H + K391A + G476K	1.6	1.6	0.9	1.1	2.0	1.6	1.1	1.0
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + V291T + D377H + K391A + G476K	1.9	1.8	1.2	1.1	2.4	1.9	1.3	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + V291A + D377H + K391A + G476K	2.4	1.9	1.2	1.2	2.9	2.1	1.5	1.3
H1* + N54S + V56T + K72R + G109A + T134E + A174S + G182* + D183* + G184T + N195F + V206L + G255A + K391A + G476K	2.1	1.9	1.0	1.1	2.1	2.0	1.4	1.2
H1* + N54S + V56T + G109A + L173H + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.0	1.8	1.2	1.1	2.6	2.0	1.4	1.2
H1* + N54S + V56T + G109A + Q125T + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.4	1.7	0.9	1.0	2.0	1.8	1.2	1.0
H1* + N16S + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.9	1.6	1.0	1.0	2.1	1.9	1.3	1.2
H1* + N54S + V56T + K72N + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.1	1.6	1.0	1.0	2.1	2.2	1.3	1.3
H1* + N54S + V56T + K72N + G109A + W167L + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.2	1.9	0.9	1.2	1.9	2.1	1.2	1.2
H1* + N54S + V56T + K72A + G109A + W167Y + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.7	1.8	0.8	1.1	1.6	2.0	1.1	1.2
H1* + N54S + V56T + G109A + Q118R + T134E + A174S + G182* + D183* + N195F + V206L + G255A + D377H + K391A + G476K	2.0	1.9	1.0	1.1	2.0	2.0	1.2	1.2
H1* + N54S + V56T + K72R + G109A + A174S + G182* + D183* + N195F + G184T + V206L + K391A + G476K	2.2	2.1	1.1	1.1	3.3	2.5	1.5	1.3

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H1* + N54S + V56T + K72R + G109A + Q118R + A174S + G182* + D183* + G184T + N195F + V206L + D377H + K391A + G476K	0.9	1.3	0.8	0.9	1.3	1.5	1.0	1.1
H1* + N54S + V56T + K72R + G109A + Q118R + A174S + G182* + D183* + N195F + V206L + G255A + D377H + K391A + G476K	2.4	2.1	1.2	1.2	2.8	2.2	1.5	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + G184T + N195F + V206L + R320A + S323N + K391A + G476K	1.9	1.8	1.2	1.1	2.5	2.0	1.5	1.4
H1* + N54S + G109A + A174S + G182* + D183* + N195F + V206L + D377H + K391A + G476K	1.5	1.7	1.1	1.1	2.3	1.8	1.4	1.3
H1* + N54S + V56T + A174S + G182* + D183* + N195F + V206L + D377H + K391A + G476K	1.8	1.7	1.1	0.9	2.1	1.7	1.4	1.2
H1* + N54S + V56T + G109A + G182* + D183* + N195F + V206L + D377H + K391A + G476K	2.0	1.8	1.2	1.2	2.6	1.7	1.3	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + D377H + K391A + G476K	1.6	1.6	1.3	1.1	2.8	2.1	1.5	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + D377H + K391A + G476K	2.0	1.6	1.1	1.1	2.6	2.0	1.3	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + D377H + G476K	2.2	2.0	1.2	1.1	2.6	1.9	1.5	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + D377H + K391A	2.0	1.9	1.2	1.1	2.5	1.8	1.6	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + D377R + K391A + G476K	1.4	1.6	1.1	1.0	1.5	1.7	1.3	1.0
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + L217T + K391A + G476K	1.5	1.7	1.1	1.0	2.3	1.2	1.4	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + D377S + K391A + G476K	2.0	1.8	1.3	1.2	2.4	1.8	1.5	1.4
H1* + N54S + V56T + G109A + R116E + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.0	1.9	1.2	1.1	2.3	2.0	1.5	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + L217Q + K391A + G476K	1.5	1.7	1.0	1.1	1.7	1.4	1.2	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + G346D + K391A + G476K	1.3	1.5	1.0	1.0	1.7	1.6	1.2	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + D377A + K391A + G476K	1.9	1.7	1.1	1.2	2.5	1.7	1.4	1.3
H1* + N54S + V56T + G109A + T134E + A174S + G182* + D183* + N195F + V206L + D377H + K391A + G476K	2.2	2.1	1.2	1.1	2.3	2.1	1.1	1.2
H1* + N54S + V56T + G109A + Q118R + T134E + A174S + G182* + D183* + G184T + N195F + V206L + K391A + G476K	1.9	1.7	1.3	1.2	2.0	1.7	1.2	1.2
H1* + N54S + V56T + K72R + G109A + Q118R + T134E + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.9	1.7	1.1	1.1	2.1	1.9	1.0	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + G346P + K391A + G476K	1.7	1.6	1.2	1.1	2.3	1.9	1.2	1.1
H1* + N54S + V56T + G109A + R116S + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.0	1.8	1.2	1.1	2.1	1.9	1.1	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + D377Q + K391A + G476K	1.7	1.8	1.2	1.1	2.0	2.0	1.2	1.2
H1* + N54S + V56T + G109A + R116I + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.2	1.8	1.2	1.1	2.1	2.0	1.3	1.3
H1* + N54S + V56T + G109A + A174F + G182* + D183* + N195F + V206L + K391A + G476K	1.7	1.5	1.1	1.0	1.5	1.6	1.3	1.2
H1* + N54S + V56T + G109A + R116G + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.0	1.9	1.2	1.2	2.1	2.0	1.0	1.2
H1* + N54S + V56T + G109A + A174T + G182* + D183* + N195F + V206L + K391A + G476K	1.9	1.8	1.2	1.2	1.9	2.0	1.1	1.2
H1* + N54S + V56T + G109A + Q118R + T134E + A174S + G182* + D183* + G184T + N195F + V206L + G255A + K391A + G476K	1.5	1.8	1.1	1.0	1.9	1.9	1.4	1.2
H1* + N54S + V56T + K72R + G109A + Q118R + T134E + A174S + G182* + D183* + N195F + V206L + G255A + K391A + G476K	1.7	1.8	1.1	1.0	2.2	1.8	1.5	1.2
H1* + T51G + N54S + V56T + G109A + Q172R + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.2	1.6	1.0	1.0	1.6	1.4	1.3	1.1
H1* + T51A + N54S + V56T + G109A + Q172D + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.5	1.7	1.0	1.0	2.3	1.9	1.4	1.2
H1* + T51S + N54S + V56T + G109A + Q172M + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.7	1.6	1.0	1.0	2.0	1.8	1.2	1.2
H1* + T51A + N54S + V56T + G109A + Q172S + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.4	1.8	1.3	1.1	2.8	2.1	1.7	1.3
H1* + T51E + N54S + V56T + G109A + Q172T + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.0	1.8	0.9	1.0	2.6	1.9	1.6	1.3
H1* + T51A + N54S + V56T + G109A + Q172A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.9	1.8	1.0	1.1	2.3	1.9	1.6	1.3
H1* + T51S + N54S + V56T + G109A + Q172N + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.2	1.9	1.1	1.1	2.4	1.9	1.5	1.3
H1* + T51G + N54S + V56T + G109A + Q172K + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.4	1.3	1.0	0.9	1.7	1.5	1.2	1.1

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H1* + T51A + N54S + V56T + G109A + Q172G + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.7	1.7	1.2	1.0	2.2	1.9	1.4	1.2
H1* + T51E + N54S + V56T + G109A + Q172L + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.0	1.9	1.0	1.1	2.2	1.8	1.2	1.1
H1* + T51E + N54S + V56T + G109A + Q172R + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.0	1.9	1.1	1.1	1.8	1.7	1.4	1.1
H1* + N54S + V56T + G109A + Q172K + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.1	1.8	1.0	1.1	2.0	1.6	1.3	1.2
H1* + T51D + N54S + V56T + G109A + Q172N + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.8	1.7	1.0	1.0	1.9	1.6	1.3	1.1
H1* + T51A + N54S + V56T + G109A + Q172K + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.7	1.7	1.2	1.1	1.8	1.4	1.3	1.2
H1* + N54S + V56T + K72R + G109A + Q172S + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.0	1.8	1.3	1.1	2.6	1.8	1.4	1.3
H1* + N54S + V56T + K72H + G109A + Q172T + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.0	1.8	1.2	1.1	2.0	1.9	1.3	1.2
H1* + N54S + V56T + K72R + G109A + Q172R + A174S + K179E + G182* + D183* + N195F + V206L + K391A + G476K	1.8	1.6	1.1	1.1	1.4	1.5	1.1	1.1
H1* + N54S + V56T + K72R + G109A + Q172M + A174S + G182* + D183* + N195F + V206L + Y295N + K391A + G476K	1.9	1.8	1.2	1.1	2.1	1.7	1.3	1.2
H1* + N54S + V56T + K72R + G109A + Q172R + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.4	1.4	1.0	1.0	1.5	1.4	1.2	0.9
H1* + N54S + V56T + K72Q + G109A + Q172R + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.0	2.0	1.0	1.0	2.6	2.1	1.3	1.2
H1* + N54S + V56T + K72R + G109A + Q172T + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.1	2.1	1.3	1.1	2.3	2.0	1.4	1.2
H1* + N54S + V56T + K72M + G109A + Q172S + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.3	1.9	1.2	1.1	2.4	2.0	1.2	1.2
H1* + N54S + V56T + K72Q + G109A + Q172M + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.1	2.1	1.3	1.1	2.8	2.2	1.5	1.3
H1* + N54S + V56T + G109A + Q172M + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.9	2.0	1.2	1.1	2.4	1.7	1.4	1.3
H1* + N54S + V56T + K72S + G109A + A174H + G182* + D183* + N195F + V206L + K391A + G476K	2.3	2.0	1.2	1.1	2.2	2.1	1.4	1.4
H1* + N54S + V56T + K72Q + G109A + A174D + G182* + D183* + N195F + V206L + K391A + G476K	2.3	2.1	1.1	1.1	2.2	1.9	1.4	1.3
H1* + N54S + V56T + K72R + G109A + A174D + G182* + D183* + N195F + V206L + K391A + G476K	2.5	2.0	1.1	1.1	2.3	2.1	1.4	1.3
H1* + N54S + V56T + K72Q + G109A + A174K + G182* + D183* + N195F + V206L + K391A + G476K	2.2	1.9	1.2	1.1	2.6	2.0	1.5	1.3
H1* + N54S + V56T + G109A + A174G + G182* + D183* + N195F + V206L + K391A + G476K	2.3	1.9	1.2	1.1	2.6	2.2	1.5	1.4
H1* + N54S + V56T + K72R + G109A + A174Q + G182* + D183* + N195F + V206L + K391A + G476K	2.5	2.1	1.2	1.2	2.3	2.1	1.5	1.4
H1* + N54S + V56T + K72Q + G109A + A174G + G182* + D183* + N195F + V206L + K391A + G476K	2.3	2.1	1.1	1.1	2.6	2.2	1.4	1.4
H1* + N54S + V56T + K72S + G109A + A174N + G182* + D183* + N195F + V206L + K391A + G476K	2.6	2.1	1.3	1.1	2.6	2.3	1.5	1.4
H1* + N54S + V56T + K72R + G109A + A174H + G182* + D183* + N195F + V206L + K391A + G476K	2.1	1.9	1.3	1.2	2.5	2.2	1.5	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + K391S + G476K	1.7	1.7	1.1	1.1	1.9	1.6	1.4	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + T444A + G476K	1.7	1.6	1.2	1.1	2.2	1.6	1.4	1.2
H1* + N54S + V56T + G109A + F113L + W167F + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.7	1.6	1.0	1.0	2.1	1.7	1.2	1.2
H1* + G50A + N54S + V56T + G109A + A174E + G182* + D183* + N195F + A204T + V206L + S323N + K391A + G476K	2.0	1.8	1.0	1.1	1.5	1.7	1.3	1.2
H1* + N54S + V56T + K72R + G109A + Q118R + A174S + G182* + D183* + G184T + N195F + V206L + G255A + K391A + G476K	1.6	1.6	1.1	1.0	1.2	1.3	1.2	1.2
H1* + N54S + V56T + M105F + G109A + A174S + G182* + D183* + N195F + V206L + D377H + K391A + G476K	1.8	1.9	1.2	1.1	2.6	1.9	1.5	1.3
H1* + N54S + V56T + K72R + G109A + Q118R + A174S + G182* + D183* + N195F + V206L + D377H + K391A + G476K	1.5	1.6	1.1	1.1	1.4	1.3	1.3	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + A204G + V206L + K391A + G476K	2.7	2.1	1.2	1.2	2.9	2.6	1.6	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + A225F + K391A + G476K	1.8	1.8	1.0	0.9	1.8	1.7	1.4	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + M246A + K391A + G476K	1.8	1.6	0.9	0.9	1.7	1.8	1.2	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + Y382I + K391A + G476K	2.1	1.7	0.9	0.9	1.8	1.5	1.3	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + S376H + K391A + G476K	1.8	1.7	1.1	0.9	2.1	2.0	1.3	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + I214E + K391A + G476K	1.7	1.5	1.1	0.9	2.1	2.1	1.2	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + I214W + K391A + G476K	2.2	1.9	1.1	1.1	2.4	2.0	1.4	1.3

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H1* + N54S + V56T + K72R + G109A + Q172K + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.7	1.4	0.9	1.0	2.1	1.7	1.2	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + A204T + V206L + D377H + K391A + G476K	1.9	1.9	1.1	1.1	2.7	2.1	1.5	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + A204S + V206L + D377H + K391A + G476K	2.1	1.6	1.2	1.1	2.2	1.8	1.3	1.3
H1* + N54S + V56T + G109A + T165S + A174S + G182* + D183* + N195F + V206L + D377H + K391A + G476K	1.9	1.8	1.1	1.1	2.3	2.0	1.4	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + T334S + D377H + K391A + G476K	2.0	1.6	1.0	1.0	2.3	1.7	1.5	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + A263G + D377H + K391A + G476K	2.0	1.6	1.0	1.0	2.1	1.6	1.4	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + M286F + D377H + K391A + G476K	2.0	1.6	1.1	1.0	2.3	1.7	1.3	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + Y267I + D377H + K391A + G476K	1.9	1.6	1.0	1.0	1.8	1.7	1.3	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + L235V + D377H + K391A + G476K	2.1	1.8	1.1	1.0	2.5	2.1	1.5	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + L235A + D377H + K391A + G476K	1.6	1.6	0.8	0.9	2.3	1.9	1.3	1.2
H1* + N54S + V56T + M105L + G109A + A174S + G182* + D183* + N195F + V206L + D377H + K391A + G476K	2.1	1.9	1.1	1.1	2.9	2.1	1.5	1.3
H1* + N54S + V56T + M105I + G109A + A174S + G182* + D183* + N195F + V206L + D377H + K391A + G476K	2.2	2.0	1.1	1.1	2.3	2.0	1.6	1.4
H1* + N54S + V56T + M105V + G109A + A174S + G182* + D183* + N195F + V206L + D377H + K391A + G476K	2.3	2.1	1.2	1.1	2.6	2.1	1.5	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + M246I + D377H + K391A + G476K	2.2	1.9	1.3	1.1	2.6	2.0	1.7	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + L250I + D377H + K391A + G476K	2.3	2.0	1.3	1.1	2.3	1.9	1.6	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + L250V + D377H + K391A + G476K	2.2	1.9	1.3	1.2	2.5	2.0	1.6	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391Y + G476K	2.2	1.9	1.2	1.1	2.1	1.9	1.6	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391V + G476K	2.1	1.9	1.0	1.0	2.1	1.8	1.4	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391M + G476K	2.2	1.8	1.3	1.1	2.6	2.1	1.5	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391E + G476K	1.8	1.6	1.2	1.0	2.1	1.9	1.3	1.1
H1* + N54S + V56T + G109A + T165G + A174S + G182* + D183* + N195F + V206L + D377H + K391A + G476K	2.3	2.0	1.3	1.1	2.9	2.0	1.7	1.3
H1* + N54R + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.3	1.4	1.1	1.1	1.1	1.1	1.1	0.8
H1* + N54L + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.6	1.4	1.1	1.0	1.3	1.6	1.3	1.2
H1* + N54S + V56T + G109S + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.4	2.4	1.4	1.2	2.3	2.0	1.4	1.3
H1* + N54S + V56T + G109A + A174V + G182* + D183* + N195F + V206L + K391A + G476K	2.2	2.2	1.3	1.3	2.3	2.0	1.5	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + M246F + D377H + K391A + G476K	2.6	2.3	1.3	1.1	2.5	2.0	1.6	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + M246V + D377H + K391A + G476K	2.2	1.8	1.3	1.2	1.9	1.8	1.4	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + M246S + D377H + K391A + G476K	2.3	1.9	1.2	1.2	1.8	1.6	1.4	1.1
H1* + N54S + V56T + G109A + A174R + G182* + D183* + N195F + V206L + K391A + G476K	1.7	1.6	1.2	1.1	1.7	1.6	1.4	1.2
H1* + N54S + V56T + G109A + A174L + G182* + D183* + N195F + V206L + K391A + G476K	1.9	1.7	1.2	1.2	2.2	1.7	1.3	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206S + K391A + G476K	1.8	1.6	1.0	1.1	2.0	1.7	1.3	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391D + G476K	1.9	2.0	1.3	1.2	2.2	2.0	1.5	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391R + G476K	2.3	1.8	1.3	1.2	2.5	1.9	1.4	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391H + G476K	2.2	1.9	1.2	1.2	2.3	2.0	1.5	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391W + G476K	2.2	1.6	1.3	1.2	2.5	2.0	1.4	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391I + G476K	2.1	1.8	1.2	1.2	2.0	1.8	1.3	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + L250T + D377H + K391A + G476K	2.0	1.6	1.1	1.1	2.6	2.0	1.4	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + L250A + D377H + K391A + G476K	2.1	1.8	1.0	1.0	2.6	2.0	1.5	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + L250F + D377H + K391A + G476K	2.2	1.7	1.1	1.1	2.4	2.0	1.6	1.2

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H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + L250M + D377H + K391A + G476K	2.2	1.7	1.1	1.1	2.0	1.9	1.4	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391Q + G476K	1.8	1.4	1.1	1.0	1.5	1.7	1.2	1.2
H1* + N54S + V56T + G109A + Q172H + L173A + A174P + G182* + D183* + N195F + V206L + K391A + G476K	2.5	1.8	1.3	1.2	2.3	2.1	1.6	1.4
H1* + N54S + V56T + G109A + Q172E + L173A + A174R + G182* + D183* + N195F + V206L + K391A + G476K	1.1	1.4	0.7	1.0	1.8	1.7	1.2	1.2
H1* + N54S + V56T + K72Q + G109A + W167R + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.9	1.7	1.2	1.1	1.6	1.6	1.0	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + G184T + N195F + V206L + G255A + K391A + G476K	1.8	1.7	1.1	1.0	2.0	1.7	1.3	1.2
H1* + N54S + V56T + K72R + G109A + A174S + G182* + D183* + G184T + N195F + V206L + D377H + K391A + G476K	1.7	1.6	1.1	1.1	1.9	1.6	1.1	1.0
H1* + N54S + V56T + G109A + A174S + G182* + D183* + G184T + N195F + V206L + G182* + D183* + G184T + N195F + V206L + G255A + D377H + K391A + G476K	2.4	2.1	1.3	1.2	2.3	2.0	1.4	1.2
H1* + N54S + V56T + G109A + Q118R + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.6	1.5	1.1	1.2	0.7	1.1	0.8	1.0
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + S376T + K391A + G476K	2.2	2.0	1.3	1.2	2.3	1.8	1.3	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + S376V + K391A + G476K	2.0	1.8	1.2	1.2	2.1	1.8	1.4	1.3
H1* + N54S + V56T + G109A + Q172E + L173I + A174N + G182* + D183* + N195F + V206L + K391A + G476K	2.3	1.9	1.1	1.1	2.1	1.9	1.1	1.2
H1* + N54S + V56T + G109A + Q172D + L173A + A174T + G182* + D183* + N195F + V206L + K391A + G476K	2.3	2.0	1.1	1.1	2.2	2.0	1.1	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + Y267M + D377H + K391A + G476K	2.1	2.0	1.2	1.2	2.6	2.0	1.5	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + Y267H + D377H + K391A + G476K	1.7	1.6	1.1	1.0	2.5	1.9	1.2	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + V238A + K391A + G476K	1.6	1.5	1.1	1.0	2.3	1.7	1.2	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + V238T + K391A + G476K	1.5	1.3	1.1	0.9	1.7	1.5	1.2	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + V238G + K391A + G476K	2.3	1.7	1.2	1.1	3.0	2.1	1.4	1.3
H1* + N54S + V56T + M105V + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.0	1.7	1.2	1.1	2.3	2.1	1.5	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + V264T + K391A + G476K	2.2	2.1	1.2	1.0	2.9	1.9	1.4	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + V264I + K391A + G476K	2.4	1.8	1.3	1.1	2.9	2.3	1.5	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + L235V + K391A + G476K	1.9	1.7	1.1	1.0	2.4	1.9	1.3	1.3
H1* + N54S + V56T + G109A + W167H + Q172N + L173A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.0	1.7	1.2	1.1	2.9	2.1	1.4	1.3
H1* + N54S + V56T + G109A + W167F + Q172E + L173P + A174K + G182* + D183* + N195F + V206L + K391A + G476K	1.7	1.9	1.2	1.1	3.3	2.2	1.5	1.3
H1* + N54S + V56T + G109A + Q172N + L173T + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.0	1.7	1.2	1.1	2.8	2.0	1.4	1.4
H1* + N54S + V56T + G109A + Q172K + L173V + A174T + G182* + D183* + N195F + V206L + K391A + G476K	1.6	1.4	1.0	1.1	1.6	1.6	1.0	1.0
H1* + N54S + V56T + G109A + Q172E + L173F + A174R + G182* + D183* + N195F + V206L + K391A + G476K	1.9	1.6	1.1	1.0	2.0	1.9	1.3	1.1
H1* + N54S + V56T + G109A + Q172E + L173T + A174R + G182* + D183* + N195F + V206L + K391A + G476K	1.9	1.8	1.2	1.2	2.5	2.1	0.9	1.3
H1* + N54S + V56T + G109A + Q172N + L173V + A174R + G182* + D183* + N195F + V206L + K391A + G476K	1.9	1.7	1.3	1.2	2.2	1.8	0.9	1.2
H1* + N54S + V56T + G109A + L173V + A174R + G182* + D183* + N195F + V206L + K391A + G476K	1.7	1.8	1.3	1.2	1.7	1.6	0.8	1.1
H1* + N54S + V56T + G109A + Q172E + L173M + A174H + G182* + D183* + N195F + V206L + K391A + G476K	2.1	2.1	1.3	1.2	2.8	2.1	1.3	1.3
H1* + N54S + V56T + G109A + Q172E + L173F + A174P + G182* + D183* + N195F + V206L + K391A + G476K	1.9	1.9	1.2	1.2	2.7	2.3	0.8	1.3
H1* + N54S + V56T + G109A + W167Y + Q172E + L173V + A174H + G182* + D183* + N195F + V206L + K391A + G476K	2.0	1.8	1.1	1.0	2.8	2.0	1.4	1.3
H1* + N54S + V56T + G109A + L173A + A174T + G182* + D183* + N195F + V206L + K391A + G476K	2.0	1.8	1.0	1.0	2.4	1.8	1.4	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + V291A + F328L + K391A + G476K	2.0	1.6	1.1	1.0	2.9	1.7	1.4	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + M246L + K391A + G476K	2.2	1.7	1.0	1.0	2.8	1.7	1.2	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + M246S + K391A + G476K	1.8	1.8	1.2	1.1	2.9	2.0	1.5	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + L250V + K391A + G476K	2.0	1.9	1.3	1.2	2.8	2.0	1.6	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + L250F + K391A + G476K	2.0	1.8	1.1	1.0	2.9	2.1	1.5	1.3

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H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + L250M + K391A + G476K	2.3	1.7	1.2	1.1	3.0	2.1	1.5	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + C474V + G476K	1.9	1.8	1.3	1.0	2.8	2.0	1.3	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + A204S + V206L + K391A + G476K	1.7	1.6	1.0	1.0	1.7	1.8	1.4	1.3
H1* + N54S + V56T + G109A + T165G + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.3	1.9	1.2	1.1	2.9	2.2	1.3	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + T334S + K391A + G476K	1.9	1.7	1.0	1.0	2.6	2.0	1.4	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + A263G + M286F + K391A + G476K	1.1	1.3	1.0	0.9	1.4	1.3	1.1	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + A263G + M286L + K391A + G476K	1.7	1.5	1.1	1.0	2.2	1.9	1.3	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + Y267H + K391A + G476K	1.9	1.9	1.2	1.2	2.7	2.2	1.5	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + Y267L + K391A + G476K	1.9	1.8	1.0	1.1	2.4	2.2	1.5	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + Y267I + K391A + G476K	1.2	1.5	0.7	0.9	1.7	1.9	1.3	1.3
H1* + T51A + N54S + V56T + G109A + Q172R + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.1	1.5	1.1	1.0	1.5	1.6	1.4	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391L + G476K	2.4	1.9	1.1	1.1	2.5	1.9	1.4	1.2
H1* + N54S + V56T + G109A + F113Q + R116H + Q172G + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.6	2.1	1.1	1.0	3.1	2.1	1.4	1.2
H1* + N54S + V56T + G109A + F113Q + Q172G + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.9	1.8	0.9	0.9	2.7	1.9	1.2	1.1
H1* + N54S + V56T + G109S + F113Q + R116Q + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.9	2.2	1.2	1.1	3.3	2.6	1.5	1.3
H1* + N54S + V56T + G109A + Q172D + A174S + G182* + D183* + N195F + A204G + V206L + K391A + P473R + G476K	2.7	2.2	1.2	1.1	3.1	2.1	1.5	1.3
H1* + N54S + V56T + K72R + G109A + T134E + A174S + G182* + D183* + N195F + A204G + V206L + G255A + K391A + P473R + G476K	2.8	2.3	1.2	1.1	3.4	2.4	1.7	1.3
H1* + N54S + V56T + K72R + G109A + W167F + A174S + G182* + D183* + N195F + A204G + V206L + K391A + G476K	2.7	2.2	1.2	1.1	3.1	2.2	1.6	1.3
H1* + N54S + V56T + K72R + G109A + A174S + G182* + D183* + G184T + N195F + A204G + V206L + K391A + P473R + G476K	2.3	2.1	1.2	1.1	2.7	2.0	1.4	1.3
H1* + N54S + V56T + G109A + W167F + Q172N + A174S + G182* + D183* + N195F + V206L + K391A + P473R + G476K	2.5	1.8	1.3	1.2	2.6	1.8	1.6	1.2
H1* + N54S + V56T + G109A + W167F + Q172G + A174S + G182* + D183* + N195F + V206L + K391A + P473G + G476K	2.4	2.0	1.3	1.2	2.8	2.0	1.6	1.2
H1* + N54S + V56T + G109A + W167F + Q172G + A174S + G182* + D183* + N195F + V206L + K391A + P473R + G476K	2.1	1.8	1.3	1.1	2.3	1.8	1.5	1.2
H1* + N54S + V56T + K72R + G109A + W167F + Q172R + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.2	1.9	1.1	1.1	2.0	1.8	1.2	1.1
H1* + N54S + V56T + K72R + G109A + R116H + W167F + Q172R + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.4	1.8	1.2	1.1	2.4	1.9	1.5	1.2
H1* + N54S + V56T + M105F + G109A + A174S + G182* + D183* + N195F + V206L + R320A + S323N + K391A + C474V + G476K	2.6	2.2	1.3	1.3	3.4	2.4	1.5	1.4
H1* + N54S + V56T + K72R + G109A + A174S + G182* + D183* + N195F + V206L + G255A + K391A + C474V + G476K	2.4	2.1	1.4	1.3	3.2	2.2	1.7	1.3
H1* + N54S + V56T + G109A + T134E + A174S + G182* + D183* + N195F + V206L + K391A + C474V + G476K	2.8	2.3	1.3	1.2	3.7	2.3	1.6	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + D377H + K391A + C474V + G476K	2.6	2.0	1.3	1.2	3.0	2.1	1.5	1.3
H1* + N54S + V56T + K72R + G109A + A174S + G182* + D183* + N195F + V206L + K391A + C474V + G476K	2.8	2.1	1.3	1.2	3.3	2.4	1.5	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + A265G + K391A + T444Q + G476K + G477A	2.1	1.9	1.0	1.1	2.6	2.0	1.5	1.3
H1* + A37V + N54S + V56T + G109A + W167F + Q172G + A174S + G182* + D183* + N195F + V206L + K391A + G346T + G477A + G476K	2.3	2.0	1.2	1.2	2.8	2.2	1.5	1.3
H1* + N54S + V56T + G109A + R116Q + Q172D + A174S + G182* + D183* + N195F + V206L + G346T + K391A + T444Q + G477A + G476K	2.3	2.1	0.9	1.1	3.1	2.4	1.4	1.3
H1* + N54S + V56T + G109A + R116Q + W167F + Q172N + A174S + G182* + D183* + N195F + V206L + G346T + K391A + T444Q + G477A + G476K	2.2	2.2	1.1	1.2	2.7	2.4	1.3	1.3
H1* + N54S + V56T + G109S + W167F + Q172R + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.9	1.4	1.2	1.1	2.1	1.6	1.3	1.2
H1* + N54S + V56T + G109S + R116Q + W167F + Q172N + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.8	2.2	1.4	1.2	3.2	3.0	1.5	1.5
H1* + N54S + V56T + G109S + W167F + Q172G + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.6	1.9	1.4	1.1	3.2	2.2	1.6	1.5

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H1* + N54S + V56T + G109A + F113Q + Q172G + A174S + G182* + D183* + N195F + V206L + K391A + T444Q + G476K	2.5	2.0	1.5	1.2	3.2	2.2	1.5	1.4
H1* + N54S + V56T + G109A + F113Q + R116H + Q172G + A174S + G182* + D183* + N195F + V206L + K391A + T444Q + G476K	2.6	2.2	1.3	1.1	3.5	2.6	1.7	1.4
H1* + N54S + V56T + G109A + F113Q + A174S + G182* + D183* + N195F + V206L + K391A + C474V + G476K	2.5	2.0	1.3	1.1	3.1	2.6	1.6	1.5
H1* + N54S + V56T + G109A + Q172N + A174S + G182* + D183* + N195F + V206L + V264I + K391A + C474V + G476K	2.4	1.7	1.4	1.1	3.2	2.4	1.5	1.4
H1* + N54S + V56T + G109A + Q172N + A174S + G182* + D183* + N195F + V206L + K391A + C474V + G476K	2.5	1.7	1.3	1.1	2.0	2.1	1.3	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + S280Q + H321Y + K391A + C474V + G476K	2.1	1.8	1.2	1.1	2.9	2.1	1.5	1.4
H1* + N54S + V56T + K72R + G109A + T134E + A174S + G182* + D183* + N195F + V206L + G255A + K391A + C474V + G476K	2.1	1.8	1.2	1.0	3.3	2.3	1.6	1.4
H1* + N54S + V56T + G109A + F113Q + R116H + Q172G + A174S + G182* + D183* + N195F + V206L + K391A + T444Q + P473R + G476K	2.4	2.0	1.2	1.1	3.3	2.3	1.7	1.3
H1* + N54S + V56T + G109A + F113Q + R116Q + Q172G + A174S + G182* + D183* + N195F + V206L + K391A + T444Q + G476K	2.8	2.1	1.3	1.1	3.1	2.3	1.5	1.3
H1* + N54S + V56T + G109A + R116H + W167F + Q172N + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.4	2.0	1.2	1.1	2.8	2.2	1.5	1.3
H1* + N54S + V56T + G109A + F113Q + R116H + T165G + Q172G + A174S + G182* + D183* + N195F + V206L + K391A + T444Q + G476K	2.4	2.0	1.2	1.1	3.1	2.2	1.6	1.2
H1* + N54S + V56T + G109A + F113Q + R116Q + T165G + Q172G + A174S + G182* + D183* + N195F + V206L + K391A + T444Q + G476K	2.5	2.1	1.3	1.2	3.4	2.3	1.7	1.4
H1* + N54S + V56T + G109A + F113Q + R116H + W167F + Q172R + A174S + G182* + D183* + N195F + V206L + F289I + K391A + G476K	2.3	1.8	1.2	1.1	2.6	2.0	1.4	1.3
H1* + N54S + V56T + G109A + F113Q + Q172N + A174S + G182* + D183* + N195F + V206L + K391A + T444Q + G476K	2.6	2.2	1.3	1.2	3.3	2.3	1.6	1.3
H1* + N54S + V56T + G109A + F113Q + R116H + Q172N + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.6	2.0	1.3	1.2	3.2	2.2	1.7	1.4
H1* + N54S + V56T + G109A + F113Q + R116Q + Q172M + A174S + G182* + D183* + N195F + V206L + K391A + T444Q + G476K	2.2	2.0	1.2	1.1	2.8	2.2	1.3	1.3
H1* + N54S + V56T + G109A + F113Q + R116H + Q172M + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.2	1.9	1.2	1.0	3.0	2.4	1.5	1.3
H1* + N54S + V56T + G109A + R116H + Q172M + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.3	2.1	1.1	1.0	2.7	2.1	1.5	1.3
H1* + N54S + V56T + G109A + F113Q + R116H + T165G + Q172M + A174S + G182* + D183* + N195F + V206L + K391A + Q395P + G476K	1.9	1.7	1.2	1.1	2.6	1.8	1.5	1.3
H1* + N54S + V56T + G109A + F113Q + A174S + G182* + D183* + N195F + V206L + K391A + T444Q + P473R + G476K	1.7	1.3	0.9	1.0	1.9	1.6	0.9	1.0
H1* + N54S + V56T + G109A + R116Q + A174S + G182* + D183* + N195F + V206L + A265G + K391A + T444Q + P473R + G476K	2.4	2.2	1.3	1.2	2.7	2.4	1.5	1.4
H1* + N54S + V56T + G109A + R116Q + A174S + G182* + D183* + N195F + V206L + K391A + T444Q + P473R + G476K	2.4	2.1	1.3	1.1	2.9	2.3	1.3	1.3
H1* + N54S + V56T + K72R + G109A + T134E + T165G + A174S + G182* + D183* + N195F + V206L + G255A + K391A + G476K	2.5	2.1	1.2	1.1	3.0	2.3	1.4	1.3
H1* + N54S + V56T + G109A + F113Q + R116H + W167F + Q172R + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.1	2.0	1.2	1.2	2.9	2.3	1.5	1.3
H1* + N54S + V56T + G109A + R116Q + W167F + Q172R + A174S + G182* + D183* + N195F + V206L + A265G + K391A + P473R + G476K	2.0	1.8	1.2	1.1	2.2	2.0	1.3	1.3
H1* + N54S + V56T + G109A + F113Q + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.1	1.9	1.0	1.0	2.6	2.0	1.4	1.2
H1* + N54S + V56T + G109A + Q172N + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.0	1.7	1.2	1.1	2.5	1.9	1.5	1.1
H1* + T51V + N54S + V56T + G109A + Q172D + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.4	1.9	1.2	1.1	2.6	1.9	1.6	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + S280Q + H321Y + K391A + G476K	2.3	1.9	1.2	1.1	2.5	1.8	1.5	1.2
H1* + N54S + V56T + G109A + F113Q + R116H + Q172N + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.6	2.3	1.3	1.2	3.1	2.4	1.7	1.4
H1* + N54S + V56T + G109A + F113Q + R116H + Q172G + L173V + A174S + G182* + D183* + N195F + V206L + K391A + T444Q + P473R + G476K	2.3	2.0	1.3	1.2	2.9	2.3	1.6	1.3
H1* + N54S + V56T + G109A + R116Q + W167F + Q172N + L173V + A174S + G182* + D183* + N195F + V206L + K391A + T444Q + G476K	2.6	2.2	1.3	1.1	3.3	2.5	1.8	1.3

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H1* + N54S + V56T + G109A + G133Q + W167F + Q172N + A174S + G182* + D183* + N195F + V206L + Q395P + K391A + G476K	2.5	2.3	1.3	1.2	3.3	2.4	1.8	1.4
H1* + N54S + V56T + G109A + F113Q + R116H + W167F + Q172N + L173V + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.3	2.0	1.2	1.2	2.8	2.3	1.6	1.3
H1* + A37V + N54S + V56T + K72R + G109A + R116H + W167F + Q172R + A174S + G182* + D183* + G184T + N195F + V206L + K391A + P473R + G476K	2.5	2.2	1.3	1.1	3.3	2.5	1.7	1.2
H1* + N54S + V56T + K72R + G109A + F113Q + R116H + W167F + Q172G + L173V + A174S + G182* + D183* + G184T + N195F + V206L + K391A + T444Q + P473R + G476K	2.2	2.2	1.2	1.1	3.1	2.4	1.7	1.3
H1* + N54S + V56T + G109S + R116Q + W167F + Q172R + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.5	2.2	1.2	1.0	3.1	2.5	1.6	1.3
H1* + N54S + V56T + G109S + R116Q + W167F + Q172G + A174S + G182* + D183* + N195F + V206L + K391A + P473R + G476K	2.5	2.1	1.2	1.1	3.0	2.3	1.6	1.2
H1* + N54S + V56T + G109S + F113Q + W167F + Q172N + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.7	2.4	1.4	1.3	3.1	2.5	1.7	1.2
H1* + A37V + N54S + V56T + G109S + F113Q + W167F + Q172R + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.8	2.4	1.3	1.2	3.5	2.5	1.7	1.4
H1* + N54S + V56T + K72R + G109A + R116H + T134E + W167F + Q172G + L173V + A174S + G182* + D183* + N195F + V206L + G255A + K391A + G476K	2.2	1.9	1.2	1.1	2.7	2.4	1.6	1.3
H1* + N54S + V56T + K72R + G109A + R116H + T134E + W167F + Q172G + L173V + A174S + G182* + D183* + N195F + V206L + G255A + K391A + Q395P + T444Q + P473R + G476K	2.5	2.0	1.2	1.1	3.5	2.5	1.6	1.3
H1* + N54S + V56T + K72R + G109A + F113Q + R116Q + W167F + Q172G + A174S + G182* + D183* + G184T + N195F + V206L + K391A + P473R + G476K	2.5	2.1	1.3	1.1	3.4	2.5	1.6	1.3
H1* + N54S + V56T + K72R + G109A + F113Q + R116Q + T134E + W167F + Q172R + A174S + G182* + D183* + N195F + V206L + G255A + K391A + T444Q + P473R + G476K	2.3	2.0	1.1	1.1	3.1	2.6	1.5	1.3
H1* + N54S + V56T + K72R + G109A + F113Q + T134E + W167F + Q172R + A174S + G182* + D183* + N195F + V206L + G255A + K391A + Q395P + A445Q + P473R + G476K	1.9	1.9	1.1	1.1	2.7	2.1	1.6	1.3
H1* + N54S + V56T + K72R + G109A + F113Q + R116H + T134E + W167F + Q172R + A174S + G182* + D183* + N195F + V206L + G255A + K391A + Q395P + G476K	2.1	2.2	1.3	1.2	2.9	2.4	1.6	1.4
H1* + N54S + V56T + K72R + G109A + T134E + W167F + Q172N + A174S + G182* + D183* + N195F + V206L + G255A + K391A + T444Q + P473R + G476K	2.6	2.4	1.3	1.2	3.2	2.5	1.7	1.4
H1* + N54S + V56T + K72R + G109A + G133Q + W167F + Q172R + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.4	2.2	1.3	1.2	3.2	2.4	1.6	1.4
H1* + N54S + V56T + K72R + G109A + F113Q + R116Q + W167F + Q172G + A174S + G182* + D183* + N195F + A204G + V206L + K391A + G476K	2.9	2.3	1.3	1.2	3.7	2.8	1.8	1.4
H1* + N54S + V56T + K72R + G109A + F113Q + R116Q + W167F + Q172N + L173V + A174S + G182* + D183* + G184T + N195F + V206L + K391A + T444Q + P473R + G476K	2.0	2.0	1.3	1.2	2.7	2.1	1.5	1.3
H1* + A37H + N54S + V56T + G109A + F113Q + W167F + Q172N + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.5	1.9	1.2	1.1	2.6	1.8	1.6	1.3
H1* + A37H + N54S + V56T + G109A + F113Q + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.9	1.7	1.3	1.2	2.8	2.0	1.3	1.3
H1* + N54S + V56T + G109A + F113Q + W167F + Q172N + A174S + G182* + D183* + N195F + V206L + K391A + T459N + G476K	2.3	1.6	1.3	1.2	2.8	1.8	1.5	1.2
H1* + N54S + V56T + G109A + F113Q + Q172R + L173V + A174S + G182* + D183* + N195F + V206L + K391A + T444Q + G476K	1.9	1.5	1.1	1.0	2.6	1.5	1.3	1.1
H1* + N54S + V56T + G109A + F113Q + R116Q + A174S + G182* + D183* + N195F + V206L + K391A + P473R + G476K	2.3	2.2	1.1	1.1	3.2	2.3	1.4	1.2
H1* + N54S + V56T + A60V + G109A + R116Q + W167F + Q172N + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.4	2.2	1.2	1.2	3.3	2.4	1.3	1.4
H1* + N54S + V56T + A60V + G109A + R116H + Q172R + L173V + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.7	2.0	1.4	1.2	3.2	2.5	1.3	1.4
H1* + A37V + N54S + V56T + A60V + G109A + R116Q + T165G + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.5	1.9	1.1	1.2	2.9	2.1	1.1	1.3
H1* + A37H + N54S + V56T + A60V + G109A + R116Q + T165G + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.7	2.2	1.2	1.2	3.6	2.5	1.7	1.4

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H1* + N54S + V56T + A60V + G109A + R116Q + W167F + Q172G + L173V + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.3	2.3	1.0	1.2	2.5	2.3	1.2	1.3
H1* + N54S + V56T + A60V + G109A + R116Q + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.3	2.1	0.9	1.1	3.1	2.4	1.4	1.3
H1* + A37H + N54S + V56T + A60V + G109A + R116Q + W167F + Q172R + A174S + G182* + D183* + N195F + V206L + K391A + G476K	3.0	2.4	1.2	1.1	3.6	2.4	1.7	1.3
H1* + A37H + N54S + V56T + A60V + G109A + R116Q + W167F + Q172N + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.7	2.7	1.1	1.1	3.0	2.4	1.4	1.3
H1* + N54S + V56T + A60V + G109A + R116Q + T165G + A174S + G182* + D183* + N195F + V206L + K391A + T444Q + P473R + G476K	2.4	2.2	1.2	1.2	3.0	2.2	1.3	1.2
H1* + N54S + V56T + A60V + G109A + R116Q + A174S + G182* + D183* + N195F + V206L + K391A + T444Q + P473R + G476K	2.5	1.8	0.9	1.0	2.2	2.0	1.1	1.2
H1* + N54S + V56T + A60V + G109A + R116Q + W167F + Q172N + L173V + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.7	2.8	1.0	1.1	3.3	2.7	1.6	1.4
H1* + N54S + V56T + A60V + G109A + R116Q + W167F + Q172R + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.8	2.5	1.3	1.2	3.6	2.7	1.7	1.4
H1* + N54S + V56T + G109A + R116A + W167F + Q172R + A174S + G182* + D183* + N195F + A204G + V206L + K391A + P473R + G476K	2.3	2.0	1.1	1.1	2.5	2.0	1.6	1.3
H1* + N54S + V56T + G109A + R116A + A174S + G182* + D183* + N195F + A204G + V206L + K391A + P473R + G476K	2.8	2.5	1.3	1.2	3.4	2.3	1.6	1.5
H1* + N54S + V56T + G109A + R116A + T165G + A174S + G182* + D183* + N195F + V206L + K391A + T444Q + P473R + G476K	2.5	2.1	1.2	1.1	2.8	2.0	1.6	1.3
H1* + N54S + V56T + G109A + G182* + D183* + N195F + K391A + G476K	2.2	2.2	1.2	1.1	2.3	2.1	1.5	1.5
H1* + N54S + V56T + G109A + G182* + D183* + N195F + V206L	2.2	2.0	1.3	1.2	2.4	1.9	1.5	1.4
N54S + V56T + G109A + G182* + D183* + N195F	2.2	2.0	1.3	1.2	2.3	2.0	1.5	1.4
N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.4	2.0	1.3	1.2	2.5	2.0	1.5	1.4
H1* + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.9	1.9	1.3	1.2	2.4	2.0	1.5	1.3
H1* + N54S + V56T + G109A + G182* + D183* + N195F + V206L + K391A + G476K	2.2	1.9	1.2	1.1	2.5	1.7	1.3	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A	2.0	1.9	1.2	1.1	2.2	1.8	1.5	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + G476K	2.0	1.9	1.2	1.1	2.5	1.9	1.4	1.3
H1* + N54S + V56T + A174S + G182* + D183* + N195F + V206L + G273R + K391A + G476K	2.3	2.0	1.2	1.1	2.6	2.1	1.4	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L	2.4	2.1	1.3	1.1	2.7	2.1	1.6	1.4
H1* + W48G + G109A + G182* + D183* + N195F + V206L + K391A + G476K	1.7	1.6	1.2	1.1	2.2	1.7	1.3	1.2
N54S + V56T + G109A + A174S + G182* + D183* + N195F + K391A + G476K	2.0	1.7	1.2	1.2	2.9	2.2	1.5	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + K391A	1.4	1.5	1.1	1.1	2.3	1.8	1.3	1.3
H1* + N54S + V56T + G109A + F113Q + Q172M + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.0	1.9	1.2	1.1	2.9	2.2	1.3	1.2
H1* + N54S + V56T + G109A + R116Q + A174S + G182* + D183* + N195F + A204G + V206L + K391A + G476K	1.8	1.6	0.9	1.0	1.7	1.8	1.0	1.1
H1* + N54S + V56T + K72R + G109A + F113Q + T134E + A174S + G182* + D183* + N195F + V206L + G255A + K391A + G476K	2.5	2.2	1.2	1.2	2.9	2.5	1.4	1.4
H1* + N54S + V56T + K72R + G109A + F113Q + T134E + A174S + G182* + D183* + N195F + V206L + G255A + K391A + P473R + G476K	2.7	2.3	1.3	1.2	3.3	2.5	1.7	1.4
H1* + N54S + V56T + K72R + G109A + R116Q + A174S + G182* + D183* + G184T + N195F + V206L + K391A + G476K	2.7	2.2	1.2	1.2	3.4	2.4	1.6	1.3
H1* + A37V + N54S + V56T + G109A + R116A + T165G + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.4	2.2	1.2	1.2	3.1	2.5	1.5	1.3
H1* + A37V + N54S + V56T + G109A + R116Q + T165G + A174S + G182* + D183* + N195F + V206L + M246F + K391A + G476K	2.5	2.2	1.2	1.2	3.0	2.3	1.5	1.3
H1* + V56A + G109A + A174S + G182* + D183* + N195F + V206L + G476K	2.0	2.0	1.1	1.1	2.5	2.1	1.5	1.3

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H1* + N54S + V56T + G109A + G182* + D183* + N195F + V206L + K391A	1.8	2.0	1.2	1.0	2.2	1.9	1.6	1.4
H1* + N54S + V56T + G109A + G182* + D183* + N195F + V206L + G476K	1.8	1.7	1.0	1.1	1.9	1.9	1.4	1.3
G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.5	1.6	1.0	1.0	1.7	1.7	1.2	1.2
N54S + V56T + G109A + G182* + D183* + N195F + V206L + K391T + G476K	1.7	1.7	1.2	1.0	1.6	1.7	1.4	1.3
H1* + N54S + V56T + K72R + G109A + F113Q + R116H + A174S + G182* + D183* + G184T + N195F + V206L + K391A + T444Q + G476K	2.4	2.2	1.2	1.2	3.4	2.3	1.4	1.4
H1* + N54S + V56T + G109S + F113Q + A174S + G182* + D183* + N195F + V206L + K391A + T444Q + G476K	2.4	2.0	1.3	1.2	3.1	2.2	1.6	1.3
H1* + N54S + V56T + G109A + F113Q + R116H + W167F + Q172G + L173V + A174S + G182* + D183* + N195F + V206L + K391A + P473R + G476K	2.9	2.3	1.3	1.2	3.8	2.6	1.8	1.4
H1* + N54S + V56T + G109A + F113Q + R116Q + Q172N + A174S + G182* + D183* + N195F + V206L + A265G + K391A + P473R + G476K	2.8	2.3	1.2	1.2	3.5	2.6	1.7	1.5
H1* + N54S + V56T + G109A + F113Q + W167F + Q172N + A174S + G182* + D183* + N195F + A204G + V206L + K391A + P473R + G476K	2.5	2.0	1.2	1.2	2.9	2.2	1.7	1.4
H1* + N54S + V56T + G109A + F113Q + R116H + W167F + Q172R + A174S + G182* + D183* + N195F + A204G + V206L + K391A + P473R + G476K	2.1	1.9	1.2	1.1	2.3	1.8	1.3	1.3
H1* + N54S + V56T + G109A + R116A + A174S + G182* + D183* + N195F + V206L + K391A + T444Q + P473R + G476K	2.1	1.9	1.1	1.1	2.7	2.2	1.5	1.3
H1* + N54S + V56T + G109A + R116H + Q172R + A174S + G182* + D183* + N195F + A204G + V206L + K391A + P473R + G476K	2.2	1.6	1.2	1.1	2.5	1.8	1.5	1.2
H1* + A37V + N54S + V56T + G109A + R116Q + T165G + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.3	2.0	1.2	1.1	3.0	2.1	1.5	1.4
H1* + A37H + N54S + V56T + G109A + R116H + T165G + A174S + G182* + D183* + N195F + V206L + M246F + K391A + G476K	2.3	2.0	1.3	1.1	3.2	2.2	1.5	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + K391A + G476K	2.0	1.9	1.1	1.1	2.1	1.9	1.2	1.2
H1* + G109A + G182* + D183* + N195F + V206L + K391A + G476K	1.7	1.6	1.1	1.1	1.8	1.5	1.3	1.2
G109A + G182* + D183* + N195F + V206L + G476K	1.9	1.7	1.2	1.1	2.1	1.5	1.2	1.1
H1* + G109A + G182* + D183* + N195F + K391A	1.9	1.9	1.2	1.2	1.6	1.7	1.3	1.3
H1* + N54S + V56T + G109A + G133S + A174S + G182* + D183* + N195F + K391A + G476H	2.2	1.9	1.2	1.1	2.4	2.0	1.3	1.3
H1* + N54S + V56T + G109A + R116A + Q172N + L173V + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.5	2.3	1.2	1.2	3.2	2.5	1.6	1.4
H1* + A37V + N54S + V56T + G109A + R116H + T165G + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.5	2.1	1.1	1.2	2.6	2.3	1.4	1.3
H1* + P45A + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + C474V + G476K	2.3	2.1	1.3	1.2	3.1	2.1	1.5	1.4
H1* + N54S + V56T + G109A + P124A + A174S + G182* + D183* + N195F + V206L + K391A + C474V + G476K	1.9	2.2	1.3	1.2	2.8	2.2	1.4	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + P364A + K391A + C474V + G476K	2.0	2.0	1.3	1.2	2.4	2.1	1.5	1.4
H1* + N54S + V56T + G109A + Q118R + T134E + A174S + G182* + D183* + N195F + V206L + K391A + G476K	3.7	2.5	1.6	1.2	3.2	2.6	1.5	1.6
H1* + N54S + V56T + G109A + A174S + G182* + D183* + G184T + N195F + V206L + K391A + G476K	3.4	2.4	1.6	1.3	3.9	2.7	1.6	1.7
H1* + G50A + T51A + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	3.2	2.2	1.5	1.2	3.1	2.4	1.8	1.5
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + G255A + Q256A + K391A + G476K	3.2	2.3	1.6	1.3	3.4	2.3	1.7	1.5
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + Y382L + K391A + G476K	3.0	2.2	1.6	1.2	2.9	2.4	1.4	1.4
H1* + G7K + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	3.7	2.2	1.5	1.1	3.7	2.6	1.7	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + G255A + Q256A + K391A + G476K	3.0	2.2	1.5	1.2	3.2	2.6	1.7	1.5
H1* + N54S + V56T + G109A + W167Y + A174S + G182* + D183* + N195F + V206L + K391A + G476K	3.4	2.3	1.4	1.3	3.0	2.4	1.8	1.5
H1* + N54S + V56T + G109A + W167H + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.9	2.0	1.4	1.1	2.5	2.1	1.8	1.5
H1* + N54S + V56T + G109A + T165V + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.6	2.0	1.4	1.2	2.5	2.2	1.6	1.5
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	3.2	2.4	1.3	1.2	3.0	2.1	1.9	1.5

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H1* + N54S + V56T + G109A + A174N + G182* + D183* + N195F + V206L + K391A + G476K	2.8	2.2	1.4	1.2	2.7	1.8	1.4	1.4
H1* + N54S + V56T + G109A + A174Q + G182* + D183* + N195F + V206L + K391A + G476K	3.5	2.0	1.6	1.2	2.4	2.1	1.8	1.5
H1* + N16H + V17L + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + Y382F + K391A + G476K	2.7	1.9	1.3	1.2	4.6	2.3	1.6	1.4
H1* + Q32S + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + Q385L + K391A + G476K	2.7	1.8	1.3	1.1	3.5	2.2	1.4	1.3
H1* + N54S + V56T + R87S + G109A + R171H + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.5	1.9	1.1	1.1	3.4	2.1	1.4	1.4
N54S + G182* + D183* + N195F	1.1	1.1	1.1	1.0	1.6	1.5	1.0	1.0
G109A + G182* + D183* + N195F	1.6	1.4	1.3	1.0	1.8	1.4	1.0	1.1
A174S + G182* + D183* + N195F	0.8	1.1	1.3	0.9	1.2	1.2	0.9	1.0
G182* + D183* + N195F + V206L	0.7	0.7	1.0	1.0	0.9	1.3	0.8	1.0
G182* + D183* + N195F + K391A	1.6	1.4	1.3	1.0	1.4	1.5	1.0	1.1
G182* + D183* + N195F + G476K	1.3	1.4	1.3	1.0	1.8	1.5	1.1	1.0
H1* + N54S + V56T + G109A + R116H + Q169E + Q172K + A174* + G182* + D183* + N195F + V206L + K391A + G476K	3.5	1.9	1.2	1.1	3.4	2.2	1.6	1.2
H1* + N54S + V56T + G109A + A174* + G182* + D183* + N195F + V206L + K391A + G476K	3.4	2.1	1.3	1.1	3.2	2.1	1.5	1.1
H1* + K72R + G109A + F113Q + R116Q + W167F + Q172G + A174S + G182* + D183* + G184T + N195F + V206L + K391A + P473R + G476K	3.1	2.0	1.1	1.0	3.0	2.1	1.7	1.4
H1* + N54S + V56T + K72R + G109A + F113Q + R116Q + W167F + Q172G + A174S + G182* + D183* + G184T + N195F + V206L + P473R + G476K	3.1	1.9	1.2	1.0	3.0	1.9	1.6	1.3
H1* + N54S + V56T + G109A + Q169E + Q172G + A174* + G182* + D183* + N195F + V206L + K391A + G476K	3.0	1.9	1.1	1.0	2.7	2.0	1.3	1.2
H1* + N54S + V56T + K72R + G109A + F113Q + R116Q + W167F + Q172G + A174S + G182* + D183* + G184T + N195F + K391A + P473R + G476K	2.9	1.8	1.1	1.0	2.9	1.9	1.4	1.1
H1* + N54S + V56T + G109A + R116W + Q169E + Q172K + A174* + G182* + D183* + N195F + V206L + K391A + G476K	2.8	1.9	1.1	1.0	3.2	2.1	1.5	1.1
H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + D183* + N195F + V206L + K391A + G476K	2.7	1.8	1.3	1.0	3.3	2.0	1.7	1.2
H1* + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.6	1.6	1.2	1.1	2.4	1.6	1.4	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + N260G + K391A + G476K	2.6	1.7	1.2	1.0	3.4	2.2	1.4	1.1
H1* + N54S + V56T + K72R + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.6	1.5	1.2	1.0	3.1	1.8	1.6	1.2
H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + D183* + G184T + N195F + V206L + K391A + G476K	2.5	1.8	1.2	1.0	2.2	1.7	1.6	1.2
H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + D183* + N195F + V206L + Y295N + K391A + G476K	2.4	1.6	1.1	1.0	2.6	1.9	1.5	1.2
H1* + N54S + V56T + G109A + W167F + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.4	1.7	1.2	1.0	2.5	1.9	1.7	1.3
H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + D183* + A186N + N195F + V206L + K391A + G476K	2.4	1.6	1.4	1.0	4.9	1.8	1.8	1.2
H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + D183* + N195F + V206L + A288V + K391A + G476K	2.4	1.6	1.1	1.0	2.4	1.6	1.3	1.1
H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + D183* + N195F + V206L + G255A + K391A + G476K	2.4	1.9	1.2	1.0	3.2	2.2	2.0	1.3
H1* + N54S + G109A + Q169E + Q172K + A174* + G182* + D183* + N195F + V206L + K391A + G476K	2.3	1.7	1.1	1.1	3.2	2.2	1.5	1.2
H1* + W48F + N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + D183* + N195F + V206L + K391A + G476K	2.3	1.6	1.1	1.0	2.3	1.6	1.4	1.2
H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + D183* + N195F + V206L + V291A + K391A + G476K	2.3	1.8	1.2	1.1	1.8	1.8	1.6	1.2
H1* + N54S + V56T + G109A + F113Q + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.3	1.7	1.3	1.0	2.3	2.0	1.7	1.3
H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + D183* + N195F + V206L + R320A + S323N + K391A + G476K	2.3	1.6	1.2	1.0	3.3	2.0	1.6	1.2
H1* + V56T + G109A + Q169E + Q172K + A174* + G182* + D183* + N195F + V206L + K391A + G476K	2.3	1.6	1.2	0.9	2.0	1.7	1.6	1.1
H1* + N54S + V56T + K72R + G109A + F113Q + R116H + W167F + Q172G + A174S + G182* + D183* + G184T + N195F + V206L + K391A + P473R + G476K	2.3	1.7	1.1	1.0	2.0	1.7	1.7	1.2

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H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + N270T + K391A + G476K	2.2	1.7	1.1	1.0	2.6	2.1	1.8	1.4
H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + D183* + N195F + V206L + G476K	2.2	1.6	1.2	1.1	2.4	1.6	1.3	1.2
H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + D183* + N195F + V206L + N260G + K391A + G476K	2.2	1.6	1.1	1.0	3.2	2.0	1.7	1.2
H1* + N54S + V56T + G109A + F113Q + R116H + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.2	1.7	1.1	1.1	3.4	2.1	1.4	1.2
H1* + N54S + V56T + G109A + Q172G + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.2	1.8	1.2	1.1	2.3	1.9	1.6	1.3
H1* + N54S + V56T + G109A + A174S + A186D + G182* + D183* + N195F + V206L + K391A + G476K	2.1	1.7	1.2	1.0	2.2	1.7	1.5	1.2
H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + D183* + N195F + V206L + A225V + K391A + G476K	2.1	1.8	1.0	0.9	2.0	2.0	1.9	1.3
H1* + G109A + Q169E + Q172K + A174* + G182* + D183* + N195F + V206L + K391A + G476K	2.1	1.5	1.1	0.9	1.8	1.4	1.3	1.1
H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + D183* + N195F + V206L + G255S + K391A + G476K	2.1	1.6	1.2	1.0	3.3	2.1	1.6	1.2
H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + D183* + N195F + V206L + K269Q + K391A + G476K	2.1	1.7	1.0	1.1	2.0	1.8	1.4	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + G255S + K391A + G476K	2.1	1.5	1.1	1.0	2.0	1.5	1.2	1.0
H1* + N54S + V56T + G109A + Q169E + Q172K + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.1	1.6	1.1	1.1	2.9	2.1	1.5	1.3
H1* + N54S + V56T + G109A + Q169E + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.0	1.7	1.2	0.9	2.9	1.8	1.8	1.2
H1* + N54S + V56T + G109A + Q169E + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.0	1.6	1.1	1.1	2.0	1.8	1.6	1.3
H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + D183* + N195F + V206L + W284H + K391A + G476K	2.0	1.5	1.1	1.0	2.8	1.9	1.4	1.2
H1* + N54S + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.0	1.5	1.1	1.0	2.3	1.8	1.4	1.2
H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + D183* + N195F + V206L + S304R + K391A + G476K	2.0	1.5	1.4	1.0	5.8	1.7	1.9	1.3
H1* + N54S + V56T + K72R + G109A + F113Q + R116Q + W167F + Q172G + A174S + G182* + D183* + G184T + N195F + V206L + K391A + P473R	2.0	1.6	1.2	1.1	2.0	1.6	1.3	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + M246V + K391A + G476K	2.0	1.7	1.2	1.1	1.9	1.9	1.5	1.3
H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + D183* + N195F + V206L + K391A	1.9	1.4	1.2	1.1	2.7	2.0	1.4	1.2
H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + D183* + N195F + V206L + N270G + K391A + G476K	1.9	1.5	1.1	1.0	2.2	1.7	1.4	1.1
H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + D183* + N195F + V206L + Y295F + K391A + G476K	1.9	1.6	1.2	1.0	2.7	1.8	1.4	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + R320A + K391A + G476K	1.9	1.5	1.2	1.1	2.9	1.8	1.3	1.3
H1* + N54S + V56T + G109A + Q169E + Q172K + A174T + G182* + D183* + N195F + V206L + K391A + G476K	1.9	1.9	1.2	1.0	2.4	2.1	1.6	1.4
H1* + N54S + V56T + G109A + Q169E + Q172K + G182* + D183* + N195F + V206L + K391A + G476K	1.9	1.7	1.1	1.0	2.6	1.7	1.8	1.2
H1* + N54S + V56T + G109A + W140Y + Q169E + Q172K + A174* + G182* + D183* + N195F + V206L + K391A + G476K	1.9	1.4	1.1	1.0	2.1	1.8	1.5	1.4
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A	1.9	1.6	1.1	1.1	2.2	1.9	1.7	1.4
H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + D183* + N195F + V206L + F328L + K391A + G476K	1.8	1.6	1.1	1.0	1.7	1.4	1.5	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + G476K	1.8	2.0	1.5	1.1	5.2	2.0	1.8	1.5
F113Q + R116H + D183* + G182* + N195F	1.8	1.7	1.1	1.0	1.9	1.9	1.7	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + A288L + K391A + G476K	1.8	1.3	1.1	1.0	2.6	1.8	1.5	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + N270G + K391A + G476K	1.8	1.6	1.2	1.1	1.9	1.8	1.6	1.2
Q169E + Q172K + A174* + D183* + G182* + N195F	1.8	1.2	1.2	1.0	3.3	1.4	1.5	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K269Q + K391A + G476K	1.8	1.0	1.1	0.9	1.8	1.4	1.5	1.1
H1* + W48F + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.8	1.4	1.1	1.0	2.2	1.7	1.5	1.2

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F113Q + R116Q + D183* + G182* + N195F	1.8	1.7	1.1	1.0	2.1	1.7	1.7	1.4
N54S + V56T + D183* + G182* + N195F	1.8	1.2	0.8	0.9	1.3	1.3	1.0	1.0
H1* + N54S + V56T + G109A + A174T + G182* + D183* + N195F + V206L + K391A + G476K	1.7	1.7	1.2	1.1	2.4	1.9	1.8	1.3
H1* + N54S + V56T + G109A + R116W + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.7	1.6	1.2	1.1	1.8	1.8	1.6	1.2
Q169E + Q172K + G182* + D183* + N195F	1.7	1.4	1.3	1.0	4.9	1.6	1.8	1.3
G182* + D183* + A186D	1.7	1.6	1.0	1.0	1.6	1.7	1.3	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + V238A + K391A + G476K	1.7	1.5	1.1	1.0	1.8	1.6	1.4	1.1
G182* + D183* + A186N	1.7	1.8	1.2	1.1	3.8	1.8	1.6	1.5
H1* + N54S + V56T + G109A + A174S + V206L + G182* + D183* + K391A + G476K	1.7	1.1	1.1	1.0	3.3	1.1	1.3	1.1
H1* + N54S + V56T + Q169E + Q172K + A174* + G182* + D183* + N195F + V206L + K391A + G476K	1.7	1.4	1.3	1.0	3.6	1.4	1.8	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206F + K391A + G476K	1.7	1.7	1.2	1.0	1.7	1.7	1.6	1.3
H1* + N54S + V56T + G109A + G182* + D183* + N195F + V206L + K391A + G476K	1.7	1.4	1.1	1.0	1.9	1.6	1.2	1.0
H1* + N54S + V56T + G109A + Q172K + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.7	1.5	1.1	1.1	3.6	1.6	1.5	1.3
H1* + N54S + V56T + G109A + Q172K + A174* + G182* + D183* + N195F + V206L + K391A + G476K	1.6	1.4	1.1	1.1	1.6	1.4	1.3	1.1
H1* + N54S + V56T + G109A + A174N + G182* + D183* + N195F + V206L + K391A + G476K	1.6	1.8	1.2	1.2	4.6	2.0	2.0	1.7
H1* + N54S + V56T + G109A + A174S + A186N + G182* + D183* + N195F + V206L + K391A + G476K	1.6	1.8	1.1	1.0	1.5	1.9	1.7	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + F289L + K391A + G476K	1.6	1.5	1.0	1.0	1.7	1.7	1.4	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + Y295F + K391A + G476K	1.6	1.2	1.0	0.8	1.2	1.2	1.3	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + V291A + K391A + G476K	1.6	1.2	1.1	1.0	1.2	1.0	1.2	1.0
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + S244Q + K391A + G476K	1.5	1.4	1.1	1.0	2.1	1.5	1.3	1.1
G182* + D183* + F328L	1.5	1.5	1.1	1.1	1.8	1.8	1.6	1.3
G182* + D183* + N195F + A288V	1.5	1.3	1.2	1.0	1.8	1.6	1.5	1.1
G182* + D183* + A186N + N195F	1.5	1.6	0.9	0.9	2.4	1.7	1.5	1.2
G182* + D183* + E190P + N195F	1.5	1.4	0.9	1.0	2.1	1.7	1.1	1.1
W140Y + G182* + D183* + N195F	1.5	1.4	1.1	1.1	1.4	1.7	1.5	1.3
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + A225V + K391A + G476K	1.5	1.0	1.1	0.9	1.5	1.0	1.5	1.0
G182* + D183* + A186D + N195F	1.5	1.7	1.4	1.1	3.3	2.0	1.8	1.6
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + P473R + G476K	1.5	1.2	1.1	1.0	1.5	1.2	1.3	1.2
H1* + W48Y + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.5	1.3	1.0	1.1	1.2	1.4	1.3	1.2
H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + D183* + N195F + V206L + K391A + G476K	1.5	1.3	1.4	1.1	3.8	1.4	1.5	1.3
G182* + D183* + N195F + V238A	1.5	1.5	1.0	1.1	1.9	1.4	1.1	1.0
G182* + D183* + Y295N	1.4	1.1	1.3	1.0	2.5	1.2	1.6	1.2
Q172G + G182* + D183*	1.4	1.4	1.1	1.0	1.4	1.3	1.5	1.3
F113Q + G182* + D183* + N195F	1.4	0.8	1.0	0.8	1.0	1.0	1.3	1.0
H1* + N54S + V56T + Q98A + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.4	1.3	1.2	1.0	2.7	1.1	1.5	1.2
G182* + D183* + N195F + Y295F	1.3	1.3	1.2	1.1	1.3	1.2	1.4	1.1
G182* + D183* + A186E	1.3	1.2	1.2	1.0	2.0	1.4	1.4	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206Y + K391A + G476K	1.3	1.1	1.2	1.1	2.8	1.2	1.3	1.2
G182* + D183* + N195F + W284H	1.3	1.6	1.1	1.2	3.3	1.8	1.6	1.6
G182* + D183* + A186Q + N195F	1.3	1.2	1.1	1.0	2.0	1.7	1.5	1.1
K72R + G182* + D183*	1.3	1.3	1.0	1.0	1.4	1.5	1.3	1.1
D183* + G182* + N195F + V326L	1.3	1.6	1.2	1.2	3.5	1.8	1.5	1.6
D183* + G182* + A186E + N195F	1.3	1.3	1.1	1.0	1.7	1.4	1.5	1.2
Q169E + G182* + D183* + N195F	1.2	1.0	1.1	1.0	1.7	0.8	1.4	1.2
D183* + G182* + N195F + K242Q	1.2	0.8	0.9	0.9	2.0	1.2	0.9	1.0
G7A + G182* + D183* + N195F	1.2	1.2	1.1	1.0	1.5	1.6	1.4	1.1
D183* + G182* + N195F + N260G	1.2	1.4	1.0	0.9	1.6	1.2	1.3	1.1
D183* + G182* + R321A + S323N	1.2	1.3	1.1	1.1	1.2	1.1	1.5	1.2
D183* + G182* + W284H	1.2	0.9	1.2	1.0	0.9	1.0	1.1	1.1
H1* + D183* + G182*	1.2	1.5	1.0	1.0	3.6	1.7	1.6	1.3
D183* + G182* + A186H + N195F	1.1	1.3	1.1	1.0	2.0	1.2	1.4	1.0
D183* + G182* + F289V	1.1	1.3	1.0	1.0	2.2	1.3	1.7	1.4
D183* + G182* + K391A	1.1	0.7	1.0	0.8	0.7	0.7	1.0	0.8
D183* + G182* + S304R	1.1	1.2	0.9	0.9	0.6	0.9	1.0	1.0
W167F + G182* + D183* + N195F	1.1	1.0	1.0	0.9	3.3	1.1	1.3	1.1
F113Q + R116H + D183* + G182*	1.1	0.7	1.0	0.8	1.2	0.8	1.3	0.9
D183* + G182* + N270G	1.1	1.0	0.9	0.9	1.0	1.3	1.0	1.0
D183* + G182* + S244Q	1.0	1.1	1.1	0.9	1.8	1.6	1.5	1.0

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H1* + N54S + V56T + G109A + A174S + G182* + D183* + G184T + N195F + V206L + K391A + G476K	1.0	1.0	1.1	1.0	0.6	0.6	0.9	0.9
D183* + G182* + N195F + Y295N	1.0	1.1	1.0	1.0	1.6	0.9	1.2	1.3
W48F + D183* + G182*	1.0	0.9	1.0	0.9	1.8	1.0	1.3	1.0
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + G255A + K391A + G476K	1.0	1.2	1.0	1.0	1.8	1.1	1.2	1.1
D183* + G182* + A186Q	1.0	0.7	1.1	0.9	0.4	0.5	1.0	0.9
W140Y + D183* + G182*	0.9	0.6	0.9	0.9	0.5	0.9	1.1	1.0
D183* + G182* + K242Q	0.9	0.8	1.0	0.9	0.8	0.9	1.0	1.1
D183* + G182* + V206L	0.9	0.8	0.9	0.9	1.1	0.9	0.9	0.9
D183* + G182* + N195F + V206F	0.9	0.9	0.9	1.1	0.5	0.8	1.1	1.0
Q169E + D183* + G182*	0.9	0.8	1.0	0.9	1.1	0.8	0.9	0.9
D183* + G182* + N195F + R320A	0.8	0.9	1.1	1.0	0.8	0.9	1.0	1.0
G182* + D183* + G184T + N195F	0.8	1.1	0.9	1.0	0.9	1.0	1.2	1.0
D183* + G182* + G184T	0.8	0.9	1.0	0.9	0.7	0.8	1.0	1.1
N54S + V56T + D183* + G182*	0.8	0.8	1.1	1.0	0.9	0.7	1.1	0.9
D183* + G182* + P473R	0.8	0.9	1.0	1.0	1.1	0.8	1.0	1.0
W48Y + D183* + G182*	0.8	0.9	1.1	1.1	1.0	0.6	1.0	0.9
D183* + G182* + R323A	0.8	0.8	0.9	1.0	1.1	0.8	1.1	0.8
G7A + D183* + G182*	0.8	0.8	1.1	0.9	1.3	1.1	1.2	1.0
G109A + D183* + G182*	0.8	0.8	0.8	0.8	0.0	0.6	1.0	1.1
W167F + D183* + G182*	0.7	0.7	1.0	0.9	1.4	0.9	1.1	1.0
H1* + N54S + V56T + G109A + Q169E + Q172K + A174S + G182* + D183* + N195F + V206L + K391A + G476K	0.7	0.7	0.8	0.9	1.1	0.5	0.8	0.8
A174S + D183* + G182*	0.7	1.2	0.8	0.9	2.2	0.9	1.1	1.3
R116Q + D183* + G182*	0.7	0.7	0.9	0.9	0.7	0.6	1.1	0.9
D183* + G182* + N195F + V291A	0.7	0.9	1.0	1.1	0.6	0.9	0.9	1.0
Q172K + D183* + G182*	0.7	0.9	0.9	1.0	0.4	0.5	1.1	0.9
N54S + D183* + G182*	0.7	1.2	1.0	1.1	1.7	1.1	1.0	1.3
D183* + G182* + N195F + S323N	0.7	0.9	0.8	0.9	1.8	1.0	1.0	1.0
D183* + G182* + N195F + M246V	0.7	0.8	1.1	1.0	0.6	0.5	0.9	0.9
W48Y + D183* + G182* + N195F	0.6	1.2	1.1	1.1	2.0	1.5	1.2	1.4
D183* + G182* + N195F + S304R	0.6	1.1	0.7	1.0	1.6	0.7	1.0	1.2
V56T + D183* + G182*	0.3	0.8	0.2	0.6	1.9	0.6	0.3	0.7

Modifications in SEQ ID NO: 14	A-15- 0.05 mg/L	A-15- 0.2 mg/L	A-40- 0.05 mg/L	A-40- 0.2 mg/L	J-15- 0.05 mg/L	J-15- 0.2 mg/L	J-30- 0.05 mg/L	J-30- 0.2 mg/L
Reference	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
G182* + D183* + N195F	1.4	1.2	1.0	1.0	1.2	1.3	1.1	1.1
H1* + G182* + D183* + N195F	1.1	1.0	1.0	1.1	1.1	1.0	1.1	1.1
N54S + G182* + D183* + N195F	1.2	1.1	0.9	1.1	1.3	1.1	1.1	1.1
V56T + G182* + D183* + N195F	1.4	1.1	1.0	1.0	1.5	1.2	1.1	1.1
G182* + D183* + N195F + I405L	1.1	1.1	1.0	1.0	1.1	1.1	1.0	1.0
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T + D476K	1.0	1.1	0.9	0.9	1.4	1.0	1.1	0.9
A174S + G182* + D183* + N195F	1.2	1.0	1.0	0.9	0.9	0.9	1.0	1.0
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T	1.4	1.3	1.1	1.0	1.6	1.3	1.3	1.1
H1* + N54S + V56T + G109A + A174S + D183* + N195F + V206L + K391A + I405L + A421H + A422P + A428T	1.2	1.3	1.0	1.0	1.6	1.3	1.2	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + I405L + A421H + A422P + A428T + D476K	1.3	1.3	1.1	1.0	1.4	1.3	1.1	1.0
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + I405L + A421H + A422P + A428T + D476G	1.2	1.1	1.1	1.0	1.4	1.2	1.2	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + I405L + A421H + A422P + A428T + D476K + G477S	1.3	1.3	1.0	1.0	1.5	1.2	1.1	1.0
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + I405L + A421H + A422P + A428T + D476K	1.3	1.2	1.0	1.0	1.5	1.2	1.2	1.0
H1* + N54S + V56T + K72R + G109A + W167F + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T	1.1	1.2	1.0	1.1	1.4	1.4	1.3	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + S280Q + H321Y + I405L + A421H + A422P + A428T	0.9	1.2	1.1	0.9	1.3	1.2	1.3	1.0
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + S304N + I405L + A421H + A422P + A428T + D476K	1.1	1.2	1.0	1.0	1.2	1.1	1.1	1.0
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + I405L + A421H + A422P + A428T + D476K + G477A	1.2	1.2	1.0	1.0	1.2	1.0	1.0	0.9

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H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + I405L + A421H + A422P + A428T + D476G + G477T	1.6	1.4	1.1	1.1	1.9	1.4	1.3	1.2
H1* + N54S + V56T + G109A + F113Q + R116Q + Q172N + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T	0.9	1.2	0.8	1.1	0.8	1.7	0.9	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + I405L + A421H + A422P + A428T + G477Q	1.7	1.3	1.1	1.0	1.9	1.4	1.3	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + I405L + A421H + A422P + A428T + G477A	1.5	1.4	1.1	1.0	1.9	1.4	1.3	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + I405L + A421H + A422P + A428T + D476K + G477Q	1.3	1.3	1.0	1.0	1.7	1.2	1.1	1.0
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + I405L + A421H + A422P + A428T + D476N	1.5	1.3	1.1	1.1	1.7	1.4	1.3	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + I405L + A421H + A422P + A428T + D476Y	1.5	1.3	1.1	1.0	1.7	1.3	1.3	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T	1.5	1.2	1.1	1.1	2.2	1.4	1.4	1.1
H1* + N54S + V56T + G109A + Q172D + A174S + G182* + D183* + N195F + A204G + V206L + I405L + A421H + A422P + A428T	1.4	1.3	1.1	1.0	2.8	2.0	1.4	1.2
H1* + N54S + V56T + K72R + G109A + T134E + A174S + G182* + D183* + N195F + A204G + V206L + G255A + I405L + A421H + A422P + A428T	1.3	1.1	1.0	1.0	2.1	2.1	1.3	1.0
H1* + N54S + V56T + G109A + F113Q + R116H + T165G + Q172G + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T	1.3	1.3	1.0	1.0	2.9	1.9	1.3	1.1
H1* + N54S + V56T + G109A + F113Q + Q172N + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T	1.3	1.3	1.1	1.1	1.9	2.2	1.5	1.3
H1* + N54S + V56T + K72R + G109A + T134E + A174S + G182* + D183* + N195F + G196R + V206L + G255A + I405L + A421H + A422P + A428T + N475S	1.4	1.3	1.1	1.1	2.5	2.3	1.5	1.3
H1* + N54S + V56T + G109A + T134E + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T	1.8	1.5	1.2	1.0	2.3	1.8	1.5	1.2
H1* + T40D + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T	2.0	1.5	1.2	1.1	1.8	1.6	1.5	1.3
H1* + N54S + V56T + G109A + F113Q + N125D + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T	2.1	1.6	1.3	1.0	2.2	1.7	1.5	1.2
H1* + N54S + V56T + A60V + G109A + R116Q + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T	1.6	1.6	1.1	1.0	1.9	1.6	1.4	1.2
H1* + N54S + V56T + G109A + R116V + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T	1.4	1.4	1.0	1.0	2.1	1.4	1.4	1.2
H1* + N54S + V56T + G109A + R116Q + G133Q + T165G + Q172G + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T	1.1	1.4	1.0	1.1	2.0	2.3	1.5	1.3
H1* + N54S + V56T + G109A + F113Q + R116H + Q172G + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T	1.7	1.6	1.2	1.1	3.0	1.9	1.5	1.2
H1* + N54S + V56T + G109A + F113Q + R116Q + Q172G + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T + G448D	1.4	1.5	1.1	1.1	2.1	1.8	1.3	1.2
H1* + N54S + V56T + G109A + F113Q + R116H + Q172N + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T	2.2	1.6	1.3	1.2	2.5	1.6	1.6	1.3
H1* + N54S + V56T + K72R + G109A + T134E + W167F + Q172N + A174S + G182* + D183* + N195F + V206L + G255A + A265G + I405L + A421H + A422P + A428T	2.2	1.6	1.3	1.2	2.4	1.6	1.7	1.2
H1* + N54S + V56T + K72R + G109A + R116Q + W167F + Q172R + L173V + A174S + G182* + D183* + N195F + V206L + A265G + I405L + A421H + A422P + A428T	2.0	1.6	1.2	1.2	2.3	1.6	1.7	1.2
H1* + N54S + V56T + K72R + G109A + R116Q + W167F + Q172R + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T	2.1	1.4	1.3	1.2	2.3	1.6	1.7	1.2
H1* + N54S + V56T + K72R + G109A + F113Q + T134E + W167F + Q172G + A174S + G182* + D183* + N195F + V206L + G255A + I405L + A421H + A422P + A428T	2.0	1.5	1.1	1.1	2.6	1.9	1.4	1.2

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H1* + N54S + V56T + K72R + G109A + F113Q + R116H + W167F + Q172G + L173V + A174S + G182* + D183* + G184T + N195F + V206L + I405L + A421H + A422P + A428T	1.7	1.6	1.0	0.9	1.8	1.5	1.1	1.1
H1* + N54S + V56T + G109A + F113Q + W167F + Q172G + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T	1.2	1.2	1.0	0.9	2.2	1.3	1.2	1.1
H1* + N54S + V56T + K72R + G109A + A174S + G182* + D183* + N195F + V206L + G255A + I405L + A421H + A422P + A428T	1.5	1.2	1.2	1.0	2.2	1.1	1.5	1.1
H1* + N54S + V56T + G109A + F113Q + R116Q + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T	1.6	1.3	1.1	1.0	1.4	1.2	1.3	1.1
H1* + N54S + V56T + K72R + G109A + F113Q + F116H + W167F + Q172G + L173V + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T	1.8	1.6	1.0	1.0	0.9	1.4	0.9	1.1
H1* + N54S + V56T + K72R + G109A + F113Q + R116Q + W167F + Q172G + A174S + G182* + D183* + G184T + N195F + V206L + I405L + A421H + A422P + A428T	1.8	1.5	1.1	1.0	1.9	1.5	1.3	1.1
H1* + N54S + V56T + G109A + F113Q + R116Q + W167F + Q172G + A174S + G182* + D183* + N195F + A204G + V206L + I405L + A421H + A422P + A428T	2.0	1.5	1.0	1.0	1.8	1.5	1.2	1.0
H1* + N54S + V56T + G109A + F113Q + R116H + W167F + Q172G + A174S + G182* + D183* + N195F + V206L + A265G + I405L + A421H + A422P + A428T	1.5	1.3	0.9	0.9	1.4	1.3	1.2	1.1
H1* + N54S + V56T + G109A + R116Q + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T	1.8	1.6	1.0	0.9	1.6	1.3	1.4	1.1
H1* + N54S + V56T + A60V + G109A + R116Q + W167F + Q172N + L173V + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T	1.4	1.4	0.7	0.9	1.1	1.3	0.8	1.0
H1* + A37H + N54S + V56T + A60V + G109A + R116Q + T165G + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T	1.9	1.6	1.1	1.1	1.7	1.6	1.4	1.2
H1* + N54S + V56T + G109A + A174S + M105F + G182* + D183* + N195F + V206L + L228I + R320A + S323N + I405L + A421H + A422P + A428T	1.4	1.2	1.0	1.0	1.5	1.2	1.2	1.0
H1* + N54S + V56T + K72R + G109A + F113Q + T134E + A174S + G182* + D183* + N195F + V206L + G255A + I405L + A421H + A422P + A428T	1.8	1.5	1.1	1.0	1.7	2.1	1.4	1.2
H1* + N54S + V56T + K72R + G109A + R116Q + V120L + A174S + G182* + D183* + G184T + N195F + V206L + I405L + A421H + A422P + A428T	2.1	1.8	1.2	1.1	2.8	2.5	1.6	1.3
H1* + N54S + V56T + G109A + F113Q + R116H + W167F + Q172G + L173V + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T	2.6	1.9	1.2	1.1	2.8	2.6	1.6	1.3
H1* + N54S + V56T + G109A + F113Q + R116Q + Q172N + A174S + G182* + D183* + N195F + V206L + A265G + I405L + A421H + A422P + A428T	2.3	1.6	1.1	1.1	3.0	2.1	1.5	1.3
H1* + A37H + N54S + V56T + A60V + G109A + R116Q + W167F + Q172R + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T	2.1	1.7	1.2	1.1	2.5	2.1	1.5	1.2

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Example 4: Wash Performance of a Subset of Variants According to the Invention (AMSA)

The following variants have been tested as described above. The color measurements were done by use of a professional flatbed scanner (Kodak iQsmart, Kodak) used to capture an image of the washed textile and with a controlled digital imaging system (DigiEye) for capture an image of the washed melamine plates.

To extract a value for the light intensity from the scanned images, 24-bit pixel values from the image are converted

into values for red, green and blue (RGB). The intensity value (Int) is calculated by adding the RGB values together as vectors and then taking the length of the resulting vector:

$$Int = \sqrt{r^2 + g^2 + b^2}$$

The results obtained were the following;

Modifications in SEQ ID NO: 13	A-15-0.05 mg/L	A-15-0.2 mg/L	A-40-0.05 mg/L	A-40-0.02 mg/L	J-15-0.05 mg/L	J-15-0.2 mg/L	J-30-0.05 mg/L	J-30-0.2 mg/L
Reference	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
G182* + D183* + N195F + V206F + N260P + L312I + L313I + R320A + T355L + L389I + T400P	0.7	1.0	0.6	0.7	1.3	0.9	0.8	1.0
G132* + D183* + N195F	0.1	0.7	1.0	1.0	1.0	0.8	1.0	1.0
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.7	1.9	1.7	1.3	2.5	1.9	1.2	1.2

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Modifications in SEQ ID NO: 13	A-15- 0.05 mg/L	A-15- 0.2 mg/L	A-40- 0.05 mg/L	A-40- 0.02 mg/L	J-15- 0.05 mg/L	J-15- 0.2 mg/L	J-30- 0.05 mg/L	J-30- 0.2 mg/L
H1* + G50A + T51A + V56T + G109A + A174S + G182* + D183* + N195F + V206L + G476K	1.5	1.6	1.5	1.4	2.5	1.6	1.3	1.2
H1* + N54S + V56T + M105F + G109A + A174S + G182* + D183* + N195F + V206L + R320A + S323N + K391A + G476K	2.5	2.2	1.6	1.4	3.8	1.9	1.2	1.1
H1* + T40K + N54S + V56T + G109A + G149H + A174S + G182* + D183* + N195F + V206L + K391A + G476K	0.9	1.2	1.0	1.0	0.8	1.2	0.9	1.1
H1* + N54S + V56T + K72R + G109A + A174S + G182* + D183* + N195F + V206L + G255A + K391A + G476K	1.9	1.7	1.4	1.4	2.8	1.5	1.2	1.1
H1* + N16H + V17L + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.6	1.8	1.7	1.4	3.6	1.8	1.3	1.1
H1* + Q22N + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.5	1.9	1.8	1.4	3.8	2.1	1.3	1.1
H1* + Q32S + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.3	1.9	1.9	1.5	3.3	2.0	1.2	1.2
H1* + H28N + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	2.7	1.7	1.6	1.5	4.2	2.0	1.3	1.1
H1* + N29S + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.6	1.3	1.3	1.4	2.6	1.6	1.2	1.0
H1* + H28N + N29S + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	1.8	1.9	1.8	1.4	3.1	1.8	1.2	1.1
H1* + N54S + V56T + G109A + A174S + R176K + N195F + V206L + K391A + G476K	2.3	2.0	1.7	1.5	5.1	1.9	1.3	1.2
H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + R176K + N195F + V206L + K391A + G476K	3.8	2.4	2.1	1.7	5.6	2.8	1.4	1.2
H1* + N54S + V56T + G109A + A174S + N175G + N195F + V206L + K391A + G476K	2.6	1.8	1.8	1.4	4.9	2.0	1.3	1.2
H1* + N54S + V56T + G109A + A174S + N195F + V206L + V291A + Y295N + Q299T + K391A + G476K	1.7	1.4	1.4	1.3	2.7	1.6	1.0	1.1
H1* + N54S + V56T + G109A + A174S + N195F + V206L + K269S + A274K + K391A + G476K	1.7	1.4	1.6	1.2	1.4	1.8	1.0	1.0
H1* + N54S + V56T + G109A + A174S + N195F + V206L + L235I + K391A + G476K	2.6	1.5	2.0	1.3	1.3	2.1	1.0	1.1
H1* + N54S + V56T + G109A + A174S + N195F + V206L + L235V + V238A + K391A + G476K	2.8	1.4	1.7	1.2	0.6	2.2	1.1	1.0
H1* + N54S + V56T + G109A + A174S + N195F + V206L + Q365S + K391A + G476K	1.9	1.5	1.9	1.3	2.0	1.7	1.2	1.0
H1* + N54S + V56T + G109A + A174S + N195F + V206L + K391A + H402R + G476K	2.2	1.4	2.0	1.2	0.6	1.2	1.1	1.1
H1* + N54S + V56T + G109A + A174S + N195F + V206L + R320S + H321N + K391A + G476K	5.6	2.7	2.5	1.6	4.2	2.9	1.2	1.2
H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + N195F + V206L + K391A + G476K	3.8	2.6	3.0	1.7	3.6	2.8	1.3	1.2
H1* + N54S + V56T + G109A + Q125S + A174S + N195F + V206L + K391A + G476K	3.1	2.2	2.4	1.4	2.0	2.3	1.3	1.1
H1* + N54S + V56T + G109A + A174S + N195F + V206L + L235V + K391A + G476K	4.2	1.9	1.9	1.5	1.8	2.4	1.0	1.0
H1* + N54S + V56T + G109A + A174S + N195F + V206L + A265G + K391A + G476K	4.2	1.5	2.2	1.4	1.2	2.6	1.1	1.1
H1* + T51Q + N54S + V56T + G109A + A174S + N195F + Y203F + V206L + K391A + G476K	3.7	2.1	2.6	1.6	3.6	2.7	1.3	1.1
H1* + N54S + V56T + G109A + Q169E + Q172K + A174* + N175Q + N195F + V206L + K391A + G476K	4.0	2.0	2.7	1.5	3.3	2.8	1.2	1.2
H1* + H28N + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	4.5	2.3	2.5	1.6	3.1	2.7	1.2	1.1
H1* + N54S + V56T + G109A + T134E + A174S + G182* + D183* + N195F + V206L + K391A + G476K	3.9	2.1	2.6	1.6	3.3	3.0	1.4	1.1
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + D377H + K391A + G476K	5.3	2.9	3.0	1.7	3.8	2.9	1.3	1.2
H1* + G50A + T51A + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	4.4	2.1	2.3	1.4	2.6	2.3	1.3	1.2
H1* + G7K + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K	4.1	2.3	2.7	1.6	2.4	2.5	1.3	1.2
H1* + N54S + V56T + G109A + A174S + G182* + D183* + N195F + M202V + V206L + K391A + G476K	2.7	1.9	1.4	1.4	1.9	2.2	0.9	1.0
H1* + N54S + V56T + M105F + G109A + A174S + N195F + V206L + M246F + K391A + G476K	3.7	2.1	2.4	1.5	2.7	2.5	1.2	1.1
H1* + N16H + V17L + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + D377H + K391A + G476K	4.5	2.7	1.8	1.4	4.3	2.2	1.5	1.3
H1* + N54S + V56T + G109A + A174S + N195F + V206L	1.1	1.9	1.3	1.1	3.0	1.9	1.3	1.1
H1* + T51G + N54S + V56T + G109A + A174S + N195F + Y203G + V206L + K391A + G476K	2.2	1.8	1.1	1.0	2.7	2.0	0.9	1.1
H1* + T51G + N54S + V56T + G109A + G110N + A174S + N195F + V206L + K391A + G476K	1.6	2.1	1.0	1.1	1.9	1.9	1.1	1.1
H1* + N54S + V56T + G109A + Q125S + A174S + N175G + N195F + V206L + K391A + G476K	2.0	2.3	1.5	1.2	2.6	1.8	1.3	1.2
H1* + N54S + V56T + G109A + A174S + N195F + V206L + A265G + V291A + F328L + K391A + G476K	0.7	2.1	1.1	1.0	1.8	1.7	1.0	1.1
H1* + N54S + V56T + G109A + A174S + N195F + V206L + M246F + A265G + V291A + F328L + Q365S + K391A + G476K	3.9	2.6	1.9	1.4	4.0	2.1	1.5	1.2

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Modifications in SEQ ID NO: 13	A-15- 0.05 mg/L	A-15- 0.2 mg/L	A-40- 0.05 mg/L	A-40- 0.02 mg/L	J-15- 0.05 mg/L	J-15- 0.2 mg/L	J-30- 0.05 mg/L	J-30- 0.2 mg/L
H1* + N16Y + N54S + V56T + G109A + R116H + A174S + N195F + V206L + G337E + D377H + K391A + G476K	4.8	3.5	2.0	1.5	4.9	2.7	1.5	1.3
H1* + I9L + N54S + V56T + G109A + Q125S + V131T + A174* + V206L + Y267F + F289H + Q361E + K391A + G476K	3.1	2.9	1.4	1.2	2.5	2.0	1.2	1.1

The invention described and claimed herein is not to be limited in scope by the specific aspects herein disclosed, since these aspects are intended as illustrations of several aspects of the invention. Any equivalent aspects are intended to be within the scope of this invention. Indeed, various modifications of the invention in addition to those shown and described herein will become apparent to those skilled in the art from the foregoing description. Such modifications are also intended to fall within the scope of the appended claims. In the case of conflict, the present disclosure including definitions will control.

SEQUENCE LISTING

Sequence total quantity: 17

SEQ ID NO: 1 moltype = DNA length = 1458
 FEATURE Location/Qualifiers
 source 1..1458
 mol_type = genomic DNA
 organism = Bacillus sp.

SEQUENCE: 1

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catcacgatg ggacgaacgg aacgattatg cagtattttg aatggaacgt tccgaatgat 60
ggacaacatt ggaaccgctt acacaacaac gctcaaaatt taaaaaatgc cggaattaca 120
gcaatctgga ttccacctgc gtggaaagga acgagccaaa atgatgtagg ctacggtgcy 180
tatgaccttt atgaccttgg tgaatttaac caaaaaggaa cggtcctgtac gaaatatgga 240
acaaaagcag aattagaacg agcgattcgt tegttaaagg cgaacgggat tcaagtgtat 300
ggcgatgttg ttatgaacca taaaggcgga gctgatttca ccgagcgtgt tcaagcgggt 360
gaagtgaacc cgcaaaacgg aaaccaagaa gtgtctggca cttatcaaat cgaagcatgg 420
acagggttca attttcctgg acgtggcaat caacattctt cgtttaaatg gcgctgggat 480
catttcgatg ggaacggatg ggaccagtct cgccaactcg caaatcgat ttataagttt 540
agaggagacg gaaaagcatg ggaactggaa gttgacactg aaaatgggaa ctatgattac 600
ttaatgtatg cagacgttga catggatcat ccagaagtga ttaacgaact aaaccgttgg 660
ggcgtctggt acgcgaatac ccttaattta gacggcttcc gactggatgc agtgaacat 720
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tcaaatagta gcggaatata cgacatgaga aacttggtta atggaacact cgttcaacgt 960
catccgagcc atgcggttac gtttgcgat aaccacgaca cacagcctgg agaagccctc 1020
gaatcggttg ttcaagcgct gtttaacca ctacgttatg caacgattct tacgagagag 1080
caaggctacc cacaagtgtt ttaacggcat tattatggca tcccaagtga cgggtgtcca 1140
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cagcacgatt accttgatca ttgggatgta attggctgga cagtggaagg aaacgcatct 1260
caccgcaact caggacttgc aaccattatg tctgatggtc caggtggatc aaaatggatg 1320
tatgttggcc gtcagaaagc tggcgaagtg tggcatgaca tgactggaaa ccgcagtggc 1380
actgtgacaa ttaatcaaga cggctgggga cacttttttg tcaacggcgg ctctgtctcc 1440
gtatgggtga aacgataa 1458

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SEQ ID NO: 2 moltype = DNA length = 1539
 FEATURE Location/Qualifiers
 source 1..1539
 mol_type = genomic DNA
 organism = Alicyclobacillus sp.

SEQUENCE: 2

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gaatgggtatc tgcccaacga cgggacgctt tggaccaaga tgggcagcga cgcgtcgac 180
ctgaagtcca tggggatcac cggcgtcttg ttcccgccgg cgtacaaagg ccaatcgag 240
tcggacgtcg gctacggcgt atacgacatg tacgacctcg gcgaattcaa ccaaaaagg 300
accgtccgca ccaagtcagg caccaaagcc cagctccaat cggcgatcac ctccctgcac 360
aacaacggca tccaagccta cggggacgtc gtccctcaacc acccgatggg cgcgagtcg 420
acggagacga tctccgcctg ggaagtcaac ccgtccaacc gcaaccaagt cacctccgg 480
gcttacaaca tctccgcttg gaccgacttc gaattcccg gcgcgggcaa cacctactcc 540
tcgtttaagt ggcactccta ctactttgac ggcgtggact gggaccaatc ccgccagctg 600
agcggcaaga tctaccagat ccaaggcacc ggcaaacgct gggactggga agtcgattcc 660

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gaaaacggca actacgacta cctgatgggc gcgacatcg actacgacca cccggacgtg 720
caaacggaag tgaagaactg gggcaagtgg ttcgtaacaa cctcaacct cgacggcgtg 780
cgctcgacg cggtcaagca catcaagtgc gactacatgt cttcctggtt gtccagcgtc 840
aaatccacga ccggcaagtc caacctgttc gccgtcggcg aatactggaa cacctcgtc 900
ggagcgctgg agaactacga gaacaaaacc aactggagca tgtcgctgtt cgacgtgccg 960
ctgcacatga acttccaagc ggacgcgaac ggccggcggt actatgatat gcgcaacctg 1020
ctcaacaaca cgatgatgaa aaatcaccgg atccaagcgg tcaccttcgt cgacaaccac 1080
gacaccgagc cgggccaagc cctgcaatcg tgggtatccg actggttcaa accgctggcc 1140
tacgcgacga tcccgaccgg tcaagaaggc taccctgtcg tgttctacgg cgactactac 1200
ggcatcccg cgcaaacggt ccccgcgaaa tccacctggt tggacaagca gctttccgca 1260
cgcaaatcct acgctgacgg caccagcac gactacttgg acaaccaaga cgtgatcgcc 1320
tggacgcgcg aagcgatct cgcgacgcg ggcctcgggtc ttgccacctg catgtcggac 1380
ggcctcggcg gctccaagac gatgtacgtc ggcaccgccc atgccggcca agtcttcaag 1440
gacatcacgc gcaaccgcgc cgacaccgtc acgatcaact ccgcaggcaa cggcaccttc 1500
ccctgcaacg gcgctcctgt ctcgatctgt gtcaacaaa 1539

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SEQ ID NO: 3      moltype = DNA length = 1560
FEATURE          Location/Qualifiers
source           1..1560
                  mol_type = genomic DNA
                  organism = Bacillus amyloliquefaciens

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SEQUENCE: 3
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acgctgatgc agtattttga atggtatcac ccgaacgacg gccagcattg gaaacgattg 180
cagaatgatg cggaacatct atcgcatatc ggaatcactg ccgtctggat tctctccgca 240
tacaaggat tgagccaatc cgataacgga tacggacctt atgatttgta tgatttagga 300
gaattccagc aaaaaggagc ggtcagaacg aaatacggca caaatcaga gcttcaagat 360
gcgatcggct cactgcattc ccggaacgct caagtatacg gagatgtggt tttgaatcat 420
aaggctggtg ctgatgcaac agaagatgta actgccgtcg aagtcacatc ggccaataga 480
aatcaggaaa cttcggagga atatcaaatc aaagcgtgga cggattttcg tttccggggc 540
cgtggaacaa cttacagtgga ttttaaatgg cattggtatc atttcgacgg agcggactgg 600
gatgaatccc ggaagatcag ccgcatcttt aagtttcgtg gggaaggaaa agcgtgggat 660
tgggaagtat caagtgaaaa cgcaactat gactatttaa tgtatgctga tgttgactac 720
gaccaccctg atgtcgtggc agagacaaaa aaatggggta tctgggatgc gaatgaactg 780
tcattagacg gcttccgtat tgatgccgcc aaacatatta aattttcatt tctgcgtgat 840
tgggttcagg ccgtcagaca ggcgacggga aaagaaatgt ttacggttgc ggagtattgg 900
cagaataatg cggggaactc cgaaaactac ttgaataaaa caagctttaa tcaatccgtg 960
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gttgaaaaatc atgacacaca gccgggacag tcatgggaat cgacagtcca aacttggttt 1140
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aatgccggcg agacatggta tgacataacg ggcaaccgtt cagatactgt aaaaaatcga 1500
tctgacggct ggggagagtt tcatgtaaac gatgggtccg tctccattta tgttcagaaa 1560

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SEQ ID NO: 4      moltype = AA length = 514
FEATURE          Location/Qualifiers
source           1..514
                  mol_type = protein
                  organism = Bacillus sp.

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SEQUENCE: 4
MNRWKAAPSW MLSLALVFTL FYTPSSASAH HDGTNGTIMQ YFEWNPNDG QHWNRLHNN 60
QNLKNAGITA IWIPPAWKG T SQNDVGYGAY DLYDLGEFNQ KGTVRTKYGT KAELEAIRS 120
LKANGIQVYG DVVMNHKGGA DFTERVQAVE VNPQNRNQEV SGTQIEAWT GFNFPGRGNQ 180
HSSFKWRWYH FDGTDWQSR QLANRIYKFR GDGKAWDEW DTENGNYDYL MYADVMDHP 240
EVINELNRWG VVYANTLNLD GFRLDAVKHI KFSFMRDLWG HVRGQTGKNL FAVAERYWKN 300
LGALENYLSK TNWMTSAFDV PLHYNLYQAS NSSGNVDMRN LLNGTLVQRH PSHAVTFVDN 360
HDTQPGEALE SFVQGWFKPL AYATILTREQ GYPQVFGDY YGIPSDGVPS YRQQIDPLLK 420
ARQQYAYGRQ HDYFDHWDVI GWTRGNASH PNSGLATIMS DPGGSKWMY VGRQKAGEVW 480
HDMTGNSRGT VTINQDQWGH FFWNGGSVSV WVKR 514

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SEQ ID NO: 5      moltype = AA length = 513
FEATURE          Location/Qualifiers
source           1..513
                  mol_type = protein
                  organism = Alicyclobacillus sp.

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SEQUENCE: 5
MKIRNGWKKT LTLFALIFL LPHSAAATFA GDNGTMMQYF EWYLPNDGTL WTKMGSDASH 60
LKSIGITGVV PPPAYKQSQ SDVGYGVYDM YDLGEFNQKG TVRTKYGTKA QLQSAITSLH 120
NNGIQAYGVD VLNRHMGADA TETISAVEVN PSNRNQVTS G AYNISAWTDF EFPGRGNTYS 180
SFKWHSYYFD GVDWDQSRQL SGKIYQIQGT GKAWDEWDS ENGNIDYLMG ADIDYDHPDV 240
QTEVKNWGWV FVNTLNLDGV RLDVAVKHIK DYMSWLLSV KSTTGKSNLF AVGEYWNTSL 300
GALENYENKT NWSMSLFQVP LHMNFQAAAN GGGYDMRNL LNNTMMKNHP IQAVTFVDNH 360
DTEPGQALQS WVSDFWFKPLA YATILTRQEG YPCVFYGDY GIPQSVSAK STWLDKQLSA 420
RKSAYGTQH DYLDNQDQVIG WTRGDSAHA GSGLATVMSD GPGGSKTMYV GTHAGQVFK 480

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DITGNRTDTV TINSAGNGTF PCNGGSVSIW VKQ 513

SEQ ID NO: 6 moltype = AA length = 514
 FEATURE Location/Qualifiers
 source 1..514
 mol_type = protein
 organism = *Bacillus amyloliquefaciens*

SEQUENCE: 6
 MIQKRKRTVS FRLVLMCTLL FVSLPITKTS AVNGTLMQYF EWYTPNDGQH WKRLQNDAEH 60
 LSDIGITAVW IPPAYKGLSQ SDNGYGPYDL YDLGEFQQKG TVRTKYGTKS ELQDAIGSLH 120
 SRNVQVYGDV VLNHKAGADA TEDVTAVEVN PANRNQETSE EYQIKAWTDF RFPGRGNTYS 180
 DFKWHWHYFD GADWDESRKI SRIFKFRGEG KAWDWEVSSE NGNYDYLMYA DVDYDHPDVV 240
 AETKKWGIWY ANELSLDGFR IDAAKHIFKS FLRDWVQAVR QATGKEMFTV AEYWQNNAGK 300
 LENYLNKTSF NQSVFVPLH FNLQAASSQG GGYDMRRLLD GTTVSRHPEK AVTFVENHDT 360
 QPGQSLESTV QTWFKPLAYA FILTRESGYP QVFGDMYGT KGTSPEKIPS LKDNIEPILK 420
 ARKEYAYGPQ HDYIDHPDVI GWTREGDSSA AKSGLAALIT DPGGSKRMY AGLKNAGETW 480
 YDITGNRSDT VKIGSDGWGE FHVNDGSVSI YVQK 514

SEQ ID NO: 7 moltype = AA length = 485
 FEATURE Location/Qualifiers
 source 1..485
 mol_type = protein
 organism = *Bacillus sp.*

SEQUENCE: 7
 HHDGTNGTIM QYFEWNVND GQHWNRLLHN AQLNKNAGIT AIWIPPAWK TSQNDVGYGA 60
 YDLYDLGEFN QKGTVRTKYG TKAELERAIR SLKANGIQVY GDVVMNHKGG ADFTERVQAV 120
 EVNPQNRNQE VSGTYQIEAW TGFNFPGRGN QHSSPKWRWY HFDGTDWDQS RQLANRIYKF 180
 RGDGKAWDWE VDTENGNYDY LMYADVMDH PEVINELNRW GVWYANTLNL DGFRLDAVKH 240
 IKFSPMRDWL GHVRGQTGKN LFAVAEYWK NLGALENYLS KTNWTMSAFD VPLHYNLYQA 300
 SNSSGNYDMR NLLNGTLVQR HPSHAVTFVD NHDTPGEAL ESFVQGWFKP LAYATILTRE 360
 QGYPQVFGYD YGIPSDGVP SYRQQIDPLL KARQQYAYGR QHDYFDHWDV IGWTRREGNAS 420
 HPNSGLATIM SDGPGGSKWM YVGRQKAGEV WHDMTGNRSG TVTINQDQGW HFFVNGGSVS 480
 VWVKR 485

SEQ ID NO: 8 moltype = AA length = 486
 FEATURE Location/Qualifiers
 source 1..486
 mol_type = protein
 organism = *Alicyclobacillus sp.*

SEQUENCE: 8
 TFAGDNGTMM QYFEWYLPND GTLWTKMGSD ASHLKSIGIT GVWPPPAYKG QSQSDVGYGV 60
 YDMYDLGEFN QKGTVRTKYG TKAQLQSAIT SLHNNGIQAY GDVVLNHRMG ADATETISAV 120
 EVNPSNRNQV TSGAYNISAW TDFEFPGRGN TYSSPKWHSY YFDGVDWDQS RQLSGKIYQI 180
 QGTGKAWDWE VDSNGNYDY LMGADIDYDH PDVQTEVKNW GKWFVNTLNL DGVRLDAVKH 240
 IKFDYMSWL SSVKSTTGKS NLFVAGEYWN TSLGALENY NKTNWSMSLF DVPLHMNFQA 300
 AANGGYYDM RNLLNNTMMK NHPIQAVTFV DNHDTEPGQA LQSWVSDWFK PLAYATILTR 360
 QEGYPCVFGY DYYGIPSQSV SAKSTWLDKQ LSARKSYAYG TQHDYLDNQD VIGWTRREGDS 420
 AHAGSGLATV MSDGPGGSKT MYVGTAGAGQ VFKDITGNRT DVTINSAGN GTFPCNGGSV 480
 SIWVKQ 486

SEQ ID NO: 9 moltype = AA length = 483
 FEATURE Location/Qualifiers
 source 1..483
 mol_type = protein
 organism = *Bacillus amyloliquefaciens*

SEQUENCE: 9
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 DLGEFQQKGT VRTKYGTKSE LQDAIGSLHS RNVQVYGDV LNHKAGADAT EDVTAVEVNP 120
 ANRNQETSEE YQIKAWTDFR FPGRGNTYS DFKWHWHYFDG ADWDESRKIS RIFKFRGEGK 180
 AWDWEVSSN GNYDYLMYAD VDYDHPDVVA ETKKWGIWYA NELSLDGFR DAACHIKFSF 240
 LRDWVQAVRQ ATGKEMFTVA EYWQNNAGKL ENYLNKTSFN QSVFVPLHF NLQAASSQGG 300
 GYDMRRLLDG TVVSRHPEKA VTFVENHDTQ PGQSLESTVQ TWFKPLAYAF ILTRESGYPQ 360
 VFGDMYGTG GTSPEKIPSL KDNIEPILKA RKEYAYGPQH DYIDHPDVIG WTREGDSSAA 420
 KSGLAALITD GPGGSKRMYA GLKNAGETWY DITGNRSDTV KIGSDGWGEF HVNDGSVSIY 480
 VQK 483

SEQ ID NO: 10 moltype = AA length = 399
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 source 1..399
 mol_type = protein
 organism = *Bacillus sp.*

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 EVNPQNRNQE VSGTYQIEAW TGFNFPGRGN QHSSPKWRWY HFDGTDWDQS RQLANRIYKF 180
 RGDGKAWDWE VDTENGNYDY LMYADVMDH PEVINELNRW GVWYANTLNL DGFRLDAVKH 240
 IKFSPMRDWL GHVRGQTGKN LFAVAEYWK NLGALENYLS KTNWTMSAFD VPLHYNLYQA 300
 SNSSGNYDMR NLLNGTLVQR HPSHAVTFVD NHDTPGEAL ESFVQGWFKP LAYATILTRE 360
 QGYPQVFGYD YGIPSDGVP SYRQQIDPLL KARQQYAYG 399

SEQ ID NO: 11	moltype = AA length = 86		
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source	1..86		
	mol_type = protein		
	organism = Alicyclobacillus sp.		
SEQUENCE: 11			
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DTVTINSAGN	GTFFPCNGGSV	SIWVKQ	86
SEQ ID NO: 12	moltype = AA length = 86		
FEATURE	Location/Qualifiers		
source	1..86		
	mol_type = protein		
	organism = Bacillus amyloliquefaciens		
SEQUENCE: 12			
PQHDYIDHPD	VIGWTREGDS	SAAKSGLAAL ITDGPGGSKR MYAGLKNAGE TWYDITGNRS	60
DTVKIGSDGW	GEFHVNDGSV	SIYVQK	86
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FEATURE	Location/Qualifiers		
REGION	1..485		
	note = Synthetic construct		
source	1..485		
	mol_type = protein		
	organism = Synthetic construct		
SEQUENCE: 13			
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YDLYDLGEFN	QKGTVRTKYG	TKAELERAIR SLKANGIQVY GDVVMNHKGG ADFTERVQAV	120
EVNPQNRNQE	VSQTYQIEAW	TGFNFPPGRGN QHSSFKWRWY HFDGTDWDQS RQLANRIYKF	180
RGDKAWDWE	VDTEGNYDYL	LMYADVDMDH PEVINELNRW GVWYANTLNL DGFRLDAVKH	240
IKFSFMRDWL	GHVRGQTGKN	LFAVAETYWKN DLGALENYLS KTNWTMSAFD VPLHYNLYQA	300
SNSSGNYDMR	NLLNGTLVQR	HPSHAVTFVD NHDTQPGEAL ESFVQGWFKP LAYATILTRE	360
QGYPQVIFYD	YYGIPSDGVP	SYRQQIDPLL KARQQYAYGT QHDYLDNQDV IGWTRREGDSA	420
HAGSGLATVM	SDGPGGSKTM	YVGTAHAGQV FKDITGNRTD TVTINSAGNG TFPCCNGGSVS	480
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SEQ ID NO: 14	moltype = AA length = 485		
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source	1..485		
	mol_type = protein		
	organism = Synthetic construct		
SEQUENCE: 14			
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YDLYDLGEFN	QKGTVRTKYG	TKAELERAIR SLKANGIQVY GDVVMNHKGG ADFTERVQAV	120
EVNPQNRNQE	VSQTYQIEAW	TGFNFPPGRGN QHSSFKWRWY HFDGTDWDQS RQLANRIYKF	180
RGDKAWDWE	VDTEGNYDYL	LMYADVDMDH PEVINELNRW GVWYANTLNL DGFRLDAVKH	240
IKFSFMRDWL	GHVRGQTGKN	LFAVAETYWKN DLGALENYLS KTNWTMSAFD VPLHYNLYQA	300
SNSSGNYDMR	NLLNGTLVQR	HPSHAVTFVD NHDTQPGEAL ESFVQGWFKP LAYATILTRE	360
QGYPQVIFYD	YYGIPSDGVP	SYRQQIDPLL KARQQYAYGP QHDYIDHPDV IGWTRREGDSS	420
AAKSGLAALI	TDGPGGSKRM	YAGLKNAGET WYDITGNRSD TVKIGSDGWG EHFHVNDGSVS	480
IYVQK			485
SEQ ID NO: 15	moltype = AA length = 483		
FEATURE	Location/Qualifiers		
source	1..483		
	mol_type = protein		
	organism = Bacillus sp.		
SEQUENCE: 15			
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YDLYDLGEFN	QKGTVRTKYG	TKAELERAIR SLKANGIQVY GDVVMNHKGG ADFTERVQAV	120
EVNPQNRNQE	VSQTYQIEAW	TGFNFPPGRGN QHSSFKWRWY HFDGTDWDQS RQLANRIYKF	180
RGKAWDWEVD	TENGNYDYLM	YADVDMDHPE VINELNRWGV WYANTLNLDG FRLDAVKHIK	240
FSFMRDWLGH	VRGQTGKNLF	AVAETYWKNDL GALENYLSKT NWTMSAFDVP LHYNLYQASN	300
SSGNYDMRNL	LNGLTLVQRHP	SHAVTFVDNH DTQPGEALES FVQGWFKPLA YATILTREQQ	360
YPQVIFYGDY	GIPSDGVPSY	RQQIDPLLKA RQQYAYGRQH DYFDHWDVIG WTRREGNASHP	420
NSGLATIMSD	PGPGSKMYMV	GRQKAGEVWH DMTGNRSSTV TINQDQGWGHF FVNGGSVSVW	480
VKR			483
SEQ ID NO: 16	moltype = DNA length = 27		
FEATURE	Location/Qualifiers		
source	1..27		
	mol_type = other DNA		
	organism = Synthetic construct		
SEQUENCE: 16			
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SEQ ID NO: 17	moltype = DNA length = 23		
FEATURE	Location/Qualifiers		

-continued

misc_feature 1..23
 note = Synthetic construct
 source 1..23
 mol_type = other DNA
 organism = Synthetic construct

SEQUENCE: 17
 cggaacgcct ggctgacaac acg

23

The invention claimed is:

1. A variant of a parent alpha-amylase having alpha-amylase activity, which:

- a) is a polypeptide having at least 85% sequence identity to SEQ ID NO: 13 or SEQ ID NO: 14, and
- b) comprises a substitution or a deletion at a position corresponding to N195, N54, V56, G109, R116, or Q172 of the amino acid sequence as set forth in SEQ ID NO: 13 or SEQ ID NO: 14, and comprises a substitution and/or deletion in two, three or four positions corresponding to positions R181, G182, D183, and G184 of the amino acid sequence as set forth in SEQ ID NO: 13 or SEQ ID NO: 14,
- c) has alpha-amylase activity, and
- d) has an improved wash performance at a low temperature of 5° C. to 40° C., wherein the improved wash performance is an Improvement Factor (IF) of >1, when compared to the parent alpha-amylase.

2. The variant of claim 1, which has a substitution at a position corresponding to N195 of the amino acid sequence as set forth in SEQ ID NO: 13 or SEQ ID NO: 14.

3. The variant of claim 1, wherein the parent alpha-amylase comprises an A and a B domain having at least 85%, but less than 100% sequence identity to SEQ ID NO: 10.

4. The variant of claim 1, wherein the parent alpha-amylase comprises a C domain having at least 85% sequence identity to SEQ ID NO: 11 or 12.

5. The variant of claim 1, which has a substitution at a position corresponding to N195F of the amino acid sequence as set forth in SEQ ID NO: 13 or SEQ ID NO: 14.

6. The variant of claim 1 wherein the parent alpha-amylase has at least 90% sequence identity to SEQ ID NO: 13 or 14.

7. The variant of claim 1 wherein the parent alpha-amylase has at least 95% sequence identity to SEQ ID NO: 13 or 14.

8. The variant of claim 3, wherein the A and B domains further comprise a modification in one or more positions corresponding to positions: H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, A60, K72, R87, Q98, M105, F113, Q118, Q125, G133, T134, W140, G142, G149, T165, W167, Q169, R171, L173, A186, E190, T193, G196, A204, V206, P211, I214, V215, L217, L219, A225, L228, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, A265, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, S304, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, and Q395 of SEQ ID NO: 13 or 14.

9. The variant of claim 8, wherein the modification is selected from the group consisting of H1*, T5K, G7K, G7A, Q11H, N16S, N16H, V17L, Q32S, A37H, A37M, A37V, A37S, A37Y, A37R, A37L, T40D, T40G, T40K, P45A, W48G, W48Y, W48F, G50A, T51K, T51E, T51G, T51A, T51S, T51G, T51D, N54S, V56T, A60V, K72R, K72H, K72S, K72Q, K72E, K72N, K72A, K72M, R87S, Q98S, Q98A, M105F, M105I, M105V, M105L, G109A, G109S,

10 F113W, F113S, F113N, F113Y, F113R, F113L, F113Q, R116Q, R116V, R116K, R116W, R116L, R116A, R116H, R116M, R116E, R116S, R116I, R116G, Q118N, Q118K, Q118G, Q118S, Q118F, Q118R, Q125P, Q125K, Q125A, Q125T, N125D, G133S, F133Q, T134E, W140Y, G142T, G149Q, T165S, T165G, T165V, W167I, W167G, W167F, W167R, W167S, W167H, W167L, W167M, W167Y, Q169E, R171H, Q172G, Q172R, Q172N, Q172D, Q172Y, Q172M, Q172S, Q172T, Q172K, Q172H, Q172E, L173V, L173G, L173H, L173A, L173I, L173P, L173T, L173F, L173M, G184T, A186D, A186N, A186E, A186Q, A186H, E190P, T193K, G196R, A204T, A204V, A204S, A204G, V206L, V206S, V206Y, P211D, I214G, I214H, I214S, I214T, I214L, I214E, I214W, V215T, L217T, L217Q, L219V, L219H, A225V, L228I, L235V, L235A, V238A, V238T, V238G, K242Q, S244Q, M246L, M246A, M246I, M246F, M246V, M246S, M248T, L250I, L250V, L250T, L250A, L250F, L250M, G255N, G255A, G255S, Q256A, N260G, A263G, V264I, V264T, A265G, Y267I, Y267M, Y267H, Y267L, K269Q, N270G, N270T, G273R, S280W, S280L, S280T, S280A, S280K, S280Q, W284H, T285L, T285Q, M286F, M286L, A288L, A288V, F289I, F289L, V291G, V291A, V291T, Y295N, Y295F, Q299V, S304N, S304R, R320A, R320V, H321Y, S323N, H324L, V326L, F328L, F328M, F328V, F328I, T334S, D337H, Q345D, G346T, G346D, G346P, G348S, G348P, T355L, T355F, S376H, S376T, S376V, D377H, D377S, D377A, D377Q, D379S, D379G, D379R, D379A, Y382M, Y382I, Y382L, Y382F, S383G, S383A, Q385L, K391A, K391Y, K391V, K391M, K391E, K391D, K391R, K391H, K391W, K391I, K391Q, K391L, K393Y, K393R, K393Q, K393S, and Q395P.

10. The variant of claim 4, wherein the C domain further comprises a modification in one or more positions corresponding to positions: T400, H402, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484 of SEQ ID NO: 13 or I405, A421, A422, A428, R439, G448, W467, D476, and G477 of SEQ ID NO: 14.

11. The variant of claim 10, wherein the modification is selected from the group consisting of:

T400P, H402R, A420Q, A420S, A420K, A420L, G423H, T444Q, T444S, T444D, T444Y, T444H, T444V, T444R, T444A, A445Q, Q449T, T459N, P473R, P473A, P473G, P473T, P473K, C474V, G476K, G477A, G477Q, K484A, K484G, K484P, K484E, K484Q, and K484S, wherein the positions correspond to SEQ ID NO: 13; or

I405L, A421H, A422P, A428T, G448D, D476K, D476G, D476N, D476Y, G477S, G477A, G477T, and G477Q, wherein the positions correspond to SEQ ID NO: 14.

12. The variant of claim 1, which comprises:

- a) a deletion and/or a substitution at two or more positions corresponding to positions R181, G182, D183, and G184 of SEQ ID NO: 13 or 14, and
- b) a further modification at one or more positions:
 - i) H1, T5, G7, Q11, N16, V17, Q32, A37, T40, P45, W48, G50, T51, K72, R87, Q98, M105, F113, Q118, Q125,

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G133, T134, W140, G142, G149, T165, W167, Q169, R171, L173, A186, E190, T193, A204, V206, P211, L214, V215, L217, L219, A225, L235, V238, K242, S244, M246, M248, L250, G255, Q256, N260, A263, V264, Y267, K269, N270, G273, S280, W284, T285, M286, F289, V291, Y295, Q299, R320, H321, S323, H324, V326, F328, T334, D337, Q345, G346, G348, T355, S376, D377, D379, Y382, S383, Q385, K391, K393, Q395, A420, G423, T444, A445, Q449, T459, P473, C474, G476, G477, and K484, wherein the positions correspond to SEQ ID NO: 13; or

ii) A37, T40, A60, K72, F113, N125, F133, T134, T165, Q169, L173, A186, G196, A204, V206, A225, L228, K242, S244, G255, N260, A265, K269, S280, W284, Y295, S304, R320, H321, S323, V326, K391, 1405, A421, A422, A428, R439, G448, W467, D476, and G477, wherein the positions correspond to SEQ ID NO: 14.

13. The variant of claim 12, wherein the deletion a) is selected from the group consisting of R181+G182, R181+D183, R181+G184, G182+D183, G182+G184, or D183+G184.

14. The variant of claim 12, which comprises one or more of the following modifications:

a) H1*, T5K, G7K, G7A, Q11H, N16S, N16H, V17L, Q32S, A37H, A37M, A37V, A37S, A37Y, A37R, A37L, T40G, T40K, P45A, W48G, W48Y, W48F, G50A, T51K, T51E, T51G, T51A, T51S, T51G, T51D, N54S, V56T, K72R, K72H, K72S, K72Q, K72E, K72N, K72A, K72M, R87S, Q98S, Q98A, M105F, M105I, M105V, M105L, G109A, G109S, F113W, F113S, F113N, F113Y, F113R, F113L, F113Q, R116Q, R116V, R116K, R116W, R116L, R116A, R116H, R116M, R116E, R116S, R116I, R116G, Q118N, Q118K, Q118G, Q118S, Q118F, Q118R, Q125P, Q125K, Q125A, Q125T, G133S, T134E, W140Y, G142T, G149Q, T165S, T165G, T165V, W167I, W167G, W167F, W167R, W167S, W167H, W167L, W167M, W167Y, Q169E, R171H, Q172G, Q172R, Q172N, Q172D, Q172Y, Q172M, Q172S, Q172T, Q172K, Q172H, Q172E, L173V, L173G, L173H, L173A, L173I, L173P, L173T, L173F, L173M, G184T, A186D, A186N, A186E, A186Q, A186H, E190P, T193K, A204T, A204V, A204S, A204G, V206L, V206S,

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V206Y, P211D, I214G, I214H, I214S, I214T, I214L, I214E, I214W, V215T, L217T, L217Q, L219V, L219H, A225V, L235V, L235A, V238A, V238T, V238G, K242Q, S244Q, M246L, M246A, M246I, M246F, M246V, M246S, M248T, L250I, L250V, L250T, L250A, L250F, L250M, G255N, G255A, G255S, Q256A, N260G, A263G, V264I, V264T, Y267I, Y267M, Y267H, Y267L, K269Q, N270G, N270T, G273R, S280W, S280L, S280T, S280A, S280K, S280Q, W284H, T285L, T285Q, M286F, M286L, A288L, A288V, F289I, F289L, V291G, V291A, V291T, Y295N, Y295F, Q299V, S340R, S304N, R320A, R320V, H321Y, S323N, H324L, V326L, F328L, F328M, F328V, F328I, T334S, D337H, Q345D, G346T, G346D, G346P, G348S, G348P, T355L, T355F, S376H, S376T, S376V, D377H, D377S, D377A, D377Q, D379S, D379G, D379R, D379A, Y382M, Y382I, Y382L, Y382F, S383G, S383A, Q385L, K391A, K391Y, K391V, K391M, K391E, K391D, K391R, K391H, K391W, K391I, K391Q, K391L, K393Y, K393R, K393Q, K393S, Q395P, T400P, H402R, A420Q, A420S, A420K, A420L, G423H, T444Q, T444S, T444D, T444Y, T444H, T444V, T444R, T444A, A445Q, Q449T, T459N, P473R, P473A, P473G, P473T, P473K, C474V, G476K, G477A, G477Q, K484A, K484G, K484P, K484E, K484Q, and K484S, wherein the positions correspond to SEQ ID NO: 13; or

b) G7A, A37H, T40D, W48Y, W48F, N54S, V56T, A60V, K72R, Q98A, G109A, F113Q, R116H, R116V, R116Q, N125D, F133Q, T134E, T165G, Q169E, Q172D, Q172G, Q172N, L173V, G184T, A186D, A186N, A186E, A186Q, A186H, E190P, G196R, A204G, V206L, V206Y, A225V, L228I, K242Q, S244O, G255A, G255S, N260G, A265G, K269Q, N270T, S280Q, W284H, A288L, A288V, F289L, Y295N, Y295F, S304N, S340R, S304N, R320A, H321Y, S323N, V326L, K391 A, 1405L, A421H, A422P, A428T, G448D, D476K, D476G, D476N, D476Y, G477S, G477A, G477T, and G477Q, wherein the positions correspond to SEQ ID NO: 14.

15. The variant of claim 1, which has an IF of >1 when compared to the parent alpha-amylase, and which comprises:

N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K,
 N54S + V56T + R87S + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K,
 T40G + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K,
 T51K + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K,
 N54S + V56T + K72R + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K,
 N54S + V56T + K72H + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K,
 A37H + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K,
 A37M + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K,
 A37V + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K,
 A37S + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K,
 A37Y + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K,
 A37R + N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K,
 N54S + V56T + G109A + F113W + A174S + G182* + D183* + N195F + V206L + K391A + G476K,
 N54S + V56T + G109A + F113S + A174S + G182* + D183* + N195F + V206L + K391A + G476K,
 N54S + V56T + G109A + F113N + A174S + G182* + D183* + N195F + V206L + K391A + G476K,
 N54S + V56T + G109A + F113Y + A174S + G182* + D183* + N195F + V206L + K391A + G476K,
 N54S + V56T + G109A + F113R + A174S + G182* + D183* + N195F + V206L + K391A + G476K,
 N54S + V56T + G109A + Q118N + A174S + G182* + D183* + N195F + V206L + K391A + G476K,
 N54S + V56T + G109A + R116Q + A174S + G182* + D183* + N195F + V206L + K391A + G476K,
 N54S + V56T + G109A + R116V + A174S + G182* + D183* + N195F + V206L + K391A + G476K,
 N54S + V56T + G109A + R116K + A174S + G182* + D183* + N195F + V206L + K391A + G476K,
 N54S + V56T + G109A + R116W + A174S + G182* + D183* + N195F + V206L + K391A + G476K,
 N54S + V56T + G109A + R116L + A174S + G182* + D183* + N195F + V206L + K391A + G476K,
 N54S + V56T + G109A + G142T + A174S + G182* + D183* + N195F + V206L + K391A + G476K,
 N54S + V56T + G109A + Q125P + A174S + G182* + D183* + N195F + V206L + K391A + G476K,

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[illegible]

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[illegible]

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[illegible]

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N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + F289L + K391A + G476K,
 N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + Y295F + K391A + G476K,
 N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + V291A + K391A + G476K,
 N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + S244Q + K391A + G476K,
 N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + A225V + K391A + G476K,
 N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + K391A + P473R + G476K,
 W48Y + N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
 N54S + V56T + G109A + Q169E + Q172K + A174* + G182* + D183* + N195F + V206L + K391A + G476K,
 N54S + V56T + Q98A + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K,
 N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206Y + K391A + G476K,
 N54S + V56T + G109A + A174S + G182* + D183* + G184T + N195F + V206L + K391A + G476K,
 N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + K391A + G476K,
 G109A + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
 A174S + D183* + G182*,
 A174S + G182* + O183* + N195F,
 N54S + V56T + G109A + A174S + G182* + D183* + N195F + V206L + G255A + K391A + G476K, or
 N54S + V56T + G109A + Q169E + Q172K + A174S + G182* + D183* + N195F + V206L + K391A + G476K,

wherein the positions correspond to SEQ ID NO: 13.

16. The variant of claim 1, which has an IF of >1 when compared to the parent alpha-amylase, and which comprises:

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N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T + D476K,
 N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T,
 N54S + V56T + G109A + A 174S + D183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T,
 N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + D476K,
 N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + D476G,
 N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + D476K + G477S,
 N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + D476K,
 N54S + V56T + K72R + G109A + W167F + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T,
 N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + S304N + I405L + A421H + A422P + A428T + 0476K,
 N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + O476K + G477A,
 N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + O476G + G477T,
 N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + G477Q,
 N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + G477A,
 N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + O476K + G477Q,
 N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + O476N,
 N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + 0476Y,
 N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T,
 N54S + V56T + G109A + Q1720 + A174S + G182* + O183* + N195F + A204G + V206L + I405L + A421H + A422P + A428T,
 N54S + V56T + K72R + G109A + T134E + A 174S + G182* + O183* + N195F + A204G + V206L + G255A + I405L + A421H + A422P + A428T,
 N54S + V56T + G109A + F113Q + R116H + T165G + Q172G + A 174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T,
 N54S + V56T + G109A + F113Q + Q172N + A 174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T,
 N54S + V56T + K72R + G109A + T134E + A174S + G182* + O183* + N195F + G196R + V206L + G255A + I405L + A421H + A422P + A428T + N475S,

N54S + V56T + G109A + T134E + A 174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T,
 T400 + N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T,
 N54S + V56T + G109A + F113Q + N1250 + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T,
 N54S + V56T + A60V + G109A + R116Q + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T,
 N54S + V56T + G109A + R116V + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T,
 N54S + V56T + G109A + R 116Q + G1330 + T165G + Q172G + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T,
 N54S + V56T + G109A + F113Q + R116H + Q172G + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T,
 N54S + V56T + G109A + F113Q + R116Q + Q172G + A174S + G182* + O183* + N195F + V206L + I405L + A421 H + A422P + A428T + G448O,
 N54S + V56T + G109A + F113Q + R116H + Q172N + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T,
 N54S + V56T + K72R + G109A + T134E + W167F + Q172N + A174S + G182* + O183* + N195F + V206L + G255A + A265G + I405L + A421H + A422P + A428T,
 N54S + V56T + K72R + G109A + R116Q + W167F + Q172R + L173V + A174S + G182* + O183* + N195F + V206L + A265G + I405L + A421H + A422P + A428T,
 N54S + V56T + K72R + G109A + R116Q + W167F + Q172R + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T,
 N54S + V56T + K72R + G109A + F113Q + T134E + W167F + Q 172G + A 174S + G182* + O183* + N195F + V206L + G255A + I405L + A421H + A422P + A428T,
 N54S + V56T + K72R + G109A + F113Q + R 116H + W167F + Q172G + L173V + A174S + G182* + O183* + G184T + N195F + V206L + I405L + A421H + A422P + A428T,
 N54S + V56T + G109A + F113Q + W167F + Q172G + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T,
 N54S + V56T + K72R + G109A + A 174S + G182* + O183* + N195F + V206L + G255A + I405L + A421H + A422P + A428T,
 N54S + V56T + G109A + F113Q + R116Q + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T,
 N54S + V56T + K72R + G109A + F113Q + R 116Q + W167F + Q172G + A174S + G182* + O183* + N195F + A204G + V206L + I405L + A421H + A422P + A428T,
 N54S + V56T + G109A + F113Q + R116Q + W167F + Q172G + A174S + G182* + O183* + N195F + A204G + V206L + I405L + A421H + A422P + A428T,
 N54S + V56T + K72R + G109A + F113Q + F116H + W167F + Q172G + L173V + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T,
 N54S + V56T + G109A + R116Q + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T,

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N54S + V56T + A60V + G109A + R116Q + W167F + Q172N + L173V + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T,
 A37H + N54S + V56T + A60V + G109A + R116Q + T165G + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T,
 N54S + V56T + G109A + A 174S + M105F + G182* + O183* + N195F + V206L + L2281 + R320A + S323N + I405L + A421H + A422P + A428T,
 N54S + V56T + K72R + G109A + F113Q + T134E + A174S + G182* + O183* + N195F + V206L + G255A + I405L + A421H + A422P + A428T,
 N54S + V56T + G109A + R116Q + V120L + A174S + G182* + D183* + K184T + N195F + V206L + I405L + A421H + A422P + A428T,
 N54S + V56T + G109A + F113Q + R116H + W167F + Q172G + L173V + A 174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T,
 N54S + V56T + G109A + F113Q + R116Q + Q172N + A174S + G182* + O183* + N195F + V206L + A265G + I405L + A421H + A422P + A428T, or
 A37H + N54S + V56T + A60V + G109A + R116Q + W167F + Q172R + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T,

wherein the positions correspond to SEQ ID NO: 14.

17. The variant of claim 1, which comprises:

G7K+N54S+V56T+G109A+A174S+G182*+O183*+
 N195F+V206L+K391A+G476K,
 I9L+N54S+V56T+G109A+Q125S+V131T+A174*+
 G182*+O183*+V206L+Y267F+F289H+Q361E+
 K391 A+G476K,
 N16H+V17L+N54S+V56T+G109A+A 174S+G182*+
 D183*+N195F+V206L+D377H+K391 A+G476K,
 N16H+V17L+N54S+V56T+G109A+A 174S+G182*+
 D183*+N195F+V206L+K391A+G476K,
 N16Y+N54S+V56T+G109A+R116H+A174S+G182*+
 O183*+N195F+V206L+G337E+O377H+K391A+
 G476K,
 Q22N+N54S+V56T+G109A+A 174S+G182*+O183*+
 N195F+V206L+K391 A+G476K,
 H28N+N29S+N54S+V56T+G109A+A174S+G182*+
 O183*+N195F+V206L+K391A+G476K,
 H28N+N54S+V56T+G109A+A 174S+G182*+O183*+
 N195F+V206L+K391A+G476K,
 N29S+N54S+V56T+G109A+A 174S+G182*+O183*+
 N195F+V206L+K391A+G476K,
 Q32S+N54S+V56T+G109A+A 174S+G182*+O183*+
 N195F+V206L+K391A+G476K,
 T40K+N54S+V56T+G109A+G149H+A174S+G182*+
 O183*+N195F+V206L+K391A+G476K,
 G50A+T51A+N54S+V56T+G109A+A174S+G182*+
 O183*+N195F+V206L+K391A+G476K,
 G50A+T51A+V56T+G109A+A174S+G182*+O183*+
 N195F+V206L+G476K,
 T51G+N54S+V56T+G109A+G11 ON+A174S+G182*+
 O183*+N195F+V206L+K391A+G476K,
 T51G+N54S+V56T+G109A+A 174S+G182*+O183*+
 N195F+Y203G+V206L+K391A+G476K,
 T51Q+N54S+V56T+G109A+A 174S+G182*+O183*+
 N195F+Y203F+V206L+K391A+G476K,
 N54S+V56T+K72R+G109A+A 174S+G182*+O183*+
 N195F+V206L+G255A+K391A+G476K,
 N54S+V56T+M105F+G109A+A174S+G182*+O183*+
 N195F+V206L+M246F+K391A+G476K,
 N54S+V56T+M105F+G109A+A174S+G182*+O183*+
 N195F+V206L+R320A+S323N+K391A+G476K,
 N54S+V56T+G109A+Q125S+A174S+N175G+G182*+
 O183*+N195F+V206L+K391A+G476K,

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N54S+V56T+G109A+Q125S+A174S+G182*+O183*+
 N195F+V206L+K391 A+G476K,
 N54S+V56T+G109A+T134E+A174S+G182*+D183*+
 N195F+V206L+K391A+G476K,
 5 N54S+V56T+G109A+Q169E+Q172K+A174*+N175Q+
 G182*+D183*+N195F+V206L+K391 A+G476K,
 N54S+V56T+G109A+Q169E+Q172K+A174*+R176K+
 G182*+D183*+N195F+V206L+K391 A+G476K,
 N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+
 D183*+N195F+V206L+K391A+G476K,
 10 N54S+V56T+G109A+A 174S+N175G+G182*+D183*+
 N195F+V206L+K391A+G476K,
 N54S+V56T+G109A+A 174S+R176K+G182*+D183*+
 N195F+V206L+K391A+G476K,
 15 N54S+V56T+G109A+A 174S+G182*+D183*+N195F+
 M202V+V206L+K391A+G476K,
 N54S+V56T+G109A+A 174S+G182*+D183*+N195F+
 V206L+L235I+K391A+G476K,
 20 N54S+V56T+G109A+A174S+G182*+D183*+N195F+
 V206L+L235V+V238A+K391A+G476K,
 N54S+V56T+G109A+A 174S+G182*+D183*+N195F+
 V206L+L235V+K391A+G476K,
 N54S+V56T+G109A+A 174S+G182*+D183*+N195F+
 V206L+M246F+A265G+V291A+F328L+Q365S+
 25 K391A+G476K,
 N54S+V56T+G109A+A 174S+G182*+D183*+N195F+
 V206L+A265G+V291A+F328L+K391A+G476K,
 N54S+V56T+G109A+A 174S+G182*+D183*+N195F+
 V206L+A265G+K391A+G476K,
 30 N54S+V56T+G109A+A 174S+G182*+D183*+N195F+
 V206L+K269S+A274K+K391A+G476K,
 N54S+V56T+G109A+A 174S+G182*+D183*+N195F+
 V206L+V291A+Y295N+Q299T+K391 A+G476K,
 N54S+V56T+G109A+A 174S+G182*+D183*+N195F+
 V206L+R320S+H321N+K391A+G476K,
 35 N54S+V56T+G109A+A 174S+G182*+D183*+N195F+
 V206L+Q365S+K391A+G476K,
 N54S+V56T+G109A+A 174S+G182*+D183*+N195F+
 V206L+D377H+K391A+G476K,
 40 N54S+V56T+G109A+A 174S+G182*+D183*+N195F+
 V206L+K391A+H402R+G476K, or
 N54S+V56T+G109A+A 174S+G182*+D183*+N195F+
 V206L+K391A+G476K,

wherein the positions correspond to SEQ ID NO: 13, and wherein the IF is determined by use of Model A detergent.

18. The variant of claim 1, which has an IF of at least >1.5 when compared to the parent alpha-amylase, and which comprises:

N54S + V56T + G109A + A 174S + G182* + D183* + N195F +
 V206L + K391A + G476K,
 N54S + V56T + R87S + G109A + A 174S + G182* + D183* +
 N195F + V206L + K391A + G476K,
 55 T40G + N54S + V56T + G109A + A 174S + G182* + D183* +
 N195F + V206L + K391A + G476K,
 T51K + N54S + V56T + G109A + A 174S + G182* + O183* +
 N195F + V206L + K391A + G476K,
 N54S + V56T + K72R + G109A + A 174S + G182* + O183* +
 N195F + V206L + K391A + G476K,
 N54S + V56T + K72H + G109A + A 174S + G182* + O183* +
 N195F + V206L + K391A + G476K,
 A37H + N54S + V56T + G109A + A 174S + G182* + O183* +
 N195F + V206L + K391A + G476K,
 A37M + N54S + V56T + G109A + A 174S + G182* + D183* +
 N195F + V206L + K391A + G476K,
 65 A37V + N54S + V56T + G109A + A 174S + G182* + D183* +
 N195F + V206L + K391A + G476K,

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V206L + K391A + A420Q + G476K,
 N54S + V56T + G109A + A 174S + G182* + O183* + N195F +
 V206L + K391A + A420S + G476K,
 N54S + V56T + G109A + A 174S + G182* + O183* + N195F +
 V206L + K391A + A420K + G476K,
 N54S + V56T + G109A + A 174S + G182* + O183* + N195F +
 V206L + K391A + A420L + G476K,
 N54S + V56T + G109A + A 174S + G182* + O183* + N195F +
 V206L + K391A + K391Y + G476K,
 N54S + V56T + G109A + A 174S + G182* + O183* + N195F +
 V206L + K391A + P473R + G476K,
 N54S + V56T + G109A + A 174S + G182* + O183* + N195F +
 V206L + K391A + P473A + G476K,
 N54S + V56T + G109A + A 174S + G182* + D183* + N195F +
 V206L + K391A + P473G + G476K,
 N54S + V56T + G109A + A 174S + G182* + D183* + N195F +
 V206L + K391A + P473T + G476K,
 N54S + V56T + G109A + A 174S + G182* + D183* + N195F +
 V206L + K391A + Q395P + G476K,
 N54S + V56T + G109A + A 174S + G182* + D183* + N195F +
 V206L + K391A + T444Q + G476K,
 N54S + V56T + G109A + A 174S + G182* + D183* + N195F +
 V206L + K391A + T444S + G476K,
 N54S + V56T + G109A + A 174S + G182* + D183* + N195F +
 V206L + K391A + T444D + G476K,
 N54S + V56T + G109A + A 174S + G182* + D183* + N195F +
 V206L + K391A + T444Y + G476K,
 N54S + V56T + G109A + A 174S + G182* + D183* + N195F +
 V206L + S280W + K391A + G476K,
 N54S + V56T + G109A + A 174S + G182* + D183* + N195F +
 V206L + S280L + K391A + G476K,
 N54S + V56T + G109A + A 174S + G182* + D183* + N195F +
 V206L + S280T + K391A + G476K,
 N54S + V56T + G109A + A 174S + G182* + D183* + N195F +
 V206L + S280A + K391A + G476K,
 N54S + V56T + G109A + A 174S + G182* + D183* + N195F +
 V206L + G255N + K391A + G476K,
 N54S + V56T + G109A + A 174S + G182* + D183* + N195F +
 V206L + T285L + K391A + G476K,
 N54S + V56T + G109A + A 174S + G182* + D183* + N195F +
 V206L + T285Q + K391A + G476K,
 N54S + V56T + G109A + A 174S + G182* + D183* + N195F +
 V206L + K391A + Q449T + G476K,
 N54S + V56T + G109A + A 174S + G182* + D183* + N195F +
 V206L + K391A + P473K + G476K,
 N54S + V56T + K72S + G109A + A 174S + G182* + D183* +
 N195F + V206L + K391A + G476K,
 T40K + N54S + V56T + G109A + A174S + G182* + D183* +
 N195F + V206L + G346T + K391A + G476K,
 G50A + N54S + V56T + G109A + Q172N + A174S + G182* +
 D183* + N195F + V206L + K391A + G476K,
 N54S + V56T + G109A + A 174S + G182* + D183* + N195F +
 V206L + K391A + G476K + K484A,
 N54S + V56T + G109A + A 174S + G182* + D183* + N195F +
 V206L + K391A + G476K + K484G,
 N54S + V56T + G109A + A 174S + G182* + D183* + N195F +
 V206L + K391A + G476K + K484P,
 N54S + V56T + G109A + A 174S + G182* + D183* + N195F +
 V206L + K391A + G476K + K484Q,
 N54S + V56T + G109A + A 174S + G182* + D183* + N195F +
 V206L + K391A + G476K + K484S,
 H1*K + G50A + N54S + V56T + G109A + W167F + A174S +
 G182* + D183* + N195F + V206L + K391A + G476K,
 N54S + V56T + G109A + L173V + A174S + G182* + D183* +
 N195F + A204T + V206L + K391A + G476K,
 N54S + V56T + G109A + W140Y + A174S + G182* + D183* +
 N195F + V206L + K391A + G476K,
 N54S + V56T + Q98S + G109A + A 174S + G182* + D183* +
 N195F + V206L + R320A + K391A + G476K,
 N54S + V56T + G109A + A 174S + G182* + O183* + N195F +
 V206L + H324L + K391A + G476K,
 N54S + V56T + G109A + G149Q + A 174S + G182* + O183* +
 N195F + V206L + K391A + G476K,
 N54S + V56T + G109A + A 174S + G182* + O183* + N195F +
 V206L + G346T + K391A + G476K + G477A.

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A37V + N54S + V56T + G109S + F113Q + W167F + Q172R + A174S + G182* + O183* + N195F + V206L + K391 A + G476K, N54S + V56T + K72R + G109A + R116H + T134E + W167F + Q172G + L173V + A174S + G182* + O183* + N195F + V206L + G255A + K391A + G476K, N54S + V56T + K72R + G109A + R116H + T134E + W167F + Q172G + L173V + A174S + G182* + O183* + N195F + V206L + G255A + K391A + Q395P + T444Q + P473R + G476K, N54S + V56T + K72R + G109A + F113Q + R 116Q + W167F + Q172G + A174S + G182* + O183* + G184T + N195F + V206L + K391A + P473R + G476K, N54S + V56T + K72R + G109A + F113Q + R116Q + T134E + W167F + Q172R + A174S + G182* + O183* + N195F + V206L + G255A + K391A + T444Q + P473R + G476K, N54S + V56T + K72R + G109A + F113Q + T134E + W167F + Q 172R + A174S + G182* + O183* + N195F + V206L + G255A + K391A + Q395P + A445Q + P473R + G476K, N54S + V56T + K72R + G109A + F113Q + R 116H + T134E + W167F + Q172R + A174S + G182* + O183* + N195F + V206L + G255A + K391A + Q395P + G476K, N54S + V56T + K72R + G109A + T134E + W167F + Q172N + A174S + G182* + O183* + N195F + V206L + G255A + A265G + K391A + T444Q + P473R + G476K, N54S + V56T + K72R + G109A + G133Q + W167F + Q172R + A174S + G182* + O183* + N195F + V206L + K391 A + G476K, N54S + V56T + K72R + G109A + F113Q + R 116Q + W167F + Q172G + A174S + G182* + O183* + N195F + A204G + V206L + K391A + G476K, N54S + V56T + K72R + G109A + F113Q + R 116Q + W167F + Q172N + L173V + A174S + G182* + O183* + G184T + N195F + V206L + K391A + T444Q + P473R + G476K, A37H + N54S + V56T + G109A + F113Q + W167F + Q172N + A174S + G182* + O183* + N195F + V206L + K391 A + G476K, A37H + N54S + V56T + G109A + F113Q + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + F113Q + W167F + Q172N + A174S + G182* + O183* + N195F + V206L + K391A + T459N + G476K, N54S + V56T + G109A + F113Q + Q172R + L173V + A 174S + G182* + O183* + N195F + V206L + K391A + T444Q + G476K, N54S + V56T + G109A + F113Q + R116Q + A174S + G182* + O183* + N195F + V206L + K391A + P473R + G476K, N54S + V56T + A60V + G109A + R116Q + W167F + Q172N + A174S + G182* + O183* + N195F + V206L + K391 A + G476K, N54S + V56T + A60V + G109A + R116H + Q172R + L173V + A 174S + G182* + O183* + N195F + V206L + K391 A + G476K, A37V + N54S + V56T + A60V + G109A + R116Q + T165G + A174S + G182* + O183* + N195F + V206L + K391A + G476K, A37H + N54S + V56T + A60V + G109A + R116Q + T165G + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + A60V + G109A + R116Q + W167F + Q172G + L173V + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + A60V + G109A + R116Q + A174S + G182* + O183* + N195F + V206L + K391A + G476K, A37H + N54S + V56T + A60V + G109A + R116Q + W167F + Q172R + A174S + G182* + O183* + N195F + V206L + K391A + G476K, A37H + N54S + V56T + A60V + G109A + R116Q + W167F + Q172N + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + A60V + G109A + R116Q + T165G + A174S + G182* + O183* + N195F + V206L + K391A + T444Q + P473R + G476K, N54S + V56T + A60V + G109A + R116Q + W167F + Q172N + L173V + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + A60V + G109A + R116Q + W167F + Q172R + A174S + G182* + O183* + N195F + V206L + K391 A + G476K, N54S + V56T + G109A + R116A + A174S + G182* + O183* + N195F + A204G + V206L + K391A + P473R + G476K, N54S + V56T + G109A + R116A + T165G + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, G109A + A174S + G182* + O183* + N195F + V206L + K391 A + G476K,

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N54S + V56T + G109A + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + G476K, N54S + V56T + A174S + G182* + O183* + N195F + V206L + G273R + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L, W48G + G109A + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + F113Q + Q172M + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + R116Q + A174S + G182* + O183* + N195F + A204G + V206L + K391 A + G476K, N54S + V56T + K72R + G109A + F113Q + T134E + A174S + G182* + O183* + N195F + V206L + G255A + K391A + G476K, N54S + V56T + K72R + G109A + F113Q + T134E + A174S + G182* + O183* + N195F + V206L + G255A + K391A + P473R + G476K, N54S + V56T + K72R + G109A + R116Q + A174S + G182* + O183* + G184T + N195F + V206L + K391A + G476K, A37V + N54S + V56T + G109A + R116A + T165G + A174S + G182* + O183* + N195F + V206L + K391A + G476K, A37V + N54S + V56T + G109A + R116Q + T165G + A174S + G182* + O183* + N195F + V206L + M246F + K391A + G476K, V56A + G109A + A 174S + G182* + O183* + N195F + V206L + G476K, N54S + V56T + K72R + G109A + F113Q + R116H + A174S + G182* + O183* + G184T + N195F + V206L + K391 A + T444Q + G476K, N54S + V56T + G109S + F113Q + A174S + G182* + O183* + N195F + V206L + K391A + T444Q + G476K, N54S + V56T + G109A + F113Q + R116H + W167F + Q172G + L173V + A 174S + G182* + O183* + N195F + V206L + K391A + P473R + G476K, N54S + V56T + G109A + F113Q + R116Q + Q172N + A174S + G182* + O183* + N195F + V206L + A265G + K391A + P473R + G476K, N54S + V56T + G109A + F113Q + W167F + Q172N + A174S + G182* + O183* + N195F + A204G + V206L + K391A + P473R + G476K, N54S + V56T + G109A + F113Q + R116H + W167F + Q172R + A174S + G182* + O183* + N195F + A204G + V206L + K391A + P473R + G476K, N54S + V56T + G109A + R116A + A174S + G182* + O183* + N195F + V206L + K391A + T444Q + P473R + G476K, N54S + V56T + G109A + R116H + Q172R + A174S + G182* + O183* + N195F + A204G + V206L + K391A + G476K, A37V + N54S + V56T + G109A + R116Q + T165G + A174S + G182* + O183* + N195F + V206L + K391A + G476K, A37H + N54S + V56T + G109A + R116H + T165G + A174S + G182* + O183* + N195F + V206L + M246F + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + K391 A + G476K, N54S + V56T + G109A + G133S + A174S + G182* + O183* + N195F + K391A + G476H, N54S + V56T + G109A + R116A + Q172N + L173V + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, A37V + N54S + V56T + G109A + R116H + T165G + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, P45A + N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + K391A + C474V + G476K, N54S + V56T + G109A + P124A + A174S + G182* + O183* + N195F + V206L + K391A + C474V + G476K, N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + P364A + K391A + C474V + G476K, N54S + V56T + G109A + Q118R + T134E + A 174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + G184T + N195F + V206L + K391A + G476K, G50A + T51A + N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + V264F + K391A + G476K, N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + Y382L + K391A + G476K, G7K + N54S + V56T + G109A + A 174S + G182* + O183* +

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N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + G476K,
G109A + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + Y295F + K391 A + G476K, or
N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + V291A + K391A + G476K,

wherein the positions correspond to SEQ ID NO: 13.

19. The variant of claim 1, which has an IF of at least >1.5 when compared to the parent alpha-amylase, and which comprises:

N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + D476G + G477T,
N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + G477Q,
N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + G477A,
N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + D476N,
N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + I405L + A421 H + A422P + A428T + D476Y,
N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T,
N54S + V56T + G109A + T134E + A 174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T,
T40O + N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T,
N54S + V56T + G109A + F113Q + N125O + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T,
N54S + V56T + A60V + G109A + R116Q + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T,
N54S + V56T + G109A + F113Q + R116H + Q172G + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T,
N54S + V56T + G109A + F113Q + R116H + Q172N + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T,
N54S + V56T + K72R + G109A + T134E + W167F + Q172N + A174S + G182* + O183* + N195F + V206L + G255A + A265G + I405L + A421H + A422P + A428T,
N54S + V56T + K72R + G109A + R116Q + W167F + Q172R + L173V + A174S + G182* + O183* + N195F + V206L + A265G + I405L + A421H + A422P + A428T,
N54S + V56T + K72R + G109A + R116Q + W167F + Q172R + A174S + G182* + O183* + N195F + V206L + I405L + A421H + A422P + A428T,
N54S + V56T + K72R + G109A + F113Q + R 116H + W167F + Q172G + L173V + A174S + G182* + O183* + G184T + N195F + V206L + I405L + A421H + A422P + A428T,
N54S + V56T + K72R + G109A + F113Q + R 116H + W167F + Q172G + A174S + G182* + O183* + G184T + N195F + V206L + I405L + A421H + A422P + A428T,
N54S + V56T + G109A + F113Q + R116Q + W167F + Q172G + A174S + G182* + O183* + N195F + A204G + V206L + I405L + A421H + A422P + A428T,
N54S + V56T + G109A + F113Q + R116H + W167F + Q172G + A174S + G182* + O183* + N195F + V206L + A265G + I405L + A421H + A422P + A428T,
N54S + V56T + G109A + R116Q + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T,
A37H + N54S + V56T + A60V + G109A + R116Q + T165G + A174S + G182* + O183* + N195F + V206L + I405L + A421H +

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A422P + A428T,
N54S + V56T + K72R + G109A + F113Q + T134E + A 174S + G182* + D183* + N195F + V206L + G255A + I405L + A421H + A422P + A428T,
N54S + V56T + K72R + G109A + R116Q + V120L + A174S + G182* + D183* + G184T + N195F + V206L + I405L + A421H + A422P + A428T,
N54S + V56T + G109A + F113Q + R116H + W167F + Q172G + L173V + A 174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T,
N54S + V56T + G109A + F113Q + R116Q + Q172N + A174S + G182* + O183* + N195F + V206L + A265G + I405L + A421H + A422P + A428T,
A37H + N54S + V56T + A60V + G109A + R116Q + W167F + Q172R + A174S + G182* + D183* + N195F + V206L + I405L + A421H + A422P + A428T,

wherein the positions correspond to SEQ ID NO: 14, and wherein the IF is determined by use of Model A detergent.

20. The variant of claim 1, which has an IF of at least >2.0 when compared to the parent alpha-amylase, and which comprises:

N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + G476K,
N54S + V56T + R87S + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + G476K,
T40G + N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + G476K,
T51K + N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + G476K,
N54S + V56T + K72R + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + G476K,
N54S + V56T + K72H + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + G476K,
A37H + N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + G476K,
A37M + N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + G476K,
A37V + N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + G476K,
A37S + N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + G476K,
A37Y + N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + G476K,
A37R + N54S + V56T + G109A + A 174S + G182* + D183* + N195F + V206L + K391A + G476K,
N54S + V56T + G109A + F113W + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
N54S + V56T + G109A + F113S + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
N54S + V56T + G109A + R116Q + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
N54S + V56T + G109A + R116V + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
N54S + V56T + G109A + R116K + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
N54S + V56T + G109A + Q125P + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + S381G + K391A + G476K,
N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + M246T + K391A + G476K,
A37L + N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + G476K,
N54S + V56T + G109A + R116A + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
N54S + V56T + G109A + R116H + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
N54S + V56T + G109A + Q125A + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
N54S + V56T + G109A + Q172G + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
N54S + V56T + G109A + Q172R + A174S + G182* + O183* +

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Q172G + L173V + A174S + G182* + O183* + N195F + V206L + G255A + K391A + Q395P + T444Q + P473R + G476K,
N54S + V56T + K72R + G109A + F113Q + R116Q + W167F + Q172G + A174S + G182* + O183* + G184T + N195F + V206L + K391A + P473R + G476K,
N54S + V56T + K72R + G109A + F113Q + R116Q + T134E + W167F + Q172R + A174S + G182* + O183* + N195F + V206L + G255A + K391A + T444Q + P473R + G476K,
N54S + V56T + K72R + G109A + F113Q + R116H + T134E + W167F + Q172R + A174S + G182* + O183* + N195F + V206L + G255A + K391A + Q395P + G476K,
N54S + V56T + K72R + G109A + T134E + W167F + Q172N + A174S + G182* + O183* + N195F + V206L + G255A + A265G + K391A + T444Q + P473R + G476K,
N54S + V56T + K72R + G109A + G133Q + W167F + Q172R + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
N54S + V56T + K72R + G109A + F113Q + R116Q + W167F + Q172G + A174S + G182* + O183* + N195F + A204G + V206L + K391A + G476K,
N54S + V56T + K72R + G109A + F113Q + R116Q + W167F + Q172N + L173V + A174S + G182* + O183* + G184T + N195F + V206L + K391A + T444Q + P473R + G476K,
A37H + N54S + V56T + G109A + F113Q + W167F + Q172N + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
N54S + V56T + G109A + F113Q + W167F + Q172N + A174S + G182* + O183* + N195F + V206L + K391A + T459N + G476K,
N54S + V56T + G109A + F113Q + R116Q + A174S + G182* + O183* + N195F + V206L + K391A + P473R + G476K,
N54S + V56T + A60V + G109A + R116Q + W167F + Q172N + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
N54S + V56T + A60V + G109A + R116H + Q172R + L173V + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
A37V + N54S + V56T + A60V + G109A + R116Q + T165G + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
A37H + N54S + V56T + A60V + G109A + R1160 + T165G + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
N54S + V56T + A60V + G109A + R116Q + W167F + Q172G + L173V + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
N54S + V56T + A60V + G109A + R116Q + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
A37H + N54S + V56T + A60V + G109A + R116Q + W167F + Q172R + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
A37H + N54S + V56T + A60V + G109A + R116Q + W167F + Q172N + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
N54S + V56T + A60V + G109A + R116Q + T165G + A174S + G182* + O183* + N195F + V206L + K391A + T444Q + P473R + G476K,
N54S + V56T + A60V + G109A + R116Q + A174S + G182* + O183* + N195F + V206L + K391A + T444Q + P473R + G476K,
N54S + V56T + A60V + G109A + R116Q + W167F + Q172R + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
N54S + V56T + G109A + R116A + W167F + Q172R + A174S + G182* + O183* + N195F + A204G + V206L + K391A + P473R + G476K,
N54S + V56T + G109A + R116A + A174S + G182* + O183* + N195F + A204G + V206L + K391A + P473R + G476K,
N54S + V56T + G109A + R116A + T165G + A174S + G182* + O183* + N195F + V206L + K391A + T444Q + P473R + G476K,
N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + K391A + G476K,

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5 N54S + V56T + K72R + G109A + R116Q + A174S + G182* + O183* + G184T + N195F + V206L + K391A + G476K,
 A37V + N54S + V56T + G109A + R116A + T165G + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
 A37V + N54S + V56T + G109A + R116Q + T165G + A174S + G182* + O183* + N195F + V206L + M246F + K391A + G476K,
 V56A + G109A + A 174S + G182* + O183* + N195F + V206L + G476K,
 10 N54S + V56T + K72R + G109A + F113Q + R116H + A174S + G182* + O183* + G184T + N195F + V206L + K391 A + T444Q + G476K,
 N54S + V56T + G109S + F113Q + A174S + G182* + O183* + N195F + V206L + K391A + T444Q + G476K,
 N54S + V56T + G109A + F113Q + R116H + W167F + Q172G + L173V + A174S + G182* + O183* + N195F + V206L + K391A + P473R + G476K,
 15 N54S + V56T + G109A + F113Q + R116Q + Q172N + A174S + G182* + O183* + N195F + V206L + A265G + K391A + P473R + G476K,
 N54S + V56T + G109A + F113Q + W167F + Q172N + A174S + G182* + O183* + N195F + A204G + V206L + K391 A + P473R + G476K,
 20 N54S + V56T + G109A + F113Q + R116H + W167F + Q172R + A174S + G182* + O183* + N195F + A204G + V206L + K391A + P473R + G476K,
 N54S + V56T + G109A + R116A + A174S + G182* + O183* + N195F + V206L + K391A + T444Q + P473R + G476K,
 25 N54S + V56T + G109A + R116H + Q172R + A174S + G182* + O183* + N195F + A204G + V206L + K391A + P473R + G476K,
 A37V + N54S + V56T + G109A + R116Q + T165G + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
 A37H + N54S + V56T + G109A + R116H + T165G + A174S + G182* + O183* + N195F + V206L + M246F + K391A + G476K,
 N54S + V56T + G109A + A 174S + G182* + O183* + N195F + K391 A + G476K,
 30 N54S + V56T + G109A + G133S + A174S + G182* + O183* + N195F + K391A + G476H,
 N54S + V56T + G109A + R116A + Q172N + L173V + A 174S + G182* + O183* + N195F + V206L + K391A + G476K,
 A37V + N54S + V56T + G109A + R116H + T165G + A 174S + G182* + O183* + N195F + V206L + K391A + G476K,
 35 P45A + N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + K391A + C474V + G476K,
 N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + P364A + K391A + C474V + G476K,
 N54S + V56T + G109A + Q118R + T134E + A 174S + G182* + O183* + N195F + V206L + K391A + G476K,
 40 N54S + V56T + G109A + A 174S + G182* + O183* + G184T + N195F + V206L + K391A + G476K,
 G50A + T51A + N54S + V56T + G109A + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
 N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + V264F + K391A + G476K,
 45 N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + Y382L + K391A + G476K,
 G7K + N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + G476K,
 N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + G255A + Q256A + K391A + G476K,
 50 N54S + V56T + G109A + W167Y + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
 N54S + V56T + G109A + W167H + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
 N54S + V56T + G109A + T165V + A 174S + G182* + O183* + N195F + V206L + K391 A + G476K,
 55 N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + K391A + G476K,
 N54S + V56T + G109A + A 174N + G182* + O183* + N195F + V206L + K391A + G476K,
 N54S + V56T + G109A + A 174Q + G182* + O183* + N195F + V206L + K391A + G476K,
 60 N16H + V17L + N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + Y382F + K391A + G476K,
 Q32S + N54S + V56T + G109A + A 174S + G182* + O183* + N195F + V206L + Q385L + K391A + G476K,
 N54S + V56T + R87S + G109A + R171H + A174S + G182* + O183* + N195F + V206L + K391A + G476K,
 65 N54S + V56T + G109A + R116H + Q169E + Q172K + A174* + G182* + O183* + N195F + V206L + K391A + G476K,

N54S+V56T+K72R+G109A+F113Q+R 116Q+W167F+
 Q172G+A174S+G182*+D183*+G184T+N195F+
 V206L+K391A+P473R+G476K,
 N54S+V56T+K72R+G109A+F113Q+R 116Q+W167F+
 Q172G+A174S+G182*+D183*+N195F+V206L+
 K391A+G476K,
 N54S+V56T+K72R+G109A+R116H+T134E+W167F+
 Q172G+L173V+A174S+G182*+O183*+N195F+
 V206L+G255A+K391A+Q395P+T444Q+P473R+
 G476K,
 N54S+V56T+K72R+G109A+R116H+T134E+W167F+
 Q172G+L173V+A174S+G182*+D183*+N195F+
 V206L+G255A+K391A+G476K,
 N54S+V56T+K72R+G109A+G133Q+W167F+Q172R+
 A174S+G182*+O183*+N195F+V206L+K391A+
 G476K,
 N54S+V56T+G109A+F113Q+R116Q+Q172N+A174S+
 G182*+O183*+N195F+V206L+A265G+K391A+
 P473R+G476K,
 N54S+V56T+G109A+R116H+A174S+G182*+D183*+
 N195F+V206L+K391A+G476K,
 N54S+V56T+G109A+R116Q+W167F+A174S+G182*+
 O183*+G184T+N195F+V206L+K391 A+G476K,
 N54S+V56T+G109A+Q169E+Q172K+A174*+G182*+
 D183*+N195F+V206L+K391A+G476K,
 and
 N54S+V56T+G109A+A 174S+G182*+D183*+N195F+
 V206L+K391A+G476K,
 wherein the positions correspond to SEQ ID NO: 13.

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23. A composition comprising the variant of claim 1.

24. The composition of claim 23, which further comprises at least one further active component.

25. The composition of claim 24, wherein the further active component is an enzyme selected from the group consisting of a cellulase, a lipase, a mannanase, a pectate lyase and a protease. 5

26. The composition of claim 23, which is a liquid laundry or liquid dish wash composition, an Automatic Dish Wash (ADW) liquid detergent composition, or a powder laundry, 10 a soap bar, or powder dish wash composition, an ADW unit dose detergent composition and a Hand Dish Wash (HOW) detergent composition.

27. The composition of claim 23, which further comprises a surfactant, builder, bleach, polymer, or which is phos- 15 phate-free.

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